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COMPOSITIONS AND METHODS FOR PRODUCING PROTEIN

Abstract

Disclosed herein is a food product comprising an animal-free composition comprising a meat protein.

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Background/Summary

CROSS-REFERENCE [0001] This application is a continuation of International Application No. PCT/US2023/028715, filed Jul. 26, 2023, which claims the benefit of U.S. Provisional Application No. 63/392,613, filed Jul. 27, 2022, and U.S. Provisional Application No. 63/402,544, filed Aug. 31, 2022, each of which is entirely incorporated herein by reference.

SEQUENCE LISTING

[0002] The instant application contains a Sequence Listing which has been submitted electronically in XML format and is hereby incorporated by reference in its entirety. Said XML copy, created on Jan. 21, 2025, is named 63116-701_601_SL.xml and is 458,343 bytes in size.

BACKGROUND

[0003] Livestock is the world's largest user of land resources on Earth. Land dedicated to the production of meat and dairy, which includes animal feed production, encompasses nearly 80% of global agricultural land. One-fifth of the world's meat production is consumed by companion animals (e.g., pets). Recombinant protein production can be used to supplement or offset meat production.

SUMMARY

[0004] An aspect of the present disclosure provides a method for producing a food product, the method comprising: a) contacting a plurality of *Pichia* cells with a recombinant nucleic acid; b) expressing a protein encoded by the recombinant nucleic acid; and c) isolating the protein from the plurality of *Pichia* cells, wherein the protein comprises an amino acid sequence that is at least 80% identical to any one of SEQ ID NOs: 54-232.

[0005] In some embodiments, the food product is a cat food product. In some embodiments, the protein is an albumin protein. In some embodiments, the albumin protein is a rabbit albumin protein. In some embodiments, the albumin protein is a duck albumin protein. In some embodiments, the duck albumin protein is a mallard duck albumin protein. In some embodiments, the duck albumin protein is a Muscovy duck albumin protein. In some embodiments, the albumin protein is a cow albumin protein. In some embodiments, the albumin protein is a sheep albumin protein. In some embodiments, the albumin protein is a pig albumin protein. In some embodiments, the albumin protein is a bison albumin protein. In some embodiments, the albumin protein is a deer albumin protein. In some embodiments, the albumin protein is a chicken albumin protein. In some embodiments, the albumin protein is a turkey albumin protein. In some embodiments, the albumin protein is a salmon albumin protein. In some embodiments, the albumin protein is a cod albumin protein. In some embodiments, the albumin protein is a tuna albumin protein.

[0006] In some embodiments, the *Pichia* cells are *Pichia pastoris* cells. In some embodiments, the protein comprises an amino acid sequence that is at least 90% identical to any one of SEQ ID NOs: 54-232. In some embodiments, the protein comprises an amino acid sequence that is at least 95% identical to any one of SEQ ID NOs: 54-232. In some embodiments, the food product is substantially animal-free.

[0007] Another aspect of the present disclosure provides a food product comprising a recombinant protein, wherein the recombinant protein comprises an amino acid sequence that is at least 80% identical to any one of SEQ ID NOs: 54-232. In some embodiments, the food product is substantially animal-free. In some embodiments, the recombinant protein comprises an amino acid sequence that is at least 90% identical to any one of SEQ ID NOs: 54-232. In some embodiments, the recombinant protein comprises an amino acid sequence that is at least 95% identical to any one of SEQ ID NOs: 54-232. In some embodiments, the food product is a cat food product.

[0008] In some embodiments, the recombinant protein is a recombinant albumin protein. In some embodiments, the recombinant albumin protein is a recombinant rabbit albumin protein. In some embodiments, the albumin protein is a recombinant duck albumin protein. In some embodiments, the recombinant duck albumin protein is a recombinant mallard duck albumin protein. In some embodiments, the recombinant duck albumin protein is a recombinant Muscovy duck albumin

protein. In some embodiments, the recombinant albumin protein is a recombinant cow albumin protein. In some embodiments, the recombinant albumin protein is a recombinant sheep albumin protein. In some embodiments, the recombinant albumin protein is a recombinant pig albumin protein. In some embodiments, the recombinant albumin protein is a recombinant bison albumin protein. In some embodiments, the recombinant albumin protein is a recombinant deer albumin protein. In some embodiments, the recombinant albumin protein is a recombinant chicken albumin protein. In some embodiments, the recombinant albumin protein is a recombinant turkey albumin protein. In some embodiments, the recombinant albumin protein is a recombinant salmon albumin protein. In some embodiments, the recombinant albumin protein is a recombinant cod albumin protein. In some embodiments, the recombinant albumin protein is a recombinant tuna albumin protein.

[0009] Additional aspects and advantages of the present disclosure will become readily apparent to those skilled in this art from the following detailed description, wherein only illustrative embodiments of the present disclosure are shown and described. As will be realized, the present disclosure is capable of other and different embodiments, and its several details are capable of modifications in various obvious respects, all without departing from the disclosure. Accordingly, the drawings and description are to be regarded as illustrative in nature, and not as restrictive.

INCORPORATION BY REFERENCE

[0010] All publications, patents, and patent applications mentioned in this specification are herein incorporated by reference to the same extent as if each individual publication, patent, or patent application was specifically and individually indicated to be incorporated by reference. To the extent publications and patents or patent applications incorporated by reference contradict the disclosure contained in the specification, the specification is intended to supersede and/or take precedence over any such contradictory material.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] The novel features of the invention are set forth with particularity in the appended claims. A better understanding of the features and advantages of the present invention will be obtained by reference to the following detailed description that sets forth illustrative embodiments, in which the principles of the invention are utilized, and the accompanying drawings (also “figure” and “FIG.” herein), of which:

[0012] FIG. 1 shows a Coomassie stained sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE) gel of an example media from *Pichia pastoris* cells including a construct for secreted recombinant rabbit albumin or wildtype cells.

[0013] FIG. 2 shows a Coomassie stained SDS-PAGE gel of an example media from *Pichia pastoris* cells including a construct for secreted recombinant rabbit Tropomyosin alpha-1 chain or wildtype cells.

[0014] FIG. 3 shows a Coomassie stained SDS-PAGE gel of an example recombinant rabbit albumin.

DETAILED DESCRIPTION

[0015] While various embodiments of the invention have been shown and described herein, it will be obvious to those skilled in the art that such embodiments are provided by way of example only. Numerous variations, changes, and substitutions may occur to those skilled in the art without departing from the invention. It should be understood that various alternatives to the embodiments of the invention described herein may be employed.

[0016] Generation of animal-free protein is a way to reduce global meat production and offset the agricultural and environmental consequences of animal farming. Animal free protein can be made

using microorganisms designed to produce meat proteins using recombinant protein expression. Foods, such as pet food (e.g., cat food, dog food, or food for other companion animals) or foods for human consumption can be developed from animal-free meat proteins as an alternative or supplement to traditional meat-based foods.

[0017] Using recombinant protein expression methods, microorganisms (e.g., a yeast or bacteria) can be contacted with a DNA expression cassette. This DNA sequence can contain a gene or genes of interest, as well as regulatory sequences which will help in the production of the gene or genes of interest. Regulatory sequences (e.g., UTRs, promoters, signal sequences, etc.) can aid in, for example, transcription or translation efficiency, protein product yield, protein expression, or protein secretion.

[0018] Microorganisms can be induced to express and secrete meat proteins of specific animals (e.g., a rabbit protein or a duck protein). Microorganisms can alternatively or additionally be induced to express and secrete specific types of meat proteins (e.g., an albumin protein). Animal-free meat proteins can be purified and formulated into food products (e.g., a pet food or food for human consumption).

Definitions

[0019] Whenever the term “at least,” “greater than,” or “greater than or equal to” precedes the first numerical value in a series of two or more numerical values, the term “at least,” “greater than” or “greater than or equal to” applies to each of the numerical values in that series of numerical values. For example, greater than or equal to 1, 2, or 3 is equivalent to greater than or equal to 1, greater than or equal to 2, or greater than or equal to 3.

[0020] Whenever the term “no more than,” “less than,” or “less than or equal to” precedes the first numerical value in a series of two or more numerical values, the term “no more than,” “less than,” or “less than or equal to” applies to each of the numerical values in that series of numerical values. For example, less than or equal to 3, 2, or 1 is equivalent to less than or equal to 3, less than or equal to 2, or less than or equal to 1.

[0021] The terms “companion animal” and “pet” are used interchangeably herein. A pet is a domestic or tamed animal that is kept for companionship. A pet can be, but is not limited to, a dog, a cat, a pig, a rat, a mouse, a snake, a turtle, a cow, a sheep, a llama, a donkey, a goat, a horse, a chicken, a parrot, a ferret, a guinea pig, a hamster, a rabbit, a fish, a frog, a toad, a salamander, an iguana, a lizard, a duck, a goose, or an insect.

[0022] The term “meat” can be defined as skeletal muscle and its associated tissues, including blood, derived from mammalian, avian, reptilian, amphibian, and aquatic species harvested for consumption. Edible offal comprising organs and non-skeletal muscle tissue can also be considered meat.

[0023] The term “animal-free” can be defined as something that is not harvested directly from an animal, living or dead (e.g., meat harvested from a dead animal).

Food Products

[0024] Described herein are food products comprising animal-free proteins. An animal-free protein can be a protein or a fragment or truncation of a protein that is not obtained from an animal (e.g., by harvesting meat from an animal). An animal-free protein can be obtained through in vitro or ex vivo methods such as cell-free protein synthesis (CFPS). Alternatively, an animal-free protein can be obtained through in vivo methods such as genetic engineering of cells for recombinant protein expression. Genetically engineered cells used for recombinant protein expression can be, but are not limited to, microorganism cells (e.g., bacteria, yeast), plant cells, insect cells, or animal cells.

[0025] In recombinant protein expression, a cell is contacted with a DNA sequence. The DNA sequence will be incorporated by the cell, and the cell's internal protein expression machinery will transcribe and translate the DNA sequence into a peptide.

[0026] A cell used in recombinant protein expression can be a microorganism cell. Microorganisms are microscopic organisms such as bacterium, viruses, or fungi (e.g., yeast). A microorganism cell

used in recombinant protein expression can be a bacterial cell. A bacterial cell can be, but is not limited to, a cell from *Escherichia coli*, *Bacillus licheniformis*, *Bacillus subtilis*, or *Lactococcus lactis*. A microorganism cell used in recombinant protein expression can be a yeast cell. A yeast cell can be, but is not limited to, a cell from *Schizosaccharomyces pombe*, *Saccharomyces cerevisiae*, *Hansenula polymorpha* (also known as *Ogataea polymorpha*), *Pichia farinosa*, *Pichia anomala*, *Pichia heedii*, *Pichia guilliermondii*, *Pichia kluyveri*, *Pichia membranifaciens*, *Pichia norvegensis*, *Pichia ohmeri*, *Pichia pastoris* (also known as *Komagataella phaffii*), *Yarrowia lipolytica*, *Pichia methanolica*, or *Pichia subpelliculosa*. A microorganism cell used in recombinant protein expression can be a filamentous fungal cell. A filamentous fungal cell can be, but is not limited to, a cell from *Trichoderma reesei* (also known as *Hypocrea jecorina*), *Aspergillus niger*, *Aspergillus oryzae*, *Aspergillus tubingensis*, or *Ashbya gossypii* (also known as *Eremothecium gossypii*).

[0027] *P. pastoris* cells can be used in recombinant protein expression methods. *P. pastoris* cells have robust protein secretion; the ability to grow to high biomass density, allowing higher yields; have a high protein yield with up to 35 g/L yield for secreted proteins; have a low endogenous secretome, allowing for less downstream purification; grow in relatively inexpensive feedstock/media (e.g., glycerol or methanol); have high genetic tractability; have a proven ability to properly express and fold distantly-related eukaryotic (e.g., plant or animal) proteins; have a 90 minute doubling time; have an ability to grow in large-scale fermenters; and have an approximately one-week fermentation time from start to harvest.

[0028] A cell used in recombinant protein expression can be chosen because it displays a desired characteristic such as having a rapid doubling time, the ability to properly fold a target recombinant protein; the ability to make appropriate post-translational modifications to a target recombinant protein; the ability to efficiently secrete the target recombinant protein; the ability to be grown cheaply; the ability to be grown on easily accessible media; or genetic tractability. Alternatively, a cell used in recombinant protein expression may not secrete or may not efficiently secrete the target recombinant protein. In an example, cells that do not secrete or do not efficiently secrete the target recombinant protein may be lysed to release the protein. Cell lysis may include mechanical disruption, liquid homogenization, sonication, freeze/thaw cycling, manual grinding, or any combination thereof. In some examples, cells that secrete the target recombinant protein may be further lysed to increase protein yield.

[0029] During recombinant protein expression, a cell is contacted with a DNA sequence that can comprise an expression cassette. An expression cassette can be a component of vector DNA which can comprise at least one gene and/or at least one regulatory sequence.

[0030] An expression cassette can comprise at least one gene of interest. An expression cassette can comprise at least 1, at least 2, at least 3, at least 4, at least 5, at least 6, at least 7, at least 8, at least 9, at least 10 or more genes of interest. Alternatively, or in addition to, an expression cassette can comprise at least one additional gene other than a gene of interest. An expression cassette can comprise at least 1, at least 2, at least 3, at least 4, at least 5, at least 6, at least 7, at least 8, at least 9, at least 10 or more additional genes other than a gene of interest. An expression cassette can be a linearized plasmid. Alternatively, an expression cassette can be a linear PCR product.

[0031] A DNA sequence (e.g., a vector or an expression cassette) can be transfected into a cell using various chemical and physical methods such as electroporation, calcium phosphate exposure, spheroplasting, polyethylene glycol (PEG) exposure, alkali cation exposure, or liposome-based transfection.

[0032] A gene of interest can be an animal protein-coding gene. An animal protein-coding gene can encode a full length or a fragment or truncated version of a naturally occurring protein.

Alternatively, or in addition to, an animal protein-coding gene can encode a modified version (e.g., a codon-optimized version) of a naturally occurring protein. Alternatively, an animal protein-coding gene can encode a synthetic or novel protein.

[0033] An animal protein-coding gene can encode an antibody, a contractile protein, an enzyme, a

hormonal protein, a structural protein, a storage protein, or a transport protein. An animal protein-coding gene can be specific to a particular type of tissue such as, but not limited to, connective tissue, epithelial tissue, muscle tissue, or nervous tissue.

[0034] An animal protein-coding gene can encode proteins represented in meat or other parts of animals that are eaten. Examples of proteins found in meat can include, but are not limited to: actin, myosin, myosin regulatory protein, albumin, carbonic anhydrase, myoglobin, collagen, elastin, titin, nebulin, sarcalumenin, creatin kinase, phosphoglucomutase, phosphoglycerate mutase, triosephosphate isomerase, alpha-actinin, glycogen phosphorylase, fructose-biphosphate adolase, glyceraldehyde-3-phosphate dehydrogenase, L-lactate dehydrogenase, tropomyosin, myosin-binding proteins, myomesin, beta-enolase, pyruvate kinase, phosphoglycerate kinase, plectin, ATP-dependent 6-phosphofructokinase, malate dehydrogenase, hemoglobin, filamin, aconitate hydratase, parvalbumin, heat shock cognate 71, troponin, calsequestrin, obscurin, glycogen debranching enzyme, elongation factor 1-alpha 2, aspartate aminotransferase, adenylate kinase isoenzyme, glucose-6-phosphate isomerase, desmin, AMP deaminase, and globulin.

[0035] An animal protein-coding gene can encode albumin. Albumin is a family of globular proteins, the most common of which are serum albumins. Albumin proteins are water-soluble and are commonly found in blood plasma. Albumin proteins differ from other blood proteins in that they are not glycosylated. Albumin proteins include, but are not limited to: serum albumin, alpha-fetoprotein, vitamin D-binding protein, afamin, ovalbumin and extracellular matrix protein 1.

[0036] An animal protein-coding gene can be a gene from a specific animal such as, but not limited to: a canine, a feline, a bird (e.g., a songbird, a pigeon, a chicken, a duck, a goose, a swan, a crow, a dove, a parrot, a turkey), a bovine (e.g., a cow, a bison), a camel, a goat, a pig, a deer, a sheep, a horse, a rodent (e.g., a mouse, a rat, a squirrel, a hamster, a guinea pig, a chipmunk, a gopher, a vole), a fish (e.g., a tuna, a salmon, a trout, a sardine, a tilefish, a halibut, a cod, a flounder), a rabbit, a lizard, a snake, or a frog.

[0037] An animal protein can comprise a mouse protein (Table 1). A mouse protein can comprise a mouse muscle protein. A mouse protein can comprise a mouse blood protein. A mouse protein can comprise a mouse albumin protein. A mouse protein can comprise a sequence that has at least or about 70%, at least or about 75%, at least or about 80%, at least or about 90%, at least or about 91%, at least or about 92%, at least or about 93%, at least or about 94%, at least or about 95%, at least or about 96%, at least or about 97% at least or about 98% at least or about 99%, or about 100% sequence identity to any one of SEQ ID NOs: 1-53.

[0038] A food product may comprise a fragment or a truncation of a mouse protein. The food product may include a fragment or truncation of any one of SEQ ID Nos: 1-53. A sequence of the fragment or truncation may be identical or substantially identical to a portion of any one of SEQ ID Nos: 1-53. A sequence of the fragment or truncation may be at least about 70%, 80%, 90%, 95%, 98%, or more identical to a portion of a sequence of any one of SEQ ID Nos: 1-53. A length of the fragment or truncation may be at least about 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or more of the length of any one of SEQ ID Nos: 1-53. A length of the fragment or truncation may be less than or equal to about 90%, 80%, 70%, 60%, 50%, 40%, 30%, 20%, 15%, 10%, or less than a length of any one of SEQ ID Nos: 1-53.

TABLE-US-00001 TABLE 1 Mouse Proteins SEQ ID NO Description Sequence 1 Mouse TITIN

MTTQAPMFTQPLQSVVVLEGSTATFEAHVSGSPVPEVSWFRDGQVISTSTLPGVQISF
SDGRARLMIPAVTKANSGRYSLRATNGSGQATSTAELLVTAETAPPNFSQRLQSMTV
RQGSQVRLQVRVTGIPTPVVKFYRDGAEIQSSLDFQISQEGDLYSLLIAEAYPEDSGT
YSVNATNSVGRATSTAELVVQGEVVPAAKTKTIVSTAQISETRQTRIEKKIEAHFDA
RSIATVEMVIDGATGQLPHKTPPRIPPKPKSRSPTPPSIAAKAQLARQQSPSPIRHSPSP
VRHVRAPTPSPVRSVSPAGRISTSPIRSVKSPLLIRKTQTTTMMATGPEVPPPWKQEGYV
ASSTEAEEMRETTMTSSTQIRREERWEGRYGVQEQTISGAAAAAASASVSSSFTAGAI

TTGTKEVQKETAQVAAVATVVAADVAAVEQAQTAAVQTATTTTAVHIQPAQE
QARKEAEKTAVTKVVVAADKAKEQELKSRTREVMVTTQEQTTHISHEQIRKETEKAF
VPKVVISATKAKEQETRITGEITTKQEQRITQETIRQETEEIAASMVVVATAKSTKLE
AAVGVQEETAIQDQDQMHLETHEQMMKETRKTVVPKVIVATPKIKEQDLVSRREAITT
KRDQVQITQEKKRKEVETTALSTIAVATAKAKEQETVLRSTREAMATRQEHQVTHG
QVGVGKKAEEAVATVVAADVQARVREPPTHVEESHQQTTLLEYGYKEHISTTKVP
EQPRRPASEPHVVPQAVKPAVIQAPSETHIKTTDQMGMHISSQVKKTTDISTERLVHV
DKRPRTASPHFTVSKISVPKTEHGYEASIAGSAIATLQKELSATSSTQKITKSVKAPT
KPGETRVRAEPTPSPQFPFADMPPPDITYKSQAGIEVKKEVGVVSISGSTVREEHFEVLR
GREAKVTETARVPAPAEVPVTPPTLVSGLKNVTVIEGESVTLECHISGYPSPKVTWYR
EDYQIESSIDFQITFQGGIARLMIREAFAEDSGRFTCSAVNEAGTVSTSCYLAVQVSEE
FDKETTLTEKFATEEKRFVESRDVVMTDTSITEEQAGPGEPAAFFISKPVVQKLVEG
GSVVFECCQIGGNPKPHVYWKKSQVPLTTGYRYKVSYNKQTGECRLVISMTFADDAG
EYTIVIRNKHGETSASASLLEEADYEALVKTQQEMLYQTMSTFIQEPKVGEIAPGFA
YSEYEKEYEKEQALIRKKMAKDTVMVRTFVEDQEFHISSFEERLIKEIEYRIIKTTLEE
LLEEDGEKMAVDISESEAIESGFDIRIKNYRILEGMGVTFHCKMSGYPLPKIAWYKD
GKRIRHGERYQMDFLQDGRASLRIPVVLPEDEGIYTAFASNIKGNAICSGKLYVEPAA
PFSAPTYMPTPEAVSRIRSVSPRSLSRSPIRMSPAMSPARMSPARMSPARMSPARMSP
GRRLEETDESQLERLYKPVFVLKPASFCKLEGQTARFDLKVVGRRPMPETFWFHNGQ
QIVNDYTHKVVIKEDGTQSLIIVPASPSDSGEWTVVAQNRAGKSTISVTLTVEAVEHQ
IKPAFVEKLKNVNIKEGSRLMKVRATGNPNPDIVWLKNSDIIVPHKYPRIRIEGTRGE
AALKIDSIIQSQSAWYTATAINKAGRDTTRCKVNVEVEFAEPEPERKLIIPRGTYRAKE
IAAPELEPLHLRYGQEQWEEGDLYDKEKQKQKPFKKKLTSLRLKRFGPAHFECRLTPI
GDPTMVVEWLHDGKPLEAANRLRMINEFGYCSLDYGAAYSRD SGVITCRATNKYG
TDHTSATLIVKDEKSLVEESQLPDGKKGLQRIEELERMAHEGALTGVTTDQKEKQKP
DIVLFPEPVRVLEGETARFRRCRVGTGYPQPKVNWYLNGLIRKSKRFRVRYDGIHYLDI
VDCKSYDTGEVKVTAENPEGVTEHKVKLEIQQREDFRSVLRRAPEPKPEFHVHEPGK
LQFEVQKVDRPVDTSSETKEVVKLKRAERITHEKVSEEESEELRSKFKRRTEEGYYEAIT
AVELKSRRKKDESYEELLKKTKDELLHWTKELT EEEKKALAE EGKITIPTFKPERIELSP
SMEAPKIFERIQSQT VGGQSDAHFRVRVVGKPDPECEWYKNGVKIERSDRIYWYWP
EDNVCELVIRDVTAEDSASIMVKAINIAGETSSHAFLLVQAKQLITFTQELQDVVAKE
KDTMATFECETSEPFIKVKWYKDGIEVHAGDKYRMHSDRKVHFLSVLTIDTSDAED
YSCVLVEDENIKTTAKLIVEGAVVEFVKELQDIEVPESYSGELECIISPENIEGKWYHN
DVELKSNGKYSITSRRGRQNLTVKDVT KEDQGEYSFVVDGKKTTCKLKMKPRPIAIL
QGLSDQKVCEGDIVQLEVKVSLENVEGVWMKDGQEVQHSDRVHIVIDKQSHMLLIE
DMTKEDAGNYSFTIPALGLSTSGNVSVYSVDVITPLKDVNVIEGTKAVLECKVSPD
VTSVKWYLNDEQIKPDDRVQSIVKGTKQRLVINRTHASDEGPYKLMVGRVETSCNL
SVEKIKIIRGLRDLTCTETQNVVFEVELSHSGIDVWNFNKGKEIKPSSKYKIEAHGKIY
KLTVLNMMKDDGEYAFYAGENTTSGKLTVAGGAISTPLTDQTVAESQEAVFECEV
ANPESEGEWLKD GKHLALSNNFRGESDGHKRRLVIAAAKLDDAGEYTYKVATSKTS
AKLKVEAVKIKKTLRNLVTETQDAVFSVELTHPDVKGVQWIKNGVVLDSNDKYEI
SVKGTLYSLKIKNCAMADESVYGFKLGRLGASARLHVETVKIHKPKDVTALENATV
TFEVSVSHDTVVPVKWFHKNVEIKPSDKHRLVSEKRVHKLMLQSI SPSDAGEYTAMV
GQLECKAKL FVETLHITKTMKSIEVPETKAASFEC EVSHFNVPSMWLKNNGVEIEMSE
KFKIVVQGKHLHQLIIMNTSTEDSAEYTFVCGNDQVSATLTVTPIMITSMLKDINAEEK
DTITFEVTVNYEGISYKWLKNNGVEIKSTDRCQMRTKKLTHSLNIRNVHFGDAADYTF
VAGKATSTATLYVEARHIEFRKHIKDIKVLEKKRAMFECEVSEPDITVQWMKDGQEL
QIADRIKIQKEKYVHRL LIPSTRMSDAGKYTVVAGGNMSTANLFVEGRDVRIRSIKKE
VQVIEKQRAVVEFEVNEDDVDAH WYKDGIEINFQVQERHQYVVERRIHRMFISETRH
SDAGEYTFVAGRNRSSVTLYVNAPEPPQVLQELQPVTVQSGKPARFCAVISGRPQPKI

WYKEEQLLSTGFKCKFLHDGQEYTLLEAFPEDAAVYTCEAKNDYGVATTSASLS
VEVPEVVSPDQEMPVYPPAIVTPLQDVTVTSEGRPARFQCQVSGTDLKVSWSYCKDKKI
KPSRFFRMTQFEDTYQLEIAEAYPEDEGTAFVANNVAVGQVSSSTATLRLEAPESILHE
RIGQQIEMEMKEIASLLSAEEDFQTYSSDLRLPNANETLELLSEPPARSTQFDSRQEGA
APVFIREISDVEISVEDVAKLSVTVTGCPKPKIQWFFNGMLLTPSADYKFVFDGDTHS
LIILFTRFQDEGEYTCCLASNEYGKAVCSAHLRISPRGERSTEMESGEKKALEKPKGPCP
PYFFKELKPVHCGPGIPAVFEYSVHGEPAPTVLWFKEDMPLYTSVCYTIHSPDGSST
FIVNDPQRGDSGLYLCKAQN LWGESTCAAELLVLPEDTDVPDASCKEESTLGVPGDF
LETSARGPLVQGVDSRQEITAFAGTISKAAALIAEETLQLSYERSVDDSEVGTGVTIGA
QKLPPVVLSTPQGTGELPSIDGAVHTQPGRGPPPTLNLQAVQAQTTLPLKEATLQFEED
EGVFPGASSAAQVSPVTIKPLITLTAEPKGNYPQSSTAAPDHALLSSVAAETLQLGEK
KIPEVDKAQRALLLSQSLAEGCVESLEVPDVAVSNMRSEPQVPPFQHTCTEGKILMAS
ADTLKSTGQDVALRTEEGKSLSFPLALEEKQVLLKEEQSEVVAVPTSQTSKSEKEPEA
IKGVKEVREQELLSKETLFPSPMPEEQRLHLKTQVRRALQAAVAREQANLFSEWLNI
DKVEVTAVNFTQEPKRILCTYLITSVSSLTEELTVTIEDIDPQMANLETGLKDALCSIV
CEERNILMAEDPRIHEEDKIDVQGGRDHLSDAQKVETVIEAEADSKYLVSKEEVSW
KVESQLKDGDNEVPQAETLKLAEESGTQKTSTEMSQEEAEGTLADLCPAVLKHLV
DTISEEGDTVHLTSSISNAKEVHWYFKGNLVPDVGKFKCLKEQNAYTLVIEAVKTED
EGEYVCEASNDSGKAKTSAKLTVGERAAPVIKRRIEPLVALGHLAKFTCEIQGAPN
VRFQWFKAGREIYESDKCSIRSSNYVSSLEILRTQVVDGCEYTCKASNEYGSVSTAT
LTVTEAYPPTFLSRPKALTTFVGKAAKFLCTVSGTPVIEIHWQKDGAALSPSPDCRVTD
ADNKHSLELSNLTQDRGIYSCKASNKFGADICQAEITIDKPHFIKELEAVQSAINKK
IHLECQVDEDRKVTTITWSKDGQKLPAGKDYKIYFEDKIASLEIPLAKLKDSGTYTCTA
SNEAGSSSSSAVAVREPPSFVKKVDP SYLMLPGESARLHCKLKGSPVIQVTWFKNN
KELSESNTVRMSFVNSEAILDITDVKVDDSGTYSCEATNDVGS DSCSTE VVIKEPPSFI
KTLEPADIVRGANALLQCEIAGTGPFVNVWFKDKKQIRSSKKYRLFTQKTFVYLEISS
FNSADVGDYECVVANEVGKCGCVATHLLKEPPTFVKKVDDFTALAGQTVTLQAAV
RGSEPI SVMWMKGQEVIKEDGKIKMSFSNGVAVLTIPDVQISLGGKYTCLAENEAGS
QTSVGELIVKEPAKIIERAELIQVTAGDPATLEYTVSGTPELKPKWYKDGRPLVASKK
YRISFKNNIAQLKFYS AELHDSGQYTFEISNEVGSSSCETTFTVLDRDIAPLFTKPLRN
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NAGGPGEPAEVP GTVKVTEMLEY PDYELDERYQEGVFVRQGGVIRLTIPIK GKPFVP
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MSSDAEMAVFGEAAPYLKSEKERIEAQNKPFDAKSSVFVDAKESYVKATVQSRE
(Myosin-4)

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QVNKL RVKSREVHTK VISEE 3 Mouse

MADEEYEEVIEYYTEETVYEEVPGETITEVYETTTTTRTSDYGQSETATPALAQPPA
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(Nebulin)

ELRRIKKVQDQLSEVKYRMDGDVAKTICHVDEKAKDIEHAKKVSQQVSKVLYKQN
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KDGDIVNAQAIDEGWMYGTVQRTGRTGMLPANYVEAI 4 Mouse ACTS
MCDEDETTALVCDNGSGLVKAGFAGDDAPRAVFPISVGRPRHQGVMVGMGQKDSY
(Actin)

VGDEAQSKRGILTLKYPIEHGIITNWDDMEKIWHHTFYNELRVAPEEHPTLLTEAPLN
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GSILASLSTFQQMWITKQEYDEAGPSIVHRKCF 5 Mouse KCRM
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(Creatine
Kinase)
NLKGGDDLDPNYVLSSRVRTGRSICKGYTLPPHCSRGERRAVEKLSVEALNSLTGEFK
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TCPSNLGTGLRGGVHVLANLSKHPKFEEILTRLRLQKRGTTGGVDTAAVGAVFDISN
ADRLGSSEVEQVQLVVDGVKLMVEMEKKLEKGQSIDDMIPAQK 6 Mouse ACTN3
MMMVMQPEGLGAGEGPFSGGGGGEYMEQEEDWDRDLLLPAWEKQQRKTFTAW
(Alpha-Actinin-

CNSHLRKAGTQIENIEEDFRNGLKLMLLLEVISGERLPRPDKGKMRFHKIANVNKAL 3)
DFIASKGVKLVSIGAEEIVDGNLKM TLMGIWTHILRFAIQDISVEETSAKEGLLLWCQR
KTAPYRNVNVQNFHTSWKDG LALCALIHRHRPDLIDYAKLRKDDPIGNLNTAFEVA
EKYLDIPKMLDAEDIVNTPKPDEKAIMTYVSCFYHAFAGAEQAETAANRICKVLAVN
QENEKLMEEYEKLASELLEWIRRTVPWLENRVGEPSMSAMQRKLED FRDYRRLHKP
PRVQEKQCQLEINFNTLQTKLRLSHRPAFMPSEGKLVSDIANAWRGLEQVEKGYEDW
LLSEIRRLQRLQHLAEKFQKASLHEAWTRGKEEMLNQHDYESASLQEV RALLRRH
EAFESDLAAHQDRVEHIAALAQELNELDYHEAASVNSRCQAICDQWDNLGTLTQKR
RDALERMEKLL ETIDQLQLEFARRAAPFNWLDGAIEDLQDVWL VHSVEETQSLT
AHEQFKATLPEADRERGAILGIQGEIQKICQTYGLRPSGNPYITLSSQDINN KWDTV
RKLVP SRDQTLQEELARQQVNERLRRQFAAQANAIGPWIQGVVEEVGRLAAGLAGS

LEEQMAGLQQEQNIYKSNDRLEGDHLQESLVFDNKHHTVYSMEHIRVGWEEQ
LLTSIARTINEVENQVLTRDAKGLSSEQQLNEFRASFNHFDRKRNGMMEPDDFRACLIS
MGYDLGEVEFARIMTMVDPNAAGVVTFQAFIDFMTRETAETDTAEQVVASFKILAG
DKNYITPEELRRELPAEQAEYCIRRMAPYKSGSGAPSGALDYVAFSSALYGESDL 7

Mouse MYH1

MSSDAEMAVFGEAAPYLKSEKERIEAQNKPFDAKSSVFVVDAKESFVKATVQSRE
(Myosin-1)

GGKVTAKTEGGTTVTVKDDQVYPMNPPKYDKIEDMAMMTHLHEPAVLNKLKERY
AAWMIYTYSGLFCVTNPNYKWLPVYNAEVVAAYRGKKRQEAPPHIFSISDNAYQFM
LTDRENQSILITGESGAGKTVNTRKVIQYFATIAVTGEKKKEEATSGKMQGTLEDQIIS
ANPLLEAFGNAKTVRNDNSSRFGKFIRIHFGTTGKLASADIETYLLEKSRVTFQLKAE
RSYHIFYQIMSNKKPDLIEMLLITTNPYDYAFVSQGEITVPSIDDQEELMATDSIDILG
FTSDERSIYKLTGAVMHYGNMKFKQKQREEQAEPDGTEDVADKAAYLQNLNSADL
LKALCYPRVKVGNEYVTKGQTVQQVYNSVGALAKAVYEKMFLWMVTRINQQLDL
KQPRQYFIGVLDIAGFEIFDFNSLEQLCINFTNEKLQQFFNHHMFVLEQEEYKKEGIE
WEFIDFGMDLAACIELIEKPMGIFSILEEECMFPKATDTSFKNKLYEQHLGKSNNFQK
PKPAKGKVEAHFSLVHYAGTVDYNIAGWLDKNKDPLNETVVGLYQKSSMKTLAYL
FSGAAAAAEAESGGGGGKKGAKKKGSSFQTVSALFRENLNKLMTNLRSTHPPHFVRC
IIPNETKTPGAMEHELVLHQLRCNGVLEGIRICRKGFPSRILYADFKQRYKVLNASAIP
EGQFIDSKKASEKLLGSIDIDHTQYKFGHTKVFFKAGLLGLLEEMRDDKLAQLITRTQ
AMCRGYLARVEYQKMVERRESIFCIQYNVRAFMNVKHWPMKLYFKIKPLLKSAE
TEKEMANMKEEFEKAKENLAKAEAKRKELEEKMVALMQEKNDLQLQVQSEADSL
ADAEEERCDQLIKTKIQLEAKIKEVTERAEDEEEINAELTAKKRKLEDECESELKKDIDD
LETLAKVEKEKHATENKVKNLTEEMAGLDETIAKLTKEKKALQEAHQQTLDLQA
EEDKVNTLTAKIKLEQQVDDLEGSLEQEKKIRMDLERAKRKLEGDLKLAQESTMD
VENDKQQLDEKLKKKEFEMSNLQSKIEDEQALGMQLQKKIKELQARIEELEEIEAE
RASRAKAEKQRSCLSRELEEISERLEEAGGATSAQIEMNKKREAEFQKMRRDLEEAT
LQHEATAATLRKKHADSVAEELGEQIDNLQRVKQKLEKEKSEMMEIDDLASNMEVI
SKSKGNLEKMCRTLEDQVSELKTKEEEQQRLINELTAQRGRLQTESGEYSRQLDEKD
SLVSQLSRGKQAFQTQIEELKRQLEEEIKAKSALAHALQSSRHDCDLLREQYEEEQEA
KAELQRAMSKANSEVAQWRTKYETDAIQRTTEELEEAKKKLAQRLQDAEEHVEAVN
AKCASLEKTKQRLQNEVEDLMIDVERTNAACAALDKKQRNFDKILAEWKQKYEET
HAELEASQKESRSLSTELFKIKNAYEESLDHLETCLKRENKNLQQEISDLTEQIAEGGK
RIHELEKIKKQIEQEKSSELQAALAEAEASLEHEEGKILRIQLELNQVKSEIDRKIAEKDE
EIDQLKRNHIRVVESMQSTLDAEIRSRNDAILRKKKMEGDLNEMEIQLNHSNRMAAE
ALRNYRNTQGILKDTQLHLDDALRGQEDLKEQLAMVERRANLLQAEIEELRATLEQ
TERSRKIAEQELLDASERVQLLHTQNTSLINTKKKLETDISQIQGEMEDIVQEARNAAE
KAKKAITDAAMMAEELKKEQDTS AHLERMKKNLEQTVKDLQHRLEAEQLALKG
GKKQIQKLEARVRELEGEVENEQKRNV EAIKGLRKHERRVKELTYQTEEDRKNVLR
LQDLVDKLQSKVKAYKRQAEEAEEQSNVNLA KFRKIQHELEEAEERADIAESQV NK
LRVKSREVHTKIISEE 8 Mouse PYGM

MSRPLSDQDKRKQISVRGLAGVENVSELKKNFNRHLHFTLVKDRNVATPRDYFAL
(glycogen

AHTVRDHLVGRWIRTQQHYYEKDPKRIYYLSLEFYMGRTLQNTMVNLALENACDE
Phosphorylase)

ATYQLGLDMEELEEIEEDAGLGNGGLGRLAACFLDSMATLGLAAYGYGIRYEFGIFN
QKICGGWQMEEADDWLRYGNPWEKARPEFTLPVHFYGRVEHTSQGAKWVDTQVV
LAMPYDTPVPGYRNNVVNTMRLWSAKAPNDFNLKDFNVGGYIQAVLDRNLAENIS
RVLYPNDNFFEGKELRLKQEYFVVAATLQDIIRRFKSSKFGSRDPVRTNFDAPDKVA
IQLNDTHPSLAIPELMRILVDLERLDWDKAWDVTVKTCAYTNHTVLPEALERWPVH

LMETLLPRHLQIYEINQRFLNRVAAAFPGDVDRLRMSLVEEGAVKRINMAHLCIA
GSHAVNGVARIHSEILKKTIFKDFYELEPHKFQNKTNGITPRRWLVLCNPGLAEVIAE
RIGEDYISDL DQLRKLLSYVDDEAFIRDVAKVKQENKLFSAYLEREYKVHINPNSLF
DVQVKRIHEYKRQLLNCLHIITLYNRIKREPNRFMVPRTIMIGGKAAPGYHMAKMIK
LITAIGDVVNHDPAVGDRLRVIFLENYRVSLAEKVIPAADLSEQISTAGTEASGTGNM
KFMLNGALTIGTMDGANVEMAEAGEENFFIFGMRVEDVERLDQRGYNAQEYYDRI
PELRQIIEQLSSGFFSPKQPD LFKDIVNMLMHHDRFKVFADYEEYIKCQDKVSELYKN
PREWTRMVIRNIATSGKFSSDRTIAQYAREIWGVEPSRQRLPAPDEKI 9 Mouse ALDOA
MPHPYPALTPEQKKELSDIAHRIVAPGKGILAADESTGSI AKRLQSIGTENTENRRFY
(fructose-
RQLLLTADDRVNPCIGGVILFHETLYQKADDGRPFPPQVIKSKGGVVGIVDKGVVPL
bisphosphate
AGTNGETTTQGLDGLSERCAQYKKDGADFAKWRCVLKIGEHTPSALAIMENANVLA
aldolase A)
RYASICQQNGIVPIVEPEILPDGDHDLKRCQYVTEKVLAAVYKALSDHHVYLEGTLL
KPNMVTTPGHACTQKFSNEEIAMATVTALRRTVPPAVTGVTF LSGGQSEEEASINLNAI
NKCPLLKPWALTFSYGRALQASALKAWGGKKENL KAAQEEYIKRALANSLACQ GK
YTPSGQSGAAASESLFISNHAY 10 Mouse MYL 1
MAPKKDVKKPAAAPAPAPAPAPAKPKEEKIDLSAIKIEFSKEQQEDFKEAFLLFDR
(myosin light
TGECKITLSQVGDVLRALGTNPTNAEVKKVLGNPSNEEMNAKKIEFEQFLPMMQAIS
chain)
NNKDQGGYEDFVEGLRVFDKEGNGTVMGAELRHVLATLGEKMKEEEVEALLAGQE
DSNGCINYEAFVKHIMSV 11 Mouse G3P
MVKVGVNGFGRIGRLVTRAAICSGKVEIVAINDPFIDLNYMVYMFQYDSTHGKFNG
(glyceraldehyde-
TVKAENGKLVINGKPITIFQERDPTNIKWGEAGAEYVVESTGVFTTMEKAG AHLKGG 3-
phosphate
AKRVIISAPSADAPMFVMGVNHEKYDNSLKIVSNASCTTNCLAPLAKVIHDNFGIVE
dehydrogenase)
GLMTTVHAITATQKTVDGPSGKLWRDGRGAAQNIIPASTGAAKAVGKVIPELNGKL
TGMAFRVPTPNVSVVDLTCRLEKPAKYDDIKKVVKQASEGPLKGILGYTEDQVVSC
DFNSNSHSSTFDAGAGIALNDNFVKLISWYDNEYGYSNRVVDLMAYMASKE 12 Mouse
TPM1 MDAIKKKMQMLKLDKENALDRAEQAEADKKAAEDRSKQLEDELVSLQKKLKGTE
(tropomyosin
DELDKYSEALKDAQEKL ELAEKKATDAEADVASLNRRIQLVEEELDRAQERLATAL alpha-
1) QKLEEA EKA ADESERGMKVIESRAQKDEEKMEIQEIQLKEAKHIAEDADRKYEEVAR
KLVIIESDLERAEEAEELSEGKCAELEEEELKTVTNNLKSLEAQAEKYSQKEDKYEEEI
KVLSDKLKEAETRAEFAERSVTKLEKSIDDELELYAQKLKYKAISEELDHALNDMT SI 13
Mouse MYPC2
MPEAKPAAKKASKGKDAPKEAPAKQTPEEPPKEAPPEDQSPTAE EPTGIFLKKPDSVS
(myosin-binding
VETGKDAVILAKVNGKELPGKPTIKWFKGKWQELGSKSGARFIFKESH DSTSNVYTV
protein C)
ELHIGKVVLGDRGDYRLEIKAKDVCDSCSFNVDVEAPRQDSSGQSLESFKRSGDGKS
EDAGELDFSGLLKKREVVEEEKKKKKDDDDLGIPPEIWELLKGAKKSEYEKIAFQYG
ITDLRGMLKRLKKAKVEVKKSAAFTKKLDPAYQVDRGNKIKLVVEISDPDLPLKWF
KNGQEIKPSSKYVFENVGKKRILTINKCTLADDAAYEVAVQDEKCFTEL FVKEPPVLI
VTPLEDQQVFVGDRVEMSVEVSEEGAQVMWMKDG VEMTRED SYKARYRFKKDGK
RHILIYSDVAQEDGGRYQVITNGGQCEAELIVEEKQLEVLQDIADLTVKAAEQAVFK

CEVDEKVTGKWYKNGVEVRPSKRITISHVGRFHKLVIDDVRPEDEGDYTFVPDGYA
LSLSAKLNFLEIKVEYVVPKQEPPKIHLD CSGKTS DNSIVV VAGNKLRLDVAITGEPPT
ATWLRGDEVFTATEGRTHIEQRPCDSSFVIESAERSDEGRYTIKVTNPVGEDVASIFLR
VVDVDPDPPEAVRVTSVGEDWAILVWEPPKYDGGQPV TGYLMERKKKGSQRWMKIN
FEVFTD TTYESTKMIEGVLYEMRVFAVNAIGVSQPSMNTKPFMPIAPTSAPQH LTVED
VTDTTTTTLKWRPPDRIGAGGIDGYLVEYCLEGSEEWVPANKEPVERCGFTVKDLPTG
ARILFRVVG VNIAGRSE PATLLQPVTIREIVEQPKIRLPRHLRQTYIRK VGEALNLVIPF
QGKPRPQVVWTKGGAPLDTSRVNVRTSDFDTVFFVRQAARSDSGEYELSVQIENMK
DTATIRIRVVEKAGPAENV MVKEVWGTNALVEWQPPKDDGNSEITGYFVQKADKK
TMEW FNVYEHN RHTSCTVSDLIVGNEY YFRIFSENICGLSDSPGVSKNTARILKTGITL
KPLEYKEHDFRTAPKFLTPLMDRVV VAGYTAALNCAVRGHPKPKV VWMK NKM EIH
EDPKFLITNYQGILT LNIRRPSPFDAGTYSCRAF NELGEALAECKLDVRVPQ 14 Mouse
MYP C3

MPEPGKKPVSAFNKKPRSAEVTAGSAAVF EAETERSGVMVRWQRDGS DITANDKY
(myosin-binding
GLAAEGKRHTLTVRDASPDDQGSYAVIAGSSKV KFDLKVTEPAPPEKA ESEVAPGAP
protein C)

EEVPAPATELEESVSSPEGSVSVTQDGSAAEHQ GAPDDPIGLFLMRPQDGEVTVGGSI
VFSARVAGASLLKPPVVKWFKGKWVDLSSKVGQHLQLHDSYDRASKVYLFELHITD
AQTT SAGGYRCEVSTKDKFDSCNFNLTVHEAIGSGDL DLRS AFRRTSLAGAGRRTSD
SHEDAGTPDFSSLLKKRDSFRRDSKLEAPAEEDVWEILRQAPPSEYERIAFQHGVEAC
HRPLKRLKGMKQDEKKSTAFQKKLEPAYQVNKGHKIRLTVELADPDAEVKWLKNG
QEIQMSGSKYIFESVGAKRTL TISQCSLADDAAYQCVVGGEKCSTELFVKEPPVLITR
SLEDQLVMVGQRVEFECEVSEEGAQVKWLKDGVELTREETF KYRFKKDGRKHHLII
NEATLEDAGHYAVRTSGGQSLAELIVQEKKLEVYQSIADLAVGAKDQAVFKCEVSD
ENVRGVWLKNGKELVPDNRIKVSHIGRVHKL TIDDVTPADEADYSFVPEGFACNLSA
KLHFMEVKIDFVPRQEPPKIHLD CPGSTPDTIVVVTGNKLRLDVPISGDPAPT VVWQK
TVTQGGKASAGPHPDAPEDAGADEEWVFDKLLCETEGRRVRVETTKDRSVFTVEGA
EKEDEGVYTVTVKNPVGEDQVNLT VKVIDVPDAPAAPKISNVGEDSCTVQWEPPAY
DGGQPV LGYILERKKKKSYRWMRLNFDLLRELSHEARRMIEGVAYEMRVYAVNAV
GMSRPSPASQPFPMPIGPPGEPTHLAVEDVSDTTVSLKWRPPERVGAGGLDGYSVEYC
QEGCSEWTPALQGLTERRSMLVKDLPTGARLLFRVRAHN VAGPGGPIVTKEPVTVQ
EILQRPRQLPRHLRQTIQKKVGEPVNLLIPFQ GKPRPQVTWTKEGQPLAGEEVSIRNS
PTDTILFIRAARRTHSGTYQVTVRIENMEDKATLILQIVDKPSPPQDIRIVETWGFNVA
LEWKPPQDDGNTEIWGYTVQKADKKTMEWFTVLEHYRRTHCVVSELIIGNGYYFRV
FSHNMVGSSDKAAATKEPVFIPRPGITYEPPKYKALDFSEAPSFTQPLANRSIIAGYNA
ILCCAVRGSPKPKISWFKNGLDLGEDARFRMFCKQGVLTLEIRKPCPYDGGVYVCRA
TNLQGEAQCECRLEVRVPQ 15 Mouse Q14BI5

MSLVAVPFYQKRHKHFDQSYRNIQTRYLLDQYALKKQATTQSSSQRS LTERSSSKRA
(myomesin 2)

SSQSSAGAMTCRLCAKRVSASEEEEEVENENRYRSQAASYGEAKRQRFLSELAQLEEN
VHLARSQARDKLDKYFMEQTVEDNLAWERHSFEDRMSRAPEILVRLRSHTIWERM S
VRLCFTVQGFPTPVVQWYKDGS LICQAGEPGKYRIESRYGVHTLEINRANFEDTATY
SAVATNSHGQVSTNAAVV VRRYRGEEEPFHSVGLPIGLPLSSVIPYTHFDVQFLEKFG
VTFRREGETVTLKCTLLVTPDLKRVQPRAEWYRDDVLLKESKWTKMFFGEGQASLS
FSHLNKDDEGLYTLRIVSRGGVSDHSAFMFVRDADPLVTGAPGAPMDLQCHDANRD
YVIVTWKPPNTTTESPVIGYFIDKCEVGTNNWVQCNDAPVKICKYPVTGLFEGRSYV
FRVRAVN NAGISRPSRISDAVAALDPVDLRR LQAIHLEGEKEIVIYQDDLEGDVQIPGP
PTNVQASEVSRNYVVL SWDPPSPRGKDPLMYFIEKS AVGSGSWQRVNAQTAVRSPR
YAVFDLAEGKSYVFRVLSANKHGLSDPSEITPPIQAQDMIVVPSAPGRVLASRNTKTS

VVYQWDRPKEEDLLGYVDCVGNHKNHPIGYNRFVHGLTTGEQYIF
RVKAVNAVGTSENSQSESEVIKVQAALTVPSHPYGITLLNCDGHSMTLGWKVPKFSG
GSAIIGYYLDKREVVHKNWHEVNSSPVKERILTVEGLTEGSLYEFKIAATNLAGIGQP
SDPSEHFKCEAWTDPEPGPAYDLTFCEVRDTSVLVWKAPVYSGSSPVSGYFVDFKEE
DSGEWKTTSEAATPNRYLKVCDLQQGKTYVFRIRAVNASGPGKPSDTSEPVLVEARP
GTKEISAGVDEEGNIYLGFDCCQEMTDASQFTWCKAYEEIADEERFEVHTEGDHSKLY
FKNPDKEDLGTYSVSVSDTDGVSSSFVLDEEELERLMALSNEIKNPTIPLKSELAYEIF
DKGQVRFWLQAEHLSPDASFRFIINDREVSDSDTHRIKCDRSTGMIEMVMDRFTIENE
GTYYTVQIQDGKAKNQSSLVLIGDAFKAVLEEAEFQRKEFLRKQGPHFAEYLHWDVT
EECEVRLVCKVANTKRETVFKWLKDDVLYETETPPDLEKGVCELLIPKLSKKDHGE
YKATLKDDRGGQDVSVLEVGGKVYEDMILAMSRVCGASASPLKVLCTPEGIRLQCFM
KYFTEEMKVSWEYHKEAKISSSEHMRIGGSEEMAWLQICEPTEKDKGKYTFEIFDGD
SHQRSDDLSGQAFDEAYAEFQQLKAAAFAEKNRGKVIGGLPDVVTIMEGKTLNLTC
TVFGNPDPEVVWFKNDKDIELSEHFLVKMEQSKYVSLTIQGVTAEEDSGKYSINVKNK
YGGEKIDVTVSVYKHGEKIPDISPPQQAQPKLIPASTSSD 16 Mouse ALBU
MKWVTFLLLLFVSGSAFSRGVFRREAHKSEIAHRYNDLGEQHFGLVLIAFSQYLQK
(albumin)
CSYDEHAKLVQEVTDFAKTCVADESAANCDKSLHTLFGDKLCAIPNLRENYGELAD
CCTKQEPERNECFLQHKDDNPSLPPFERPEAEAMCTSFKENPTTFMGHYLVHEVARRH
PYFYAPELLYYAEQYNEILTQCCAEADKESCLTPKLDGVKEKALVSSVRQRMKCSS
MQKFGERAFKAWAVARLSQTFPNADFAEITKLATDLTKVNKECCHGDLLECADDR
AELAKYMCENQATISSKLQTCCKDLLKKAHCLSEVEHDTMPADLPAIAADFVEDQ
EVCKNYAEAKDVFLGTFLYEYSRRHPDYSVSLLLRLAKKYEATLEKCCAEANPPAC
YGTVLAEFQPLVEEPKNLVKTNCDLYEKLGEYGFQNAILVRYTQKAPQVSTPTLVEA
ARNLGRVGTKCCTLPEDQRLPCVEDYLSAILNRVCLLHEKTPVSEHVTKCCSGSLVE
RRPCFSALTVDETYVPKEFKAETFTFHSDICTLPEKEKQIKKQTALAELVKHKPKATA
EQLKTVMDDDFAQFLDTCKKAADKDTCTFSTEGPNLVTRCKDALA 17 Mouse ENOB
MAMQKIFAREILDSRGNPTVEVDLHTAKGRFRAAVPSGASTGIYEALRLDGDKARY (beta-
enolase)
LGKGVLKAVEHINKTLGPALLEKKLSVVDQEKVDKFMIELDGTENKSKFGANAILG
VSLAVCKAGAAEKGVPLYRHIADLAGNPDLVLPVPAFNVINGGSHAGNKLAMQEF
MILPVGASSFKEAMRIGAEVYHHLKGVKAKYKDATNVGDEGGFAPNILENNEAL
ELLKTAIQAAGYPDKVVIGMDVAASEFYRNGKYDLDFKSPDDPARHISGEKLGELYK
NFIQNYPPVVSIEDPFDQDDWATWTSFLSGVDIQIVGDDLTVTNPKRIAQAVEKKACN
CLLLKVNQIGSVTESIQACKLAQSNGWGMVSHRSGETEDTFIADLVVGLCTGQIKT
GAPCRSERLAKYNQLMRIEEALGDKAVFAGRKFRNPKAK 18 Mouse MLRS
MAPKKAKRRAGAEGSSNVFSMFDQTQIQEFKEAFTVIDQNRDGIIDKEDLRDTFAAM
(myosin
GRLNVKNEELDAMMKEASGPINFTVFLTMFGEKLGADPEDVITGAFKVLDPEGKG
regulatory light
TIKKQFLEELLTTQCDRFSQEEIKNMWAAFPDPVGGNVVDYKNICYVITHGDAKDQE
chain 2) 19 Mouse KPYR
MSVQENELPQQLWPWIFKSQKDLAKSALSGAPGGPAGYLRRASVAQLTQELGTAFF
(pyruvate kinase
QQQQLPAAMADTFLEHLCLLDIDSEPVAARSTSIATIGPASRSVDRLKEMIKAGMNI PKLR)
ARLNFSHGSHEYHAESIANIREAAESFATSPLSYRPVAIALDTKGPEIRTGVLQGGPES
EVEIVKGSQVLVTVDPKFRTRGDAKTWVVDYHNITQVVAVGGRIYIDDGLISLVVRK
IGPEGLVTEVEHGGFLGNRKGVNLPNAEVDLPGLSEQDLLDLRFGVEHYVDIIFASV
RKASDVAVRDALGPEGRGIKIISKIENHEGVKKFDEILEVSDGIMMARGDLGIEIPAE
KVFLAQKMMIGRCNLAGKPVVCATQMLESMITKARPTRAETSDVANAVLDGADCI

MLSGETAKGSFPVEAVKMQHAIAREAEAAVYHRQLFEELRRAAPLSRDPTEVTAIGA
VEASFKCCAAAIIVLTKTGRSAQLLSRYRPRAAVIAVTRSAQAARQVHLSRGVFPLLY
REPPEAVWADDVDRRVQFGIESGKLGRGFLRVGDLVIVVTGWRPGSGYTNIMRVLTIS 20
Mouse CAH3
MAKEWGYASHNGPDHWHELYPIAKGDNQSPIELHTKDIKHDPSSLQPWSASYDPGSA
(carbonic
KTILNNGKTCRVVFDDTYDRSMLRGGPLSGPYRLRQFHLHWGSSDDHGSEHTVDGV
anhydrase 3)
KYAAELHLVHWNPKYNTFGEALKQPDGIAVVGIFLKIGREKGEFQILLDALDKIKTK
GKEAPFTHFDPSCLFPACRDYWTYHGSFTTPPCEECIVWLLLKEPMTVSSDQMAKLR
SLFSSAENEPVPLVGNWRPPQPVKGRVVRAFSK 21 Mouse LDHA
MATLKDQLIVNLLKEEQAPQNKITVVGVGAVGMACAISILMKDLADELALVDVMED (L-
lactate
KLKGEMMDLQHGSFLKTPKIVSSKDYCVTANSKLVIITAGARQQEGESRLNLVQRN
dehydrogenase
VNIFKFIIPNIVKYSPHCKLLIVSNPVDILTYVAWKISGFPKNRVIGSGCNLDSARFRYL A)
MGERLGVHALSCHGWVLGEHGDSSVPVWSGVNVAGVSLKSLNPELGTADADKEQW
KEVHKQVVDSAYEVIKLGYSWAIGLSVADLAESIMKNLRRVHPISTMIKGLYGIN
EDVFLSVPCILGQNGISDVVKVTLTPEEEEARLKKSADTLWGIQKELQF 22 Mouse
MSSDAEMAVFGEAAPYLKSEKERIEAQNRPFDAKTSVFVAEPKESFVKGTIQSKDA
G3UW82
GKVTVKTEAGATLTVKEDQIFPMNPPKYDKIEDMAMMTHLHEPAVLYNLKERYAA
(myosin, heavy
WMIYTYSGLFCVTVNPKWLPVYNPEVVAAYRGKKRQEAPPHIFSISDNAYQFMLT
polypeptide 2)
DRENQSILITGESGAGKTVNTKRVIQYFATIAVTGDKKKKEEATSGKMQGTLEDQIISA
NPLLEAFGNAKTVRNDNSSRFGKFIRIHFGTTGKLASADIETYLLEKSRVTFQLKAER
SYHIFYQITSNKKPELIEMLLITTNPYDYPFVSQGEISVASIDDQEELMATDSADILGFT
NDEKVSIIYKLTGAVMHYGNMKFKQKQREEQAEPDGTETVADKAAYLQGLNSADLLK
ALCYPRVKVGNEYVTKGQTVEQVTNAV GALAKAMY EKMF LWMVTRINQQLDTKQ
PRQYFIGVLDIAGFEIFDFNSLEQLCINFTNEKLQQFFNHHMFVLEQEEYKKEGIEWTF
IDFGMDLAACIELIEKPMGIFSILEEECMFPKATDTSFKNKLYEQHLGKSANFQKPKV
VKGKAEAHFSLIHYAGTVDYNITGWLDKNKDPLNETVVGLYQKSSVKTLAYLFSGA
QTAEAEASSGGAACKKGAKKKGSSFQTVSALFRENLNKLMTNLRSTHPPHFVRCIIPNE
TKTPGAMEHELVLHQLRCNGVLEGIRICRKGFPSRILYADFKQRYKVLNASAIPEGQY
IDSKKASEKLLGSIDIDHTQYKFGHTKVFFKAGLLGLEEMRDDKLAQLITRTQAMC
RGFLARVEYQKMVERRESIFCIQYNIRAFMNVKHWPWMKLFFKIKPLLKSAETEKE
MATMKEEFQKTKDDLAKSEAKRKELEEKMVSLLEKNDLQLQVQAEAEGLADAE
RCDQLIKTKIQLEAKIKEVTERAEDEEEINAELTAKKRKLEDECSELKKDIDDLELTLA
KVEKEKHATENKVKNLTEEMAGLDETI AKLTKEKKALQEAHQQTLDLQAEEDKV
NTLTAKIKLEQQVDDLEGSLEQEKKL RMDLERAKRKLE GDLKLAQESIMDIENEKQ
QLDERLKKKEFEMS NLQSKIEDEQAIGIQLQKKIKELQARIEELEEEIEAERASRAKAE
KQRSDLSRELEEISERLEEAGGATSAQIEMNKKREAEFQKMRRDLEEATLQHEATAA
TLRKKHADSV AELGEQIDNLQRVKQKLEKEKSEM KMEIDDLASN VETVSKAKGNLE
KMCRTLEDQVSELKSKEEEQQRLINDLT TQRGRLQTESGEFSRQLDEKEALVSQLSR
GKQAF TQQIEELKRQLEEEVKAKNALAHALQSSRHDCDLLREQYEEEQESKAELQR
ALSKANSEVAQWRTKYETDAIQRT EELEEAKKKLAQRLQA AEEHVEAVNAK CASLE
KTKQRLQNEVEDLMLDVERTNAACAALDKKQRNFDKILAEWKQKYEETHAELEAS
QKEARSLGTELFKMKNAYEESLDQLETLKRENKNLQQEISDLTEQIAEGGKRIHELEK
IKKQVEQEKC ELQAAL EEA EASLEHEEGKILRIQLELNQVKSEIDRKIAEKDEEIDQLK

RNHIRVESMQSTLDAEIRSRNDAIRIKKKMEGDLNEMEIQLNHSNRMAAEALRNYR
NTQGILKDTQLHLDDALRGQEDLKEQLAMVERRANLLQAEIEELRATLEQTERSRI
AEQELLDASERVQLLHTQNTSLINTKKKLETDISQIQGEMEDIVQEARNAAEKAKKAI
TDAAMMAEELKKEQDTS AHLERMKKNMEQTVKDLQLRLDEAEQLALKGGKKQIQ
KLEARVRELEGEVESEQKRNAEAVKGLRKHERRVKELTYQTEEDRKNILRLQDLVD
KLQAKVKSYSYKRQAEAEAEQSNNTLSKFRKIQHELEEAERADIAESQVNKL RVKSRE
VHTKIISEE 23 Mouse Q6P6L5

MPEPTKKEENEVSAPAPPPEEPNKEKEAGTAPAKEWSL GESPAGGEEQDKQNANSQL
(myosin-binding
STLFVEKPQTGSVKVGANITFVAKVRAEDLLRKPTVKWFKGKWMDLASKAGKHLQ
protein C)

LKETFERQTRVYTFEMQIIKAKENYAGNYRCEV TYKDKFDSCSFDLEVHESTGTT PNI
DIRSAFKRSGEGQEDAGELDFSGLLKRREVKQQEEPEIDVWELLKNANPNEYEKIAF
QYGITDLRGMLKRLKRMRRVEKKSAFAKILDPAYQVDKGGKVR FVVELADPKLE
VKWFKNGQEIRPSTKYIFEHKGNERIMFINNCSLTDDSEYYVTAGDEK CSTE L FVREP
PIMVT KQLED MNAYCGERVEMEVEVSEDDANVKWFKNGEEIFPGPKSRYKIKVEGK
KHTLIIEGATKADSAEYSAMTTGGQSSAKLSVDLRPLKIT TPLTDQTVKLGKEVCLKC
EISENVPGKWTKNGLPVQEGERLKVVHKGRIHKLVIANALIEDEGEYVFTP DAYNVP
LSAKVHVIDPPKIILDGLDADNTVTVIAGSKLRLEIPVTGEPPP KAIWSRADKAIMEGS
GRIRAESYPDSSTLVIDVAERDDSGVYNINLKNEAGEAHASIKIKVVDIPDP P VAPNVT
EVGDDWCIMNWEPPAYDGGSPILGYFIERKKKQSSRWMLNFDLCKETTFEPKKMIE
GVAYEVRIFAVNAIGISKPSMPSKPFVPLAVTSPPTLLAVDSVTDSSVTMKWRPPDQI
GAAGLDGYVLEYCFEGTEDWIVANTDLIDKTKFTINGLPTDAKIFVRVKAINAAGAS
EPKYYSQPILVKEIIEPPKIRIPRHLKQTYIRRVGEAVNLVIPFQGKPRPELTWKKDGEE
IDKNQINIRNSETDTIIFIRKAERSHSGKYDLQVKVDKYVENASIDIQIVDRPGPPQTVT
IEDVWGENVALTWTPPKDDGNAAITGYTIQKADKKSMEWFTVIEHYHRTNATITEL
VIGNEY YFRVFAENMCGLSEDATMTKESAVIAKD GKIYKNPVYEDFN FTEAPMFTQP
LVNTYAIAGYNATLNC SVRGNP KPKITWMKNKVAIVDDPRYRMFSNQGVCTLEIRK
PSPYDGGTYCCKAVNDLGTVEIECKLEVKVVAQ 24 Mouse PGM1

MVKIVTVKTQAYPDQKPGTSGLRKRVKVFQSNANYAENFIQSIVSTVEPALRQEATL
(phosphogluco-
VVGGDGRFYMT EAIQLIVRIAAANGIGRLVIGQNGILSTPAVSCIIRKIKAI GGIILTASH
mutase-1)

NPGGPNGDFGIKFNISNGGPAPEAITDKIFQISK TIEEYAICPDLKVDLGV LGKQQFDLE
NKFKPFTVEIVDSVEAYATMLRNIFDFNALKELLSGPNRLKIRIDAMHG VVGPPYVKKI
LCEELGAPANS AVNCVPLEDFGGHHPDPNLTYAADLVETMKSGEHDFGA AFDGDG
DRNMILGKHGFFVNPSDSVAVIAANIFSIPYFQQTGVRGFARSMPTSGALDRVANAT
KIALYETPTGWKFFGNLMDASKLSLCGEESFGTGS DHIREDGLWAVLAWLSILATR
KQSVEDILKDHWQKFG RNFFTRYDYEEVEAEGANKMMKDLEALMLDRSFVGKQFS
ANDKVYTVEKADNFEYSDPVDGSISK NQGLRLIFADGSRIIFRLSGTGSAGATIRLYID
SYEKDVAKINQDPQVMLAPLISIALKVSQ LQERTGRTAPT VIT 25 Mouse TPM2

MDAIKKKMQMLKLDKENAIDRAEQAEADKKQAEDRCKQLEEEQQALQKKLKGTE
(tropomyosin
DEVEKYSES VKDAQEKLEQA EKKATDAEADVASLNRR IQLVEEELDRAQERLATAL beta
chain)

QKLEEA EKA ADESERGMKVIENRAMKDEEKMELQEMQLKEAKHIAEDSDRKYE EV
ARKLVILEGELERSEERA EVAESKCGDLEELKIVTN NLKSLEAQADKYSTKEDKYE
EEIKLLEEK LKEAETRAEFAERSVAKLEKTIDDE VYAQKMKYKAISEELDNALN DITSL
26 Mouse TPIS

MAPTRKFFVGGNWKMN GRKKCLGELICTLNAANVPAGTEVVCAPPTAYIDFARQKL

(triosephosphate
DPKIAVAAQNCYKVTNGAFTGEISPGMIKDLGATWVVLGHSERRHVFGESDELIGQK
isomerase)
VSHALAEGLGVIACIGEKLDEREAGITEKVVFQTKVIADNVKDWSKVVLAYEPVW
AIGTGKTATPQQAQEVHEKLRGWLKSNVNDGVAQSTRIIYGGSVTGATCKELASQP
DVDGFLVGGASLKPEFVDIINAKQ 27 Mouse PGAM2
MTTHRLVMVRHGESLWNQENRFCGWFDAELSEKGAEAAKRGATAIKDAKIEFDICY
(phosphogly-
TSVLKRAIRTLWTILDVTDQMWPVVRTWRLNERHYGGTGLNKAETA AKHGEEQ cerate
VKIWRRSFDTPPPPMDEKHNYT SISKDRRYAGLKPEELPTCESLKDTIARALPFWNE
mutase 2)
EIAPKIKAGQRVLIAAHGNSLRGIVKHLEGMSDQAIMELNLPTGIPIVYELDQNLKPT
KPMRFLGDEETVRKAMEAVAAQ GKAK 28 Mouse SRCA
MKALLLLCCFLASLLLSGQAEVEDASEEAPLRDRSHIDKTLMLNEDKPADDYSAVLQ
(sarcalumenin)
RLRKIYHTSIKPLEQSYKYNELRQHEITDGEITSKPMVFLG PWSVGKSTMINYLLGL
EDTRYQLYTGA EPTTSEFTVLMHGPKLKTIEGIVMAADSARSFSPLEKFGQNFLEKLI
GIEVPHKLLERVTFVDTPGHIENRKQQERGYPFNDVCQWFIDRADLIFVVFDP TKLDV
GLELEMLFRQLKGRESQIRIILNKADNLATQMLMRVYGALFWSLAPLINVTEPPRVY
VSSFWPQDYKPDTHRELFLKEEISLLEDLNQVIENRLENKIAFIRQHAIRVRIHALLVD
RYLQTYKDKMTFFSDGELVFKDIVEDPDKFYIFKTILAKTNVSKFDLPNREAYKDFFG
INPISNFKLLSQQCSYMG GCFLEKIERAITQELPSLLGSIGLGKNPGAPNCDKTGCGET
PKNRYKKH 29 Mouse PGK 1
MSLSNKLTLDKLDVKGKRVVMRVDFNVPMKNNQITNNQRIKAAVPSIKFCLDNGAK
(phosphoglycerate
SVVLMSHLGRPDGVPMPDKYSLEPVAAELKSLLGKDVFLKDCVGPEVENACANPA
kinase 1)
AGTVILLENLRFHVEEEGKGKDASGNKVKAEP AKIDAFRASLSKLGDVYVND AFGT
AHRAHSSMVG VNL P QKAGGFLMKKELNYFAKALESPERPFLAILGGAKVADKIQLIN
NMLDKVNEMIIGGGMAFTFLKVLNNMEIGTSLYDEEGAKIVKDLMSKA EKNGVKIT
LPVDFVTADKFDENAKTGQATVASGIPAGWMGLDCGTESSKKYAEAVGRAKQIVW
NGPVGVFEWEAFARGTKSLMDEVVKATSRGCITIIGGGDTATCCA KWNTEDKVSHV
STGGGASLELLEGKVLPGVDALSNV 30 Mouse PLEC
MVAGMLMPLDRLRAIYEVLFREGVMVAKKDRRPRSLHPHVPGV TNLQVMRAMAS
(plectin)
LKARGLVRETFAWCHFYWYLTNEGIDHLRQYLHLPPEIVPASLQRVRRPVAMVIPAR
RRSPHVQTMQGPLGCPPKRGPLPAEDPAREERQVYRRKEREEGAPETPVVSATT VGT
LARGPEPAPATDERDRVQKKTFTK WVNKHLIKHWRAEAQRHISDLYEDLRDGHNL
ISLLEVLSGD SLPREKGRMRFHKLQNVQIALDYL RHRQVKLVNIRNDDIADGNPKLT
LGLIWTIILHFQISDIQVSGQSEDMTAKEKLLLWSQRMVEGYQGLRCDNFTTSWRDG
RLFNAIIHRHKPMLIDMNKVYRQTNLENLDQAFSVAERDLGVTRLLDPEDVDVPQPD
EKSIITYVSSLYDAMPRVPGAQDGV RANELQLRWQEYRELVL LLLQWIRHHTAA FEE
RKFPSSFEEIEILWCQFLKFKETELPAKEADKNRSKVIYQSLEGAVQAGQLKIPPGYHP
LDVEKEWGKLHVAILEREKQLRSEFERLECLQRIVSKLQMEAGLCEEQLNQADALLQ
SDIRLLASGKVAQRAGEVERDLDKADGMIRLLFNDVQTLKDGRHPQGEQMYRRVY
RLHERLVAIRTEYNLRLKAGVGAPVTQVTLQSTQRRPELEDSTLRYLQDLLAWVEEN
QRRIDSAEWGVDLPSVEAQLGSHRGMHQ SIEEFRAKIERARNDESQ LSPATRGAYRD
CLGRDLQYAKLLNSSKARLSLES LHGFVAAATKELMWLNEKEEEEVGFWDSDRN
TNMAAKKESYSALMRELEMKEKKIKEIQNTGD RLLREDHPARPTVESFQAALQTQW
SWMLQLCCCIEAHLKENTAYFQFFSDVREAEELQKLQETLRRKYSCDRTITVTRLE

LLQDAQDEKEQLNEYKYGHLGSLAKRAKAIIVQLKPRNPAHPVRGHVPLIAVCDYKQ
 VEVTVHKGDQCQLVGPAQPSHWKVLSGSSSEAAVPSVCFLVPPPNQEAQEAVARLE
 AQHQALVTLWHQLHVD MKSLLAWQSLSRDIQLIRSWSLVTFRTLKPEEQRQALRNL
 ELHYQAF LRDSQDAGGFGPEDRLVAEREY GSCSRHYQQLLSLEQGEQEESRCQRCI
 SELKDIRLQLEACETRTRTVHRLRLPLDKDPARECAQRIAEQQKAQAEVEGLGKGVARL
 SAAEAEKVLALPEPSPAAPT LRSELELT LGKLEQVRSLSAIYLEKLKTISLVIRSTQGAEE
 VLKTHEEQLKEAQAVPATLQELEATKASLKKLRAQAEAAQQPVFNTLRDEL RGAQEV
 GERLQQRHGERDVEVERWRERV TQLLERWQAVLAQTDVVRQRELEQLGRQLRYRE
 SADPLSAWLQDAKRRQEIQAVPIANCQAAREQLRQE KALLEEIERHGEKVEECQKF
 AKQYINAIKDYELQLITYKAQLEPVASPAKKPKVQSGSESVIQEYVDLRTRYSEL TTL
 TSQYIKFISETLRRMEEEEERLAEQQRAEERERLAEVEAALEKQRQLAEAHAQAKAQA
 ELEAQELQRRMQEEVARREEAAVDAQQQKRSIQEELQHLRQSSEAEIQAKAQQVEA
 AERSRMRIEEEEIRVVRLQLETTERQRGGAEGELQALRARAEAEAAQKRQAQEEAERL
 RRQVQDESQRKRQAEAEALRVKAEAEAAAREKQRALQALDELRLQAEAEAERRLRQ
 AEAERARQVQVALETAQRSAEVELQSKRASFAEKTAQLERTLQEEHVTVAQLREEA
 ERRAQQQAEAEERAREEAERELERWQLKANEALRLRLQAE EVAQQKSLAQADA EKQ
 KEEAEREARRRGKAE EQAVRQRELAEQELEKQRQLAEGTAQQRLAAEQELIRLRAE
 TEQGEQQRQLLEELARLQHEATAATQKRQELEAE LAKVRAEMEVLLASKARAE EEE
 SRSTSEKSKQRLEAEAGRFRELAEEAARLRALAE EAKRQRQLAEEDAARQRAEAEER
 VLTEKLA AIS EATRLKTEAEIALKEKEAENERLRRLAEDEAFQRRRLEEQAALHKADI
 EERLAQLRKASESELERQKGLVEDTLRQRRQVEEEIMALKVSFEKAAAGKAELELEL
 GRIRSNAEDTMRSKEQAELEAARQRQLAEEEEQRRREAEERVQRSLAEEEEAARQR
 KVALEEVERL KAKVEEARRLRERAEQESARQLQLAQEAAQKRLQAE EKAHAFVVQ
 QREEELQQT LQQEQNMLDRLRSEAEAAARRAAEEAEAREQAEREAAQSRKQVEEAE
 RLKQSAEEQAQAQAQAQAAAEKLRKEAEQEAARRAQAEQAALKQKQAADAEMEK
 HKKFAEQTLRQKAQVEQELTTLRLQLEETDHQKSILDEELQRLKA EVTEAARQRSQV
 EEELFSVRVQMEELGKLKARIEAENRALILRDKDNTQRFLEEEAEKMKQVAEEAARL
 SVAAQEAARLRQLAEEDLAQQRALAEKMLKEKMQAVQEATRLKAEAE LLQQQKEL
 AQEQARRLQEDKEQMAQQLVEETQGFQRTLEAERQRQLEMSAEAEERLKL RMAEMS
 RAQARAEDDAQRFRKQAE EIGEKLHRT ELATQEKVTLVQTLEIQRQQSDHDAERLRE
 AIAELEREKEKLKQEAKLLQLKSEEMQTVQQEQILQETQALQKSFLSEKDSLLQRERF
 IEQEKAKLEQLFQDEVAKAKQLREEQQRQQQQMEQE KQELMASMEEARRRQREAE
 EGVRRKQEELQHLEQQRQQQEKLLAEENQRLRERLQRLEEEHRAALAHSEIATTQA
 ASTKALPNGRDAPDGPSVEAEPEYTFEGLRQKVPAQQLQEAGILS QEELQRLAQGHT
 TVAELTQREDVYRYLKGRSSIAGLLLKPTNEKLSVYTALQRQLLSPGTALILLEAQAA
 SGFLLDPVNRRLTVNEAVKEGVVGPELHHKLLSAERAVTGYKDPYTGEQISL FQA
 MKKDLIVRDHGVRLLEAQIATGGIIDPVHSHRVPVDVAYKRGYFDEEMNRILSDPSD
 DTKGFFDPNTHENLTYLQLLERCVEDPETGLRLLPLTDKAAKG GELVYTDTEARDVF
 EKATVSAPFGKFQGRVTIWEIINSEYFTA EQRRDLLQQFRTGHITVEKIIKIVITVVEE
 HERKGQLCFEGLRALVPAAELLDSGVISHELYQQLQRGERSVREVAEADSVRQALR
 GTNVIAGVWLEEAGQKLSIYEALKKDLLQPEVAVALLEAQAGTGH IIDPAT SARLTV
 DEAVRAGLVGPELHEKLLSAEKAVTGYRDPYSGQSVSLFQALKKGLIPREQGLRLLD
 AQLSTGGIVDP SKSHRVPLDVAYARGYLDKETNRALTSPRDDARVYHDPSTQEPVTY
 SQLQQRCRSDQLTGLSLLPLSEKAVRARQEEVYSELQARETLEQAKVEVPVGSFKGR
 AMTVWELISSEYFTEEQRQELLRQFRTGKVTVEKVIKIVITIVEEVETR RQERLSFSGL
 RAPVPASELLDAKILSRAQFDQLKD GKT SVKELSEVGSVRTLLQGSGCLAGIYLED SK
 EKVTIYEAMRRGLLRPSTATLLLEAQAA TGFLVDPVRNQRLYVHEAVKAGVVGPEL
 HEKLLSAEKAVTGYKDPYSGNTISL FQAMKKGLVLRDH AIRLLEAQVATGGIIDPVH
 SHRLPVDVAYQRGYFDEEMNRVLADPSDDTKGFFDPNTHENLTYLQLLERCVEDPE
 TGLRLLPLKGA EKTEVVETTQVYTEEETRRAFEETQIDIPGGGSHGGSSMSLWEVMQ

SNMIPEDQARLMADDFQAGRVTKERMIHIEIEKTEIIRQQNLASYDYVRRRLTAEDL
YEARIISLETYNLFREGTKNLREVLEMESAWRYLYGTGAVAGVYLPGSRQTLTIYQA
LKKGLLSAEVARLLLEAQAATGFLDPVKGERLTVDEAVRKGLVGPHELHDLRLSAE
RAVTGYRDPYTEQTISLFQAMKKELIPAEALRLLEDAQLATGGIVDPRLGFHLPLEVA
YQRGYLNKDTHTDQLSEPSEVRSYVDPSTDERLSYTQLLKRCRRDDPSGQMLLLLSDA
RKLTFRGLRKQITVEELVRSQVMDEATALQLQEGLTSIEEVTKNLQKFLEGTSCIAGV
FVDATKERLSVYQAMKKGIIRPGTAFELLEAQAATGYVIDPIKGLKLTVEEAVRMGI
VGPEFKDKLLSAERA VTGYKDPYSGKLISLFQAMKKGLILKDHGIRLLEAQIATGGII
DPEESHRLPVEVAYKRGLFDEEMNEILTDPSDDTKGFFDPNTEENLTYLQLMERCITD
PQTGLCLLPLKEKKRERKTSSKSSVRKRRRVIVDPETGKEMSVYEAYRKGLIDHQT
LELSEQECEWEEITISSSDGVVKSMIIDRRSGRQYDIDDAITKNLIDRSALDQYRAGTL
SITEFADMLSGNAGGFRSRSSSVGSSSSYPISAGPRTQLASWSDPTEETGPVAGILDT
ETLEKVSITEAMHRNLVDNITGQRLLEAQA ACTGGIIPSTGERFPVTEAVNKGLVDKI
MVDRLINLAQKAFCGFEDPRTKTKMSAAQALKKGWLYYEAGQRFLEVQYLTGGLIEP
DTPGRVSLDEALQRGTVDARTAKLRDVSAYS SKYLTCPKTKLKISYKDALDRSMVE
EGTGLRLLEAAAQSSKGYYSYPSVSGSGSTAGSRTGSRTGSRAGSRRGSF DATGSGFS
MTFSSSSYSSSGYGRRYASGPSASLGGPESAVA 31 Mouse ACTN2

MNQIEPGVQYNYVYDEDEYMIQEEEWDRDLLLLDPAWEKQQRKTFTAWCNSHLRKA
(alpha-actinin-2)
GTQIENIEEDFRNGLKLMMLLEVISGERL PKPDRGKMRFHKIANVNKALDYIASKGVK
LVSIGAEIIVDGNVKMTLGMWTHILRFAIQDISVEETSAKEGLLLWCQRKTAPYRNV
NIQNFHTSWKDGLGLCALIHRHRPDLIDYSKLNKDDPIGNINLAMEIAEKHLDIPKML
DAEDIVNTPKPDERAIMTYVSCFYHAFAGAEQAETAANRICKVLAVNQENERLMEE
YERLASELLEWIRRTIPWLENRTPEKTMQAMQKKLEDFRDYRRKHKPPKVQEKCQL
EINFNTLQTKLRISNRPAFMPSEGKMOVSDIAGAWQRLEQA EKG YEEWLLNEIRRLER
LEHLAEKFRQKASTHETWAYGKEQILLQKDYESASL TEVRALLRKHEAFESDLAAH
QDRVEQIAAIAQELNELDYHDAVNVDNRCQKICDQWDRLGTLTQKRREALERTEKL
LETIDQLHLEFAKRAAPFNNWMEGAMEDLQDMFIVHSIEEIQSLITAHEQFKATLPEA
DGERQSILAIQNEVEKVIQSYSIRISSSNPYSTVTMDEL RNKWDKVKQLVPVRDQSLQ
EELARQHANERLRRQFAAQANAIGPWIQNKMEEIARSSIQITGALEDQMNQLKQYEH
NIINYKNNIDKLEGDHQLIQEALVFDNKHTNYTMEHIRVGWELLTTIARTINEVETQI
LTRDAKGITQEQMNEFRASFNFDRRKNGLMDHEDFRACLISMGYDLGEAEFARIM
TLVDPNGQGT VTFQSFIDFMTRETADTD TAEQVIASFRILASDKPYILAEELRRELPPD
QAQYCIKRMPPYSGPGSVPGALDYTA FSSALYGESDL 32 Mouse PFKAM

MTHEEHHA AKTLGIGKAI AVLTS GGDAQGMNA AVRAVVRVGIFTGARVFFVHEGY (ATP-
dependent

QGLVDGGEHIREATWESVSMMLQLGGTVIGSARCKDFREREGRLRAAHNLVKRGIT 6-
NLCVIGGDGSLTGADTFRSEWSDLLNDLQKD GKITAE EATKSSYLNIVGLVGSIDNDF
phosphofru-

CGTDMTIGTDSALHRIVEIVDAITTTAQSHQRTFVLEVMGRHCGYLALVTSLS CGAD
nctokias e)

WVFIPECPPDDDWEHL CRRLSETRTRGSRLNIIIVAEGAIDKNGKPITSEDIKNLVVK
RLGYDTRVTVLGHVQRGGTPSAFDRILGSRMGVEAVMALLEGTPDTPACVVSLSGN
QAVRLPLMECVQVTKDVT KAMDEKRFDEAIKLGRSFMNNWEVYKLLAHVRPPVS
KGGLHTVAVMNVGAPAAAGMNA AVRSTVRIGLIQGNRVLVVDHGFEG LAKGQIEEA
GWSYVGGWTGQGGSKLGTKRTL PKNLEQISANITKFNIQGLVIIGGFEAYTG GLEL
MEGRKQFDEL CIPFVVIPATVSNNVPGSDFSIGADTALNTICTTCDRIKQSAAGTKRR
VFIIETMGGYCGYLATMAGLAAGADAAYIFEEPFTIRDLQVNVEHLVQKMKTTVKR
GLVLRNEKCNENYTTDFIFNLYSEEGKGIFDSRKNVLGHMQGGGSPTPFDRNFATKM
GAKAMNWMSGKIKESYRN GRIFANTPDSGCVLGMRKRALVFQPVTELKDQTD FEH

RIKPEQWWLKLRLKILAKYEIDLDTSDHAHLEHISRKSGEAAV 33 Mouse MDHM
MLSALARPAGAALRRSFSTSAQNNAKVAVLGASGGIGQPLSLLLKNSPLVSRLTLYDI
(malate
AHTPGVAADLSHIETRANVKGYLGPQLPDCLKGCDVVVIPAGVPRKPGMTRDDLF
dehydrogenase)
NTNATIVATLTAACAQHCPEAMVCIIANPVNSTIPITAEVFKKHGVYNPNKIFGVTTL
DIVRANTFVAELKGLDPARVNVVPVIGGHAGKTIPLISQCTPKVDFPQDQLATLTGRIQ
EAGTEVVKAKAGAGSATLSMAYAGARFVFSLVDAMNGKEGVVECSFVQSKETECT
YFSTPLLLGKKGLEKNLGIGKITPFEEKMIAEAIPELKASIKKGEDFVKNMK 34 Mouse
MYG MGLSDGEWQLVLNVWGKVEADLAGHGQEVLIGLFKTHPETLDKFDKFKNLKSEED
(myoglobin)
MKGSEDLKKHGCTVLTALGTILKKKGQHAAEIQPLAQSHATKHKIPVKYLEFISEIIIE
VLKKRHSGDFGADAQGAMSKALELFRNDIAAKYKELGFQG 35 Mouse HBBB1
MVHLTDAEKA AVSCLWGK VNSDEVGGEALGRLLVVYPWTQRYFDSFGDLSSASAI
(hemoglobin
MGNAKVKAHGGKVITAFNDGLNHLDSLKGTFASLSELHCDKLHVDPENFRLLGNMI
subunit beta-1) VIVLGHHLGKDFTPAAQA AFQKV VAGVATALAHKYH 36 Mouse FLNC
MMNNSNYSDASGLGLVDEADEMPSTEKDLAEDAPWKKIQQNTFTRWCNEHLKCVG
(filamin-C)
KRLTDLQRDLSDGLRLIALLEVL SQKRM YRKFHPRPNFRQM KLENVSVALEFLEREH
IKLVSIDSKAIVDGNLKLILGLIWTLILHYSISM PMWEDEDEDARKQTPKQRL LGWI
QNKVPQLPITNFNRDWQDGKALGALVDNCAPGLCPDWEAWDPNQPVQNAREAMQ
QADDWLGV PQVIAPEEIVDPNVDEHSVM TYLSQFPKAKLKPGAPVRSKQLNPKKAIA
YGP GIEPQGNTVLQPAHFTVQTV DAGVGEVLVYIEDPEGHTEEAKVVPNNDKDRTY
AVSYVPKVAGLHKVTVLFAGQNIERSPF EVNVGMALGDANKVSARGPGLEPVGNV
ANKPTYFDIYTAGAGTG DVAVVIVDPQGRRTVEVALEDKGDNTFRCTYRPMMEGP
HTVHVAFAGAPITRSPFPVHVAEACNPNACRASGRGLQPKGVRVKEVADFKVFTKG
AGSGELKVTVKGPKGTEEPVKVREAGDGVFECEYYPVVPGKYVVTITWGGYAIPRS
PFEVQVSPEAGA QKVRAWGPGL ETGQVGKSAD FVVEAIGTEVGT LGFSIEGPSQAKI
ECDDKGDGSCDVRYWPTEPGEYAVHVICDDEDIRDS PFIAHIQPAPPD CFPDKVKAF
GPGLEPTGCIVDRPAEFTIDARAAGKGDLKLYAQDADGCPIDIKVIPNGDGTFRCSYV
PTKPIKHTIIVSWGGVNV PKSPFRVNVGEGSHPERVKVYGP GVEKTGLKAN EPTYFT
VDCSEAGQGDV SIGIKCAPGVVGPVEADIDFDI IKNDNDTFTVKYTPPGAGHYTIMVL
FANQEIPASPFHIKVDP SHDASKVKAEGPGLSRTGVEVGKPTHFTVLT KGAGKAKLD
VHFAGAAKGEAVRDFEIIDNHDYSYTVKYTAVQQGNMAVTVTYGGDPVPKSPFVV
NVAPPLDL SKVKVQGLNSKVAVGQEQA FSVNTRGAGGQGQLDVRMTSPSRRPIPCCK
LEPGGGAEAAQAVRYMPPEEGPYKVDITYDGHPVPGSPFAVEGVLPPDPSKVCAYGPG
LKGGLVGT PAPFSIDTKGAGTGGLGLTVEGPCEAKIECQDNGDGSCAVSYLPTEPGE
YTINILFAEAHIPGSPFKATIQPVFDPSKVRASGPGLERGKAGEAATFTVDCSEAGEAE
LTIEILSDAGVKA EVLIQNNADGTYHITYSPA FPGTYTITIKYGGHPIPKFPTRVHVQPA
VDTSGIKVSGPGVEPHGVLREVTTEFTVDARSLTATGGNHVTARVLNPSGAKTDTYV
TDNGDGTYRVQYTAYEEGVHLVEVLYDEVAVPKSPFRVGVTEGCDPTRVRAFGPGL
EGGLVNKANRFTVETR GAGTGGLGLAIEGPSEAKMSCKDNKDGSC TVEYIPFTPGDY
DVNITFGGQPIPGSPFRVPVKDV VDPGKVKCSGPGLGTGVRARVPQTFTVDCSQAGR
APLQVAVLGP TGVAEPVEVRDNGDGTHTVHYTPATDGPYTVAVKYADQEVPRSPFK
IKVLP SHDASKVRASGPGLNASGIPASLPVEFTIDARDAGQGLLTVQILDPEGKPKKA
NIRDNGDGTYTVSYLPDMSGRYTITIKYGGDEIPYSPFRIHALPTGDASKCLVTVSIGG
HGLGACLGPRIQIGEETVITVDAKAAGKGKVTCTVSTPDGAELDVDVVENHDGTFDI
YYTAPEPGKYVITIRFGGEHIPNSPFHVLACDPLPHVEEPAEMLQMRQPYAPLRPGTC
PTHWATEEPVVPVEPLESMLRPFNLVIPFTVQKGELTGEVRMPSGKTARPNITDNKD

GTITVRYAPTEKGLHQMGIKYDGNHIPGSPLOQFYVDAINSRHVSAYGPGLSHGMV NK
PATFTIVTKDAGEGGLSLAVEGPSKAEITCKDNKDGTCTVSYLPTAPGDYSIIVRFDD
KHIPGSPFTAKITGDDSMRTSQLNVGTSTDVSLKITEGDLSQLTASIRAPSGNEEPCLL
KRLPNRHIGISFTPKEVGEHVSVVRKSGKHVTNSPFKILVGPSEIGDASKVRVWGKGL
SEGQTFQVAEFIVDTRNAGYGGLGLSIEGPSKVDINCEMEDGTCKVTYCPTEPGTYI
INIKFADKHVPGSPFTVKVTGEGRMKESITRRRQAPS IATIGSTCDLNLKIPGNWFQM
VSAQERLRTFTRSSHTYTRTERTEISKTRGGETKREVRVEESTQVGGDPFPAVFGDF
LGRERLGSFGSITRQQEGEASSQDMTAQVTSPSGKTEAAEIVEGEDSAYSVRFPVQE
MGPHTVTVKYRGQHVPGSPFQFTVGPLGEGGAHKVRAGGTGLERGVAGVPAEFSIW
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GNYLIAIKYGGPQHIVGSPFKAKVTGPRLSGGHS LHETSTVLVETVTKSSSSRGASYSS
IPKFSSDASKVVTRGPGLSQA FVGQKNSFTVDCSKAGTNMMMVG VHGPKTPCEEVY
VKHMGNRVYNVTYTVKEKGDYILIVKWGDESVP GSPFKVNVP 37 Mouse ACON
MAPYSLLVTRLQKALGVRQYHVASVLCQRAKVAMSHFEPSEYIRYDLLEKNINIVRK
(aconitate
RLNRPLTLSEKIVYGHLDDPANQEIERGKTYLRLRPDRVAMQDATAQMAMLQFISSG
hydratase)
LPKVAVPSTIHCDHLIEAQVGGEKDLRRAKDINQEVYNFLATAGAKYGVGFWRPGS
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GTQETILLNHTFNETQIEWFRAGSALNRMKELQQ 38 Mouse PRVA
MSMTDVL SAEDIKKAIGAFAAAADSFDHKKFFQMVGLKKKNPDEVKKVFHILDKDKS
(parvalbumin GFIEEDELGSILKGFSSDARDLSAKETKTLLAAGDKDGDGKIGVEEFSTLVAES
alpha) 39 Mouse HBA
MVLSGEDKSNIAAWGKIGGHGA EYGAEALERMFA SFPTTKTYFPHFDVSHGSAQV
(hemoglobin
KGHGKKVADALASAAGHLDDLPGALSALSDLHA HKLRVDPVNFKLLSHCLLVTLAS
subunit alpha) HHPADFTPAVHASLDKFLASVSTVLTSKYR 40 Mouse HBAZ
MSLMKNERAIIMSMWEKMAAQAEP IGTETLERLFC SYPQTKTYFPHFDLHHGSQQLR
(hemoglobin
AHGFKIMTAVGDAVKSIDNLSSALT KLSELHAYILRVDPVNFKLLSHCLLV TMAARF
subunit zeta) PADFTPEVHEAWDKFMSILSSILTEKYR 41 Mouse HSP7C
MSKGPAVGIDLGTTYSCVGVFQH GKVEI IANDQGNRTTPSYVAFTDTERLIGDAAKN
(heat shock
QVAMNPTNTVFDAKRLIGRRFDDAVVQSDMKHWP FMVVNDAGRPKVQVEYKGET
cognate 71)
KSFYPEEVSSMVLTKMKEIAEAYLGKTVTNAVVTVPAYFNDSQRQATKDAGTIAGL
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APPSGGASSGPTIEEVD 42 Mouse TNNI2

MGDEEKRNRAITARRQHLKSVMLQIAATELEKEESRRESEKENYLSEHCPPLHIPGSM
(troponin I)
SEVQELCKQLHAKIDVAEEEEKYDMEVKVQKSSKELEDMNQKLFDLRGKFKRPPLRR
VRMSADAMLKALLGSKHKVCMDLRANLKQVKKEDTEKERDLRDVGDWRKNIEEK
SGMEGRKKMFESES 43 Mouse CASQ1

MRATDRMGARAVSELRLALLFVLVLGTPRLGVQGEDGLDFPEYDGVDRVINVNAK
(calsequestrin-1)
NYKNVFKKYEVLLALLYHEPPEDDKASQRQFEMEELILELAAQVLEDKGVGFGLVDS
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EIDFYEA FMEEPMTIPDKPNSEEEIVSFVEEHRRSTLRKLPESMYETWEDDL DGIHIV
AFAEEADPDGYEFLETLKAVAQDNTENPDLSIIWIDPDDFLLVPYWEKTFDIDL SAP
QIGVVNVTDADSIWMEMDNEEDLPSADELEDWLEDVLEGEINTEDDDDDDDDDDDDD
DDDDDD 44 Mouse OBSCN

MDHSFSGAPRFLTRPKAFVVS VGKDATLSCQIVGNPTPHVSWEKDRQPVEAGARFRL
(obscurin)

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AQVR 45 Mouse F8VPN4

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(glycogen

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debranching

GVDNHVLPLDCVTLQTFLAKCLGPFDEWESRLRVAKESGYNMIHFTPLQTLGLSRSC
enzyme)

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GVYDNALDNDNYNLARGFNYHQGPEWLWPIGYFLRAKLYFSKMMGPETAAKTVFL
VKNVLSRHYVHLERSPWKGLPELTNENGQYCPFSCETQAWSMAVVLETLYDL 46 Mouse
EF1A2 MGKEKTHINIVVIGHVDSGKSTTTGHLIYKCGGIDKRTIEKFEKEAAEMGKGSFKYA
(elongation

WVLDKKAERERGITIDISLWKFETTKYYITIIDAPGHRDFIKNMITGTSQADCAVLIV
factor 1-alpha 2)

AAGVGEFEAGISKNGQTREHALLAYTLGVKQLIVGVNKM DSTEPAYSEKRYDEIVK
EVSAYIKKIGYNPATVPFVPISGWHGDNMLEPSPNMPWFKGWKVERKEGNASGVSL
LEALDTILPPTRPDKPLRLPLQDVYKIGGIGTVPVGRVETGILRPGMVVTFAPVNITT
EVKSVEMHHEALSEALPGDNVGFNVKNVSVKDIRRGNVCGDSKADPPQEAAQFTSQ
VIILNHPGQISAGYSPVIDCHTAHIACKFAELKEKIDRRSGKKLEDNPKSLKSGDAIIV
EMVPGKPMCIVESFSQYPPPLGRFAVRDMRQTVAVGVKNVEKKSSGGAGKVTKSAQK
AQKAGK 47 Mouse AATC

MAPPSVFAQVPQAPPVLVFKLTADFRDDPDPRKVNLGVGAYRTDESQPWVLPVVRK
(aspartate

VEQKIANDNSLNHEYLPILGLAEFRSCASRLVLGDNSLAIRENRVGGVQSLGGTGAL
aminotransferase

RIGADFLGRWYNGTDNKNTPIIYVSSPTWENHNAVFSAAAGFKDIRPYCYWDAEKRL
DLQGFLNDLENAPEFSIFVLHACAHNPTGTDPTPEQWKQIAAVMQRRFLFPFFDSAY
QGFASGDLEKDAWAIRYFVSEGFELFCAQSFSKNFGLYNERVGNLTVVGKESDSVLR
VLSQMEKIVRITWSNPPAQGARIVAATLSDPELFKEWKGNVKT MADRILTMRSELRA
RLEALKTPGTWWHITEQIGMFSFTGLNPKQVEYLVNEKHIYLLPSGRINMCGLTTKNL
DYVATSIHEAVTKIQ 48 Mouse KAD1

MEEKLKKAKIIFVVGPGSGKGTQCEKIVQKYGYTHLSTGDLLRAEVSSGSERGKKL
(adeny late

SAIMEKGELVPLDTVLDMLRDAMLAKVDSSNGFLIDGYPREVKQGEEFEQKIGQPTL
kinase LLYVDAGAETMTQRLLRGETSGRVDDNEETIKKRLETYYNATEPVISFYDKRGIVR
isoenzyme 1) KVNAEGTVDTVFSEVCTYLD SLK 49 Mouse G6PI

MAALTRNPQFQKLLEWHRANSANLKLRELFEADPERFNNFSLNLNTNHGHILVDYS
(glucose-6-

KNLVNKEVMQMLVELAKSRGVEAARDNMFSGSKINYTENRAVLHVALRNRSTPI
phosphate

KVDGKDVMPEVNRVLDKMKSFQQRVRS GDWKG YTGKSITDIINIGIGGSDLGPLMV
isomerase)

TEALKPYSKGGPRVWFVSNIDGTHIAKTLASLSPETSLFIIASKTFTTQETITNAETAKE
WFLEAAKDPSAVAKHFVALSTNTAKVKEFGIDPQNMFEFWDWVGGRYSLWSAIGL
SIALHVGFDHFEQLLSGAHWMDQHFLKTPLEKNAPVLLALLGIWYINCYGCETHALL
PYDQYMHRFAAYFQQGD MESNGKYITKSGARVDHQTGPIVWGEPGTNGQHAFYQL
IHQGTKMIPCDFLIPVQTQHPIRKGLHHKILLANFLAQTEALMKGKLPEEARKE LQAA
GKSPEDLEKLLPHKVFEGNRPTNSIVFTKLTPFILGALIAMYEHKIFVQGIMWDINSFD
QWGVELGKQLAKKIEPELEGSSAVTSHDSSSTNGLISFIKQQRDTKLE 50 Mouse DESM
MSQAYSSSQRVSSYRRTFGGAPGFSLSPLSSPVFPRAGFGTKGSSSSMTSRVYQVSR

(desmin)

TSGGAGGLGSLRSSRLGTTRAPSYGAGELLDFSLADAVNQEFLATRTNEKVELQELN
DRFANYIEKVRFLQEQNAALAAEVNRLKGREPTRVAELYEEEMRELRRQVEVLTNQ
RARVDVERDNLIDDLQRLKAKLQEEIQLREEAENNLAAFRADVDAATLARIDLERRI
ESLNEEIAFLKKVHEEEIRELQAQLQEQQVQVEMDMSKPDLTAAALRDIRAQYETIAA
KNISEAAEWYKSKVSDLTQAANKNNDALRQAKQEMMEYRHQIQSYTCEIDALKGT
NDSLMRQMRELEDRFASEANGYQDNARLEEEIRHLKDEMARHLREYQDLLNVKM
ALDVEIATYRKLLERGEESRINLPIQTF SALNFRETSPEQRGSEVHTKKTVMIKTIETRD
GEVVSEATQQQHEVL 51 Mouse KCRS

MASAFSKLLTGRNASLLFTTLGT SALTTGYLLNRQKVSADAREQHKLFPSPADYPDL
(creatine kinase

RKHNNCMAECLTPTIYAKLRNKMTPSGYTLDQCIQTGVDPNGHPFIKTVGMVAGDE S-
type) ESYEVFADLFDPVIKLRHNGYDPRVMKHPTDL DASKITHGQFDERYVLSSRVRTGRSI
RGLSLPPACSRARERREVENVAITALEGLKGDLAGRYYKLSEMTEQDQQLIDDHFLF
DKPVSPLTLCAGMARDWPDARGIWHNYDKTFLIWINEEDHTRVISMEEKGGNMKRVF
ERFCRGLKEVERLIQERGWEFMWNERLGYILTCPSNLGTGLRAGVHVRIPKLSKDP
FSKILENLRLQKRGTTGGVDTAAVADVYDISNIDRIGRSEVELVQIVIDGVNYLVDCEK
KLERGQDIKVPPLPQFGRK 52 Mouse

MRKHRHLPLVAVFSLLSGIATTHAQHQGDVKNNGAAADIVFLVDSSWSAGKDRFLL
E9PWQ3

VQEFLSDVVESLAVGDNDHFALVRLNGNPHTEFLLNTYHSKQEVLSHIVNMSYIGG
(collagen, type

SNQTGKGLEEYVIHSHLTEASGSRAADGVPQVIIVLTDGQSEDGFALPSAELKSADVNI,
alpha 3)

VFAVGVEGADERALGEVASEPLSMHVFNLENVTSLHGLVGNLVSCIHSSVNPERAG
DKESLKDITAQDSADIIFLIDGSQNTGNANFDVIRDFLVNVLERLSVGNQQVQVGVV
QYSEEPITMFSLSNSYPSKAAVLDAVKGLSLVGGESANIGQALDFVVENHFTRAGGSR
VEEGVPQVLVLISAGPSSDEIRDSVVALKQASVFSFGLGAQAASRAELQHIATDDSLV
FTVPEFRSFGDLQEQLPYLVGVAQRHIVLQPPAIVTQVMEVNKRDIVFLVDGSSSLG
PSNFNAIRDFVTRVIQRLEIGQDLVQVSVAQYADTVKPEFYLSYTNKRDAITAVRK
MRALNGSALYTGSSLDVFRNNLFTSSAGHRAAEGVPKLLVLITGGKSLDEVSQPAQE
LKRGSIMALAVGSKAADEDELKEIAFDSSLVFIPAEFRPAPLQNMLPSLMAPLRTLGT
TTEVHVNKRDIIFLLDGSDNVGKNNFPYVRDFVTNLVNSLDVGSNDIRVGLVQFSDT
PVTEFSLDTYQTKSELLAHLRRLQLKGGSGLNAGSALS YIHANHFTEAGGSRTREHV
PQLLLLLMAGPSEDAYLQAANALVRSGLVTFCTVGTNRADKAELEHIAFNPSLVYLM
DDFRSLPSLPQQLIQPLTTYVSGGVQEVPLSQPESKRDILFLFDGSVNVLGQFPAVRDF
LYRIIEELDVKPDGTRVAIAQFSDDVRLESRFSEHQTKAEILNLVKKMKLKTGKALNL
GYALDYALRNIFVRSAGSRIEDNVQQFLVLLVAGRSSDAVAGPASSLKQRGVVPFIF
QAKNANPSELEQIVLSPAFILAAESLPKIGDLQSQIVSLLKAEQSGPVSGEKDVVFLI
DGSEGVRS GFPLLKDFVQRVVESLDVGPDRVRVALVQYSDRTRPEFYLSHMDQQG
VISAIRRLTLLGGPTPNTGAALFVLRNLTSSSTGSRIAEGVPQLLIVLTAEPSGDDVRG
PSVVLKQGGAVPIGIGIGNADISEMQTISFIPDFAVAIPTFRELGTIQQVISERVIQLNRE
ELSSLKPILTPSTGAGVGSKKDVVFLIDGSRNAGPEFQYIRTLIERIVEYLDIGFDTTRV
AVIQFSEDSKMEFPLNAHFSKDEVQNAVRRRLRPKGGSQVYIGNALEYVLKNIFQRPL
GSRIIEGVPPQLVLISSGKSDDEVDDSAVELKQFGVAPLTIARHTDQEELVKISLSPEY
VYSVSTFRELPRLEQKLLTPITTLTSQQIHQILASTRYPPSVVESDAADIVFLIDSSDAV
KPDGIAHIRDFVSRIVRRLNIGPSKVRIGVVQFSNDVFPEFYLKTHKSQSSVLEAIRRLR
FKGGSP LNTGRALEFVARNLFVKSAGSRIEDGVPQHLVFLGGSQDDVARHAQVIS
SSGIVSLGIGDRNIDRTDLQTITNDPRLVFTVREFREL PNIEERVMLSFGPSGATPQPPG
VDLPSPSRPEKKKADIVFLLDGSINFRRDSFQEVLRFASEIVDTVYEDGDSIRVGLVQY

NSDPTDFLDFSTKRQIIDAINKVYKGGRIHANTRVGFIHLLRNHFVPEAGSRLDE
RVPQIAFVITGGKSVEDAQDVSLALTQKGVKVFVAVGVNRNIDSEEVGKIASNSATAFR
VGSVQELSELSETVLETLHDAMHETLCPGVTDVSKACNLEVILGFDGSRDQNVFVSQ
KGLESKVDIILNRISQIQRISCSGNQLPTVRVSVMANTPSGPVEAFDFAEYQPELFEKF
RNMRSQRPHYVLTADTLKLYQNKFRQSSPDTVKVVIHFTDGADGDMADLYRASEELR
QAGAQUALILVGLERVANLERLMHLEFGRGFMYDRPLRLNLLDLDYELAEQLDNIAE
KACCGVPCKCSGERGDRGPIGSIGPKGISGEDGYRGYPGDEGGPGERGPPGVNGTQG
FQGCPCGQRGVKGSRGFPGEKGEIGLDGLDGEEDKGLPGSSGEKGSPPGRRGDK
GPKGDKGERGDVGIRGDPGDSGRDSQQRGPKGETGDIGPMGLPGRDGIPGSPGDPGK
DGGSGRRGPAGAKGNRGGPGQPGFEQEQTRGSQGPPIGPPGLIGEQQIPGPRGG
GGTAGAPGERGRTGPLGRKGEPGEPGPKGSIGNRGPRGETGDDGRDGVGSEGRRGK
KGERGFPGYPGPKGTPGEPGADGPPGPKGIRGRRGNSGPPGATGQKGDPGYPGPSGH
KGNRGDSVDQCALIQSIRDKCPCCYGPLECPVFPTTELAFALDTSEGVTQDTFSRMREV
LLGIVGDLTIAESNCPRGARVAVVTYNNEVTTEIRFADSKKKSALLDSIQNLQVALTS
KQQSLETAMSFVARNTFKRVRSGFLMRKVAVFFSNKPTRASPLREAVLKLSDAGIT
PLFLTSQEDRQLINALQINNTAVGHALVLPARRDLTDFLKNVLTCHVCLDICNIDPSC
GFGSWRPSFRDRRAAGSDVDIDLAFILDSSEATTLFQFNEMKKYIGYVIRQLDLSPDP
KASQHFARVAVVQQSTYESVDNASVPPVKVEFSLTDYGAKEKLLDFLSRRMTQLQG
TMGLGNAIEYTIENIFESAPNPRDLKIMVLMLTGDMQRQQLEEAQRAILQAKCKGYF
FVVLGIGRKVNIKEVYSFASEPNDVFFKFVDKSTELNEEPLMRFGRLPSFVSSENAF
YLPPDLRKQCDWFQGDQPVKNGVKFGHKQINPPHTANSSLTSKVVTTMKPVTTTKP
TAIVNLPAPAKPAPAPAPAPQVLA KPDPAPAKPAQARPAKPAKPAKPAKPAKPAKPA
PAQTASVRPAPAKPAPPQPA AAKPVPAKPAVPAQPAAPPQPA AAKPVPAKPAVPAQPA
PPQPA AAKPVPAKPAVPAQPA AQAQPMMPAQPVLT KSAVKPASANKPVAAKPVATNT
ATATARPALAAKPA AAKPAATRPLAAAIRPVATKPEAPRQQA KPAATKPAATKPLAR
VSREVQVSEVTENSARLHWERPEPSSSFFYDLTVTSAHDQSLVLRQNLTVTDRVIGG
LLAGQLYHVVVVSYLQSQVRAIYQGSFNTKKTQPPPLQAAHRASSSTINLMVNTEPL
FLTCTDICKLSRDAGTCVDFKLLWHYDLESKSKCRFWYGGCGGNENRFHSQEECEK
MCSPDLLV 53 Mouse AMD1

MPLFKLTGQGKQIDDAMRSFAEKVFASEVKDEGGRHEISPFDVDEICPISLHEMQAHI (AMP
FHMENLSMDGRRKRRFQGRKTVNLSIPQSETSSTKLSHIEEFISSPTYESVPDFQRVQI
deaminase 1)

TGDYASGVTVEDFEVCKGLYRALCIREKYMQKSFRFPKTPSKYLNRNIDGEALVG
NESFYVPVFTPPPKKGEDPFRTEDL PANLGYHLKMKAGVIYIYPDEAAANRDEPKPYP
YPNLDDFLDDMNFLALIAQGPVKTYAHRRLKFLSSKFQVHQMLNEMDELKELKNN
PHRDFYNCRKVDTHIHAAACMNQKHLLRFIKKSYHIDADRVVYSTKEKSLTLKELFA
KLNMHYPYDLTVDSL DVHAGRQTFQRFDKFNDKYNPVGASELRDLYLKT DNYINGE
YFATIIKEVGADLVEAKYQHAEPRLSIYGRSPDEWNKLSSWFVCNRIYCPNMTWMIQ
VPRIYDVFRSKNFLPHFGKMLENIFLPVFEATINPQAHPDLSVFLKHITGFDSVDDESK
HSGHMFSSKSPKPEEWTMENNPSTYTYAYYMYANITVLNSLRKERGMNTFLFRPHC
GEAGALTHLMTAFMIADNISHGLNLKKSPVLQYLFFLAQIPIAMSPLSNNSLFLEYAK
NPFLDFLQKGLMISLSTDDPMQFHFTKEPLMEEYAIAAQVFKLSTCDMCEVARNSVL
QCGISHEEKAKFLGNNYLEEGPVGNDIRRTNVAQIRMAYRYETWCYELNLIAEGLKA TE

[0039] An animal protein can comprise a rabbit protein (Table 2). A rabbit protein can comprise a rabbit muscle protein. A rabbit protein can comprise a rabbit blood protein. A rabbit protein can comprise a rabbit albumin protein. A rabbit protein can comprise a sequence that has at least or about 70%, at least or about 75%, at least or about 80%, at least or about 90%, at least or about 91%, at least or about 92%, at least or about 93%, at least or about 94%, at least or about 95%, at least or about 96%, at least or about 97% at least or about 98% at least or about 99%, or about 100% sequence identity to any one of SEQ ID NOs: 54-78. A rabbit albumin protein can comprise

a sequence that has at least or about 70%, at least or about 75%, at least or about 80%, at least or about 90%, at least or about 91%, at least or about 92%, at least or about 93%, at least or about 94%, at least or about 95%, at least or about 96%, at least or about 97% at least or about 98% at least or about 99%, or about 100% sequence identity to SEQ ID NO: 54-78.

[0040] A food product may comprise a fragment or a truncation of a rabbit protein. The food product may include a fragment or truncation of any one of SEQ ID Nos: 54-78. A sequence of the fragment or truncation may be identical or substantially identical to a portion of any one of SEQ ID Nos: 54-78. A sequence of the fragment or truncation may be at least about 70%, 80%, 90%, 95%, 98%, or more identical to a portion of a sequence of any one of SEQ ID Nos: 54-78. A length of the fragment or truncation may be at least about 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or more of a length of any one of SEQ ID Nos: 54-78. A length of the fragment or truncation may be less than or equal to about 90%, 80%, 70%, 60%, 50%, 40%, 30%, 20%, 15%, 10%, or less than a length of any one of SEQ ID Nos: 54-78.

TABLE-US-00002	TABLE	2	Rabbit	Proteins	SEQ ID	NO	Description	Amino	Acid
Sequence 54	Rabbit	KCRM							
FGNTHNKYKLN	YKSEEEYPDL	SKHNNHMAKV	LTPDLYKKLR	DKETPSGFTL	(Creatine				
DDVIQTGVDNP	GHFIMTVGCV	AGDEESYTVF	KDLFDPIIQD	RHGGFKPTDK	H Kinase	M-			
type) KTDLN	HENLKGDDLD	PHYVLSSRV	RTGRSIKGYT	LPPHCSRGER	RAVEKLS				
VEALNSLTGE	FKGKYYP	LKSMTEQE	QQQLIDDH	FLFDKPVSP	LLLASGMARD				
WPDARGIWH	NDNKSFLV	WVNEEDHL	RVISMEKGG	NMKEVFRR	FCVGLQKIE				
EIFKKAGHP	FMWNEHLG	YVLTCPSN	LGTGLRGG	VHVKLAHLS	KHPKFEEILT				
RLRLQKRGT	TGGVDTAA	VGSVFDIS	NADRLGSSE	VEQVQLVVD	GVKLMVEME				
KKLEKGQS	IDDMIPAQ	K	55 Rabbit						
MMMVMQPEGL	GAREGPFAG	GGGGGGEY	MEQEEDWDR	DLDDPAWEK	QQRK	G1U4H8			
TFTAWCNSHL	RKAGTQIEN	IEEDFRNGL	KLMLLLEVI	SIGERLPRPD	KGKMRFH	(Actinin			
Alpha KIANV	NKALDFIASK	GVKLVSIGA	EEIVDGNL	KMTLGMWTH	ILRFAIQDIS	VE 3)			
ETSAKEGLLL	WCQRKTAPY	RNVNVQNF	HTSWKDGLA	LALCALIHR	HRPDLIDY				
AKLRKDDPI	GNLNTAFE	VAEKYLDIP	KMLDAEDIV	NTPKPDEKA	IMTYVSCFY				
HAFAGAEQA	ETAANRICK	VLAVNQEN	EKLMEEYE	KLASELLEW	IRRTVPWL				
ENRVGEPSM	SAMQRKLED	FRDYRRLH	KPPRVQEK	CQLEINFNT	LQTKLRLSH				
RPAFMPSEG	KLVS	DIANA	WRGLEQVE	KGYEDWLL	SEIRRLQRL	QHLAEKFQQ			
KASLHETW	TRGKEDML	TQRDYESA	SLQEVRA	LLRRHEAF	ESDLAAHQ	DRVE			
HIAALAQEL	NELDYHEA	ASVNSRC	QAICDQWD	NLGTLTQK	RRDALER	MEKL			
LETIDQLQL	EFARRAAP	FNNWLDGA	IEDLQDVW	LVHSVEET	QSLTAHD	QFK			
ATLPEADR	ERGAILGI	QGEIQKIC	QTYGLRPS	STNPNYIN	LSPQDINT	KWDTVRK			
LVPSRDQTL	QEELARQ	QMNERLRR	QFAAQAN	AVGPWIAK	VVEVGR	LAAGL			
AGSLEEQMA	GLRQQEQ	NIINYKSN	IDRLEGDH	QLLQESLV	FDNKHTV	YSMEH			
IRVGWEQL	LSIARTIN	EVENQVL	TRDAKGL	SQEQLNEF	RASFNHF	DRKRNG			
MMEPDDFRA	CLISMYDL	GEVEFAR	IMTMVDP	NAAGVVTF	QAFIDFM	TRETAE			
TDTAEQVVA	SFKILAGD	KNYITPEE	LRELPAEQ	AEYCIHR	MAPYKG	GAGAPA			
GALDYVAF	S	SALYGES	DL	56 Rabbit	PYGM				
SRPLSDQE	KRKQISVR	GLAGVEN	VT	ELKKNFNR	HLHFTLVK	DRNVATPR	DYY	(Glycogen	
FALAHTVR	DHLVGRW	IRTQQHY	EKDPKRI	YLSLEFY	MGR	TLQNTMV	NLA		
Phosphorylase,									
LENACDEAT	YQLGLDME	EELEEIEE	DAGLGNGG	LGR	LAAACFLD	SMATLGL	AA	muscle	
form) YGYG	IRYEF	GIFNQKIC	GGWQMEE	ADDWLR	YGNPWEK	ARPEFTLP	VHFYGR		
VEHTSQGA	KWVDTQV	VLAMPYD	TPVPGYR	NNVVNT	MRLWSA	KAPNDFNL			
KDFNVGGY	IQAVLDR	NLAENIS	RVLYPN	DNFFEG	KELRLKQ	EYFVVAAT	LQD		
IIRRFKSS	KFGCRDP	VRTNFDA	FPDKVAI	QLNDTHP	SLAIPEL	MRVLVD	LRLD		
WDKAWEV	TVKTCAY	TNHTVL	PEALER	WPVHL	LETLLPR	HLQIIEIN	QRFLN		

RVAAAGDVPDRLRMSLVEEGAVKRYNMAHLCLAGSHAVNGVARIHSEILK
KTIFKDFYELEPHKFQNKTNGITPRRWLVLCNPGLAEIIAERIGEEYISDLDQLR
KLLSYVDDEAFIRDVAKVKQENKLFKFAAYLEREYKVHINPNSLFDVQVKRIH
EYKRQLLNCLHVITLYNRIKKEPNKFVVPRTVMIGGKAAPGYHMAKMIIKLIT
AIGDVVNHDPVVGDRLRVIFLENYRVSLAEKVIPAADLSEQISTAGTEASGTG
NMKFMLNGALTIGTMDGANVEMAEAGEENFFIFGMRVEDVDRLDQRGYN
AQEYYDRIPELRQIIEQLSSGFFSPKQPDFKDIVNMLMHHDRFKVFADYEEY
VKCQERSALYKNPREWTRMVIRNIATSGKFSSDRTIAQYAREIWGVEPSRQR LPAPDEKIP
57 Rabbit ALDOA
HSHPALTPSEQKELSDIAHRIVAPGKGILAADESTGSIKRLQSIGTENTENR (Fructose-
RFYRQLLLTADDRVNPCIGGVILFHETLYQKADDGRPFQVIKSKGGVVGIV Bisphosphate
DKGVVPLAGTNGETTTQGLDGLSERCAQYKKDGADFAKWRCVLKIGEHTPS Aldolase
A) ALAIMENANVLARYASICQQNGIVPIVEPEILPDGDHDLKRCQYVTEKVLAAY
YKALSDHHIYLEGTLLKPNMVTTPGHACTQKYSHEEIAMATVTALRRTVPPAV
TGVTFLSGGQSEEEASINLNAINKCPLLKPWALTFSYGRALQASALKAWGGK
KENLKAQEEYVKRALANSLACQGKYTPSGQAGAAASESLFISNHAY 58 Rabbit MYL1
APKKDVKKPAAAAAPAPAPAPAPAPAKPKKEEKIDLSAIKIEFSKEQQDEFK (Myosin
Light EAFLLYDRTGDSKITLSQVGDVLRALGTNPTNAEVKKVLGNPSNEEMNAKKI
Chain, Skeletal
EFEQFLPMLQAISNNKDQGTIEDFVEGLRVFDKEGNGTVMGAELRHVLATL Muscle
GEKMKEEEVEALMAGQEDSNGCINYEAFVKHIMSI Isoform) 59 Rabbit G3P
MVKVGVNGFGRIGRLVTRAAFN SGKVDVVAINDPFIDLHYMVYMFQYDSTH
(Glyceraldehyde-
GKFHGTVKAENGKLVINGKAITIFQERDPANIKWGDAGAEYVVESTGVFTTM 3-Phosphate
EKAGAHKGGAKRVIISAPSADAPMFVMGVNHEKYDNSLKVSNASCTTNCL
Dehydrogenase)
APLAKVIHDHFGIVEGLMTTVHAITATQKTVDGPSGKLWRDGRGAAQNIIPAS
TGAAKAVGKVIPELNGKLTGMAFRVPTPNVSVVDLTCRLEKAAKYDDIKKV
VKQASEGPLKGILGYTEDQVVSCDFNSATHSSTFDAGAGIALNDHFVKLISWY
DNEFGYSNRVVDLMVHMASKE 60 Rabbit TPM1
MDAIKKKMQLKLDKENALDRAEQAEADKKAEDRSKQLEDELVSLQKKL
(Tropomyosin
KGTEDELDKYSEALKDAQEKLLEAEKKATDAEADVASLNRRIQLVEEELDRA Alpha-1
Chain) QERLATALQKLEEAKEKADESERGMKVIESRAQKDEEKMEIQEIQLKEAKHI
AEDADRKYEEVARKLVIIESDLERAEEAEELSEGKCAELEELKTVINNLKSL
EAQAEKYSQKEDKYEEEIKVLSDKLKEAETRAEFAERSVTKLEKSIDDEDEL
YAQKLKYKAISEELDHALNDMTSI 61 Rabbit Myosin-
EAKPAARKAPKGKDAPKEEPPKEAAAEPPEAPPEDQSPTAEPTGIFLKKPD Binding
protein SVSVETGKDAVIMAKLNGKELPAKPTIKWFKGKWLELGSKSGARFSFKETHD C
(fast-type) AASNVSLELHIGKVVLGDRGDYRLEVKAKDVCDSCSFNIDVEAPRQDSAGQ
SLESFKRAGDAKPDGAGELDFSGLLKKREVVEEEKKKKKDDDDDLGIPPEIWE
LLKGAKKSEYERIAFQYGITDLRGMLKRLKKAKVEVKKSAAFTKKLDPAYQ
VDRGNKVKLVVEISDPDLPLKWFKNGQEIKPSLKYVFENVGKKRILTINKCTL
ADDAAYEVVNDEKCFTEL FVKEPPVLIVTPLEDQQVFVGDVMSVEVSEE
GAQVMWMKDGVELTREGSFKSNYRFKKDGKRHILYSEVNLEDGRGHYQVM
TNGGQCEAELIVEEKQLEVLQDIADLTVKASEQAMFKCEVSDEKVTGKWKY
NGVEVRPSKRITISHVGRFHKLVIDDVRPEDEGDYTFVPDGYALSLSAKLNFL
EIKVEYVPKQEPPIHLDCSGKTSDNAIVVAGNKLRLDV SITGEPPP VATWM
KGDEVFTVTEGRVRIEQRPDSSSFVIESAERGDEGRYTIKVTNPAGEDVATVM
LRVVDVPDPPEAVRVT SVGEDWAILVWDPPKYDGGQPVTGYLLERKKKGSQ

RWMKLFNEVFTEETTYESTKMIIEGILYEMRVFAVNAIGVVSQPSMNTKPFMPIAP
TSAPQHLLTVEDVTDTTTLKWRPPDRIGAGGIDGYLVEYCVESGSEWIPANK
EPVERCGFTVKGLPTGAKILFRVIGVNIAGRSEPATLVQPVTIREIVEQPKIRLP
RHLRQTYVRKVGEHINLVIPFQGKPRPQVWVTKGGAPVDTSRVHVRTSDFDT
VFFVREAARSDSGEYELSVQIENMKDTATVRIRVVEKAGPAENVMVKEVWG
TNALVEWQPPKDNNGNSEITGYLVQKADKKTMEWFTVYEHNRHTSCTVSDLI
VGNEYFYFRVYTENVCGLSDSPGVSKNTAHILKTGITFKPFYKEHDFRSPPKF
LTPLIDRVVAGYAAALNCAVRGHPKPKVWWMKNKVEIHEDPKFLMTNHQ
GVLTLNIRRPSPFDAGTYSCRAFNELGEALAECLDVRVPQ 62 Rabbit G1T4T6
MSLVAVPFYQKRHKHFDQSYRNIQTKYLMDEYTAKKRASTQMSTHKSASTQ (Myomesin
2) MATHKSALQKASSQTSSGGTTCRLCAKRLSAAEEEEQESKNRYRSLVASIGE
AKRQRFLSELAQLEESVHLARSQARDVVDKYALQQMVEDKMAWERHVFEE
RLSRAPEILVQLRSHTVWEKMSVKLCFTVQGFPTPVVQWYKDGSLICQAGEP
GKYRIDSKYGVHTLEISRASFDDTATYSAVATNAHGQVSTNAAVVVRRYRG
DEEPFYSVGLPIGYVLVKESKWTRMFFGEGQASLSFSHLNKDDEGLYTLRIVS
RGGVSDHSAFLFVRDADPLVTGAPGAPMDLQCHDANRDYVIVTWKPPNTTT
ESPVIGYFVDRCEVGTNNWVQCNDAPVKICKYPTGLYEGRSYIFRVRAVNS
AGISRPSRVSDAVAALDPADLRRLQAIHLEGEKEVVYQDDLEGDVQVPGPPT
NVHASEISRNYVVLWEPPVPRGKEPLMYFIEKSVVGSGSWQRVNAQTAVRS
PRYAVFDLAEGKPYVFRVLSANKHGLSDPSEITSPIQAQDTIAVPSAPGRVLAS
RNTKTSVVVQWDRPKHEEDLLGYVVDCCVAGTNAWEPCNHKPIGYNRFVV
HGLTTGEQYIFRVKAVNALGTSENSQESEVIKVQAALTVPSHPYGVTLNCD
GHSMTLGWKVPKFSGGSPILGYVVDKREVHHKNWHEVNSSPIQERILTVEGL
TEGSFYEFKIAATNLAGIGQPSDPSEHFKCEAWTAPEPGPAYDLTFCEVRDTS
VMLWKAPVYSGSSPVSGYFVDFKEEDSGEWSTVNEAATPNRYLKVSDLHQG
RTYVFRVRVAVNASGVGKPSDVSEPVLVEARPGTKEISAGVDEEGNIYLGFD
QEMTDASHFTWCKSYEEIADPERFHVETKGDHSLYFKNPDKADLGTYSVSV
SDTDGVSSSFVLDEEELERLLALSNEIKNPTIPLKSELAYEIFDKGQVRFWLQT
EPLSPDASFRFVINDREVSDSETHRMKCDKASGIIEMVMMDRFTIENEGTYTVQI
HDGKAKNQSSLVLIGDAFRKVLLEAEFQRKEFLRKQGPFAEYHLHWDVTEEC
EVRICKVANTKKETVFKWLKDDVLYETETPPDLEKGVCELLIPKLSKKDHG
EYKATLKDERGQDASILEIAGKVYEDMILAMSRVCGMSASPLKILCTPEGIRL
QCFMKYFTEEMRVNWyHKDAKISSSEHMRIGGSSEDMAWLQICEPTEKDKGK
YTFEIFDGKDNHQRSLLSGQAFDEAFAEFQRLKAAAFKAEKNRGKVIGGLPD
VVTIMEGKTLNLTCTVFGNPDPVWVWFKNDQDIELSEHFSVKVEQAKYVSLTI
KGVTSSEDSGKYSINVKNKYGGEEKIDVTVSVYKHGEAIPVAPPQQAQPKLIPA SASATAP 63
Rabbit ENOB

AMQKIFAREILDSRGNPTVEVDLHTAKGRFRAAVPSGASTGIYEALRLDGDGK (Beta-
enolase) SRYLGKGVLKAVEHINKTLGPALLEKKLSVVDQEKVDKFMIELDGTENKSKF
GANAILGVSLAVCKAGAAEKGVPLYRHIADLAGNHDLVLPVPAFNVINGGSH
AGNKLAMQEFMILPVGASSFREAMRIGAEVYHHLKGVKVIKAKYGKDATNVGD
EGGFAPNILENNEALELLKTAIQAAGYPDKVVIGMDVAASEFHRNGKYDLDF
KSPDDPARHITGQKLGEYKSFYKYNPVVSIEDPFDQDDWATWTSFLSGVDIQI
VGDDLTVTNPKRIAQAVEKKACNCLLLKVNQIGSVTESIQACKLAQSNWGV
MVSHRSGETEDTFIADLVVGLCTGQIKTGAPCRSERLAKYNQLMRIEALGD
KAVFAGRKFRNP KAK 64 Rabbit MLRS
APKKAKRRRAAAEGGSSNVFSMFDQTQIQEFKEAFTVIDQNRDGIIDKEDLRDT (Myosin
FAAMGRLNVKNEELDAMMKEASGPINFTVFLTMFGEKLGADPEDVITGAF Regulatory
KVLDPEGKGTIKKQFLEELLTTQCDRFSQEEIKNMWAAFPDPVGGNVYDKNI Light
Chain 2, CYVITHGDAKDQE Skeletal Muscle Isoform Type 2) 65 Truncation of

IFSIDNAYQFMQDRENQSIGSGAGTIVNTKRVIQYFATIAITGDKKKEE Rabbit
G1SJN7 PTPGKMQGTLEDQIISANPLLEAFGNAKTVRNDNSSRFGKFIRIHFGATGKLAS
(Myosin 4) ADIETYLLEKSRVTFQLKAERSYHIFYQIMSNKKPELIEMLLITTNPYDFAFVS
QGEITVPSIDDDQEELMATDSAVDILGFTSDEKVAIYKLTGAVMHYGNLKFQK
KQREEQAEPDGTVEADKAAYLTSLNSADLLKALCYPRVKVGNEYVTKGQTV
QQVYNAV GALAKAIYEKMF LWMVTRINQQLDTKQPRQYFIGVLDIAGFEIFD
FNSLEQLCINFTNEKLQQFFNHHMFVLEQEEYKKEGIEWEFIDFGMDLAACIE
LIEKPMGIFSILEEECMFPKATDTSFKNKLYEQHLGKSNNFQKPKPAKGKAEA
HFSLVHYAGTVDYNITGWLDKNKDPLNETVVGLYQKSGMKTLAFLFSGAQT
GEEGGGGGGKKGKGGKGGSSFQTVSALFRENLNKLMTNLRSTHPHFVRCLIPNE
TKTPGAMEHELVLHQLRCNGVLEGIRICRKGFP SRILYADFKQRYKVLNASAI
PEGQFIDSKKASEKLLGSIDVDHTQYKFGHTKVFFKAGLLGLLEEMRDEKLA
QLITRTQAICRGYLTRVEFKMMERRESIFCIQYNIRAFMNVKHWPWMKLYF
KIKPLLKSAETEKEMANMKEEFKTKHEELAKSEAKRKELEEKMVALMQEKN
DLQLQVQAEADGLADAEERCDQLIKTKIQLEAKIKEVTERAEDEEEINAELTA
KKRKLEDECESELKKDIDDLELTAKVEKEKHATENKVKNLTEEMAGLDENIA
KLTKEKKALQEAHQQTLDLQAEEDKVNTLTAKATKLEQQVDDLEGSLEQE
KKLRMDLERAKRKLEGDLKLAQESTMDIENDKQQLDEKLKKKEFEMTNLQS
KIEDEQALAMQLQKKIKELQARIEELEEIEAERASRAKAEKQRS DLSRELEEI
SERLEEAGGATSAQIEMNKKREAEFQKMRRDLEEATLQHEATAAALRKKHA
DSVAELGEQIDNLQRVKQKLEKEKSELKMEIDDLAGNMETVSKAKGNLEKM
CRTLEDQLSEVKTKEEEHQRLINELSAQKARLHTESGEFSRQLDEKDAMVSQ
LSRGKQAFTQQIEELKRQLEEETKAKSALAHALQSSRHDCDLLREQYEEEQE
AKAELGRAMSKANSEVSQWRTKYETDAIQRTEELEEA KKKLAQRLQDAEEH
VEAVNSK CASLEKTKQRLQN 66 Truncation of

IFYQIMSNKKPELIEMLLITTNPYDFAFVSQGEITVPSIDDDQEELMATDSAVDIL Rabbit
G1SJN7 GFTSDEKVAIYKLTGAVMHYGNLKFQKQKQREEQAEPDGTVEADKAAYLTSL
(Myosin 4) NSADLLKALCYPRVKVGNEYVTKGQTVQQVYNAV GALAKAIYEKMF LWM
VTRINQQLDTKQPRQYFIGVLDIAGFEIFDFNSLEQLCINFTNEKLQQFFNHHM
FVLEQEEYKKEGIEWEFIDFGMDLAACIELIEKPMGIFSILEEECMFPKATDTSF
KNKLYEQHLGKSNNFQKPKPAKGKAEAHFSLVHYAGTVDYNITGWLDKNK
DPLNETVVGLYQKSGMKTLAFLFSGAQTGEEGGGGGGKKGKGGKGGSSFQTVS
ALFRENLNKLMTNLRSTHPHFVRCLIPNETKTPGAMEHELVLHQLRCNGVLE
GIRICRKGFP SRILYADFKQRYKVLNASAIPEGQFIDSKKASEKLLGSIDVDHT
QYKFGHTKVFFKAGLLGLLEEMRDEKLAQLITRTQAICRGYLTRVEFKMME
RRESIFCIQYNIRAFMNVKHWPWMKLYFKIKPLLKSAETEKEMANMKEEFK
TKEELAKSEAKRKELEEKMVALMQEKN DLQLQVQAEADGLADAEERCDQLI
KTKIQLEAKIKEVTERAEDEEEINAELTAKRKLEDECESELKKDIDDLELTAK
VEKEKHATENKVKNLTEEMAGLDENIAKLTKEKKALQEAHQQTLDLQAE
DKVNTLTAKATKLEQQVDDLEGSLEQEKKLRMDLERAKRKLEGDLKLAQES
TMDIENDKQQLDEKLKKKEFEMTNLQSKIEDEQALAMQLQKKIKELQARIEE
LEEIEAERASRAKAEKQRS DLSRELEEISERLEEAGGATSAQIEMNKKREAEF
QKMRRDLEEATLQHEATAAALRKKHADSV AELGEQIDNLQRVKQKLEKEKS
ELKMEIDDLAGNMETVSKAKGNLEKMCRTLEDQLSEVKTKEEEHQRLINELS
AQKARLHTESGEFSRQLDEKDAMVSQ LSRGKQAFTQQIEELKRQLEEETKAK
SALAHALQSSRHDCDLLREQYEEEQEAKAELGRAMSKANSEVSQWRTKYET
DAIQRTEELEEA KKKLAQRLQDAEEHVEAVNSK CASLEKTKQRLQNEVEDL
MIDVERSNAACAALDK 67 Truncation of

TLDDLQAEEDKVNTLTAKATKLEQQVDDLEGSLEQEKKLRMDLERAKRKLE Rabbit
G1SJN7 GDLKLAQESTMDIENDKQQLDEKLKKKEFEMTNLQSKIEDEQALAMQLQKKI

(Myosin 4) MELQARIELEEIEAERASRAKAQKRSDDLSELEEISERLEEAGGATSAQIE
MNKKREAEFQKMRRDLEEATLQHEATAAALRKKHADSV AELGEQIDNLQRV
KQKLEKEKSELKMEIDDLAGNMETVSKAKGNLEKMCRTLEDQLSEVKTKEE
EHQRLINELSAQKARLHTESGEFSRQLDEKDAMVSQLSRGKQAFTQQIEELKR
QLEEETKAKSALAHALQSSRHDCDLLREQYEEEEQEAKAELGRAMSKANSEVS
QWRTKYETDAIQRTEEELEEAKKKLAQRLQDAEEHVEAVNSK CASLEKTKQR
LQNEVEDLMIDVERSNAACAALDKKQRNFDKVLAEWKHKYEETQAELEASQ
KESRSLSTEVFKVKNAYEESLDHLETLKRENKNLQQEISDLTEQIAESGKHIHE
LEKVKKQIDQEKSELQAAL EEAEGSLEHEEGKILRIQLELNQVKSEIDRKIAEK
DEEIDQLKRNHLRVVESMQSTLDAEIRSRNDALRIKKKMEGDLNEMEIQLNH
ANRQAAEAIKNLRNTQGILKDTQLHLDDAVRGQDDLKEQLA 68 Truncation of
LAQESTMDIENDKQQLEKLKKKEFEMTNLQSKIEDEQALAMQLQKKIKELQ Rabbit
G1SJN7 ARIEELEEEIEAERASRAKAQKRSDDLSELEEISERLEEAGGATSAQIEMNKK
(Myosin 4) REAEFQKMRRDLEEATLQHEATAAALRKKHADSV AELGEQIDNLQRVKQKL
EKEKSELKMEIDDLAGNMETVSKAKGNLEKMCRTLEDQLSEVKTKEEEHQR
LINELSAQKARLHTESGEFSRQLDEKDAMVSQLSRGKQAFTQQIEELKRQLEE
ETKAKSALAHALQSSRHDCDLLREQYEEEEQEAKAELGRAMSKANSEVSQWR
TKYETDAIQRTEEELEEAKKKLAQRLQDAEEHVEAVNSK CASLEKTKQRLQNE
VEDLMIDVERSNAACAALDKKQRNFDKVLAEWKHKYEETQAELEASQKESR
SLSTEVFKVKNAYEESLDHLETLKRENKNLQQEISDLTEQIAESGKHIHELEKV
KKQIDQEKSELQAAL EEAEGSLEHEEGKILRIQLELNQVKSEIDRKIAEKDEEI
DQLKRNHLRVVESMQSTLDAEIRSRNDALRIKKKMEGDLNEMEIQLNHANR
QAAEAIKNLRNTQGILKDTQLHLDDAVRGQDDLKEQLAMVERRANLMQAEI
EELRASLEQTERSRRVAEQELLDASERVQLLHTQNTSLINTKKKLETDISQIQG
EMEDIVQEARN AEEKAKKAITDAAMMAEELKKEQD TSAHLERMKKNMEQT
VKDLQHRLDEAEQLALKGGKKQIQKLEARVRELENEVESEQKRNV EAVKGL
RKHERRVKELTYQTEEDRKNVLRQLDLVDKLQSKVKAYKRQAEEAEEQSN 69
Truncation of
VFVADPKESFVKATVQSREGGKVTAKTEAGATVTVKEDQVFPMNPPKYDKI Rabbit
MYH4 EDMAMMTHLHEPAVLYNLKERYAAWMIYTYSGLFCVTVPYKWLVPVYNAE
(Myosin 4) VVTAYRGKKRQEAPPHIFSISDNAYQFMLTDRENQSILITGESGAGKTVNTKR
VIQYFATIAVTGDKKKKEEATSGKMQGTLEDQIISANPLLEAFGNAKTVRNDNS
SRFGKFIRIHFGTTGKLASADIETYLLEKSRVTFQLKAERSYHIFYQIMSNKKP
DLIEMLLITTNPYDYAFVSQGEITVPSIDDQEELMATDS AIDILGFTSDERSIY
KLTGAVMHYGNMKFKQKQREEQAEPDGTEVADKAAYLQSLNSADLLKALC
YPRVKVGNEYVTKGQTVQQVYNAV GALAKAVYEKMFLWMVTRINQQLDT
KQPRQYFIGVLDIAGFEIFDFNSLEQLCINFTNEKLQQFFNHMHMFVLEQEEYKK
EGIEWEFIDFGMDLAACIELIEKPMGIFSILEEECMFPKATDTSFKNKLYEQHL
GKSNNFQKPKPAKGKVEAHFSLVHYAGTVDY NITGWLDKNKDPLNETVVGL
YQKSAMKTLAFLFTGTAAAEAEAGGGKKGGKKKGSSFQTVSALFRENLNKLM
TNLRSTHPPHFVRCIIPNETKTPGAMEHELVLHQLRCNGVLEGIRICRKGFP SRIL
YADFKQRYKVLNASAIPEGQFIDSKKASEKLLGSIDVDHTQYKFGHTKVFFKA
GLLGLLEEMRDDKLAQLITRTQAMCRGFLARVEYKKMVERRESIFCIQYNIR
AFMNVKHWPWMKLYFKIKPLLKSAETEKEMANMKEEFEKTKESLAKAEAK
RKELEEKMVALMQEKNDLQLQVQAEADSLADAEERCDQLIKTKIQLEAKIKE
VTERAEDEEEINAELTAKKRKLEDECESELKKDIDDL ETLAKVEKEKHATENK
VKNLTEEMAGLDETI AKLTKEKKALQEAHQQTLDLQAEEDKVNTLTAKT
KLEQQVDDLEGSLEQEKKIRMDLERAKRKLEGDLKLAQESTMDIENDKQQLE
DEKLKKKEFEMSNLQSKIEDEQALAMQLQKKIKELQARIEELEEIEAERASR
AKAEKQRSDDLSELEEISERLEEAGGATSAQIEMNKKREAEFQKMRRDLEEA

TLQHEATLAATRAADSLAEQIDNLRVKKLEKEKSELKMEIDDLAS
NMETVSKAKGNLEKMCRTL 70 Truncation of
TLVSDVDYRNYLHQWTCMPDQNDVIQAKKAYELQSDAVYKADLEWLRGIG Rabbit
Nebulin WMPSDSVSVNHAKHAADIFSEKKYRTKIETLNFPTVDDRVDYVTAKQSGEIL
DDIKYRKDWNSIKSKYTLTETPLLHTVQEAARILDQYLYKESWEKQKATGYI
LPPDAVPFVHAHHSSDVQSELKYKAEHV KQRGHYVGVPTMRDDPKLVWFE
HAGQIQNDRLYKEDYHKTAKINIPADMVSVVAAKEGQNLVSDIDYRNYLH
QWTCHPDQNDVIQARKAYDLQSDNVYKADLEWLRGIGWIPLDSVDHVRVT
KNQEMVNQIKYKKDALDNYPNFTSVVDPPAIVLAKINAVNQSDVKYKETFN
KAKGKYTFSPDTPHISHSKDMGKLYSTILYKGAWEGTKAYGYTLDERYIPIVG
AKHAD FVNSELKYKETYEKQKGHYLAGK VISEFPGVVHCLDFQKMRSVLNY
RRDYENTKANVHIPNDMMNHVLAKKCQYILSDLEYRHYFHQWTS LPDEPKV
LLARNAQEILSDNVYKDDLNLWLKGIGCYVWDTPQILHAKKSYDLQSQLQYT
AAGKENLQNYNLVTDTPLYVTALQSGINASEVKYKENYHQIKDKYTTVLETV
DYDRTRNLKDLYSSNLYKEAWN KVKATSYILPTSTLSLTHAKNQKHLASHIK
YREEYEKFKALYTLPKSVEDDPNTARCLRVGKLNIDRLYRSVYEKNKMKIHI
VPDMVEMVTAKDTQKKVSEIDYRLHLHEWICH PDLQVNSHVRKVTDQISDIV
YKDDLNLWLKGIGCYVWDTP EILHAKHAYDLRNDIKYKAHMMKTRNDYKLV
TDTPVYVQAVKSGKQLSDAVYHHDYVHSVRGKVAPT TKTVDLERALHAYK
LQSENLYKTSRLTLPTGYRLPGDTPHFKHNDIRYMSSYFKYKEVYEHKAN
GYTLGPNDVPFVNVRRVNKVT SERLYRELYHKLKDKIH TTPDTPEIRQVKKT
QEA VSELIYKSDFFKMQGHMISLPYTPHVLHCRYVGDITS DIKYKEDLQILKG
LGCFLYDTPDMVRSRHLRKLWSNYLYTDNARKMRDKYKVVLDTPEYRKVQ
ELKTHLSELVYRASGRKQKSIFTSVPDTPDLLRAKRGQKLQSQYLYVELATKE
RPHHHAGNQTTALKHARDVKDMVSEKKYKVQYEKMKDKYTPIPDTPILIRA
KNAYWNASDLRYKET FQKTGKYHTVKDALDIVYHRKVTDDISKVKYKEN
YMRHLGIWRSIPDRPEHFHRAVTD AVSDIKYKEDLTWLKGIGCYAYDTPDF
TLAEKNKTLYSKYKYKEVFERTK SDFKYVADCPINRHF KYATQLMNEKKYR
ADYEQRKDKYQLVVDEPRHLLAKTAGDQISQIKYRKQYEKSKDKFTSIVDTP
EHLRTTKVNKQISDILYKLEYNKAKPRGYTTIHDT PMLLHVRKVKDEVSDLK
YKEVYQRNKS NCTIKPDAVHIKAAKDAYKVNTNLDYKKQYEANKAHWRWT
PDRPDFIQA AKSSLQQSDFEYKLDREFLKGCKLSVTDDKDMVLALKNSLIESD
LKYKEKHVKERGSCHAVPDTPQILLAKTVSNLV 71 Truncation of

EHTRNVNQIQSDNVYKDEYNNFFKGIGWIPIGSLEVEKVKKAGDALNEKKYR Rabbit
Nebulin QHPDTIKFTSVPDSMGMVLAQHNTKQLSDLNYKVEGEKVKHKYTMDPEVPQ
FIQARVNAFNLSDAH YKADWKKTLAKGYDLRPDAIPIVAAKSSRN IASDCKY
KEAYEKAKGKQIGFLSLQDDPKLVHYMNVAKIQSDREYRKGYEASKTRYHT
PLDMLSVTAAKKSQEVATNANYKQPFHNYTLLPDALPVEHSRNAMQIQSDN
LYKSDFTGWMKGIGWIPIQSLEVEKAKKAGEILSEKRYRQHPEK LKFTYAMD
TMEQALNKANKLNMDKRLYTKKW DDKTTIHVMPDTPDILLSRVNQITMSD
KLYRAGWEEEEKKGYDLRPDAISIKAAKASRD IASDYKYKQAYEQAKGKHI
GFRSLEDDPKLVHFMQVAKMQSDREYKKAYEKS KTSFHTPVDMLSVVAAK
KSQEVATNANYRNVIHTYNMLPDAMGFELAKNMMQIQSDN QYKADYADFM
KGIGWLPLGSLEAEKNKKAMEI ISEKKYRQH PDTLKYSTLMDSMNMVLAQN
NAKIMNEHLYKQAW EADKTKVHIMPDI PQIVLAKANAINMSDKLYKLSMEE
AKKKGYDLRTDAIPIKAAKASRD IASDYKYKYNYEKQKGKMVGFRSLEDDP
KLVHSMQVAKMQSDREYKKNYEKT KTSYHTPADMLSVTAAKDAQANITNT
NYKHLIHKYILLPDAMSLELSRNMNRIQSDNE YKQDYNEWYKGLGWSPAGS
LDVEKAKKATEYASDQKYRQHPSNFQFKKLTDSMDMVLAQNAHTMNKHL
YTIDWNKDKTKIHVMPDTPDILQAKQNQTLFSQKFYKLGWEEALKKGYDLP

VDAISVQLAKADISVQKQYGRSLQDDPKLVLSMNVA
KMQSEREYKKDFEKWKTKFSSPVDMLGVVLAKKCCQALVSDVDYKNYLHQ
WTCLPDQNDVIHAKKAYELQSENNMYKSDLEWLKGIGWSPLGSLEAEKNKRA
SEILSEKKYRQPPDTNKFSTIPDAMDIVLAKTNAKNRSDRLYREAWDKDKTQ
VHIMPDTPDILAKANLINTSDKLYRMGYEELKKKGYDLPVDAIPIKAAKASR
EIASEYKYKEGFRKQLGHHIGARNIKDDPKMMWSMHVAKIQSDREYKKDFE
KWKTKFSSPVDMLGVVLAKKCCQTLVSDIDYKNYLHQWTCLPDQNDVTHAR
HAYDLQSDNLYKSDLQWLKGIGWIPTGSIEDEKNKRATQILSDHVYRQHDPK
FKFSILMDSIPMVLAKNNATNMNHRLYTEAWDKDKTTVHIMPDTPEVLLAK
QNQVNCSEKLYKLGLEETKRKGYDMRVDAIPIKAAKASRDIASEFKYKEGYR
KQLGHHIGARAIHDDPKMMWSMHVAKIQSDREYKKDFEKWKTKFSSPVDML
LGVVLAKKCCQALVSDVDYKNYLHQWTCLPDQNDVMHARQAYDLQSDNLY
KSDLQWLRGIGWVPIGSLDVEKC 72 Truncation of
DVTHARHAYDLQSDNLYKSDLQWLKGIGWIPTGSIEDEKNKRATQILSDHVY Rabbit
Nebulin RQHDPKFKFSILMDSIPMVLAKNNATNMNHRLYTEAWDKDKTTVHIMPDTPEVLLAKQNQVNCSEKLYKLGLEETKRKGYDMRVDAIPIKAAKASRDIASEFK
YKEGYRKQLGHHIGARAIHDDPKMMWSMHVAKIQSDREYKKDFEKWKTKF
SSPVDMLGVVLAKKCCQALVSDVDYKNYLHQWTCLPDQNDVMHARQAYDL
QSDNLYKSDLQWLRGIGWVPIGSLDVEKCKKATEILSDKIYRQPPDRFKFTSV
TDSLEQVLAKNNAINMNKRLYTEAWDKDKTQIHIMPDTPEIMLARMNKINYS
ESLYKLANEEAKKKGYDLRSDAIPIVAASRDIAASYKYKDG YRKQLGHHI
GARDIKDDPKMMWSMHVAKIQSDREYKKDFEKWKTKFSSPVDMLGVVLAK
KCQTLVSDIDYKNYLHQWTCLPDQNDVMHARQAYDLQSDNIYKSDLQWLR
GIGWVPIGSLEVAKCKRAAEILSDNIYRQPPDKLKFTSV PDSLEQVLAKNNAIN
MNKRLYTEAWDKDKTQIHIMPDTPEIMLARQNKINYS ESLYKLANEEAKKKG
YDLRSDAIPIVAASRDIAASYKYKDG YRRQLGHHIGARDIKDDPKMMWS
MHVAKIQSDREYKKDFEKWKTKFSSPVDMLGVVLAKKCCQTLVSDIDYKNYL
HQWTCLPDQNDVMHARQAYDLQSDNIYKSDLQWLRGIGWVPIGSVDLVKC
KRAAEILSDNIYRQSPDKLKFTSV TDSLEQVLAKNNAINMNKRLYTEAWDKD
KTQIHMPDTPEIMLARQNMNYS ENFYRQAMEEAKKEGYDLRSDAIPIVA
AKASRDIAASYKYKEAYRKQLGHHIGARAIHDDPKMMWSLHIAKVQSDREY
KKEFEKYKTRYSSPVDMLGIVLAKKCCQTLVSDVDYKNLLHEWTCCLPDQNDV
IQARKAYDLQSDNLYKSDLEWLRGIGWVPIGSLEVVKAKRATEILSDNLYRT
RPDTLKFTSV TDSLEQVLAKNNAINMNKRLYTEAWENDKKTIVHMPDTPEIM
LAKLNKINYS DKLYKLAL EESKKEGYDLRVDAIPIQAAKASRDIAASYKYKE
GYRKQLGHHIGARNIKDDPKMMWSMHVAKIQSDREYKKDFEKWKTKFSSP
VDMLGVVLAKKCCQILVSDIDYKHPLHEWICLPDQNDVIQARKAYDLQSDAV
YKADLEWLRGIGWVPIGSVEVEKVKRAGEILSDRK YRQPADKLKFTSITDTPE
IVLAKSNALTM SKHLYTEAWDADKTSIHVMPDTPEILLAKSNSANISQKLYTK
GWDESKMKDYDLKADA IPIKSAKASRDIAASYKYKEAYEKQKGHHIGAQSIE
DDPKIMCAIHA 73 Truncation of
DKRETSRPNWAQVSANVPITSCSVEK LIEGHEYQFRICAENKYGVGDPVFTEP Rabbit
TTN AIAKNPYDPPGRCDPPVISNVTKDHMTVSWKPPADDGGSPITGYLLEKRETHA
VNWTKVNRKPV IERTIKATGLQEGTEYEFRTAINKAGPGKPSDASKAVYAAQ
DPLYPPGPPAFPKVYDTTRSSVSLSWGKPAYDGGSPIIGYLVEVKRADTDNW
VRCNLPQKLQKTRFEVTGLMENTEYQFRVYAVNKVGYSDPSDVPDKHCPKD
ILIPPEGELDADLRKTLILRAGVTMRLYVPVKGRPPPKITWSKPNVNL RERIGL
DIKSTDFDTFLRCENVNKYDAGKYILTLENSCGKKEYTIVVKVLDTPGPPVNV
TVKEISRDSAYITWDPPIVDGGSPIINYVVEKRD AERKSWSTVTTECSKTSFRV
SNLEEGKSYFFRVFAENEY GIGDPGETRDAVKASETPGPVVDLKVLT VTKSSC

NIGWKKPRSDYGGVDFLSTGYVDFLQVDMKSLSLQYSTKDLNEGKQY
TFRVSAENENGEGTPSEITVVAKDDVVAPDLDLKDLPLDCYLAKENSNFRLKI
PIKGKPAPSVTWKKGEDPLATDTRVSVESSAVNTTLVVYDCQKSDAGKYTIT
LKNVAGTKEGTLSEIKVVVGKPGIPTGPIKFDEVTAEAITLKWGPPKDDGGSEITN
YILEKRDSVNNKWVTCASAVQKTTFRVTRLHEGMEYTFRVSAENKYGVGEG
LKSEPIVAKHPFDVDPAPPPNIVDVRHDSVSLTWTDPRKTGGSPITGYHIEFK
ERNLLWKRANKTPIRMKDFKVTGLTEGLEYEFRVMAINLAGVGKPSLPSEP
VVALDPIDPPGKPEVINVTRNSVTLIWTEPKYDGGHKLTYIVEKRDLPSKTW
MKANHINVPDCAFTVTDLVEGGKYEFRIRAKNTAGAISAPSESTGTIICKDEYE
APTIVLDPTIKDGLTIKAGDTIVLNAISILGKPLPKSSWSKAGKDIRPSDITQITS
TPTSSMLTVKYASRKDAGEYTITATNPFGTKEEHVRVTVLDVPGPPGPIEISNV
SAEKATLTWTPPLEDGGSPIKSYVLEKRETSRLLWTVVAEDIQSCRHVVTCLI
QGNEYLFVSAVNHYGKGEPVQSEPVKMVDRFGPPGPPGKPEVSNVTKNTA
TVSWKRPTDDGGSEITGYVERREKKGLRWVRATKTPVSDLRCKVTGLQEG
NTYEFRVSAENRAGIGPPSDASNYVLMKDVAYPPGPPSNARVTDTTKKSASL
AWGKPHYDGGLEITGYVVEHQKVGDETWVKDTTGPALRITEFVVPDLHTKE
KYNFRISAINDAGVGEPVDPVEIVEREMAPDFELDAELRRTLVRAGLSIRI
FVPIKGRPAPEVTWTKDDINLKTRANIENTESFTLLIPECNRYDTGKFVMTIEN
PAGKKSGFVNVRVLDTPGPVLNLRPTDITKDSVTLHWDLPLIDGGSRTNYIV
EKREATRKSYSVTTKCHKCT 74 Truncation of
DKETPSGFTLDDVIQTGVDNPGHPFIMTVGCVAGDEESYTVFKDLFDPIIQDR Rabbit
KCRM HGGFKPTDKHKTDLNHENLKGDDLDPHYVLSSRVRTGRSIKGYTLPPHCSR
(Creatine GERRAVEKLSVEALNSLTGEFKGKYYPKSMTEQEQQLIDDHFLFDKPV SPL
Kinase M-LLASGMARDWPDARGIWHNDNKSFLVWVNEEDHLRVISMEKGGNMKEVFR
Type) RFCVGLQKIEEIFKKAGHPFMWNEHLGYVLTCPSNLGTGLRGGVHVKLAHLS
KHPKFEEILTRLRLQKRGTTGGVDTAAVGSVFDISNADRLGSSEVEQVQLVVD
GVKLMVEMEKKLEKQSIDDMIPAQK 75 Truncation of
GGGGGGEYMEQEEDWDRDLLDPAWEKQQRKTFTAWCNSHLRKAGTQIEN Rabbit
IEEDFRNGLKLMILLEVISGERLPRPDKGKMRFHKIANVNKALDFIASKGVKL G1U4H8
VSIGAEIIVDGNLKM TLMGIWTLRFAIQDISVEETSAKEGLLLWCQRKTAPY (Actinin
Alpha RNVNVQNFHTSWKDGLALCALIHRHRPDLIDYAKLRKDDPIGNLNTAFEVAE 3
KYLDIPKMLDAEDIVNTPKPDEKAIMTYVSCFYHAFAGAEQAETAANRICKV
LAVNQENELMEEYEKLASELLEWIRRTVPWLENRVGEPSMSAMQRKLEDF
RDYRRLHKPPRVQEKCLEINFNTLQTKLRLSHRPAFMPSEGKLVSDIANAW
RGLEQVEKGYEDWLLSEIRRLQRLQHLAEKFQQKASLHETWTRGKEDMLTQ
RDYESASLQEVRA LLRRHEAFESDLAAHQDRVEHIAALAEQELNELDYHEAAS
VNSRCQAICDQWDNLGTLTQKRRDALERMEKLLETIDQLQLEFARRAAPENN
WLDGAIEDLQDVWLVSVEETQSLLTAHDQFKATLPEADRERGAILGIQGEIQ
KICQTYGLRPSSTNPYINLSPQDINTKWDTVRKLVPSRDQTLQEELARQQMNE
RLRRQFAAQANAVGPWIQAKVEEVGRLAAGLAGSLEEQMAGLRQQEQNIIN
YKSNIDRLEGDHQLLQESLVFDNKHTVYSMEHIRVGWEQLLTSIARTINEVEN
QVLTRDAKGLSQEQLNEFRASFNHFDRKRNGMMEPDDFRACLISMYDLGEV
EFARIMTMVDPNAAGVVTFQAFIDFMTR 76 Truncation of
IYYLSLEFYMGRTLQNTMVNLALENACDEATYQLGLDMEELEEIEEDAGLGN Rabbit
PYGM GGLGRLAACFLDSMATLGLAAYGYGIRYEFGIFNQKICGGWQMEEADDWLR
(Glycogen YGNPWEKARPEFTLPVHFYGRVEHTSQGAKWVDTQVVLAMPYDTPVPGYR
Phosphorylase,
NNVVNTMRLWSAKAPNDFNLKDFNVGGYIQAVLDRNLAENISRVLYPNDFN Muscle
Form) FEGKELRLKQEYFVVAATLQDIIRRFKSSKFGCRDPVRTNFDAPDKVAIQLN
DTHPSLAIPELMRVLVDLERLDWDKAWEVTVKTCAYTNHTVLPEALERWPV

HLLETLRQLIYRINQRFLNRVAAAFPGDVEDRLRRMSLVEEGAVKRINMA
 HLCIAGSHAVNGVARIHSEILKKTIFKDFYELEPHKFQNKTNGITPRRWLVLCN
 PGLAEIIAERIGEEYISDLQRLKLLSYVDDEAFIRDVAKVKQENKLKFAAYLE
 REYKVHINPNSLFDVQVKRIHEYKRQLLNCLHVITLYNRIKKEPNKFVVPRTV
 MIGGKAAPGYHMAKMIKLITAIGDVVNHDPPVGDRLRVIFLENYRVSLAEK
 VIPAADLSEQISTAGTEASGTGNMKFMLNGALTIGTMDGANVEMAEAGEEN
 FFIFGMRVEDVDRLDQRGYNAQEYYDRIPELRQIIEQLSSGFFSPKQPDFKDI
 VNMLMHDRFKVFADYEEYVKCQERVSAKYKNPREWTRMVIRNIATSGKFS
 SDRTIAQYAREIWGVPEPSRQRLPAPDEKIP 77 Truncation of
 SHPALTPQKKELSAIAHRIVAPGKGILAADESTGSIKRLQSIGTENTEENRRF Rabbit
 ALDOA YRQLLLTADDRVNPCIGGVILFHETLYQKADDGRPFQVIKSKGGVVGIVDK
 (Fructose- GVVPLAGTNGETTTQGLDGLSERCAQYKKDGADFAKWRCVLKIGEHTPSAL
 Bisphosphate AIMENANVLARYASICQNGIVPIVEPEILPDGDHDLKRCQYVTEKVLAAVYK
 Aldolase A) ALSDDHHIYLEGTLLKPNMVTTPGHACTQKYSHEEIAMATVTALRRRTVPPAVTG
 VTFLSGGQSEEEASINLNAINKCPLLKPWALTFSYGRALQASALKAWGGKKE
 NLKAAQEEYVKRALANSLACQGKYTPSGQAGAAAS 78 Rabbit ALBU
 EAHKSEIAHRENDVGEEHFIGLVLITFSQYLQKCPYEEHAKLVKEVTDLAKAC (Albumin)
 VADESAANCDKSLHDIFGDKICALPSLRDITYGDVADCCEKKEPERNECFLHH
 KDDKPDLPFARPEADVLCFAHDDDEKAFFGHYLYEVARRHPYFYAPELLYY
 AQKYKAILTECCEAADKGACLTPLKDALEGKSLISAAQERLRCASIQKFGDRA
 YKAWALVRLSQRFPKADFTDISKIVTDLTKVHKECCHGDLLECADDRADLAK
 YMCEHQETISSHLKECCDKPILEKAHCIYGLHNDETTPAGLPAAVEEFVEDKDV
 CKNYEEAKDLFLGKFLYEYSRRHPDYSVVLRLGKAYEATLKKCCATDDP
 HACYAKVLDEFQPLVDEPKNLVKQNCELYEQLGDYNFQNALLVRYTKKVPQ
 VSTPTLVEISRSLGKVGSKCKHPEAERLPCVEDYLSVVLNRLCVLHEKTPVS
 EKVTKCCSESLVDRRPCFSALGPDETYVPKEFNAETFTFHADICTLPETERKIK
 KQTALVELVKHKPHATNDQLKTVVGFEFTALLDKCCSAEDKEACFAVEGPKL VESSKATLG

[0041] An animal protein can comprise a cow protein (Table 3). A cow protein can comprise a cow muscle protein. A cow protein can comprise a cow blood protein. A cow protein can comprise a cow albumin protein. A cow protein can comprise a sequence that has at least or about 70%, at least or about 75%, at least or about 80%, at least or about 90%, at least or about 91%, at least or about 92%, at least or about 93%, at least or about 94%, at least or about 95%, at least or about 96%, at least or about 97% at least or about 98% at least or about 99% or about 100% sequence identity to any one of SEQ ID NO: 79-93.

[0042] A food product may comprise a fragment or a truncation of a cow protein. The food product may include a fragment or truncation of any one of SEQ ID Nos: 79-93. A sequence of the fragment or truncation may be identical or substantially identical to a portion of any one of SEQ ID Nos: 79-93. A sequence of the fragment or truncation may be at least about 70%, 80%, 90%, 95%, 98%, or more identical to a portion of a sequence of any one of SEQ ID Nos: 79-93. A length of the fragment or truncation may be at least about 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or more of a length of any one of SEQ ID Nos: 79-93. A length of the fragment or truncation may be less than or equal to about 90%, 80%, 70%, 60%, 50%, 40%, 30%, 20%, 15%, 10%, or less than a length of any one of SEQ ID Nos: 79-93.

TABLE-US-00003 TABLE 3 Cow Proteins SEQ ID NO Description Amino Acid
 Sequence 79 Cow KCRM
 MPFGNTHNKHKLNFKAEEEEYPDL SKHNNHMAKALTLEIYKKLRDKETPSG (Creatine
 kinase FTLDDVIQTGVDNPGHPFIMTVGCVAGDEESYTVFKDLFDPIIQDRHGGFKP M-type)
 TDKHKTDNLNHNKLGDDLDPNYVLSSRVRTGRSIKGYALPPHCSRGER
 AVEKLSVEALNSLTGEFKGKYYP LKSMTEQEQQQLIDDHFLFDKPVSPLLL
 ASGMARDWPDARGIWHNDNKSFLVWVNEEDHLRVISMEKGGNMKEVFR

RFCVGLQKIEEIFKKAGHPFMWNEHLYGVLTCPSNLGTGLRGGVHVKLAH
LSKHPKFEEILTRLRLQKRGTTGGVDTAAVGSVFDVSNADRLGSSEVEQVQL
VVDGVKLMVEMEKKLEKGQSIDDMIPAQK 80 Cow ACTN3
MMMVLQPEGLGTGEGPFAGGRGGGEYMEQEEDWDRDLLDPAWEKQQR (Alpha-actinin-
3) KTFTAWCNSHLRKAGTQIENIEEDFRNGLKLMLLLEVISGERLPRPDKGKM
RFHKIANVVKALDFIASKGVKLVSIGAEIIVDGNLKM T LGM IWTIILRFAIQ
DISVEETSAKEGLLLWCQRKTAPYRNVNVQNFHTSWKDGLALCALIHRHR
PDLIDYAKLRKDDPIGNLNTAFEVAEKYLDIPKMLDAEDIVNTPKPDEKAIM
TYVSCFYHAFAGAEQAETAANRICKVLAVNQENEKLMEEYEKLASELLEW
IRRTVPWLENRVGEPSSMSAMQRKLED FRDYRRLHKPPRVQEK CQLEINFNT
LQTKLRLSHRPAFMPSEGKLVSDIANAWRGLEQAEKGYEDWLLSEIRRLQR
LQH LA EK FQ QKASLHEAWTRGKEDMLSQRDYETASLQEV RALLRRHEAFE
SDLA AHQDRVEHIAALAAQELNELDYHEAASVNSRCQAICDQWDNLGTLTQ
KRRDALERMEKLLETIDQLQLEFARRAAPFNWLDGAVEDLQDVWLVS
VEETQSLVTAHDQFKATLPEADRERGAILGIQGEIQKICQTYGLRPSSTNPYI
TLTPQDINTKWDTVRKLVPSRDQMLQEELTRQQVNRLRRQFAAQANAIG
PWIQGKVEEVGRLAAGMAGSLEE QMAGLRQQEQNIINYKSNIDRLEGDHQ
LLQESLVFDNKHTVYSMEHIRVGWEQLLTSIARTINEVENQVLTRDAKGLS
QEQLNEFRASFNHFD RKRNGMMEPD DFRACLISMGYDLGEVEFARIMTMV
DPNAAGVVTFQAFIDFMTRETAETDTAEQVVASFKILAGDKNYITAEELRR
ELPAEQAEYCI RRM APYKGAGAPAGALDYVAFSSALYGESDL 81 Cow PYGM
MSRPLTDQEK RKQISVRGLAGVENVT ELKKNFNRHLHFTLVKD RNVATPR (Glycogen
DYYFALAYTVRDHLVGRWIRTQQHYE KDPKRIYYLSLEFYIGRTLQNTM phosphorylase,
VNLALENACDEATYQLGLDMEELEEIEEDAGLGNGGLGRLAACFLDSMAT muscle form)
LGLAAYGYGIRYEF GIFNQKISGGWQMEEADDWLRYGNPWEKARPEFTLP
VHFYGRVEHTSQGAKWVD TQVVLAMPYDTPVPGYRNNVVNTMRLWSAK
APNDFNLKDFNVGGYIQA VLD RNL AENISRVLYPN D NFFEGKELRLKQEYF
VVAATLQDIIRRFKSSKFGCLDPVRTNFDAFPDKVAIQLNDTHPSLAIP ELM
RILVDQERLEWEKAW EVTVKTCAYTNHTVLPEALERWPVHLIETLLPRHLQ
IIEINQRFLNRVAAAFPGD VDR LRRMSLVEEGAVKRINMAHL CIAGSHAV
NGVARIHSEILKKTIFKDFYELEPHKFQNK TNGITPRRWLVMCNPGLAEIIAE
RIGEEYIADLDQLRKLLSYVDDES FIRDVAKVKQENKLKFSAYLEKEYKVHI
NPNSLFDIQVKRIHEYKRQLLNCLHVITLYNRIKKEPNKFFVPRTVMIGGKA
APGYHMAKMIKLITAIGDVVNHD PVVGDRLRVIFLENYRVSLAEKVIPAA
DLSEQISTAGTEASGTGNMKFMLNGALTIGTMDGANVEMAE EAGEENFFIF
GMRVEDVERLDQKGYNAQEYYDRIPELRHVIDQLSSGFFSPKQPDLFKDIV
NMLMHHDRFKVFADYEEYIKCQERVSALYKNPREWTRMVIRNIATSGKFS
SDRTIAQYAREIWGVEPTRQRMPAPDEKI 82 Cow A6QLL8
MPHQYPALTPEQKKELCDIAHRIVAPGKGILAADESTGSI AKRLQSIGTENTE (Fructose-
ENRRFYRQLLL TADDRVNPCIGGVILFHETLYQKADDGRPPQVIKAKGGV bisphosphate
VGIKVDKGVVPLAGTNGETTTQGLDGLSERCAQYKKDGADFAKWRCVLKI aldolase)
GEHTPSSLAIMENANVLARYASICQQNGIVPIVEPEILPDGDHDLKRCQYVT
EKVLA AVYKALSDHHIYLEGTLLKPNMVTPGHACTQKYSHEEIAMATVTA
LRRTVPPAVPGITFLSGGQSEEEASINLNAINKCPLLKPWALTFSYGRALQAS
ALKAWGGKKENL KAAQEEYVKRALANSLACQGKYTPSGKAGAAASESLFI SNHAY 83
Cow MYL1 MAPKKDVKKPAAAAAPAPAPAPAPAPAPAPPKEEKIDLSAIKIEFSKQQQDE
(Myosin light
FKEAFLLFDR TGECKITLSQVGDVLRALGTNPTNAEVKKVLGNPSNEEMNA chain 1/3,
skeletal KKIEFEQFLPMLQAISNNKDQGT YEDFVEGLRVFDKEGNGTVMGAELRHV
muscle isoform) LATLGEKMKEEEVEALMAGQEDSNGCINYEAFVKHIMSN 84 Cow G3P

MVKVGVNGVGRIGRLVTRAAFN SGKVDIVAINDPFIDLHYMVYMFQYDST
(Glyceraldehyde-
HGKFNGTVKAENGKLVINGKAITIFQERDPANIKWGDAGA EYVVESTGVFT 3-phosphate
TMEKAG AHLKGGAKRVIISAPSADAPMFVMGVNHEKYNN TLKIVSNASCT dehydrogenase)
TNCLAPLAKVIHDHFGIVEGLMTTVHAITATQKTVDGPSGKLWRDGRGAA
QNIIPASTGAAKAVGKVIPELNGKLTGMAFRVPTPNVSVVDLTCRLEKPAK
YDEIKKVVKQASEGPLKGILGYTEDQVVSCDFNSDTHSSTFDAGAGIALND
HFVKLISWYDNEFGYSNRVVDLMVHMASKE 85 Cow TPM1
MDAIKKKMQLKLDKENALDRAEQAEADKKA AEDRSKQLEDELVS LQKK (Tropomyosin
LKATEDEL DKYSEALKDAQE KLELA EK KATDAEADVASLNRR IQLVEEEL alpha-1 chain)
DRAQERLATALQKLEEA EKAADESERGMKVIESRAQKDEEKMEIQEIQLKE
AKHIAEDADRKYEEVARKLVIIESDLERA EERAELSEGKCAELEEEELKTVTN
NLKSLEAQAEKYSQKEDKYEEEIKVLSDKLKEAETRAEFAERSVT KLEKSID
DLEDELYAQKLKYKAISEELDHALNDMTSI 86 Cow E1BNV1
MPEAKPAAKKAPAGKDAAAKPAPKEATPQEAPAAPTEAPPEDQSPTAE EP (Myosin
binding TGVFLKKPDSVSVETGKDTVIVAKLNGKELPAKPAVKWFKGKWLELGSKS
protein C2) GARFSFKESHDAASN VYTIELHITKVVLGDRGDYRIEVKAKDFCDSCAFNID
VEAPRQDSAGQSLESFKRSGEAKSDTAGELDFSGLLKKRQVV EEEKKKKK
DDDDL GIPPEIWELLKGAKKSEYERIAFYGITDLRGMLKRLKKAKVEVKK
SAAFTKKLDPAYQVDRGNKIKLVVEISDPDLPLKWFKNGQEIKPSSKYVFE
NVGKKRILTINKCTLADDAAYEVVKDEKCFTEL FVKEPPVLIVTPLEDQQ
VFGDRVEMAVEVSEEGAQVMWLKDGVELTRED SFKARYRLKKDGKRHI
LIYSEVTMEDKGHYQVMTNGGQCEALIVEEKQLEVLQDIADLTVKASEQ
AVFKCEVSDEKVTGKWKNGVEVRPSKRITISHTGRFHKLVIDDVRPEDEG
DYTFVPDGYALSLSAKLK FLEIKVEYVPKQEPPKIHLDCSGQTSENAIVVVA
GNKL RMDVSITGEPRPVATWMKEDEVFTGT EGRVRIEQRGDISSFVIESAER
GDEGRYTIKVTNPVGEDVASILLKVVDVPDPPEAVRVT SVGEDWAILVWEP
PKYDGGRPVTGYLLERKKKGSQRWMKLNFEVFTETTYESTNMIEGILYEM
RVFAVNVIGVSQPSMNTKPFMPIAPTSEPQH LTVEDVTDTTTTLKW RPPDKI
GAGGIDGYLIEYCVESGSEWIPANTEPVERCGFTVKNLPTGAKITFRVVG V
NIAGRSQPATLAQPVTIREIVEQPKIRLPRHLRQTYVRKVGEHLNLVIPFQ GK
PRPQVVWTKGGAPIDTSHVHVRTSDFDTVFFVRQAARSDSGEYELTVQIEN
MKDTATVCIRVVEKAGPPQNMVKEVWGTNALVEWQPPKDDGNSEITGY
FVQKADKKTMEWFTVYERNRHTSCTVSDLIMGNEY YFRVYSENVCGLS DL
PGVSKNTARIVKTGMTLKLPEYKEHDFRTPPKFLTPLPDRVVVAGYAAALN
CAVRGHPKPKVVMKNKMEIREDPKFLMTNQQGVLT LNIRRPSPFDSGT Y
SCRAVNELGEALAECKLEVRVPQ 87 Cow E1BF23
MSLVAVPFYQKRHKHFDQSYRNIQTRYLLDEYS AKKKRASAQSSLQRSVT (Myomesin 2)
RKSSSQRESSRAALGESTLGETTLGETSLGDTTLGR TTCRLCAKRMSSALEE
EAQERKRRYQSMVAAYGEAKRERYLSELAQLEEDVHVARTHARDQLDRL
ALQHAGRHLHHA VDDRLAWERHSFEERISRAPEILVRLRSHTVWERM SV
KLCFTVQGFPTPVVQWYKDGS LICQAGEPGKYRIDSKYGVHTLEINRADFD
DTATYS AVATNVHGVSTNAAVVRRFRGDEEPFHSVGLPIGLPLSSVIPYT
HFDVQFIEKFGVTFRREGETLTLKCTLLVTPDLKRVQPRAEWYRDDVLLKE
SKWTKMFFGEGQASLSF SHLNKDDEGLYTLRIVSRGGITDHSAFLFVRDAD
PLVTGAPGAPMDVHCHDANRDYVIVTWKPPNATKESPVIGYFIDRCEVGTD
NWIQCNDAPVKICKYPVTGLFEGRSYIFRVRAVNSAGISRPSRVSEAVAALD
PVDLRR LQAVHLEGEKDIVVYQEELEGEVQIPGPPTNVHASSETSRTYVVLS
WDPPEPRGKEPLMYFIEKSMVGS GSWQRVNAQTAVRSPRYAVFDLAEGKP
YVFRVLSANKHGLSDPSEITAPIQAQDTIVVPSAPGRVLASRNTRTSVVVQW

DRPKHEEDYGYNDRCVHGLTTFGEQYIF
RVKAVNAVGTSENSQESDVIKVQAALTVPSHPYGITLLNCDGHSMILGWK
VPKFSGGSPILGYYVDKREAAHKNWHEVNSSPLKERILTVEGLTEGSLYEF
KIAAANMAGIGQPSDPSSELFKCEAWTMPEPGPAYDLTFCEVRDTSVLVLLWK
PPVYSGSSPVSGYFVDYKEEDSGEWLTVHETATPHRYLKVCDLHQGKTYV
FRVRAVNASGVGRPSDTSEPVLEARPGTKEVSAGVDEEGNVYLSFDCPEV
TDASHFTWCKSYEDIADDDRFRKVETAGDHSKLYFKNLDKEDLGTYSVSVS
DTDGVSSSFVLDEEELERLKALSNEIKNPTIPLKSELAYEIFDKGQVRFWLQ
AEHLSPDSSYRFIINDREVADSETHRIKCDKSTGIEMVMMDRFTIDNEGTYTV
QIQDGKAKSQSSLVLIGDAFKAVLKEAEFQRKEFLRKQGPHFAEYLHWDVT
EECEVRLVCKVANTKKETVFKWLKDDVLYETEKMPDLEKGVCELLIPKLS
KKDHGEYKATLKDDRGQDVSTLEIAGKVYEDMILAMSRVCGASASPLKIL
CTPEGIRLQCFMKYFTEEMKVNWYHKDAKITSSEHMRIGGNEQMAWLQIC
EPTEKDKGKYTFEILDGKKNHQRSDDLQAFDEAFAEFQQLKAAAFAEKN
RGKVIGGLPDVVTIMEGKTLNLTCTVFGNPDPEVVWFKNDKDIELSDHFSV
KVEQGKYVSMTIKGVTSSEDSGKYSIHVKNKYGGEKIDVTVSVYKHGEEIPD
VVLPQQAQPKLLPAAAPSPAH 88 Cow ENOB

MAMQKIFAREILDSRGNPTVEVDLHTAKGRFRAAVPSGASTGIYEALRLD (Beta-enolase)
GDKSRYLGKGVLKAVEHINKTLGPALLEKKLSVVDQEKVDKFMIELDGTE
NKSKFGANAILGVSLAVCKAGAAEKGVPLYRHIADLAGNPELILPVPFNV
NGGSHAGNKLAMQEFMILPVGASSFREAMRIGAEVYHHLKGVKAKYVKD
ATNVGDEGGFAPNILENNEALELLKTAIQAAAGYPDKVVIGMDVAASEFYRN
GKYDLDFKSPDDPARHISGEKLGELYKNFIKNYPVVSIEDPFDQDDWATWT
SFLSGVNIQIVGDDLTVTNPKRIAQAVEKKACNCLLLKVNQIGSVTESIQAC
KLAQSNGWGVMVSHRSGETEDTFIADLVVGLCTGQIKTGAPCRSERLAKY
NQLMRIEALGDKAVFAGRKFRNPKAK 89 Cow MLRS

MAPKKAKRRAAAEGGSSSVFSMFDQTQIQEFKEAFTVIDQNRDGIIDKEDL (Myosin
RDTFAAMGRLNVKNEELDAMMKEASGPINFTVFLNMFGEKLGADPEDVI regulatory
light TGAFKVLDPEGKGTIKKKFLEELLTTQCDRFSQEEIKNMWAAFPDPVGGNV chain 2)
DYKNICYVITHGDAKDQE 90 Cow MYH1

MSSDQEMAVFGEAAPYLKSEKERIEAQNKPFDAKTSVVFVADPKESFVKAT (Myosin-1)
VQSREGGKVTAKTEAGATVTVKEDQVFPMNPPKFDKIEDMAMMTHLHEP
AVLYNLKERYAAWMIYTYSGLFCVTVNPKWLPVYNAEVVTAYRGKKRQ
EAPPHIFSISDNAYQFMLTDRENQSILITGESGAGKTVNTKRVIQYFATIAVT
GEKKKEEPTSGKMQGTLEDQIISANPLLEAFGNAKTVRNDNSSRFGKFIRIH
FGTTGKLASADIETYLLEKSRVTFQLKAERSYHIFYQIMSNKKPELIEMLLIT
TNPYDYAYVSQGEITVPSIDDQEELMATDSAIEILGFTSDERSIYKLTGAV
MHYGNLKFQKQKREEQAEPDGTEVADKAAYLQGLNSADLLKALCYPRVK
VGNEFVTKGQTVEQVYNAV GALAKAVYDKMFLWMVARINQQLDTKQPR
QYFIGVLDIAGFEIFDFNSLEQLCINFTNEKLQQFFNHMHMFVLEQEEYKKEGI
EWEFIDFGMDLAACIELIEKPMGIFSILEEECMFPKATDMSFKNKLYEQHLG
KSNNFQKPKPAKGKAEAHFSLIHYAGTVDYNITGWLDKNKDPLNETVVGL
YQKSSVKTLALLFSGPASGEAEGGPKKGGKKKGSSSFQTVSALFRENLNKLM
TNLRSTHPPHFVRCIIPNETKTPGAMEHELVLHQLRCNGVLEGIRICRKGFP
SRILYADFKQRYKVLNASAIPEGQFIDSKKASEKLLASIDVDHTQYKFGHTKVF
FKAGLLGLLEEMRDEKLAQLITRTQARCRGFLARVEYQKMVERRESIFCIQ
YNVRAFMNVKHWPWMKLYFKIKPLLKSAETEKEMANMKEEFEKTKEELA
KSEAKRKELEEKMTLTQEKNDLQLQVQSEADALADAEERCDQLIKTKIQL
EAKIKEVTERAEDEEEINAELTAKKRKLEDECSELKKDIDDLELTAKVEKE
KHATENKVKNLTEEMAGLDETIAKLTKEKKALQEAHQQTLDLQAEEDK

VNTLTKAKTKTQVDDLEQVDDLEQKLEKRLMDLERAKRKLKGLKLAQEST
MDIENDKQQLDEKLKKKEFEMSNLQSKIEDEQALAMQLQKKIKELQARIEE
LEEEIEAERASRAKAEKQRSDLSRELEEISERLEEAGGATSAQIEMNKKREA
EFQKMRRDLEEATLQHEATAAALRKKHADSVaelGEQIDNLQRVKQKLEK
EKSEMMEIDDLASNMETVSKAKGNLEKMCRALEDQLSELKTKEDEQQRL
INDLTTQRARLQTESGEFSRQLDEKDALVSQLSRGKQAFQTQQIEELKRQLEE
EIKAKSALAHALQSARHDCDLLREQYEEEEQEGKAELGRAMSKANSEVAQ
WRTKYETDAIQRTEELEEAKKKLAQRLQDAEEHVEAVNAKASLEKTKQR
LQNEVEDLMIDVERTNAACAALDKKQRNFDKILSEWKQKYEETHAELEAS
QKESRSLSTELFKIKNAYEESLDQLETLKRENKNLQQEISDLTEQIAEGGKRI
HELEKVKKQVEQEKSEIQAALAEAASLEHEEGKILRIQLELNQVKSEIDRKI
AEKDEEIDQLKRNHIRIVESMQSTLDAEIRSRNDAIRLKKKMEGDLNEMEIQ
LNHANRMAAEALKNYRSTQAILKDTQIHLDDALRGQEDLKEQLAMVERR
ANLLQAEIEELRATLEQTERSARKIAEQELLDASERVQLLHTQNTSLINTKKK
LETDITQIQGEMEDIIEARNAEKAKKAITDAAMMAEELKKEQDTSAHLE
RMKKNLEQTVKDLQHRLDEAEQLALKGGKKQIQKLEARVRELEGEVESEQ
KRNVEAVKGLRKHERRVKELTYQTEEDRKNILRLQDLVDKLQAKVKS YKR
QAEAEQSNVNLSKFRKLQHELEEAERADIAESQVNKL RVKSREVHTKII SEE 91 Cow
MPNFQTRTRFYETTTTTRTSDYEQSETSRPALAQPVPEKPVVERKRVIRKKVDP A0A3Q1M707
SKFMTPTYIAHSQKMQNLFSTNKYKENYEKAKGKPYAITTDTPELRRIKKVQ (Nebulin)
DQLSEVKYRVDGDVAKTICHVDEKAKDIEHAKKVSQQVSKVLYKQNWED
TKDKYLLPPDAPELVQAIKNTAMFSKKLYTEDWEADKTMFYYPYNDSPELR
RVAQAQKALS DIAYKKGLAEQQTQFTSLPDPDIEFAKKVTNQVSKQKYKE
DYENKVKGKWSETPCFEIATARMNSNNISTKKYQEDFEHMKDQIYFMQTE
TPEYKVNKQAGVAASKVKYKQDYEKNKGKADYNVLPASENPLLRLKAA
GDALSDKLYKENYEKTKAKSINYCETPKFKLDTVLHNFSSDTKYKDSYLKD
ILGHYVGSYEDPYHTHCMRVSAQNSDKNYKAEYEEDRGKGFFPQTITQEY
EAIKKLDQCKDHTYKVHPDKTKFTQVTDSPVLMQAQVNSKQLSDLNYKA
KHENEKFKCHIPPDTPAFIQHKL NAYNLSDN IYKHDWEKTKAKKFDIKVDA
IPLLA AKANTKIASDV MYKKDY EKSKGKMIGALSINDDPKMLHSLKTAKN
QSDRLYKENYEKTKAKSMNYCETPKYQLDTLLKNFSEAKYKDSYVQNVL
GHYIGSFEDPYQVHCLKISAQNSDKNYKAEYEEDKGKCYFPQTITQEY EAIK
KLDQCKDHTYKVHPDKTKFTAVTDTPVLLQAQLNTKQLSDLNYKAKHEG
EKFKCHIPADAPQFIQHRVNAYNLSDN IYKHDWEKTKAKKFDIKVDAIPLL
AAKANTKIASDV MYKKDY EKSKGKMIGALSINDDPKMLHSLKTAKNQSD
HEYRKDY EKSKTIYTAPLDMPLTHAKKSQAIASDV DYKHLLHNYSYPPDS
VNVDLAKKAYALQSDVEYKADYN SWMKGCGWMPFGSLEMEKAKRASDI
LNEKKYRQH PDTLKF TSIEDAPIIVQSKINQAQRSDVAYKAKGEEVIHKYSL
PADLPQFIQAKVNAYNISENLYKADLKDLSKKGYDLRIDAPIKA AKAARQ
AASDVQYKKDY EKAKGKMVG FQSLQDDPKLVHYMNVAKIQSDREYKKA
YEKTKTRHNTPHDMVNIVA AKKAQDVASNVNYKHSLH HYTYLPDAMDLE
LSKNMMHIQSDN VYKEDYNNWMKGIGWIPIGSLEVEKVKKAGDALNEKK
YRQH PDTLKF TSI VDSPVMVQAKQNTQQVSDILYKAKGEDVKHKYTMSPD
LPQFLQAKCNAYNLS DVCYKRDWHDLIAKGTNVLGDAIPITAAKASRN IAS
DYKYKEAYEKAKGKQVGFRSLQDDPKLVHYMNVAKLQSDREYKKNYEN
TKTSYHTPGDMVSITAAKMAQDVATNVNYKQPIHHYTYLPDALSLEHIRN
VNQIQSDN VYKDEYNHFFKGMGWIPIGSLEVEKVKKAGDALNEKKYRQHP
DTIKFTSV PDSMGMVLAQHNTKQLSDLNYKVEGEKVVKHKYTM DDPV PQFI
QAKVNAYNMSDSHYKADWKKTLAKGYDLRPDAIPIVA AKSSRN IASDFKY
KEAYEKTGKGQIGFRSLQDDPKLVHFMNVAKMQSDREYKKG YEASKTKY

HTPLDMLSVTA AKKQSEVATNANYRNVIHTYNMLPDAMSLELAKNMMQIQSDNQYK
SDNLYKSDFTNWMKGIGWLPLESLEVEKAKKAGEILSEKKYRQHPEKLKFT
YAMDTMEQALNKS NKLIMDKRLYTEKWNKDKTTHVMPDTPDILLSRVNQ
ITMSDKLYKAGWEEEEKKKGYDLRPDAISIKAAARASRDIA SDYKYKQAYEQ
AKGKQIGFRSLEDDPKLVHFMQVAKMQSDREYKKAYEKS KTSFQTPVDML
SVVAAKKSQEVATNANYRNVIHTYNMLPDAMSLELAKNMMQIQSDNQYK
ADYADFMKGIGWLP LGSLEAEKNKKAMEI ISEKKYRQH PDTLKYSTLMDS
MNMVLAKNNAKIMNEHLYKQAW EADKTKVHIMPDI PQIILAKANAINISD
KLYKLSLEEAKKKGYDLRTDAIPIKA AKA SRDIA SDYKYKHSY EKERGKM
VGFRSLEDDPKLVHSMQVAKMQSDREYKKNYEKT KTSYHTPADMLS VTA
AKDAQANITNTNYKHLIHKYILLPDAMNIQLSKNMNRIQSDNEYKQDYNE
WYKGLGWSPAGSLEVEKAKKATEYASDQKYRQHPSNFQFTKL NDSMDMV
LAKQNAHTMNKYLYTVDWNKDKT KIHVMPDTPDILQAKQNQTMYSQKL
YKLGWEEALKKGYDLPVDAISVQAAKASRDIVSDFKYKQGYRKQIGHIG
FRSLQDDPKLVLSMNVAKMQSEREYKKDFE KWKTKFSSPVDMLGVVLAK
KCQALVSDVDYKNYLHGWTCLPDQNDVIHAKKAYDLQSENLYKSDLEWL
KGIGWSPLGSLEAEKNKRASEI ISEKKYRQPPDRNKFTSIPDAMDIVLAKTN
AKNRSDILYREAWDKDKTQIHIMPDTPD IILAKANLINTSDKFYRMGYEELR
KKGYDLPVDAIPIKA AKA SREIASEYKYKEGFRMQLGHHIGARNIKDDPKM
MWSMHVAKIQSDREYKKDFE KWKTKFSSPVDMLGVVLAKK CQTLVSDID
YKNYLHQWTCLPDQSDVIHARRAYDLQSDNLYKSDLQWLRGIGWLPSGSL
EDEKNKRASQILSDHVYRQH PDQFKFSSLMDSIPMVLAKNNAITMNHRLYT
EAWDKDKTTVHIMPDTPEVLLAKQ NQIN YSEKLYKLGLDEAKRKGYDMRI
DAIPIRAAKASRDIA SEFKYKEGYRRQLGHHIGARAIHDDPKMMWSMHVA
KIQSDREYKKDFE KWKTKFSSPVDMLGVVLAKK CQTLVSDIDYKNYLHQ
WTCLPDQSDVIHARQAYDLQSDNVYKSDLQWLRGIGWVPIGSLDVEKSKR
ASEILSDKLYRQPPDKFKFTSVTDSLEQVLAKNNAINM NKRLYTEAWDKD
KTQVHIMPDTPEITLARTNKVNYSENLYKLAHEEAKKKGYDLRSDAIPIVA
AKASRDII SDYKYKDG YRKQLGHHIGARDIKDDPKMMWSMHVAKIQSDRE
YKKDFE KWKTKFSSPVDMLGVVLAKK CQTLVSDIDYKNYLHQWTCLPDQ
SDVIHARQAYDLQSDNVYKSDLQWLRGIGWVPIGSVDVVKCKRAAEILSD
NLYRQPPDTLKF TSV PDSLEQVLAKNNAINM NKRLYTEAWDKDKTQIHIMP
DTPEIMLARQNKLNYSETLYKL ANEEAKKKGYDLRSDAIPIVA AKA SRDIIS
DYKYKDG YRKQLGHHIGARDIKDDPKMMWSMHVAKIQSDREYKKDFE K
WKTKFSSPVDMLGVVLAKK CQTLVSDIDYKNYLHQWTCLPDQSDVIHARR
AYDLQSDNIYKSDLQWLRGIGWVPIGSVDVVKCKRAAEILSDNLYRQR PDT
LKFTSV PDSLEQVLAKNNAINM NKRLYTEAWDKDKTQIH MMPDTPEITLA
RQNKIN YSENLYRQAMEEAKKEGYDLRSDAIPIVA AKA SREIASDYKYKEA
YRKQLGHHIGARAIHDDPKMMWSVHVAKMQSDREYKKDFE KYKTRFSSP
VDMLGIVLAKK CQTLVSDVDYKHPLHEWTCLPDQNDIIHARKAYDLQSDN
LYKSDLEWLKGIGWVPIGSVEVLKAKRAGEILSDNIYRQR PDTLKFTSVTDS
PEQVLAKNNAINM NKRLYTEAWDNDKKTTHVMPDTPEIMLAKLNRINYS D
KLYKLAL EESRKEGYDLRLDAIPIQA AKA SREIASDYKYKEGYRKQLGHHI
GARNIKDDPKMMWSIHAGKLQSDLEYKKDFE KWKTKFSSPVDMMGLVQA
KKCQILVSDIDYKNLLHEWTCLPDQNDIIQARKAYDLQSDAIYKADLEWLR
GIGWVPIDSVGVEHAKRAGEILSERKYRQPAVQLKFTSITDTPEIVLAKNNA
LNVSKHLYTEAWDADKTSIHVMPDTPEILLAKSNSANISHKLYTKGWDESK
MKDYDLRAD AISIKSAKASRDIA SDYKYKEAYEKHKGHHIGARSVEDDPRI
MCAMNAGRIQSEREYKKEFQKWKTKFSSPVDMLGILLAKK CQTLVSDIDY
RNYLHHWTCLPDQNDIIQARKAYDLQSDNVYKSDLEWLRGIGWMPEGSVE

MNRVKN AQD LN KANSLQMTPEV V LAKANS LQ M SEKLYQE
AWN KDKTNISIPSDT PLILQAQINAMQISNKLYQKDWDEAKQKGYDLRAD
AIEIKHAKASREIASEYKYKEGYRKQLGHHVGFRTLQDDPKLVWSIHA AKI
QSDREYRKAYEKS KGIHNTPLDMMSIVQAKKCQVLVSDIDYRNYLHQWIC
LPDQNDVIQAKKAYELQSDAIYKADLEWL RGIGWMPEGSPEVLRV KNAQH
IFRDSVYRTPVVKLKYTSIVDTPEVVLAKSNAENISIPKYREVWDKDKTSIHI
MPDTPEINLARTNALNVSNKLYREGWDEMKMSCDVRLDAIPIQAAKASREI
ASDYKYKLDHEKQKGHYVGTRTARDDNKIRWALIAGKIQNEREYRLHWA
KWKSKFQSPDMLSIEHSKNSQTLVSDIDYRHYLHQWTCMPDQNDVIQAR
KAYDLQSDAIYKADLEWL RGIGWMPADSVSVNHAKHMADIFNEKKYRTKI
ETLSFTPVDDRVDYVTAKHSGEILNDIKYRKDWNDTKSKYTLTETPQLHTA
QEAARILDQYLYKESWEKQKATGYILPPDAVPFVHAHHS GDVQSELKYKA
EHVKQKGHYVGVP TMRDDPKLVWFEHAGQIQNDRLYKENYHKTKAKIHI
PPDMVSVLAAKEGQALASDIDYRNYLHQWICH PDQNDVIQARKAYDLQSD
NIYKADLEWL RGIGWIPLDSVDHVRVTRNQEMMNQIKYKKDALANYPNFT
SVVDPPEIVLAKINSVNQSDVKYKETFNKLIKGYIFSPDTPYITHSKDMEKL
YSTILYKRAWDGTKAYGYTLDERYIPIVGAKHAD FVNSELKYKETYEK LK
GHYLAGKEISEFPNVVHCLDFQKMRSLLNYRRHYEDTKANVHIPQDMMNH
VLAKRCQYILSDLEYRHYFHQWTS LPEEPNVVRVRHAQEILSDNVYKDDL
NWLKGIGCYVWDTPQILHAKKSYDLQSQILYTAAGKENLKNYNLVTDTPL
YVTALQSGINASEVKYKENYHQTKDKYTTVLETVDYDRIKNLKD L FSSNLY
KEAWDKVKATSYILPPNTVSLTHAKNQKYMASHIKYREEYEKFKALYTLP
RSVEDDPNTARCLRVGKF NIDRLYRSVYEKNKMKIHI VPD MVEMVTAKDS
QKKVSEIDYRLHLHEWICH PDLQVNSHVRKVTDQISDIVYKDDLTWLKGIG
CYVWDTP EILHAKHAYDLRNDIKYKAHV LKTRNNYKLVTDTPVYVQAVK
SGKQLSDAVYHYDYVHSIRGRVAPTTKTVDLDRALHAYKLQSENLYRKAG
LHALPTGYRLPVDTPHFKHTKDTRYMSSYFKYKEVYEHMKAYGYTLGPN
DVPFVNVRVN NITSERLYRQLYHKLKDKIHTTPDTPEIRQVKKTQEAVSEL
IYKSDFFKMQGHMISLPYTPQVLHCRYVG DITS DIKYKEDLQVL RGMGCFL
YDTPDMVRSRHLRKLWSHYLYTDKARKMRDKYKVVLDTPEYRKVQELKT
HLSELVYRAAGRKQKSIFTSVPDTPDLTRAKRGQKLQSQYLYVELATKERP
HHHAGNQTTALKHARHV KDMVSENKYKIQYEKMKDKYTPVPDTPILIRAK
RAYWNASDLRYKET FQKTKGKYHTVKDALDIVYHRTVTDHISKIKYKENY
MSQLGIWRSIPDRPEHFHRAVTD AVSDVKYKQDLTWLKGIGCYAYDTPD
FTLAEKNKTLYSKYKYKEVFERTKS NFKYVADCPINRHFKFATQLMNEKK
YRADYEQRKDKYHLVVDEPRHLLAKIAGDQISQIKYRK NYEETKDKFTSIV
DTPEHLRTTKV NKQISDILYKLEYNKAKPRGYTTIHDTPMLLHVRKV KDEV
SDLKYKEVYQRTKS NCTIEPDAVHIKAAKDAYKVNTNLDYKKKYEATKAY
WKWTPDRPDFIQA AKSTLQQSDFEYKLDREYLKGCKLSVTDDKDMVLAL
KNSIIESDLKYKEKHVKERGSCHAVPDTPQILLAKTVSSLVSENKYKSYVKK
HLAQGSYTTLPETRDTIHVKEVTKNVSDTNYKKKFVKEKGKSNYSIMLEPP
DVKHAMDVAKKQSNVAYKKDAKENLHYTTVADRPDIKKATQAAKQASE
VEYRAKHRKEGSHGLSMLGRPDIEMAKKA AKLSSQVQYRENFNKEKGKTP
KYNPKDSQLYKVMKDANTLASEVKYKADLKKLHKPV TDMKESLIMNHVL
NTSHLASSYQYKKNYEKS KGHYHTIPDNLEQLHLKEATELQSIVKYKEKYE
KERGKPM LDFETPTYITAKESQQMQSGKEYRKDYEESIKGRNLTGLEVTPA
LLHVKYATKIASEKEYRKDLEESIRGKGLSEMEDTPDMLRAKNATQILNEK
EYKRDLELEV KGRGLNAMANETP DFLRARNATDIASQIKYKQSAEMEKAN
FTSVVDTP EIIHAQQVKNLSSQKKYKEDAEKCMSYYETVLDTP EMQRVREN
QKNFSL LQYQYDLKNSKGKITVVQDTPEILRVKENQKNFSSVLYKEDVSPG

TAIGTPMEMPMRQQTQDHISSVKKYVQGTPIPDLPVKKRVKQTKHQISS
VMYKENLGTGIPTPVTPEIERVKRNQENFSSNLSLKHEVTSKSIILTKYNSLK
LPHTPLPPVLYKENLGKGTPTPITPEMERVKRNQENFSSVLYKENMGKGT
LPVTPEMERIKHNQENISSVLYKENMRKATPTPVTPEMERAKRNQENISSV
YSDSFRKQIQGKAAYVLDTPEMRRVRETQRHISTVKYHEDFEKHKGCFTPV
VTDPIERVKKNTQDFSDICYRGIQRKVVEMEQKRNDQDQETITGLRVWRT
NPGSVFDYDPAEDNIQSRSLHMINAQAQRRSREQRSASGLSISGGEEKSEH
SEAAHLSTYSDGGVFFSAASTGYKHARTTELPPQQRSSSVATQQTTVSSIPSH
PSTAGKIFRAMYDYMAADADEVSKFDGDAIVNVQAIDEGWMYGTVQRTG
RTGMLPANYVEAI 92 Cow E1BP87

MSSDQEMAIFGEAAPYLKSEKERIEAQNKPFDAKTSVFVVDPKESFVKAT (Myosin
heavy VQSREGGKVTAKTEAGATVTVKEDQVFPMNPPKYDKIEDMAMMTHLHEP chain

4) AVLYNLKERYAAWMIYTYSGLFCVTVNPNYKWLPVYNAEVVTAYRGKKRQ
EAPPHIFSISDNAYQFMLTGELDVSIGESGAGKTVNTRKVIQYFATIAVTGEK
KKEEPTSGKMQGTLEDQIISANPLLEAFGNAKTVRNDNSSRFGKFIRIHFGTT
GKLASADIETYLLEKSRVTFQLKAERSYHIFYQIMSNKKPELIEMLLITNPY
DYAYVSQGEITVPSIDDELIATDSIDILGFTSDERSIYKLTGAVMHYGN
LKFKQKQREEQAEPDGTEDVADKAAYLQSLNSADLLKALCYPRVKVGNEFV
TKGQTVQQVYNAV GALAKAVYEKMF LWMVTRINQQLDTKQPRQYFIGVL
DIAGFEIFDFNSLEQLCINFTNEKLQQFFNHMHMFVLEQEEYKKEGIEWEFIDF
GMDLAACIELIEKPMGIFSILEEECMFPKATDTSFKNKLYEQHLGKSNNFQK
PKPAKGKAEAHFSLVHYAGTVDYNIA GWLDKNKDPLNETVVGLYQKSAM
KTLAFLFAGGPSAEEGGGGKKGKGGKSSSFQTVSALFRENLNKLMTNLRS
THPHFVRCIIPNETKTPGAMEHELVLHQLRCNGVLEGIRICRKGFP SRILYAD
FKQRYKVLNASAIPEGQFIDSKKASEKLLGSIDIDHTQYKFGHTKVFFKAGL
LGILEEMRDEKLAQLITRTQAICRGFLMRVEFRKMMERRESIFCIQYNVRAF
MNVKHWPWMKLYFKIKPLLKSAETEKEMANMKEEFEKTKEELAKSEAKR
KELEEKMVTLMQEKNDLQLQVQSEADGLADAEERC DQLIKTKIQLEAKIKE
LTERAEDEEEINAELTAKKRKLEDECESELKKDIDDLELTLAKVEKEKHATE
NKVKNLTEEMAGLDENIAKLTKEKKALQEAHQQTLDLQAEEDKVNTLT
KSKTKLEQQVEDLEGSLEQEKKL RMDLERAKRKLEGDLKLAQESTMDIEN
DKQQLDDEKLKKKEFEMSNLQSKIEDEQALAMQLQKKIKELQARIEELEEEIE
AERASRAKAEKL RSDLSRELEEISERLEEAGGATSAQIEMNKKREAEFQKM
RRDLEEATLQHEATAAALRKKHADSV AELGEQIDNLQRVKQKLEKEKSEM
KMEIDDLASNMETVSKAKGNLEKMCRTLEDQLSEVKSKEEEHQRLINELSV
QKAHLHTESGEFSRQLEEKDALVSQLSRGKQAFQTQQIEELKRQLEEETKAK
SALAHALQSARHDCDLLREQYEEEQEGKAELQRALSKANSEVAQWRTKYE
TDAIQRTEELEEA KKKLAQRLQDAEEHVEVVNAK CASLEKTKQRLQNEVE
DLMLDVERSNAACAALDKKQRNFDKVLSEWKQKYEETQADLEASQKESR
SLSTELFKVKNAYEESLDQLETLKRENKNLQQEISDLTEQIAEGGKHIHELE
KVKKQIEQEKGELQAAL EEA EASLEHEEGKILRIQLELNQVKSEIDRKIAEK
DEEIDQLKRNHLRVVESMQSTLDAEIRSRNDALRIKKKMEGDLNEMEIQLN
HANRQATEAIKNLRNTQGVLKDTQLHLDDAIRGQDDLKEQLAMVERRAN
LMQAEIEELRASLEQTERSRRVAEQELLDASERVQLLHTQNTSLINTKKKLE
TDITQIQGEMEDIVQEARNAEEKAKKAITDAAMMAEELKKEQD TSAHLER
MKKNMEQTVKDLQHRLDEAEQLALKGGKKQIQKLEARVRELENEVENEQ
KRNAEAVKGLRKHERRVKELTYQTEEDRK NVLRLQDLVDKLQSKVKAYK
RQAEAEAEQSNVNLAKFRKLQHELEEA EERADIAESQVNKL RVKSRE VHTK VISEE 93
Cow ALBU MKWVTFISLLLLFSSAYS RGVFRRDTHKSEIAHRFKDLGEEHFKGLVLIAFS
(Albumin) QYLQQCPFDEHV KLVNELTEFAKTCVADESHAGCEKSLHTLFGDELCKVA

SLRETYGMDYGDMEACQPERNECEPCHEPDLPLKLPDNPNTLCDEFKAD
EKKFWGKYLYEIAARRHPYFYAPELLYYANKYNGVFQECCQAEDKGACLLP
KIETMREKVLASSARQRLRCASIQKFGERALKAWSVARLSQKFPKAEFVEV
TKLVTDLTKVHKECCHGDLLECADDRADLAKYICDNQDTISSKLKECCDKP
LLEKSHCIAEVEKDAIPENLPPLTADFAEDKDVCKNYQEAKDAFLGSFLYE
YSRRHPEYAVSVLLRLAKEYEATLEECCAADDPHACYSTVFDKLKHLVDE
PQNLIKQNCDDQFEKLGEYGFQNALIVRYTRKVPQVSTPTLVEVSRSLGKVG
TRCCTKPESERMPCTEDYLSLILNRLCVLHEKTPVSEKVTKCCTESLVNRRP
CFSALTPDETYVPKAFDEKLFTFHADICTLPDTEKQIKKQTALVELLKHKPK
ATEEQLKTVMENFVAFVDKCCAADDKEACFAVEGPKLVVSTQTALA

[0043] An animal protein can comprise a sheep protein (Table 4). A sheep protein can comprise a sheep muscle protein. A sheep protein can comprise a sheep blood protein. A sheep protein can comprise a sheep albumin protein. A sheep protein can comprise a sequence that has at least or about 70%, at least or about 75%, at least or about 80%, at least or about 90%, at least or about 91%, at least or about 92%, at least or about 93%, at least or about 94%, at least or about 95%, at least or about 96%, at least or about 97% at least or about 98% at least or about 99% or about 100% sequence identity to any one of SEQ ID NO: 94-108.

[0044] A food product may comprise a fragment or a truncation of a sheep protein. The food product may include a fragment or truncation of any one of SEQ ID Nos: 94-108. A sequence of the fragment or truncation may be identical or substantially identical to a portion of any one of SEQ ID Nos: 94-108. A sequence of the fragment or truncation may be at least about 70%, 80%, 90%, 95%, 98%, or more identical to a portion of a sequence of any one of SEQ ID Nos: 94-108. A length of the fragment or truncation may be at least about 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or more of a length of any one of SEQ ID Nos: 94-108. A length of the fragment or truncation may be less than or equal to about 90%, 80%, 70%, 60%, 50%, 40%, 30%, 20%, 15%, 10%, or less than a length of any one of SEQ ID Nos: 94-108.

TABLE-US-00004 TABLE 4 Sheep Proteins SEQ ID NO Description Amino Acid
Sequence 94 Sheep W5PJ69

MPFGNTHNKHKLNFKAEEEEYPDL SKHNNHMAKALTLEIYKKLRDKETPSG (Creatine
kinase) FTLDDVIQTGVDNPGHPFIMTVGCVAGDEESYTVFKDLFDPIIQDRHGGFKP
TDKHKTDLNHENLKGDDLDPHYVLSSRVRTGRS IKG YALPPHCSRGERR
AVEKLSVEALNSLTGEFKGKYYP LKSMTEQEQQQLIDDHFLFDKPVSPLLL
ASGMARDWPDARGIWHNDNKSFLVWVNEEDHLRVISMEKGGNMKEVFR
RFCVGLQKIEEIFKKAGHPFMWNEHLGYVLTCPSNLGTGLRGGVHV KLAH
LSKHPKFEEILTRLRLQKRGTGGVDTAAVGSVFDVSNADRLGSSEVEQVQL
VVDGVKLMVEMEKKLEKGQSIDDMIPAQK 95 Sheep W5P9P1

MMMVLPQPEGLGTGEGPFAGGGGRGGGEYMEQEEDWDRDLLLDP AWEKQQ (Actinin
alpha 3) RKTFTAWCNSHLRKAGTQIENIEEDFRNGLKLMLLLEVISGERLPRPDKGK
MRFHKIANVNKALDFIASKGVKLVSIGAE EIVDGNLKM T LGM IWTIILRFAI
QDISVEETSAKEGLLLWCQRKTAPYRNVNVQNFHTSWKDGLALCALIHRH
RPDLIDYAKLRKDDPIGNLNTAFEVAEKYLDIPKMLDAEDIVNTPKPDEKAI
MTYVSCFYHAFAGAEQAETAANRICKVLAVNQEN EKLME EYEKLASELLE
WIRRTVPWLENRVGEP SMSAMQRKLED FRDYRRLHKPPRVQEK CQLEINF
NTLQTKLRLSHRPAFMPSEGKLVSDIANAWRGLEQA EKG YEDWLLSEIRRL
QRLQH LAEK FQQKASLHEAWTRGKEEMLSQRDYETASLQEVRALLRRHEA
FESDLAAHQDRVEHIAALAQELNELDYHEAASVNSRCQAICDQWDNLGTL
TQKRRDALERMEKLLETIDQLQLEFARRAAPFN NWLDGAVEDLQDVWLV
HSVEETQSLVTAHDQFKATLPEADRERGAILGIQGEIQKICQTYGLRPSSTNP
YIALTPQDINTKWDTVRKLVP SRDQMLQEELTRQQMNERLRRQFAAQANA
IGPWIQGKVEEGGRLAAGMAGSLEE QMAGLRQQEQNIINYKSNIDRLEGDH

QLLQESLVFSLVDTVYSMHIRVGVGEQLTSTIARTINEVENQVLTRDAKGL
SQEQLNEFRASFNHFDRKRNGMMEPDDFRACLISMGYDLGEVEFARIMTM
VDPNAAGVVTFQAFIDFMTRETAETDTAEQVVASFILAGDKNYITAEELR
RELPAEQARRAPDRGAGAPAGALDYVAFSSALYGESDL 96 Sheep PYGM
MSRPLTDQEKRKQISVRGLAGVENVTELKKNFNRHLHFTLVKDRNVATPR (Glycogen
DYYFALAYTVRDHLVGRWIRTQQHYEYKDPKRIYYLSLEFYIGRTLQNTM phosphorylase)
VNLALENACDEATYQLGLDMEELEEIEEDAGLGNGGLGRLAACFLDSMAT
LGLAAYGYGIRYEFGIFNQKISGGWQMEEADDWLRYGNPWEKARPEFTLP
VHFYGRVEHTSQGAKWVDTQVVLAMPYDTPVPGYRNNVVNTMRLWSAK
APNDFNLKDFNVGGYIQAFLDRNLAENISRVLYPNDNFFEGKELRLKQEYF
VVAATLQDIIRRFKSSKFGCLDPVRTNFDAPDKVAIQLNDTHPSLAIPELM
RILVDQERLEWEKAWEVTVKTCAYTNHTVLPEALERWPVHLIETLLPRHLQ
IIYEINQRFLNRVAAAAPGDVDRLRRMSLVEEGAVKRINMAHLCIAGSHAV
NGVARIHSEILKKTIFKDFYELEPHKFQNKTNGITPRRWLVMCNPGLAEVIA
ERIGEEYIADLDQLRKLLSYVDDESFIRDVAKVKQENKLFSAYLEKEYKV
HINPNSLFDIQVKRIHEYKRQLLNCLHVITLYNRIKKEPNKFFVPRTVMIGGK
AAPGYHMAKMIIRLITAIGDVVNHDPPVGDRLRVIFLENYRVSLAEKVIPAA
DLSEQISTAGTEASGTGNMKFMLNGALTIGTMDGANVEMAEAGEENFFIF
GMRVEDVERLDQKGYNAQEYYDRIPELRHIIDQLSSGFFSPKQPDLFKDIVN
MLMHDRFKVFADYEEYVKCQERVSAKYKNPREWTRMVIRNIATSGKFSS
DRTIAQYAREIWGVEPTRQRMPAPDEKI 97 Sheep
MPHQYPALTPEQKKELCDIAHRIVAPGKGILAADESTGSIKRLQSIGTENTE
A0A6P7DGM3 ENRRFYRQLLLTADDRVNPCIGGVILFHETLYQKADDGRPFQVIKAKGGV
(Fructose- VGIKVDKGVVPLAGTNGETTTQGLDGLSERCAQYKKDGADFAKWRCVLKI
bisphosphate GEHTPSSLAIMENANVLARYASICQQNGIVPIVEPEILPDGDHDLKRCQYVT
aldolase) EKVLAAVYKALSDHHIYLEGTLLKPNMVTTPGHACTQKYSHEEIAMATVTA
LRRTVPPAVPGITFLSGGQSEEEASINLNAINKCPLLKPWALTFSYGRALQAS
ALKAWGGKKENLKAQEEYVKRALANSLACQGKYTPSGKAGAAASESLFI SNHAY 98
Sheep W5QD16
ILLNKMAPKKDVKKPAAAAAPAPAPAPAPAPAPPKEEKIDLSAIKIEFSK (Myosin light
QQQDEFKEAFLLYDRTGDGKITLSQVGDVLRALGTNPTNAEVKKVMGNPS chain 1)
NEEMNAKKIEFEQFLPMLQAISNNKDQGTIEDFVEGLRVFDKEGNGTVMG
AELRHVLATLGEKMKEEEVEALMAGQEDSNGCINYEAFVKHIMSN 99 Fragment of
IGRLVTRAAFNTGKVDIVAINDPFIDLHYMVYMFQYDSTHGKFHGTVKAEN Sheep G3P
GKLVINGKAITIFQERDPANIKWGDAGAEYVVESTGVFTTMEKAGAHKLG
(Glyceraldehyde-
GAKKVIISAPSADAPMFVMGVNHEKYNNTLKIVSNASCTTNCLAPLAKVIH 3-phosphate
DHFGIVEGLMTTVHAITATQKTVDGPSGLWRDGRGAAQNIIPASTGAAKA dehydrogenase)
VGKVIPELNGKLTGMAFRVPTPNVSVDLTCLREKPAKYDEIKKVVKQASE
GPLKGILGYTEDQVVSCDFNSDTHSSTFDAGAGIALNDHFVKLISWYDNEF
GYSNRVVDLMVHMASKE 100 Sheep W5QI03
MDAIKKKMQLKLDKENALDRAEQAEADKKAEDRSKQLEDELVSLQKK
(Tropomyosin 1)
LKATEDELDKYSEALKDAQEKLLEAEKKATDAEADVASLNRRIQLVEEEL
DRAQERLATALQKLEEAKEAADESERGMKVIESRAQKDEEKMEIQEIQLKE
AKHIAEDADRKYEEVARKLVIIESDLERAEEAEELSEGQVRQLEEQLRIMDQ
TLKALMAAEDKYSQKEDKYEEEIKVLSDKLKEAETRAEFAERSVTKLEKSI
DDLEDELYAQKLKYKAISEELDHALNDMTSIESGSCQRRKP 101 Sheep W5PX04
MPEAKPAAKKAPAGKDAAKAPAPKEAPPQEAPAAPSTEAPPEDQSPTAEAP (Myosin
binding TGVFLKKPDSVSVENGKDTVIVAKLNGKELPAKPAVKWFKGKWLELGSKS

protein C2) GARHSHFADNVAASVNYVVELGDRGDYRIVLGDGDFCDSCSTFNI

DVESPRQDSAGQSLESFKRSEAKSDTAGELDFSGLLKKRQVVEEEKKKKK
DDDDLGIPEIWEELLKGAKKSEYERIAFYQGITDLRGMLKRLKKAKVEVKK
SAAFTKKLDPAYQVDKGNKIKLVVEISDPDLPLKWFKNQGEIKPSSKYVFE
NVGKKRILTINKCTLADDAAYEVVVKDEKCFTELFVKVEPPVLIVTPLEDQ
QVFVGNRVEMAVEVSEEGAQVMWLKDGVELTREDSEFKSRYRLKKDGKRH
ILIYAEVALEDKGHYKVMTNGDQCEAELIVEEKQLEVLQDIADLTVKASEQ
AVFKCEVSDEKVTGKWKNGVEVRPSKRITISHTGRIHKLVIDDVRPEDEG
DYTFVPDGYALSLSAKLNFLGAGLSPLPPEPPKIHLDSCSGQTSSENSVVVAG
NKLRLDVSITGEPRPVATWMKEDEVFTGTTEGRVRIEQRGDISSFVIESAERG
DEGRYTIKVTNPAGEDVASILLKVVDVPDPPEAVRVTSSVGEDWAVLVWEP
PKYDGGRPVTGYLLERKKKKSQRWMKLNFEVFTETTYESTNMIEGILYEM
RVFAVNAIGVSQPSMNTKPFMPIAPTSEPQHLLTVEDVTDTTTTLKWVPPEKI
GAGGIDGYLIEYCVESGSEWIPANTEPVERCGFTVKNLPTGAKITFRVVG
NIAGRSQPATLAQPVITIREIVEQPKIRLPRHLRQTYVRKVGEHLNLVIPFQGK
PRPQVVWTKGGAPIDTNHVHVRTSDFDTVFFVRQAARSDSGEYELTVQIEN
MKDTATIRIRVVEKAGPPENVMVKEVWGTNALVEWQPPRDDGNSEITGYL
VQKADKKTMEWFTVYERNRHTSCTVSDLIMGNEYFRVYSENICGLSDLP
GVSKNTARIVKTGMTLKLPEYKEHDFRTPPKFLTPLFDRVVAGYAAALNC
AVRGHPKPKVVWMKNKMEIREDPKFLMTNHQGVLTNLNRRPSPFDSGTYS
CRAVNELGEALAECKLEVRGEP 102 Sheep W5P6W4

MSLVAVPFYQKRHKHFDQSYRNIQTRYLLDEYAAKKKRASAQSSLQRSVT (Myomesin 2)

RKSSSQRESSRAALGETTLGETTLGRTTCPLCAKRMSSALEEEAQRKR
QSMVAAYGEAKRERYLSELAQLEEDVHVARTHAREELDRLALQHAGRHG
LHHAVDDRLAWERHWFEDRISRPEILVRLRSHTVWERMSVKLCFTVQGF
PTPVVQWYKDGSLICQAGEPGKYRIDSKYGVHTLEINRADFDDTATYSAV
TNVHGVSTNAAVVRRFRGDEEPPFHSVGLPIGLPLSSVIPYTHFDVQFIEK
FGVTFRREGETLTLKCTLLVTPDLKRVQPRAEWYRDDVLLKESKWTMFF
GEGQASLSFSHLNKDDEGLYTLRIVSRGGVTDHSAFLFVRDADPLVTGAPG
APMDVHCHDANRDYVIVTWKPPNTTKESPVIGYFIDRCEVGTDNWVQCND
APVKICKYPVTGLFEGRSYIFRVRAVNSAGISRPSRVSEAVAALDPVDLRL
QAVHLEGEKDVIYQEDLEGEVQIPGPPTNVHASSETSRTYVVLSWDPPEPRG
KEPLMYFIEKSMVGSGSWQRVNAQTAVRSPRYAVFDLAEGKPYVFRVLSA
NKHGLSEPSEITAPIQAQDTIVVPSAPGRVLASRNTRTSVVVQWDRPKHEED
LLGYYVDCSVAGSNVWEP CNHKPIGYNRFFVHGLTTGEQYIFRVKAVNAV
GTSENSQESDVIKVQAALTVPSHPYGITLLNCDGHSMILGWKVPKFSGGSPI
LGYIIDKREAHHKNWHEVNSSPLKERILKVEGLTEGSLYEFKIAAANMAG
MGSQQTELFKCEAWTMPEPGPAYDLTFCEVRDTSVLVLLWKPPVYSGSSPVS
GYFVDYKEEDSGEWLTVNEVTTPHRYLKVCDLHQGKTYVFRVRAVNASG
VGRPSDTSEPVLVEARPGTKEVSAGVDEEGNIYLSFDCPEVTDASHFTWCK
SYEDIADDDRFRKVETAGDHSKLYFKNLDKEDLGTYSVSVSDTDGVSSSFVL
DEEELERLKLALSNEIKNPTIPLKSELAYEIFDKGQVRFWLQAEHLSPDSSYRF
IINDREVADSETHRIKCDKSTGVIEMVMMDRFTVDNEGTYTVQIQDGKAKSQ
SSLVLIGDAFKAVLEEAQEFQRKEFLRKQGPFAEYLHWDVTEECEVRLVCK
VANTKKETVFKWLKDDVLYETEKMPDLEKGVCELLIPKLSKKDHGEYKAT
LKDDRQGDVSTLEIAGKVYEDMILAMSRVCGASASPLKILCTPEGIRLQCF
MKYFTEEMKVNWYHKDAKIASSEHMRIGGSEMAWLQICEPTEKDKGKY
TFEILDGKKNHQRSLDLGQAFDEAFQQLKAAAFKAEKNRGKVIGGLPD
VVTIMEGKTLNLTCTVFGNPDPEVVWFKNDKDIELSDHFSVKVEQAKYVS
MTIKGVTSEDSGKYSMHVKNKYGGGEKIDVTVSVYKHGEEIPDVVLPQQAK

PKLLPAAAPSAH 103 Sheep W5P663

MAMQKIFAREILDSRGNPTVEVDLHTAKGRFRAAVPSGASTGIYEALRLD

(Phosphopyruvate

GDKSRYLGKGVLKAVEHINKTLGPALLEKKLSVVDQEKVDKFMIELDGTE hydratase)

NKSKFGANAILGVSLAVCKAGAAEKGVPLYRHIADLAGNPELILPVPAFNVI

NGGSHAGNKLAMQEFMILPVGASSFREAMRIGAEVYHHLKGVIAKAKYGKD

ATNVGDEGGFAPNILENNEALELLKTAIQAAGYPDKVVIGMDVAASEFYRN

GKYDLDFKSPDDPARHISGEKLGELYKSFIKNYPVVSIEDPFDQDDWATWT

SFLSGVNIQIVGDDLTVTNPKRIAQAVEKKACNCLLLKVNQIGSVTESIQAC

KLAQSNGWGVMVSHRSGETEDTFIADLVVGLCTGQIKTGAPCRSERLAKY

NQLMRIEEALGDKAVFAGRKFERNPKAK 104 Sheep W5P7V0

QAPKKAKRRAAAEGGSSNVFSMFDQTQIQEFKEAFTVIDQNRDGIIDKEDL (Myosin light

RDFAAMGRLNVKNEELDAMMKEASGPINFVFLTMFGEKLGADPEDVI chain)

TGAFKVLDPGKGTIKKQFLEELLTTQCDRFSQEEIRNMWAAFPDPVGGNV

DYKNICYVITHGDAKDQE 105 Sheep W5PPG6

MSSDQEMAIFGEAAPYLKSEKERIEAQNKPFDAKTSVFVADPKESFVKAT (Myosin heavy VQSREGGKVTAKTEAGATVTVKEDQVFPMNPPKFDKIEDMAMMTHLHEP chain 1)

AVLYNLKERYAAWMIYTYSGLFCVTVNPKWLPVYNAEVVAAYRGKKR

QEAPPHIFSISDNAYQFMLTDRENQSILITGESGAGKTVNTKRVIQYFATIAV

TGEKKKEEPTSGKMQGTLEDQIISANPLLEAFGNAKTVRNDNSSRFGKFIRI

HFGTTGKLASADIETYLLEKSRVTFQLKAERSYHIFYQIMSNKKPELIEMLLI

TTNPYDYAFVSQGEITVPSIDDQEELMATDSAIEILGFTSDERSIYKLTGAV

MHYGNLKFQKQREEQAEPDGTEVADKAAYLQGLNSADLLKALCYPRVK

VGNEFVTKGQTVEQVYNAV GALAKAIYDKMFLWMVTRINQQLDTKQPRQ

YFIGVLDIAGFEIFDFNSLEQLCINFNNEKLQQFFNHHMFVLEQEEYKKEGIE

WEFIDFGMDLAACIELIEKPMGIFSILEEECMFPKATDTSFKNKLYEQHLGKS

NNFQKPKPAKGKAEAHFSLIHYAGTVDYNITGWLDKNKDPLNETVVGLYQ

KSSVKTLAFLFSGAASAEAEAGGAKKGGKKGSSSFQTVSALFRENLNKLM

TNLRSTHPPHFVRCIIPNETKTPGAMEHELVHLQRCNGVLEGIRICRKGFP SR

ILYADFKQRYKVLNASAIPEGQFIDSKKASEKLLGSIDVDHTQYKFGHTKVF

FKAGLLGLLEEMRDEKLAQLITRTQARCRGFLARVEYQRMVERRESIFCIQ

YNVRAFMNVKHWPMKLYFKIKPLLKSAETEKEMANMKEEFEKTKEELA

KSEAKRKELEEKMTLTQEKNDLQLQVQSEADALADAEERCDQLIKTKIQL

EAKIKEVTERAEDEEEINAELTAKKRKLEDECESELKKDIDDLELTLAKVEKE

KHATENKVKNLTEEMAGLDETIAKLTKEKKALQEAHQQTLDLQAEEDK

VNTLTAKTKLEQQVDDLEGSLEQEKKLRMDLERAKRKLEGDLKLAQEST

MDIENDKQQLEKLKKKEFEMSNLQSKIEDEQALGMQLQKKIKELQARIEE

LEEEIEAERASRAKAEKQRS DLSRELEEISERLEEAGGATSAQIEMNKKREA

EFQKMRRDLEEATLQHEATAAALRKKQGGVGEQLDNLQRVKQKLEKEKS

EMKMEIDDLASNMETVSKAKGNLEKMCRTLEDQLSELKTKEDEQQRLIND

LTQQRARLQTESGEFSRQLDEKDALVSQLSRGKQAFQTQQIEELKRQLEEEIK

AKSALAHALQSARHDCDLLREQYEEEEEQAKAELQRALSKANSEVAQWRT

KYETDAIQRTTEELEEAKKKLAQRLQDAEEHVEAVNAKCASLEKTKQRLQN

EVEDLMIDVERTNAACAALDKKQRNFDKILSEWKQKYEETHAELEASQKE

SRSLS TELFKIKNAYEESLDQLETLKRENKNLQPTTEEISDLTEQIAEGGKRIH

ELEKVKKQVEQE KSEIQAAL EEA EASLEHEEGKILRIQLELNQVKSEIDRKIA

EKDEEIDQLKRNHIRVVESMQSTLDAEIRSRNDAIRLKKKMEGDLNEMEIQ

LNHANRMAAEALKNYRNTQAILKDTQLHLDDALRGQEDLKEQLAMVERR

ANLLQAEIEELRATLEQTERS RKIAEQELLDASERVQLLHTQNTSLINTKKK

LETDITQIQGEMEDI IQEARN AEEKAKKAITDAAMMAEELKKEQD TSAHLE

RMKNEQTVKDLQHRLKLEAQKLEARVRELEAEVESEQ
KRNVE TVKGLRKHERRVKELTYQTEEDRKNILRLQDLVDKLQAKVKS YKR
QAEAE EQSNVNL SKFRKLQHELEEA EERADIAESQVNKL RVKSREVHTKII SEE 106
Sheep W5PFV9

MADDEEYEEVVEYYTEETVYEEVPGETITEFYETTTTTRTSDYEQSDTSRPAL (Nebulin)
AKPVPEKPVERKRVIRKKVDPSKFMTPYIAHSQKMQDLFSTNKYKENYEK
AKGQPYAITTDTPELRRIKKVQDQLSEVKYRTDGDVAKTICHVDEKAKDIE
HAKKVSQQVSKVLYKQNWEDTKDKYLLPPDAPELVQAIKNTAMFSKKLY
TEDWEADKTLFYPPYNDSPELRRVAQAQKALSDITYKKGLAEQQTQFTSLPD
PPDIEFAKKVTNQVSKQKYKEDYENKVKGKWSETPCFEIATARMNSQNIST
KKYQEDFENTKDQIYFMQTETPEYKVNKQAGVAASKVKYKQDYEKNKGK
ADYNVLPASENPLLRLQKAAGDALSDKLYKENYEKTKAKSINYCETPKFKL
DTVLHNFSSDTKYKDSYLNILGHYVGSYEDPYHTHCMRVSAQNSDKNYK
AEYEEDRGKGFFPQTITQEYEAIKKLDQCKDHTYKVHPDKTKFTQVTDSPV
LVQAQVNSKQLSDLNYKAKHENEKFKCHIPPDTPAFIQHKVNAYNLSDNIY
KHDWEKTKAKKFDIKVDAIPLLA AKANTKIASDVMYKKDYEKSKGKMIG
ALSINDDPKMLHSLKTAKNQSDRLYKENYEKTKAKSMNYCETPKYQLDTL
LKNFSEAKYKDSYVQNVLGHYIGSFEDPYREHCMKVAAQNSDKNYKAEY
EEDKGKCYFPQTITQEYEAIKKLDQCKDHTYKVHPDKTKFTAVTDTPVLLQ
AQLNTKQLSDLNYKAKHEGEKFKCHIPADAPQFIQHRVNAYNLSDNIYKH
DWEKTKAKKFDIKVDAIPLLA AKANTKIASDVMYKKDYEKSKGKMIGALS
INDDPKMLHSLKTAKNQSDREYRKDYEKSKTIYTAPLDMLPLTHAKKSQEI
ASDV DYKHL LHNYSYPPDSVNVDLAKKAYALQSDVEYKADYNSWMKGC
GWVPFGSLEMEKAKRASDILNEKKYRQHPDTLKFTSIEDAPIIVQSKINQAQ
RSDVAYKAKGEEVIHKYSLPADLPQFIQAKVNAYNISENMYKADLKDLSK
KGYDLRIDAIPIKAAKAAARQAASDVQYKKDYEKAKGKMVG FQSLQDDPKL
VHYMNVAKIQSDREYKKDYEKTKTRHNTPHDMVNIVA AKKAQDVASNV
NYKHS LHHTYLPDAMDLELSKNMMHIQSDNVYKEDYNNWMKGIGWIP
GSLEVEKVKKAGDALNEKKYRQHPDTLKFTSIVDSPVMVQAKQNTQQVSD
ILYKAKGEDVKHKYTMSPDLPQFLQAKCNAYNLSDVCYKRDWHDLIAKG
TNVLGDAIPITA AKASRN IASDYKYKEAYEKAKGKQVGFRSLQDDPKLVH
YMNVAKLQSDREYKKNYENTKTSYHTPGDMVSITA AKMAQDVATNVNY
KQPIHHYTYLPDALSLHIRNVNQIQSDKVYKDEYNHFFKGIGWIPIGSLEV
EKVKKAGDALNEKKYRQHPDTIKFTSVPDSMGMVLAQHNTKQLSDLNYK
VEGEKVKHKYTLDPDVPQFIQAKVNAFNMSDSHYKADWKKTLAKGYDLR
PDAIPIVA AKSSRN IASDFKYKEGYEKAKGKQIGFRSLQDDPKLVHFMNVA
KMQSDREYKKGYEASKTKYHTPLDMLSVTAAKKSQEVATNTNYKQPFHH
YTLLPDALNVEHSRNAMQIQSDNLYKSDFTSWMKGIGWMPLESLEVEKAK
KAGEILSEKKYRQHPEKLKFTYAMDTMEQALNKS NKLIMDKRRYTEKWN
KDKTTHVMPDTPDILLSRVNQITMSDKLYKAGWEEEEKKGYDLR PDAIPI
KAARASRD IASDYKYKQAYEQAKGKQIGFRSLEDDPKLVHFMQVAKMQS
DREYKKAYEKS KTSFHTPV DMLSVVAAKKSQEVATNANYRNVIHNYNML
PDAMSLELAKNMMQIQSDNQYKADYADFMKGIGWLPLGSLEAEKNKKA
MEIISEKKYRQHPDTLKYSTLMDSMNMLAKNNAKIMNEHLYKQAW EAD
KTKVHIMPDI PQIILAKANAINMSDKLYKLSLEEAKKKGYDLRTDAIPIKAA
KASRD IASDYKYKHNYEKERGMVGFRSLEDDPKLVHSMQVAKMQSDRE
YKKNYEKTKTSYHTPADMLSVTAAKDAQANITNTNYKHLIHKYILLPDAM
NIQLSKNMNRLQSDNEYKQDYNEWYKGLGWSPAGSLEVEKAKRATEYAS
DQKYRQHPSNFQFSKLNDSMDMVLAKQNAHTMNKYLYTVDWNKD KTKI
HVMPDTPDILQAKQNQTIYSQKLYKLGWEEALKKGYDLPVDAISVQAAKA

SRDIVSKYDFKQGYRQKQIGLQDDPKLVLSMNVAKMQSEREYKK
DFEKWKTKFSSPVDMLGVVLAKKCQALVSDVDYKNYLHGWTCLPDQND
VIHAKKAYDLQSENLYKSDLEWLKGIGWSPLGSLEAEKNKRASEIIEKKY
RQPPDRNKFTSIPDAMDIVLAKTNAKNRSDILYREAWDKDKTQVHIMPDT
DIILAKANLINASDKFYRMGYEELRKKGYDLPVDAIPIKAAKASREIASEYK
YKEGFRKQLGHHIGARNIKDDPKMMWSMHVAKIQSDREYKKDFEKWKTK
FSSPVDMLGVVLAKKCQTLVSDIDYKNYLHQWTCLPDQSDVIHARRAYDL
QSDNLYKSDLQWLKGIGWLPSGSLEDEKNKRATQILSDHVYRQHPDKFKFS
SLMDSIPMVLAKNNAITMNHRLYTEAWDKDKTTVHIMPDTPGVLLAKQNN
INYSEKLYKLGLDEAKRKGYDMRIDAIPIRAASRDIASEFKYKEGYRKQ
LGHHIGARAIHDDPKMMWSMHVAKIQSDREYKKDFEKWKTKFSSPVDML
GVVLAKKCQTLVSDIDYKNYLHQWTCLPDQSDVIHARQAYDLQSDNVYK
SDLQWLKGIGWVPIGSLDVEKCKRASEILSDKLYRQPPDKFKFTSVTDSLEQ
VLAKNNAINMNKRLYTEAWDKDKTQVHIMPDTPEITLARMNKVNYSES
LYKLAHEEAKKKGYDLRSDAIPIVAASRDIIISDYKYKDG YRKQLGHHIGAR
DIQDDPKMMWSMHVAKIQSDREYKKDFEKWKTKFSSPVDMLGVVLAKKC
QTLVSDIDYKNYLHQWTCLPDQSDVIHARRAYDLQSDNVYKSDLQWLRGI
GWVPIGSVDVVKCKRAAEILSDNLYRQPPDTLKFTSVPSLEQVLAKNNAI
NMNKRLYTEAWDKDKTQVHIMPDTPEIMLARQNKLNYSETLYKLANEEA
KKKGYDLRSDAIPIVAASRDIIISDYKYKDG YRKQLGHHIGARDIKDDPK
MMWSMHVAKIQSDREYKKDFEKWKTKFSSPVDMLGVVLAKKCQTLVSDI
DYKNYLHQWTCLPDQSDVIHARQAYDLQSDNIYKSDLQWLRGIGWVPIGS
VDVVKCKRAAEILSDNLYRQRPDTLKFTSVPSLEQVLAKNNAINMNKRL
YTEAWDKDKTQIHMMPTPEITLARQNKINYSNLYRQAMEEAKKEGYDL
RSDAIPIVAASREIASDYKYKEAYRKQLGHHIGARAIHDDPKMMWSVH
VAKMQSDREYKKDFEKYKTRFSSPVDMLGIVLAKKCQTLVSDVDYKHPLH
EWTCLPDQNDIIHARKAYDLQSDNLYKSDLEWLKGIGWVPIGSFDVVKAK
RAGEILSDNIYRQRPDTLKFTSVTDSPEQVLAKNNAINMSKRLYTEAWDND
KKTIHVMPDTPEIMLAKLNRINYSKLYKLAL EESKKEGYDLRLDAIPIQAA
KASREIASDYKYKEGYRKQLGHHIGARNIKDDPKMMWSIHATKLQSDREY
KKDFEKWKTKFSSPVDMMGLVQAKKCQILVSDIDYKNLLHEWICLPDQND
VIQARKAYDLQSDAIYKADLEWLRGIGWVPIGSVGVENVKRAGEILSERKY
RQPAVQLKFTSITDTPEIVLAKNNALNMSKHLYTEAWDADKTSIHVMPDTP
EILLAKSNSANISHKLYTKGWDESKMKDYDLRADAIKSAKASRDIASDY
KYKEAYEKQKGHHIGARSVEDDPKIMCAMNAGKIQSEREYKKEFQKWKT
KFSSPVDMLGILQAKKCQILVSDIDYRNYLHHWTCLPDQNDIIQARKAYDL
QSDAIYKADLEWLRGIGWMPEGSP EVL RVKNAQHIFRDSVYRTPVVKLKY
TSIVDTPEVVLAKSNAENISIPKYREVWDKDKTSIHIMPDTPEINLARTNALN
VSNKLYREGWDEMKMSCDVRLDAIPIQAAKASREIASDYKYKLDHEKQKG
HYVGTRTARDDNKIRWALIAGKIQNEREYRMHWAKWKSKFQSPDMLSIE
HSKNSQTLVSDIDYRHYLHQWTCMPDQNDVIQARKAYDLQSDAIYKADLE
WLRGIGWMPADSVSVNHAKHMGDIFNEKKYRTKIETLNFTPVDDRVDYVT
AKHSGEILNDIKYRKDWNDTKSKYTLTETPQLHAAQEAARILDQYLYKES
WEKQKATGYILPPDAVPFVHAHHS GDVQSELKYKAEHV KQKGHYVGVPT
MRDDPKLVWFEHAGQIQNDRLYKESYHKTKAKIHPPDMVSVLAAKEGQA
LASDIDYRTYLHQWICH PDQNDVIQARKAYDLQSDNIYKADLEWLRGIGW
MPLDSVDHVRVTRNQEMVNQIKYKKDALANYPNFTSVVDPPEIVLAKINA
VNQSDVKYKETFNKLIKGYIFSPDTPYITHSKDMEKLYSTILYKRAWEGT
KAYGYTLDERYIPIVGAKHAD FVNSELKYKETYEK LKGHYLAGKEISEFPN
VVHCLDFQKMRSLLNYRRHYEDTKANVHIPQDMMNHVLAKRCQYILSDL

EYRHYFHQWTPQILHAKKSYDLQSQILYTAAGKENLKNYNLVTDTPLYVTALQSGINASE
VKYKENYHQIKDKYTTVLETVDYDRTRNLKDLFSSNLYKEAWDKVKATS
YILPPNTVSLTHAKNQKHMASYVKYREEYDKFKALYTLPRSVEDDPNTAR
CLRVGKFNIDRLYRSVYEKNKMNIHIVPDMVEMVTAKDSQKKVSEIDYRL
HLHEWICHPLDLQVNSHVRKVTDQISDIVYKDDLTWLKGIGCYVWDTPEILH
AKHAYDLRNDIKYKAHVLRNRYKLVTDTPVYVQAVKSGKQLSDAVYH
YDYVHSIRGRVAPTTKTVDLDRALHAYKLQSENLYRKAGLHALPTGYRLP
VDTPHFKHTKDTRYMSSYFKYKEVYEHMKAYGYTLGPNDVPFVNVRRVN
NITSERLYRQLYHKLKDKIHHTPDTPAIRQVKKTQEAVSELIYKSDFFKMQG
LRKLWSHYLYTDKARKMRDKYQVVLDTPPEYRTVQELKTHLSELVYRASG
RKQKSIFTSVPDTPDLIRAKRGQKLQSQYLYVELATKERPHHHAGNQTTAL
KHARHVKDMVSENKYKIQYEEKMKDKYTPVPDTPILIRAKRAYWNASDLR
YKETFQKTKGRYHTVKDALDIVYHRTVTDHISKIKYKENYMSQLGIWRSIP
DRPEHFHHRVTDVSDVKYKADLTWLKGIGCYAYDTPDFTLAEKNKTLY
SKYKYKEVFERTKSNFKYVADCPINRHFKFATQLMNEKKYRADYEQRKDK
YHLVVDEPRHLLAKIAGDQISQIRYRKNYEKTCKDKFTSIVDTPEHLRTTKVN
KQISDILYKLEYNKAAPRGYTTIHDTPMLLHVRKVKDEVSDLKYKEVYQRT
KSNCTIEPDAVHIKAAKDAYKVNTNLDYKKQYEATKAYWKWTPDRPDFIQ
AAKSTLQQSDFEYKLDREYLKGCKLSVTDDKDMVLALKNSIIESDLKYKEK
HVKERGSCHAVPDTPQILLAKTVSSLVSENKYKQYVKKHLAQGSYTTLPET
RDTIHVKEVTKNVSDTNYKKKFVKEKGSNYSIMLEPPDVKHAMDVAKK
QSNMKVAYKKDAKENLHYTTVADRPDIKKATQAAKQASEVEYRAKHRKE
GSHGLSMLGRPDIEMAKKAACLSSQVKYRENFNKEKGKTPKYNPKDSQLY
KVMKDANTLASEVKYKADLKKLHKPVTDMMKESLIMNHVLNTSHLASSYQ
YKKNYEKSCKGHYHTIPDNLEQLHLKEATELQSIVKYKEKEYEKERGKPLD
FETPTYITAKESQQMQSGKEYRRDYEESIKGRNLTGLEVTPALLHVKYATKI
ASEKEYRKDLEESIRGKGLSEMEDTPDMLRAKNATQILNEKEYKRDLELEV
KGRGLNAMANETPDFLRARNATDIASQIKYKQSAEMEKANFTSVVDTPEII
HAQQVKNLSSQKKYKEDAEKCMSYYETVLDTPEMQRVRENQKNFSLQY
QYDLKNSKGKITVVQDTPEILRVKENQKNFSSVLYKEDVSPGTAIGKTPEM
MRVKQTQDHISSVKYKEVIGQGTPIPDLPEVKRVKQTQKHISSVMYKENLG
TGIPTPVTPEIERVKRNQENFSSVLYKENLGKGTPTPTPEMERVKRNQENFS
SILYKENLSKGTPLPVTPEMERVKRNQENFSSVLYKENVGKGTPTPTPEM
QRVKRNQENISSVLYKENLGKATPTPTPEMERVKRNQENFSSVLYKENMR
KATPTPTPEMERAKRNQENISSVLYSDSFRKQIQGKAAYVLDTPEMRRVR
ETQRHISTVKYHEDFEKHKGCFTPVVTDPITERVKKNMQDFSDISYRGIQRK
VVEMEQKRNDQDQETVTGLRVWRTNPGSVFDYDPAEDNIQSRSLHMINAQ
AQRRSREQRSASGLSVSGGEEKSEHSEAAHLSTYSDDGGVFFSATSTGYKH
ARTTELPQQRSSSVATQQTTVSSIPSHPSTAGKIFRAMYDYMAADADEVSF
KDGDIVNVQAIDEGWMYGTVQRTGRTGMLPANYVEAI 107 Sheep W5PIU1
MSSDQEMAIFGEAAPYLKSEKERIEAQNKPFDAKTSVVFADPKESFVKAT (Myosin-4)
VQSREGGKVTAKEAGATVTVKEDQVFPMNPPKFDKIEDMAMMTHLHEP
GVLYNLKERYAAWMIYTYSGLFCVTNPNYKWLPVYNAEVVTAYRGKKRQ
EAPPHIFSISDNAYQFMLTDRENQSILITGESGAGKTVNTKRVIQYFATIAVT
GEKKKEEPTSGKMQGTLEDQIISANPLLEAFGNAKTVRNDNSSRFGKFIRIH
FGTTGKLASADIETYLLEKSRVTFQLKAERSYHIFYQIMSNNKKPELIEMLLIT
TNPYDYAFVSQGEITVPSIDDQEELMATDSAIEILGFTSDERSIYKLTGAVM
HYGNLKFQKQKQREEQAEPDGTEVADKAAYLQSLNSADLLKALCYPRVKV
GNEFVTKGQTVQQVYNAV GALAKAVYEKMFLWMVTRINQQLDTKQPRQ

YFIGVLDIAGLEIFDNFNEKLQQFFNHMFVLEQEEYKKEGIE
WEFIDFGMDLAACIELIEKPMGIFSILEEECMFPKATDTSFKNKLYEQHLGKS
NNFQKPKPAKGKAEAHFSLVHYAGTVDYNIAGWLDKNKDPLNETVVGLY
QKSAMKTLAFLFAGGPSAEVHVSEGGGGGKKGGKKKGSSSFQTVSALFRENL
NKLMTNLRSTHPPHFVRCIIPNETKTPGAMEHELVLHQLRCNGVLEGIRICRK
GFPSRILYADFKQRYKVLNASAIPEGQFIDSKKASEKLLGSIDIDHTQYKFGH
TKVFFKAGLLGILEEMRDEKLAQLITRTQAICRGFLTRVEFRKMMERSRESI
FCIQYNVRAFMNVKHWPWMKLYFKIKPLLKSAETEKEMANMKEEFEKTK
EELAKSEAKRKELEEKMTLMQEKNDLQLQVQSEADGLADAEERCDQLIK
TKIQLEAKIKELTERAEDEEEINAELTAKKRKLEDECESELKKDIDDLELTAK
VEKEKHATENKVKNLTEEMAGLDENIAKLTKEKKALQEAHQQTLDLQA
EEDKVNTLTLSKTKLEQQVDDLEGSLEQEKKLRMDLERAKRKLEGDLKLA
QESTMDIENDKQQLDEKLKKKEFEMSNLQSKIEDEQALGMQLQKKIKELQ
ARIEELEEIEAERASRAKAEKQRSDLRELEEISERLEEAGGATSAQIEMNK
KREAEFQKMRRDLEEATLQHEATAALRKKHADSVAEELSEQIDNLQRVKQ
KLEEEGSLNLTVQALDIISNFKIINRYYYIEGNLEKMCRTLEDQLSEVKSKEE
HQRLINELSAQKARLHTESGEFSRQLDEKDALVSQLSRGKQAFQTQQIEELKR
QLEETKAKSALAHALQSARHDCDLLREQYEEEQEAKAELQRSMSKANSE
VAQWRTKYETDAIQRTEELEEAKKKLAQRLQEAEEHVEAVNAKCASLEKT
KQRLQNEVEDLMLDVERSNAACAALDKKQRNFDKVLSEWKQKYEETQAE
LEASQKESRSLSTELFKVKNAYEESLDQLETLKRENKNLQQEISDLTEQIAE
GGKHIHELEKVKKQIEQEKGELQAALAEAEASLEHEEGKILRIQLELNQVKS
EIDRKIAEKDEEIDQLKRNHLRVVESMQSTLDAEIRSRNDALRIKKKMEGDL
NEMEIQLNHANRQATEATKNLRNTQGVVKDTQLHLDDAIRGQDDLKEQLA
MVERRANLMQAEIEELRASLEQTERSRRVAEQELLDASERVQLLHTQNTSL
INTKKKLETDITRVHGSVVGSGIISKHLTFVYISVGCAFQAAMMAEELKKEQ
D TSAHLERMKKNLEQTVKDLQHRLDEAEQLALKGGKKQIQKLEARVRELE
NEVENEQKRNAEAVKGLRKHERRVKELTYQTEEDRKNVLRQLQDLVDKLQ
SKVKAYKRQAEAEQANVNLSKFRKLQHELEAEERADIAESQVNKL RV
KSREVHTK VISEE 108 Sheep ALBU

MKWVTFISLLLLFSSAYSRGVFRRDTHKSEIAHRENDLGEENFQGLVLIAFS (Albumin)

QYLQQCPFDEHVKLKELTEFAKTCVADESHAGCDKSLHTLFGDELCKVA
TLRETYGDMADCCEKQEPERNECFLNHKDDSPDLPLKPEPDTLCAEFKAD
EKKFWGKYLYEVARRHPYFYAPELLYYANKYNGVFQECCQAEDKGACLL
PKIDAMREKVLASSARQRLRCASIQKFGERALKAWSVARLSQKFPKADFTD
VTKIVTDLT KVHKECCHGDLLECADDRADLAKYICDHQDALSSKLKECCD
KPVLEKSHCIAEVDKDAVPENLPPLTADFAEDKEVCKNYQEAKDVFLGSFL
YEYSRRHPEYAVSVLLRLAKEYEATLEDCCAKEDPHACYATVFDKLKHLV
DEPQNLIKKNCELFEKHGEYGFQNALIVRYTRKAPQVSTPTLVEISRSLGKV
GTKCCAKPESERMPCTEDYLSLILNRLCVLHEKTPVSEKVTKCCTESLVNRR
PCFSDLTLDETYVPKPFDEKFFTFHADICTLPDTEKQIKKQTALVELLKHKP
KATDEQLKTVMENFVAFVDKCCAADDKEGCFVLEGPKL VASTQAALA

[0045] An animal protein can comprise a pig protein (Table 5). A pig protein can comprise a pig muscle protein. A pig protein can comprise a pig blood protein. A pig protein can comprise a pig albumin protein. A pig protein can comprise a sequence that has at least or about 70%, at least or about 75%, at least or about 80%, at least or about 90%, at least or about 91%, at least or about 92%, at least or about 93%, at least or about 94%, at least or about 95%, at least or about 96%, at least or about 97% at least or about 98% at least or about 99% or about 100% sequence identity to any one of SEQ ID NO: 109-123.

[0046] A food product may comprise a fragment or a truncation of a pig protein. The food product

may include a fragment or truncation of any one of SEQ ID Nos: 109-123. A sequence of the fragment or truncation may be identical or substantially identical to a portion of any one of SEQ ID Nos: 109-123. A sequence of the fragment or truncation may be at least about 70%, 80%, 90%, 95%, 98%, or more identical to a portion of a sequence of any one of SEQ ID Nos: 109-123. A length of the fragment or truncation may be at least about 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or more of a length of any one of SEQ ID Nos: 109-123. A length of the fragment or truncation may be less than or equal to about 90%, 80%, 70%, 60%, 50%, 40%, 30%, 20%, 15%, 10%, or less than a length of any one of SEQ ID Nos: 109-123.

TABLE-US-00005 TABLE5 Pig Proteins SEQ ID NO Description Amino Acid Sequence
109 Pig KCRM

MPFGNTHNKYKLNFKAE E EYPDL SKHNNHMAKALTLEIYKKLRDKETPSGF (Creatine kinase TLDDVIQTGVDNPGHPFIMTVGCVAGDEESYVVF KD LFDPIIQDRHGGYKPT M-type) DKHKTDL NHENLKG GDDLDP NYVLSSRVRTGRSIKGYTLPPHCSRGERRAV EKLSVEALNSLTGEFKGKY YPLKSMTEQEQQQLIDDHFLFDKPVSP LLLASG MARDWPDARGIWHNDNKSFLVWVNEEDHLRVISMEKGGNMKEVFRRFCV GLQKIEEIFKKAGHPFMWNEHLGYVLTCP SNLGTGLRGGVHV KLAHLSKHP KFEEILTRLRLQKRGTGGVDTAAVGSVFDVSNADRLGSSEVEQVQLVVDGV KLMVEMEKKLEKGGQSIDDMIPA QK 110 Pig

MMMVMQPEGLGTREGPFAGGSGGGGEYMEQEEDWDRDLLLDP AWEKQQ A0A480REY2 RKTFTAWCNSHLRKAGTQIENIEEDFRNGLKLMLLLEVISGERLPRPDKGKM (Actinin alpha 3) RFHKIANVNKALDFIASKGVKLVSIGAE EIVDGNLKM T LGM IWTIILRFAIQDI SVEETSAKEGLLLWCQRKTAPYRNVNVQN FHTSWKDGLALCALIHRHRPDL IDYAKLRKDDPIGNLNTAFEVAEKYLDIPKMLDAEDIVNTPKPDEKAIMTYV SCFYHAFAGAEQAETAANRICKVLAVNQEN EKLME EYEKLASELLEWIRRT VPWLENRVGEP SMSAMQRKLEDFRDYRRLHKPPRVQEK CQLEINFNTLQTK LRLSHRPAFMPSEGKLVSDIANAWRGLEQVEKGYEDWLLSEIRRLQRLQHL AEKFQQKASLHEAWTRGKEDMLSQRDYESASLQEV RALLRRHEAFESDLA AHQDRVEHIAALA QELNELDYHEAASVNSRCQAICDQWDNLGTLTQKR RD ALERMEKLLETIDQLQLEFARRAAPFN NWLDGAVEDLQDVWL VHSVEETQS LVTAEHEQFKATLPEADRERGAILGIQGEIQKICQTYGLRPSSTNPYIALSPQDI NTKWDTVRKLVPSRDQMLQEELTRQQVNERLRRQFAAQANAIGPWIQGV EEVGR LAAGMAGSLEE QMAGLRQQEQNIINYKSNIDRLEGDHQLLQESLVF DNKHTVYSMEHIRVGWEQLLTSIARTINEVENQVLTRDAKGLS QEQLNEFR ASFNHFDRKRNGMMEPDDFRACLISMGYDLGEVEFARIMTMVDPNAAGVV TFQAFIDFMTRETAETDTAEQVVASFKILAGDKNYITAEELRRELPAEQAEYC IRRMAPYKGAGAPAGALDYVAFSSALYGESDL 111 Pig

MAMSRPLTDQEK RKQISVRGLAGVENVAELKKNFN RHLHFTLVKDRNVAN A0A5G2QU59 ARDY YFALAYTVRDHLVGRWIRTQQHYYEKDPKRIYYLSLEFYMGRTLQN (Alpha-1,4 TMVNLALENACDEATYQLGLDMEELEEMEEDAGLGNGGLGRLAACFLDSM glucan ATLG LAAYGYGIRYEF G IFN QKITGGWQMEEADDWLRYGNAWEKARPEFT phosphorylase) LPVHFYGRVEHTNQGAKWVD TQVVLAMPYDTPVPGYRNNVVNTMRLWSA KAPNDFNLKDFNVGGYIQAVLDRNLAENISRVLYPNDNFFEGKELRLKQEYF VVAATLQDIIRRFKSSKFGCRDPVRTSFD A FPDKVAIQLNDTHPSLAIP ELMRI LVDEERLEWEKAWDVTVRTCAYTNHTVLPEALERWPVHLMETLLPRHLQII YEINQRFLNRVAAAYPGDV DRLRRMSLVEEGAVKRINMAHL CIAGSHAVNG VARIHSEILKKTIFKDFYELEPHKFQNK TNGITPRRWLVMCNPGLAEVIAERI GEEYIADLDQLRKLLSYVDDEAFIRDVAKVKQENKLKFSAYLEREYKVHINP NSLFDIQVKRIHEYKRQLLNCLHVITLYNRIKKEPNKFVVPRTVMIGGKAAP GYHMAKMIIKLITAIGDVVNHD PVVGDRLRVIFLENYRVSLAEKVIPAADLS EQISTAGTEASGTGNMKFMLNGALTIGTMDGANVEMAE EAGEENFFIFGMR

VEDVERLDQKGYNAQEYYDRIPELRHVIDQLSSGFFSPKQPD LFKDIVNMLM
HHDRFKVVFADYEDYIKCQERVSA LYKNPREWTRMVIRNIATSGKFSSDRTIA
QYAREIWGVEPSRQRLPAPDEKI 112 Pig
MPHQYPALTPEQKKELSDIAHRIVAPGKGILAADESTGSI AKRLQSIGTENTE A0A286ZYX8
ENRRFYRQLLLTADDRVNPCIGGVILFHETLYQKADDGRFPFQVIKAKGGVV (Fructose-
GIKVDKGVVPLAGTNGETTTQGLDGLSERCAQYKKDGADFAKWRCVLKIG bisphosphate
EHTPSSLAIMENANVLARYASICQQVGLQNGIVPIVEPEILPDGDHDLKRCQY aldolase)
VTEKVLAAVYKALSDHHIYLEGTLLKPNMVTPGHACTQKYSHEEIAMATVT
ALRRTVPPAVPGITFLSGGQSEEEASINLNAINKCPLLKPWALTFSYGRALQA
SALKAWGGKKENLKAAQEEYVKRALANSLACQGKYTPSGKAGAAASESLF ISNHAY 113
Pig MAPKKDVKKPAAAAAPAPAPAPAPAPAPAPPKEEKIDLSA IKSFSADQIAEFK
A0A287BJF1 EAFLLFDRGTGECKITLSQVGDVLRALGTNPTNAEVKKVLGNPSNEEMNAKKI
(Myosin light
EFEQFLPMLQAISNNKDQGSYEDFVEGLRVFDKEGNGTVMGAELRHVLATL chain 1)
GEKMKEEEVEALMAGQEDSNGCINYEAFVKHIMSI 114 Pig G3P
MVKVG VNGFGRIGRLVTRAAFN SGKVDIVAINDPFIDLHYMVYMFQYDSTH
(Glyceraldehyde-
GKFHGTVKAENGKLVINGKAITIFQERDPANIKWGDAGATYVVESTGVFTT 3-phosphate
MEKAGAH LKGGAKRVIISAPSADAPMFVMGVNHEKYDNSL KIVSNASCTTN
dehydrogenase)
CLAPLAKVIHDHFGIVEGLMTTVHAITATQKTVDGPSGKLWRDGRGAAQNII
PASTGAAKAVGKVIPELNGKLTGMAFRVPTPNVSVVDLTCRLEKPAKYDDI
KKVVKQASEGPLKGILGYTEDQVVSCDFNSDTHSSTFDAGAGIALNDHFVK
LISWYDNEFGYSNRVVDLMVHMASKE 115 Pig TPM1
MDAIKKKMQLKLDKENALDRAEQAEADKKA AEDRSKRLEDELVSLQKK (Tropomyosin
LKATEDEL DKYSEAPKDAQEKLELA EK KATDAEADVASLNRRIQLVEEELD alpha-1
chain) RAQERLATALQKLEEA EKA ADESERGMKVIESRAQKDEEKMEIQEIQLKEA
KHIAEDADRKYEEVARKLVIIESDLERAEEERAELSEGKCAELEELKTVINNL
KSLEAQAEKYSQKEDKYEEEIKVLSDKLKEAETRAEFAERSVTKLEKSIDDL
EDEL YAQKLKYKAISEELDHALNDMTSI 116 Pig
MPEAKPATKKAPAGKDAAKPAPKEAPPKGAP EEPPEAPPEDQSPTAE EPTG A0A5G2QB13
IFLKKPDSVSVENGKDALIVAKVNGKELPGKPEIKWFKGKWLELGSKSGARF (Myosin
binding SFKESHDSASNVYTLELRIGKVVLGDRGDYRLEVKA KDFCDSCAFNIDVEAP
protein C2) RQDSAGQSLESFKRS GEGKPEGAGELDFSGLLKKRQVVEEEKKKKKKDDDDL
GIPPEIWELLKGAKKSEYEKIAFQYGITDLRGMLKRLKKAKVEVKKSAAFTK
KLDPAYQVDQGNKIKLVVEISDPDLPLKWKNGQEIKPSSKYVFENVGKKRI
LCINKCTLADDAAYEVAVKDEKCFTEL FVREPPVLIVTPLEDQQVYVGDRVE
MTVEVSEEGAQVMWLKDGVELTRED SFKSRYRFKKDGKRHILIYSDVSLED
RGRYQVMTNGGQCEAELSVEEKQLEVLQDIADLTVKASEQAVFKCEVSDEK
VTGKWKYKNGVEVRPSKRITISHVGRFHKLVIDDVRPEDEGDYTFVPDGYALS
LSAKLNFLEIKVEYVPKQEPPKIHLDCSGKTS ENAVVVVAGNKLRLDVSITG
EPPP VATWMKEDEVFSGTEGRVRLEQRADISSFVIESAERGDEGRYTIKVTNP
AGEDVASIFLRVVDVPDPPEAVRVTTVGEDWAILVWEPPKYDGGQPVTGYL
LERKKKGSQRWMKLNFEVFTETTYESTNMIEGILYEMRVFAVNAIGVSQPSV
NTKPFMPIAPTSEPQH LTVEDVTDTTTTLKW RPPDRIGAGGIDGYLVEYCVE
GSDEWIPANTEPVERCGFTVKNLPTGARITFRVVG VNIAGRSQPATLAQPVTI
REIVGEPPRQSSVLIRASPD SLVATTLSVPLRRVPQPYAEPAPGIVSPEQPKIRL
PRHLRQTYIRKVG EHLNLVIPFQ GKPRPQV VWTGKGAPIDTSRVHVRTSNFD
TVFFVRQAARSDSGEYELSVQIENMKDTATVHIRVVEKAGPPQNMVKEV
WGTNALVEWQPPKDDGNTEITGYFVQKADKKTMEWFNVYEHN RHTSCTV

SDLMIGNEYYFRVYSENICGLSDSPGVSKNTARILKTGITFKLPEFKEHDFTRTP
PKFLTPLPDRVVVAGYTAALNCAVRGHPKPKVWWMKNQMEIGEDPKFLMT
NHQGVLTNLIRRPSPFDSGTYSRCASTSWGRPWPASWKSECRSEAPASTQG
SWLLSPDPDLPEPLSLSRVITRFLSRTQ 117 Pig
TMQHVYTKMSLVAVPFYQKRHRHFDQSYRNIQTRYLLDEYAAKKRASTQS A0A287BSB6
SSQKSVTQESSSQRESSSERASSQTLLGGTTCRLCAKRVSTALEEEEEQERKRR
(Uncharacterized
YHSLLAAYGEAKRERYLSELAQLEEDVHLARTHARDKLDRLFALQHVDTHA protein)
LRHAVGDRLAWERHSFEDRISRAPERILVRLRSHTVWERMSVKLCFTVQGFPT
PVVQWYKDGSLICQAGEPGKYRIESKYGVHTLEINRADFDDTATYSAVATN
VHGQVSTNAAVVRRFHHGDEEPPHFSVGLPIRLPLSSVIPYTHFDVQFLEKFGV
TFRREGETLTLKCTLLVTPDLKRVQPRAEWYRDDVLLKESKWTKMFFGEGQ
ASLSFSLNKKDDEGLYTLRIVSRGGISDHS AFLFVRDADPLVMGAPGAPMDV
HCHDANRDYVIVTWKPPNTTKE SPVLGYFIDRCEVGTDNWVQCNDAPVKIC
KYPVSGLVEGRSYTFRVRAVNSAGVSRPSRASEAVAALDPLDLKRLQGIHW
EGDKDIVIYQDDLEGEVQIPRPPTNVHASEISRTYVVL SWDPPTPRGKEPLMY
FIEKSVVSGGTWQRENAQTAVRSPRYAVFDLAEGKSYVFRVLSANKHGLSD
PSEITSPIQAQDTIVVPSAPGRVLASRNTKTSVVVQWDQPKHGEDLLGY YVD
CCVAGSNVWEP CNHKPIGYNR FVVHGLTTGEQYIFRVKAVNAVGTSENSQE
SDVIKVQAALTVP SHPYGITLLNCDGHSM TLGWKVPKFSGGSP IIGYYVDKR
EGHHKNWHEVNSSPLKERILTVEGLTEGSTYEFKIAAPTWWGSGCRRIPVSS
SSVRPGQRLS QVPTSGWVGTLGFFLVLLSLYRRYLLGS GEHGD S 118 Pig ENOB
MAMQKIFAREILDSRGNP TVEVDLHTAKGRFRAAVPSGASTGIYEAL ELRDG (Beta-enolase)
DKSRYLGKGVLKAVEHINKTLGPALLEKKLSVVDQEKVDKFMIELDGTENK
SKFGANAILGVSLAVCKAGAAEKGVPLYRHIADLAGNPDLVLPVPAFN VING
GSHAGNKLAMQEFMILPVGASSFRE AMRIGAEVYHHLKGVIKGKYGKDAT
NVGDEGGFAPNILENNEALELLKTAIQAAGYPDKVVIGMDVAASEFYRNGK
YDLDFKSPDDPSRHITGEKLGELYKSF IKNYPVVSIEDPFDQDDWKTWTSF LS
GVNIQIVGDDLTVTNLKRIAQAVEKKACNCLLLKVNQIGSVTESIQACKLAQ
SNGWGV MVSHRSGETEDTFIADLVVGLCTGQIKTGAPCRSERLAKYNQLMR
IEEALGDKAVFAGRKFRNPKAK 119 Pig Q5XLD2
MAPKKAKRRAAAEGSSNVFSMFDQTQIQEFKEAFTVIDQNRDGIIDKEDLRD (Myosin
light TFAAMGRLNVKNEELDAMMKEASGPINF TVFLT MFGEKLGADPEDVITGA chain)
FKVLDPEGKGTIKKH FLEELLTTQCDRFSQEEIKNMWAAFPDPVGGNV DYK
NICYVITHGDAKDQE 120 Pig MYH1
MSSDQEMAIFGEAAPYLRKSEKERIEAQNKPFDAKTSVFVAEPKESFVKGT V (Myosin-1)
QSREGGKVTVKTEAGATLTVKEDQVFPMNP PKFDKIEDMAMMTHLHEPAV
LYNLKERYAAWMIYTYSGLFCVTVPYKWL PVYNAEVVTAYRGKKRQEAP
PHIFSISDNAYQFMLTDRENQSILITGESGAGKTVNTKRVIQYFATIAVTGEKK
KEPTSGKMQGTLEDQIISANPLLEAFGNAKTVRNDNSSRFGKFIRIHFGTTG
KLASADIETYLLEKSRVTFQLKAERSYHIFYQIMSNKKPELIEMLLITNPYDY
AFVSQGEITVPSIDDQEELMATDSAIEILGFTSDERVS IYKLTGAVMHYGNLK
FKQKQREEQAEPDGT EVADKAAYLQGLNSADLLKALCYPRVKVGN EFVTK
GQTVQQVYNAV GALAKAVYDKMFLWMVTRINQQLDTKQPRQYFIGVLDIA
GFEIFDFNSLEQLCINF TNEKLQQFFNHMMFVLEQEEYKKEGIEWEFIDFGMD
LAACIELIEKPMGIFSILEEECMFPKATDTSFKNKLYEQHLGKSNNFQKPKPA
KGKVEAHFSLIHYAGTVDYNITGWLDKNKDPLNETVVGLYQKSSVKTLAFL
FTGAAGADAEAGGGKKGKGGKGGSSFQTVSALFRENLNKLMTNLRSTHPHF
VRCIIPNETKTPGAMEHELV LHQLRCNGVLEGIRICRKGFP SRILYADFKQRY
KVLNASAIPEGQFIDSKKASEKLLGSIDIDHTQYKFGHTKVFFKAGLLGLLEE

MRDEKLAQLTRTQARFVLRVERRESIFCIQYNIRAFMNVVKH
WPWMKLYFKIKPLLKSAETEKEMANMKEEFEKTKESLAKAEAKRKELEEK
MVALMQEKNDLQLQVQAEADSLADAEERCDQLIKTKIQLEAKIKEVTERAE
DEEEINAELTAKKRKLEDECESELKKDIDDLELTLAKVEKEKHATENKVKNLT
EEMAGLDETIAKLTKEKKALQEAHQQTLDLQAEEDKVNTLTAKATKLEQ
QVDDLEGSLEQEKKLRLMDLERAKRKLEGDLKLAQESTMDIENDKQQQLDEK
LKKKEFEMSNLQSKIEDEQALAMQLQKKIKELQARIEELEEIEAERASRAK
AEKQRSDSLRELEEISERLEEAGGATSAQIEMNKKREAEFQKMRRDLEEATL
QHEATAATLRKKHADSVAEELGEQIDNLQRVKQKLEKEKSEMMEIDDLAS
NMETVSKAKGNLEKMCRTLEDQSELKTKEEEQQRLINDLTAQRARLQTES
GEYSRQLDEKDTLVSQLSRGKQAFQTQIEELKRQLEEEIKAKSALAHAVQSS
RHDCDLLREQYEEEQEAKAELGRAMSKANSEVAQWRTKYETDAIQRTEELE
EAKKKLAQRLQDAEEHVEAVNAKCASLEKTKQRLQNEVEDLMIDVERSNA
ACAALDKKQRNFDKILAEWKQKYEETHAELEASQKESRSLSTELFKVKNAY
EESLDQLETLKRENKNLQQEISDLTEQIAEGGKRIHELEKIKKQVEQEKSEIQ
AALEEAEASLEHEEGKILRIQLELNQVKSEVDRKIAEKDEEIDQLKRNHVRV
VESMQSMLDAEIRSRNDAIRLKKKMEGDLNEMEIQLNHANRMAAEALRNY
RNTQGILKDTQIHLDDALRSQEDLKEQLAMVERRANLLQAEIEELRATLEQT
ERSRKVAEQELLDASERVQLLHTQNTSLINTKKKLETDISQIQGEMEDIIEA
RNAEEKAKKAITDAAMMAEELKKEQDTS AHLERMKKNLEQTVKDLQHRL
DEAEQLALKGGKKQIQKLEARVRELEGEVESEQKRVETVKGLRKHERRVK
ELTYQTEEDRKNILRLQDLVDKLQAKVKS YKRQAEEAEEQSNVNLSKFRKL
QHELEEAERADIAESQVNKL RVKSREVHTKIIEE 121 Pig F1SHX0
MADDEEYEEVVEYYTEETVYEEVPGETITEVYETTTTTRTSGYEQSDTSQPAV (Nebulin)
AQPMPSKPVERKKVIRKKVDPSPKFMTPIYIAHSQKMQDLFSTNKKYKKNYEKA
KGKPYAITTDTPELRRIKKVQDQLSEVKYRIDGDVAKTICHVDEKAKDIEHA
KKVSQQVSKVLYKQNWEDTKDKYLLPPDAPELVQAIKNTAMFSKKLYTED
WEADKSLFYYPNDSPELRRVAQAQKALSDIAYKKGHAEQQTQFTSLPDPPI
EFAKKVTNQVSKQKYKEDYENKIKGKWSETPCFEIATARMNAKNISTRKYQ
EDFENMKDQIYFMQTETPEYKVNKQAGTAASKVKYKEDYEKNKGKADYN
VLPASENPLLRLHLKAAGDALSDKRYKENYEKTKAKSINYCETPKFKLDTVL
HNFSSDTKYKDSYLKNILGHYVGSFEDPYHTHCMKVAAENSDKNYKA EYE
EDRGKGFFPQTITQEYEAIKKLDQCKDH TYKVHPDKTKFTQVTDSPVLVQA
QVNSKQLSDLNYKAKHENEKFKCHIPPDTPAFIQHKVNAYNLSDNVYKHD
WEKTKAKKFDIKVDAIPLLA AKANTKNTSDVMYKKDYEKSKGKMIGALSIN
DDPKMLHSLKTAKNQSDRLYKENYEKTKAKSMNYCETPKYQLDTLLKNFS
EAKYKDSYVQNVLGHYIGSFEDPYQLHCMKVSAQNSDKNYKA EYEEDKGK
CYFPQTITQEYEAIKKLDQCKDH TYKVHPDKTKFTAVTDTPVLLQAQLNTK
QLSDLNYKAKHEGEKFKCNIPADAPQFIQHRVNAYNLSDNVYKHDWEKTK
AKKFDIKVDAIPLLA AKANTKNTSDVMYKKDYEKSKGKMIGALSINDDPKM
LHSLKTAKNQSDREYRKDYEKSKTIYTAPLDM LQVTQAKKSQAIASDV DYK
HLLHNYSYPPDSINVDLAKKAYALQSDVEYKADYNSWMKGC GWIPFGSLE
MEKAKRASDILNEKKYRQHPETLKFTSIEDAPIIVQSKINQAQRSDIAYKAKG
EEIIHKYSLPADLPFIQAKVNAYNISESMYKADLKDLSKKGYDLRTDAIPIKA
AKAARQAASDVQYKKDYEKAKGKMVG FQSLQDDPKLVHYMNVAKIQSDR
EYKKDYEKTKTKYNTPHDMFNVVAAKKAQDVASNVNYRHSLHHYTYLPD
AMDLELSKNMMHIQSDNAYKEDYNTWMKGIGWIPLGSLEVEKVKKAGDA
LNEKKYRQHPDTLKFTSIVDSPVMVQAKQNT RQVSDILYKAKGEDVKHKYT
MGPDLPQFLQAKCNAYNISDV CYKRDWHDLIAKGNNVLGDAIPITAAKASR
NIASDYKYKEAYEKAKGKHVGFRSLQDDPKLVHYMNVAKLQSDREYKKN

YENTKTSYHTPDGMSITAHMAQDVATNMNYKQPIHHTYTLPLDSTHT
RNVNQQSDNVYKDEYNHFFKGIGWIPLGSLEVEKVKKAGDALNERKYRQH
PDTIKFTSVPDSMGMVLAQHNTKQLSDLNKVEGEKCLKHKYTMDPDVPQFI
QAKVNAINMSDAHAKADWKKTLAKGYDLTPDAIPIVAAKSSRNIAADFKEYK
EAYEKAKGKQVGFRLQDDPKLVHYMNVAKIQSDREYKKGYEASKTRYHT
PLDMLSVTAAKKSQEVATNTNYKQPFHHYTLLPDALNVEHSRNAMQIQSDN
LYKSDFTNWMKGIGWIPMESLEVEKAKKAGEILSEKKYRQHPEKCLKFTYAM
DTMEQALNKSNNKLIMDKRLYTEKWNKDKTTHVMPDTPDILLSRVNQITLS
DKLYKAGWEEEEKKGYDLRPDAISIKAAARASRDIAADYKYKQAYEQAKGK
HIGFRSLEDDPKLVHFMQVAKMQSDREYKKAYEKSSTSFTHTPVDMLSVVA
AKKSQEVATNANYRNVIHTYNMLPDAMSFELAKNMMQLQSDNQYKADYA
DFMKGIGWLPLGSLEAEKNKKAMEIIEKKYRQHHPDTLKYSTLMDSMNML
AKNNAKIMNEHLYKQAWADKTKVHIMPDIPIILAKANAVNMSDKLYKLS
LEESKKKGYDLRTDAISIQAAKASRDIAADYKYKHSYEEKERGKMVGFRSLED
DPKLVHSMQVAKMQSDREYKKSYEKTTSYHTPADMLSVTAARDAQANIT
NTNYKHLIHKYILLPDAMNIQLSKNMNHIQSDNEYKQDYNECYKGLGWSPA
GSLEVEKAKKATEYASDQKYRQHPSNFQFQKLNDSDMDMVLAKQNAHTMN
KYLYTADWNKDKTTHVMPDTPDILQAKQNQTLSQKLYKLGWEEALKKG
YDLPVDAISVQLAKVSRDNASDFKYKQGYRKQLGHHIGFRNLQDDPKLVLS
MNVAKMQSEREYKKDFEKWKTKFTSPVDMLGVVLAKKCCQALVSDVDYKN
YLHQWTCLPDQNDVIQAKRSYELQSENLYKSDLEWLRGIGWSPLGSLEAEK
NKRASEIIEKKYRQPPDRNKFTSIPDAMDIVLAKTNAKNRSDRLYREAWDK
DKTQVHIMPDTPIILAKANLINSSDKLYRMGYEELKKKGYDLPVDAIPIKA
AKASREIASEYKYKEGFRKQLGHHVVGARNIEDDPKMMWSMHVAKIQSDRE
YKKDFEKWKTKFSSPVDMLGVVLAKKCCQTLVSDIDYKNYLHQWTCLPDQS
DVIHARRAYDLQSDNLYKSDLQWLKGIGWLPSGSLEDEKNKRATQILSDHV
YRQHPPDKFKFSSLMDSIPMVLAKNNAITMNHRLYTEAWDKDKTTVHIMPDT
PEVLLAKQNKINYSEKLYKLGLEEAKRKGDMRLDAIPRAAKASRDIASEF
KYKEGYRKQLGHHIGARAIHDDPKMMWSMHVAKVQSDREYKKDFEKWKTK
KFSSPVDMLGVVLAKKCCQTLVSDIDYKNYLHQWTCLPDQSDVIHARRAYDL
QSDNMYKSDLQWLRGIGWVPIGSLDVEKCKRATEILSDKIYRQPPDTFKFTS
VPDSLEQVLAKNNAINMKNRLYTEAWDKDKTQVHIMPDTPEIMLARQNRL
NYSETLYKLANEAEAKKGYDLRSDAIPVAAKASRDIVSDYKYKDGYRKQL
GHHIGARNIEDDPKMMWSMHVAKIQSDREYKKDFEKWKTKFSSPVDMLGL
VQAKKCCQILVSDVDYKNYLHQWTCLPDQNDVIHARQAYDLQSDNIYKSDL
QWLKGIGWVPIGSVDVVKCKRAAEILSDHIYRQPPDKLKFTSVPDSLEQVLA
KNNAINMKNRLYTEAWDKDKTQIHIMPDTPEIMLARQNKINYSESLYRQAM
EEAKKEGYDLRSDAIPVAAKASRDIAADYKYKEAYRKQLGHHIGARAVHD
DPKMMWSLHVAKMQSDREYKKDFEKYKTRFSSPVDMLGIVLAKKCCQTLVS
VDYKHPLHEWTCLPDQNDIIHARKAYDLQSDNLYKSDLEWMKGIGWIPLG
SVEVVKAKRAGEILSDNIYRQRPDTLKFTSVTDSPELVLAKNNALNMKNRL
YTEAWDNDKKTTHVMPDTPEIMLAKLNRINYSKLYRLALEESKKEGCDLR
VDAIPIQAAKASREIASDYKYKEAYRKQLGHHIGARNIEDDPKMMWSIHAA
KIQSDREYKKDFEKWKTRFSSPVDMLGLVQAKKCCQILVSDIDYKHPLHEWIC
LPDQNDIIQARKAYDLQSDAIYKADLEWLRGIGWVPLGSVEVEKVKRAGEIL
SERKYRQPADQLKFTCITDTPEIVLAKNNALTMKHLYTEAWDTDKTSIHV
MPDTPEILLAKSNSANISQKLYTKGWDESKMKDYDLKADAISIKASAKASRDI
ASDYKYKEAYEKQKGHHIGAQSVEDDPKIMCAIHAGKIQSEREYKKDFQKW
KTKFTSPVDMLGILLAKKCCQTLVSDIDYRNYLHHWTCLPDQNDIIQARKAYE
LQSDALYKADLEWLRGIGWMPQGSPEVLRVKNQAQEILRDSVYRTPVVKLK

YTSIVDTPEVVLAKSNAENISIPKYREVWWDKDKTSHVIMHPDTPPEINLARANAL
NVSNNKIYREGWDEMKMSCDVRLDAIPIQAAKASREIASDYKYKLDHEKQKG
HYVGTLTARDDNKIRWALIAGKIQNEREYRLHWAKWWSKQSPADMLSITH
SKNCQALVSDIDYRNYLHQWTCLPDQHDVIQARKAYDLQSDAIYKADLEW
LRGIGWMPSDSVSVNHAKHAADIFSEKKYRTKIETLNFTPVDDRVDYVTAK
QSGEILNDIKYRKDWNDTKSKYTLTETPQLHAAQEAARILDQYLYKESWEK
QKATGYILPPDAVPFVHAHHSSDVQSELKYKADHVKKQKGHYVGVPTMRDD
PKLVWFEHAGQIQNDRLYKEHYHKTKAKINIPDMVSVVAAKEGQILVSDID
YRNYLHQWTCHPDQNDVIQARKAYDLQSDNVYKADLEWLRGIGWIPLDSV
DHVRVTRNQEMVNIQIKYKKDALANYPNFTSVVDPPAVVLAKINSVNQSDV
KYKETFNKLLKGKYIFSPDTPYITHSKDMGKLYSTILYKGAWEGTKAYGYTL
DERYIPIVGAKHADFVNSELKYKETYEKQKGHYLAGKEISEFPNVVHCLDFQ
KMRSVLNYYRRHYEDTKANVHIPNDMMNHVLAKRCQYILSDLEYRHYFHQ
WTSLEPEPNVIRARNAQEILSDNVYKDDLNLWLKGIGCYVWDTPQILHAKKS
YDLQSQILYTADGKENLKNYNMVTDTPLYVTALQSGINASEVKYKENYHQI
KDKYTTVLETVDYDRTRNLKDLYSSNVYKEAWDKVKATSILPSNTVSLTH
AKNQKHLASHIKYREEYEKFKALYTLPKSVDDDPNTARCLRVGKFNIDRLY
RSVYEKNKMKIHMVPDMVEMVTAKDSQKKVSEIDYRLHLHEWICHPLQV
NSHVRKVTDQISDIVYKDDLNLWLKGIGCYVWDTPCILHAKHAYDLRNDIKY
QAHVKKTRNNYKLVTDTPLYVQAVKSGKQLSDAVYHYDYVHSSVRGKVAP
TTKTVDLDRALHAYKLQSENLYRKDGLNSLPTGYRLPGDTPHFKHNKDIRY
MSSYFKYKEVYEHFKANGYTLGPKDVPFVHVRVNNVTSERLYRQLYHKL
KDKIHPTPDTPAIRQVKKTQEAVSELIYKSDFFKMQGHMISMPYTPHVLHCR
YVGDITSIDIKYKEDLQILRGLGCFLYDTPDMVRSRHLRKLWSNYLYTDNAR
KMRDKYKVVLDTPEYRKVQELKTHLSELVYRAAGKKQKSIFTSVPDTPDLL
RAKRGQKLQSQYLYVELATKERPHHHAGNQTTALKHARDVKDMVSEKKY
KVQYEKMKDKYTPVPDTPILIRAKKAYWNASDLRYKETFQKTGKYHTVK
DALDIVYHRKVTDHISKIKYKENYMSQLGIWRSIPDRPEHFHRAVTDVSD
VKYKEDLTWLKGIGCYAYDTPDFTLAEKNKTLYSKYKYKEVFERTKSDFKY
EADCPINRHFKFATQLMNEKKYRADYEQRKDKYQLVVDEPRHLLAKIAGD
QISQIKYRKNYEKTDKFTSVVDTPEHLRTTKVKNQISDILYKLEYNKAKPR
GYTTIHDTPMLLHVRKVKDEVSDLKYKEVYQRTKSNTCTIEPDAVHIKAAKD
AYKVNTNLDYKKQYEANKAHWRWTPDRPDFIQAAKSSLQQSDFEYKLDRE
FLKGCKLSVTDDKDMVLALRNSVIESDLKYKEKHVKERGTCHAVPDTPQIL
LAKTVSNLVSENKYKRHVKKHLAQGSYTTLPETRDTVHVKEVTKHVSDTN
YKKKFVKERGKSNYSIMLEPPEVKHAMEVAKKQSDVAYKKDAKENLHYTT
VADRPDIKKATQAAKQASEVEYRAKHRKEGSHGLSMLGRPDIEMAKKAAT
LSSQVKYRENFNKEKGKTPKYNPKESQLYKIMKDANNLASEVKYKADLKK
LHKPVTDMKESLIMNHVLNTSHLASSYQYKKNYEKSCKGHYHTIPDNLEQLH
LKEATELQSIVKYKEKYEKERGKPMLEDFETPTYITAKESQQMQSGKEYRKD
YEESIKGRNLTGLEVTPALLHVKYATKIASSEKEYRKDLEESIRGKGLSEMEDT
PDMLRAKNATQILNEKEYKRDLELEVKGRGLNAMANETPDFMRARNATDI
ASQIKYKQLAEMEKANFTSVVDTPEIIHAQQVKNLSSQKKYKEDAEKCMSY
YETVLDTPEMQRVRENQKNFSLQYQCDLKNSSKGKITVVQDTPEILRVKEN
QKNFSSVLYKEDVSPGTAIGKTPEMMRVKQTQDHISSVKYKETIGQGTPIPD
LPEVKRVKETQKHISSVMYKENLGTGIPTPVTPEIERVKRNQENFSSVLYKEN
LGKGTPTPITPEMERVKRNQENFSSVLYKENMGKGTPLPVTPEMERVKHNQ
ENISSILYKENVGKGTPTPVTPEMQRVKRNQENISSVIYKENLGKATPTPFTPE
MERVKRNQENFSSVLYKENMRKATPTPVTPEMERAKRNQENISSVLYSDSF
RKQIQGKAAYVLDTPEMRRVRETQRHISTVKYHEDFEKHKGCFPTPVVTDPTIT

ERVKKNTQDFKSTQVIRKRVVMEQKRNQDQETITGLRVWRTNPGSVF
DYDPAEDNIQSRSLHMLNAQAQRRSREQSRASALSISGGEEKSEHSEAARL
STYSDGGVFFSATSAAAYKHAKTTELPQQRSSSVATQQTTVSSIPSHPSTAGKI
FRAMYDYMAADADEVSRFDGDAIVNVQAIDEGWMYGTVQRTGRTGMLPA NYVEAI 122
Pig MYH4 MSSDQEMAIFGEAAPYLKSEKERIEAQNKPFDAKTSVFVAEPKESFVKGTV
(Myosin-4) QSREGGKVTVKTEAGATLTVKEDQVFPMNPPKFDKIEDMAMMTHLHEPAV
LYNLKERYAAWMIYTYSGLFCVTVNPNYKWLVPVYNAEVVTAYRGKKRQEAP
PHIFSISDNAYQFMLTDRENQSILITGESGAGKTVNTKRVIQYFATIAVTGEKK
KEEPTPGKMQGTLEDQIISANPLLEAFGNAKTVRNDNSSRFGKFIRIHFGTTG
KLASADIETYLLEKSRVTFQLKAERSYHIFYQIMSNKKPELIEMLLITTNPYDY
AFVSQGEITVPSIDDQEELMATDSAIEILGFTSDERSIYKLTGAVMHYGNLK
FKQKQREEQAEPDGTEVADKAAAYLQGLNSADLLKALCYPRVKVGNEFVTK
GQTVQQVYNVAVGALAKAVYDKMFLWMVTRINQQLDTKQPRQYFIGVLDIA
GFEIFDFNSLEQLCINFTNEKLQQFFNHMHMFVLEQEEYKKEGIEWEFIDFGMD
LAACIELIEKPMGIFSILEEECMFPKATDTSFKNKLYEQHLGKSNNFQKPKPA
KGKAEAHFSLIHYAGTVDYNITGWLDKNKDPLNETVVGLYQKSSVKTALAFL
FAERQSSEEGGTKKGGKKKGSSFQTVSALFRENLNKLMTNLRSTHPHFVRCI
IPNETKTPGAMEHELVLHQLRCNGVLEGIRICRKGFP SRILYADFKQRYKVLN
ASAIPEGQFIDSKKASEKLLGSIDIDHTQYKFGHTKVFFKAGLLGTLEEMRDE
KLAQLITRTQAMCRGFLMRVEFRKMMERRESIFCIQYNIRAFMNVKHWPW
MKLYFKIKPLLKSAETEKEMANMKEEFEKTKEDLAKSEAKRKELEEKMVAL
MQEKNDLQLQVQAEADGLADAERCDQLIKTKIQLEAKIKEVTERAEDEEEEI
NAELTAKKRKLEDECESELKKDIDDLTLAKVEKEKHATENKVKNLTEEMA
GLDENIAKLTKEKKALQEAHQQTLDLQAEEDKVNTLTAKATKLEQQVDD
LEGSLEQEKKLRLMDLERAKRKLEGLDLKLAQESTMDIENDKQQLDEKLKKKE
FEMSNLQSKIEDEQALAMQLQKKIKELQARTEELEEEIEAERASRAKAEKQR
SDLSRELEEISERLEEAGGATSAQIEMNKKREAEFQKMRRDLEEATLQHEAT
AAALRKKHADSVAEELGEQIDNLQRVKQKLEKEKSELKMEIDDLASNMETVS
KAKGNLEKMCRTLEDQLSEVKTKEEEHQRLINELSAQKARLQTESGEFSRQL
DEKEALVSQLSRGKQAFQTQQIEELKRQLEEETKAKSALAHAVQSSRHDCDLL
REQYEEEQEAKAELGRAMSKANSEVAQWRTKYETDAIQRTEELEEAKKKL
AQRLQDAEEHVEAVNAKCALEKTKQRLQNEVEDLMLDVERSNAACAALD
KKQRNFDKILAEWKHKYEETQAELEASQKESRSLSTELFKVKNAYEESLDQ
LETLKRENKNLQQEISDLTEQIAEGGKHIHELEKVKKQIEQESELQAALAEA
EASLEHEEGKILRIQLELNQVKSEIDRKIAEKDEEIDQMKNRHNIRVVESMQST
LDAEIRSRNDALRIKKKMEGDLNEMEIQLNHANRQATEAIRNLRNNTQGVLK
DTQLHLDDAIRGQDDLKEQLAMVERRANLMQAEIEELRASLEQTERSRRVA
EQELLDASERVQLLHTQNTSLINTKKKLETDISQIQGEMEDIVQEARNAEKA
KKAITDAAMMAEELKKEQDTS AHLERMKNMEQTVKDLQHRLDEAEQLA
LKGGKKQIQKLEARVRELENEVENEQKRNVEAVKGLRKHERRVKELTYQT
EEDRKNVLRQLQDLVDKLQSKVKAYKRQAEEAEQSNVNL SKFRKLQHELEE
AEERADIAESQVNKL RVKSREVHTK VISEE 123 Pig ALBU

MKWVTFISLLFLFSSAYSRGVFRRTYKSEIAHRFKDLGEQYFKGLVLIAFSQ (Albumin)
HLQQCPYEEHVKLVRVTEFAKTCVADESAENC DKSIHTLFGDKLCAIPSLR
EHYGDLADCCEKEEPPERNECFLQHKNNDNP DIPKLKPD PVALCADFQEDEQK
FWGKYLYEIAARRHPYFYAPELLY YAIYKDV FSECCQAADKAACLLPKIEHL
REKVL TSAAKQRLK CASIQKFG ERAFKAWSLARLSQRFPKADFTEISKIVTDL
AKVHKECCHGDLLECADDRADLAKYICENQDTISTKLKECCDKPLLEKSHCI
AEAKRDEL PADLNP LEHDFVEDKEVCKNYKEAKHVFLGTFLYEYSRRHPDY
SVSLLLRIAKIYEATLEDCCAKEDPPACYATVFDKFQPLVDEPKNLIKQNC

FEKLGEYGFQNALRYTSTYQVSTPLVEVARKLGLVGSRCCKRPEEERL
SCAEDYLSLVNRLCVLHEKTPVSEKVTKCCTESLVNRRPCFSALTPDETYK
PKEFVEGTFTFHADLCTLPEDEKQIKKQ TALVELLKHKPHATEEQLRTVLGN
FAAFVQKCCAAPDHEACFAVEGPKFVIEIRGILA

[0047] An animal protein can comprise a bison protein (Table 6). A bison protein can comprise a bison muscle protein. A bison protein can comprise a bison blood protein. A bison protein can comprise a bison albumin protein. A bison protein can comprise a sequence that has at least or about 70%, at least or about 75%, at least or about 80%, at least or about 90%, at least or about 91%, at least or about 92%, at least or about 93%, at least or about 94%, at least or about 95%, at least or about 96%, at least or about 97% at least or about 98% at least or about 99% or about 100% sequence identity to any one of SEQ ID NO: 124-138.

[0048] A food product may comprise a fragment or a truncation of a bison protein. The food product may include a fragment or truncation of any one of SEQ ID Nos: 124-138. A sequence of the fragment or truncation may be identical or substantially identical to a portion of any one of SEQ ID Nos: 124-138. A sequence of the fragment or truncation may be at least about 70%, 80%, 90%, 95%, 98%, or more identical to a portion of a sequence of any one of SEQ ID Nos: 124-138. A length of the fragment or truncation may be at least about 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or more of a length of any one of SEQ ID Nos: 124-138. A length of the fragment or truncation may be less than or equal to about 90%, 80%, 70%, 60%, 50%, 40%, 30%, 20%, 15%, 10%, or less than a length of any one of SEQ ID Nos: 124-138.

TABLE-US-00006 TABLE6 Bison Proteins SEQ ID NO Description Amino Acid Sequence
124 Bison MPFGNTHNKHKLNFKAEEEEYPDL SKHNNHMAKALTLEIYKKLRDKETPSG
A0A6P3IIV2 FTLDDVIQTGV DNP GHPFIMTVGCVAGDEESYTVFKDLFDPIIQDRHGGGFKP
(Creatine kinase)

TDKHKTDL NHENLKG GDDLDP NYVLSSRVRTGRS IKG YALPPHCSR GERR
AVEKLSVEALNSLTGEFKGKYYP LKSMTEQEQQQLIDDHFLFDKPVSP LLL
ASGMARDWPDARGIWHNDNKSFLVWVNEEDHLRVISMEKGGNMKEVFR
RFCVGLQKIEEIFKKAGHPFMWNEHLGYVLTCPSNLGTGLRGGVHV KLAH
LSKHPKFEEILTRLRLQKRGTGGVDTA AVGSVFDVSNADRLGSSEVEQVQL
VVDGVKLMVEMEKKLEKGQSIDDMIPA QK 125 Bison

MNQIEPGVQYNYVYEDDEYMIQEEEWDRDLLDPAWEKQQRKTFTAWCN A0A6P3J4W6
SHLRKAGTQIENIEEDFRNGLKLMLLLEVISGERLPKPDRGKMR FHKIANVN (Alpha-
actinin-2) KALDYIASKGVKLVSIGAE EIVDGNVKMTLGM IWTIILRFAIQDISVEETSAK
EGLLLWCQRKTAPYRNVNIQNFHTSWKDGLGLCALIHRHRPDLIDYSKLNK
DDPIGNINLAMEIAEKHLDIPKMLDAEDIVNTPKPDERAIMTYVSCFYHAF A
GAEQAETAANRICKVLAVNQENERLMEEYERLASELLEWIRRTIPWLENRT
PEKTMQAMQKKLEDFRDYRRKHKPPKVQEK CQLEINFNTLQTKLRISNRPA
FMPSEGKMVSDIAGAWQRLEQA EKG YEEWLLNEIRRLERVEHLAEKFRQK
ASTHETWAYGKEQILLQKDYESSTLTEVRALLRKHEAFESDLAAHQDRVE
QIAAIAQELNELDYHDAVNVNDRCQKICDQWDRLGTLTQKRREALERTEK
LLETIDQLHLEFAKRAAPFNNWMEGAMEDLQDMFIVHSIEEIQSLITAHEQF
KATLPEADGERQSILAIQNEVEKVIQSY SIRISSNPYSTVTVD EIRSKWDKV
KQLVPIRDQSLQEELARQHANERLRRQFAAQANAIGPWIQNKMEEIARSSIQ
ITGALEDQMNQLKQYEHNIINYKNNIDKLEGDHQLIQEALVFDNKHTNYTM
EHIRVGWELLTTIARTINEVETQILTRDAKGITQEQMNEFRASFNFHFD RRK
NGLMDHEDFRACLISMGYDLGEAEFARIMTLVDPNGQGTVTFQSFIDFMTR
ETADTDTAEQVIASFRILASDKPYILAEELRRELPPDQAQYCIKRMPAYSGP
GSVPGALDYTAFSSALYGESDL 126 Bison

MSRPLTDQEKRKQISVRGLAGVENVT ELKKNFNRHLHFTLVKDRNVATPR A0A6P3HAI4
DYYFALAYTVRDHLVGRWIRTQQHY YEKDPKRIYYLSLEFYIGRTLQNTM (Alpha-1,4

glucan VNLALENACLYEYQGLDMELEEEDAGLNGGLGRLAACFLDSMAT
phosphorylase) LGLAAYGYGIRYEFGIFNQKISGGWQMEEADDWLRYGNPWEEKARPEFTLP
VHFYGRVEHTSQGAKWVDTQVVLAMPYDTPVPGYRNNVNTMRLWSAK
APNDFNLKDFNVGGYIQAVLDRNLAENISRVLYPNDNFFEGKELRLKQEYF
VVAATLQDIIRRFKSSKFGCLDPVRTNFDAPDKVAIQLNDTHPSLAIPELM
RILVDQERLEWEKAW EVT VKTCAYTNHTVLPEALERWPVHLIETLLPRHLQ
IIYEINQRFLNRVAAAFPGDVDRLRRMSLVEEGAVKRINMAHL CIAGSHAV
NGVARIHSEILKKTIFKDFYELEPHKFQNKTNGITPRRWLVMCNPGLAEIIAE
RIGEEYIADLDQLRKLLSYVDDES FIRDVAKVKQENKLKFSAYLEKEYKVHI
NPNSLFDIQVKRIHEYKRQLLNCLHVITLYNRIKKEPNKFFVPRTVMIGGKA
APGYHMAKMIIKLITAIGDVVNHDPPVVGDRLRVIFLENYRVSLAEKVIPAA
DLSEQISTAGTEASGTGNMKFMLNGALTIGTMDGANVEMAE EAGEENFFIF
GMRVEDVERLDQKGYNAQEYYDRIPELRHIIDQLSSGFFSPKQPD LFKDIVN
MLMHDRFKVFADYEEYVKCQERV SALYKNPREWTRMVIRNIATSGKFSS
DRTIAQYAREIWGVEPTRQRMPAPDEKI 127 Bison
MPHQYPALTPEQKKELCDIAHRIVAPGKGILAADESTGSI AKRLQSIGTENTE A0A6P3H8G9
ENRRFYRQLLLTADDRVNPCIGGVILFHETLYQKADDGRFPQVIKAKGGV (Fructose-
VGIKVDKGVVPLAGTNGETTTQGLDGLSERCAQYKKDGADFAKWRCVLKI bisphosphate
GEHTPSSLAIMENANVLARYASICQQNGIVPIVEPEILPDGDHDLKRCQYVT aldolase)
EKVLA AVYKALSDHHIYLEGTLLKPNMVT PGHACTQKYSHEEIAMATVTA
LRRTVPPAVPGITFLSGGQSEEEASINLNAINKCPLLKPWALTFSYGRALQAS
ALKAWGGKKENL KAAQEEYVKRALANSLACQGKYTPSGKAGAAASESLFI SNHAY 128
Bison MAPKKDVKNPAAAAAPAPAPAPAPAPAPAPPKEEKIDLSAIIKIEFSKQQQDE
A0A6P3J7N1 FKEAFLLFDR TGECKITLSQVGDVLRALGTNPTNAEVKKV LGNPSNEEMNA
(Myosin light KKIEFEQFLPMLQAISNNKDQGT YEDFVEGLRVFDKEGNGTVMGAELRHV
chain 1/3) LATLGEKMKEEEVEALMAGQEDSNGCINYEAFVKHIMSN 129 Bison
MVKVG VNGFGHIGRLVTRAAFNSGKVDVIAINDPFIDLHYIVYMFQYDTTH A0A6P3J4R7
GKFHGTVKAENGKLIINGKAVTIFQERDPASIKWGDAGAEYVVESTGVFTT
(Glyceraldehyde-
MEKAG AHLKGGAKSVIISAPSADAPMFVLGMNHEKYNNNLKIVSNASCTT 3-phosphate
NCLAPXAKVIHDHFGIVEGLMTTFHAITVTQKTVDGPSGKLWHDGGGAAQ
dehydrogenase) NIIPASTGAAKAVGKVIPELNRKLTGMAFCVPTPNVSVM DLTYHLEKPAKY
DEIKKVVKQTSEGPL EGILGYTQDQVVSCDFNSDSYXSTFDAGAGIALNDH
FVKLISWYDNEFGYSNRVVDLMVHMASKE 130 Bison
MDAIKKKMQLKLDKENALDRAEQAEADKXPVPHGHSQLEDEL VSLQKK A0A6P3GY04
LKATEDEL DKYSEALKDAQE KLELA EK KATDAEADVASLNRRIQLVEEEL (Tropomyosin
DRAQERLATALQKLEEA EKAADESERGMKVIESRAQKDEEKMEIQEIQLKE alpha-1 chain
AKHIAEDADRKYEEVARKLVIIESDLERAEEAE LSEGKCAELEEEELKTVTN isoform X2)
NLKSLEAQA EKYSQKEDKYEEEIKVLSDKLKEAETRAEFAERSVT KLEKSID
DLEDELYAQKLKYKAISEELDHALNDMTSM 131 Bison
MPEAKPAAKKAPAGKDAAAKPAPKEATPQEAPAAPPTGEIDGAPAGRAEP A0A6P3IB01
GGSSAASRVVPPRPLLRSVFPTAPGLTALPYSQPCRAPQRFQGGGLTVPLPP (Myosin-binding
HSPLLPGKDTVIVAKLNGKELPAKPAVKWFKGKWLELGSKSGARFSFKESH protein C,
fast DAASNVYTIELHITKVVLGDRGDYRIEVKAKDFCDSCAFNIDVEAPRQDSA type)
GQSLESFKRSGEAKSDTAGELDFSGLLKKRQVVEEEKKKKKDDDDDLGIPPEI
WELLKGAKKSEYERIAFQYGITDLRGMLKRLKKAKVEVKKSAAFTKKLDP
AYQVDRGNKIKLVVEISDPDLPLKWFKNGQEIKPSSKYVFENVGKKRILTIN
KCTLADDAAYEVVVKDEKCFTELFVKEPPVLIVTPLEDQQVFVGD RVEMA
VEVSEEGAQVMWLKDGVELTREDSFKARYRLKKDGKRHILYSEVTMEDK
GHYQVMTNGGQCEAELIVEEKQLEVLQDIADLTVKASEQAVFKCEVSDEK

VTGKYNKGWKNYKNGEYVTPEDKIDVRFPEDEGDYTFVPDGYAL
SLSAKLKFLKVEYVVPKQEPPKIHLD CSGQTSENAIVV VAGNKL RMDVSIT
GEPRPVATWMKEDEVFTGTEGRVHIEQRGDISSFVIESAERGDEGRYTIKVT
NPVGEDVASILLKVVDVPDPPEAVRVT SVGEDWAVLVWEPPKYDGGRPVT
GYLLERKKKGSQRWMKLNFEVFTETTYESTNMIEGILYEMRVFAVNVIGVS
QPSMNTKPFMPIAPTSEPRHLTVEDVTD TTTTLKWRPPDKIGAGGIDGYLIE
YCVEGSDEWIPANTEPVERCGFTVKNLPTGAKITFRVVG VNIAGRSQPATL
AQPVTIREIVEQPKIRLPRHLRQTYVRKVGEHLNLVIPFQGKPRPQV VWTKG
GAPIDTSHVHVRTSDFDTVFFVRQAARSDSGEYELTVQIENMKDTATVCIR
VVEKAGPPQNMVKEVWGTNALVWQPPKDDGNSEITGYFVQKADKKT
MEWFTVYERNRHTSCTVSDLIMGNEYFRVYSENVCGLSDLPGVSKNTARI
VKTGMTLKLPEYKEHDFRTPPKFLTPLPDRVVVAGYAAALNCAVRGHPKP
KVVWMKNKMEIREDPKFLMTNQQGVLT LNIRRPSPFDSGTYS CRAVNELG
EALAECKLEVRVPQ 132 Bison
MSLVAVPFYQKRHKHFDQSYRNIQTRYLLDEYS AKKKRASAQSSLQRSVT A0A6P3GV86
RKSSSQRESSRAALGESTLGETTLGETSLGDTTLGR TTCRLCAKRMSSALEE (Myomesin-2)
EAQERKRRYQSMVAAYGEILVRLRSHTV WERMSVKLCFTVQGFPTPVVQ
WYKDGS LICQAGEPGKYRIDSKYGVHTLEINRADFDDTATYSAVATNVHG
QVSTNAAVVRRFRGDEEPFHSVGLPIGLPLSSVIPYTHFDVQFIEKFGVTFR
REGETLTLKCTLLVTPDLKRVQPRAEWYRDDVLLKESK WTKMFFGEGQAS
LSFSHLNKDDEGLYTLRIVSRGGITDHS AFLFVRDADPLVTGAPGAPMDVH
CHDANRDYVIVTWKPPNTTKESPVIGYFIDRCEVGT DNWIQCNDAPVKICK
YPVTGLFEGRSYIFRMRAVNSAGISRPSRVSEAVAALDPVDLRRLQAVHLE
GEKDVIYQEELEGEVQIPGPPTNVHASETSRTYVVL SWDPPEPRGKEPLMY
FIEKSMVGSGSWQRVNAQTAVRSPRYAVFDLAEGKPYVFRVLSANKHGLS
DPSEITAPIQAQDTIVVPSAPGRVLASRNTRTSVV VQWDRPKHEEDLLGY
VDCSVAGSNVGEPCNHKPIGYNR FVVHGLTTGEQYIFRVKAVNAVGTSENS
QESDVIKVQAALTVPSHPYGITLLNCDGHSMILGWKVPKFSGGSPILGY YV
DKREAHHKNWHEVNSSPLKERILTVEGLTEGSLYEFKIAAANMAGIGQPSD
PSELFKCEAWTMPEPGPAYDLTFCEVRDTSLVLLWKPPVYSGSSPVSGYFV
DYKEEDSGEWLTVHETATPHRYLKVCDLHQGKTYVFRVRAVNASGVGRP
SDTSEPVLVEARPGTKEVSAGVDEEGNVYLSFDCPEVTDASHFTWCKSYED
IADDDRFBK VETAGDHSKLYFKNLDKEDLGTYSVSVSDTDGVSSSVLDEEE
LERLKALSNEIKNPTIPLKSELAYEIFDKGQVRFWLQAEHLSPDSSYRFIIND
REVADSETHRIKCDKSTGIIEMVM DRFTIDNEGTYTVQIQDGKAKSQSSLVL
IGDAFKAVLKEAEFQRKEFLRKQGP HFAEYLHWDVTEECEVRLVCKVANT
KKETVFKWLKDDVLYETEKMPDLEKGVCELLIPKLSKKDHGEYKATLKDD
RGQDVSTLEIAGKVYEDMILAMSRVCGASASPLKILCTPEGIRLQCFMKYFT
EEMKVNWYHKDAKITSSEHMRIGGSEQMAWLQICEPTEKDKGKYTFEILD
DKKNHQ RSLDLSGQAFDEAFAEFQQLKAAAF AEKNRGKVIGGLPDVVTIM
EGKTLNLTCTVFGNPDPEV VWFKNDKDIELSDHFSVKVEQAKYVSMTIKG
VTSEDSGKYSIHVKNKYGG EKIDVTVSVYKHGEEIPDVVLPQQA KPKLLPA AAPSPAH 133
Bison MRTLRAMAMQKIFAREILDSRGNPTVEVDLHTAKGRFRAAVPSGASTGIYE
A0A6P3HSX8 ALELRDGDKSRYLGKGV LKAVEHINKTLGPALLEKKLSVVDQEKVDKFMI
(Phosphoryruvate
ELDGTENKSKFGANAILGVSLAVCKAGAAEKGVPLYRHIADLAGNPELILP hydratase)
VPAFN VINGGSHAGNKLAMQEFMILPVGASSFRE AMRIGAEVYHHLKGVIK
AKYGKDATNVGDEGGFAPNILENNEALELLKTA IQAAGYPDKVVIGMDVA
ASEFYRNGKYDLDFKSPDDPARHISGEKLGELYKNFIKNYPVVSIEDPFDQD
DWATWTSFLSGVNIQIVGDDLTVTNPKRIAQAVEKKACNCLLLKVNQIGSV

TESIQACLAQSNQSGVMVSGHSTGCTGFIADLVVGLCTGQIKTGAPCRS
ERLAKYNQLMRIEEALGDKAVFAGRKFRNPKAK 134 Bison
MAPKKAKRRAAAEGGSSSVFSMFDQTQIQEFKEAFTVIDQNRDGIIDKEDL A0A6P3H0Z6
RDTFAAMGRLNVKNEELDAMMKEASGPINFTVFLNMFGEKLGADPEDVI (Myosin
TGAFKVLDPGKGTIKKQFLEELLTTQCDRFSHEEIKNMWAAFPDPVGGNV regulatory
light DYKNICYVITHGDAKDQE chain 2) 135 Bison
MSSDQEMAIFGEAAPYLKSEKERIEAQNKPFDAKTSVVFVADPKESFVKAT A0A6P3HQ42
VQSREGGKVTAKTEAGATVTVKEDQVFPMNPPKFDKIEDMAMMTHLHEP (Myosin-1)
AVLYNLKERYAAWMIYTYSGLFCVTVNPKWLPVYNAEVVTAYRGKKRQ
EAPPHIFSISDNAYQFMLTDRENQSILITGESGAGKTVNTKRVIQYFATIAVT
GEKKKEEPTSGKMQGTLEDQIISANPLLEAFGNAKTVRNDNSSRFGKFIRIH
FGTTGKLASADIETYLLEKSRVTFQLKAERSYHIFYQIMSNKKPELIEMLLIT
TNPYDYAYVSQGEITVPSIDDEELMATDSAIEILGFTSDERSIYKLTGAV
MHYGNLKFQKQKQREEQAEPDGTEVADKAAYLQGLNSADLLKALCYPRVK
VGNEFVTKGQTVEQVYNAV GALAKAVYDKMFLWMVARINQQLDTKQPR
QYFIGVLDIAGFEIFDFNSLEQLCINFTNEKLQQFFNHMHMFVLEQEEYKKEGI
EWEFIDFGMDLAACIELIEKPMGIFSILEEECMFPKATDMSFKNKLYEQHLG
KSNNFQKPKPAKGKAEAHFSLIHYAGTVDYNITGWLDKNKDPLNETVVGL
YQKSSVKTLLALLFSGPASGEAEGGPKKGGKKKGGSSFQTVSALFRENLNKLM
TNLRSTHPPHFVRCIIPNETKTPGAMEHELVHLHQLRCNGVLEGIRICRKGFP
SRILYADFKQRYKVLNASAIPEGQFIDSKKASEKLLASIDVDHTQYKFGHTKVF
FKAGLLGLLEEMRDEKLAQLITRTQARCRGFLARVEYQKMVERRESIFCIQ
YNVRAFMNVKHWPMKLYFKIKPLLKSAETEKEMANMKEEFEKTKEELA
KSEAKRKELEEKMTLTQEKNDLQLQVQSEADALADAEERCDQLIKTKIQL
EAKIKEVTERAEDEEEINAELTAKKRKLEDECESELKKDIDDLELTLAKVEKE
KHATENKVKNLTEEMAGLDETIAKLTKEKKALQEAHQQTLDLQAEEDK
VNTLTAKTKLEQQVDDLEGSLEQEKKLRMDLERAKRKLEGDLKLAQEST
MDIENDKQQLDEKLKKKEFEMSNLQSKIEDEQALAMQLQKKIKELQARIEE
LEEEIEAERASRAKAEKQRS DLSRELEEISERLEEAGGATSAQIEMNKKREA
EFQKMRRDLEEATLQHEATAAALRKKHADSVAELGEQIDNLQRVKQKLEK
EKSEMMEIDDLASNMETVSKAKGNLEKMCRALEDQLSELKTKEDEQQRL
INDLTTQRARLQTESGEFSRQLDEKDALVSQLSRGKQAFQTQQIEELKRQLEE
EIKAKSALAHALQSARHDCDLLREQYEEEEEQEGKAELGRAMSKANSEVAQ
WRTKYETDAIQRT EELEEAKKKLAQRLQDAEEHVEAVNAKCASLEKTKQR
LQNEVEDLMIDVERTNAACAALDKKQRNFDKILSEWKQKYEETHAELEAS
QKESRSLSTELFKIKNAYEESLDQLETLKRENKNLQQEISDLTEQIAEGGKRI
HELEKVKKQVEQE KSEIQAAL EEA EASLEHEEGKILRIQLELNQVKSEIDRKI
AEKDEEIDQLKRNHIRVVESMQSTLDAEIRSRND A IRLKKKMEGDLNEMEI
QLNHANRMAAEALKNYRNTQAILKDTQIHLDDALRGQEDLKEQLAMVER
RANLLQAEIEELRATLEQTERS RKIAEQELLDASERVQLLHTQNTSLINTKK
KLETDITQIQGEMEDI IQEARN AEEKAKKAITDAAMMAEELKKEQD TSAHL
ERMKKNLEQTVKDLQHRLDEAEQLALKGGKKQIQKLEARVRELEGEVESE
QKRNVEAVKGLRKHERRVKELTYQTEEDRKNILRLQDLVDKLQAKVKSY
KRQAE EAE EQSNVNLSKFRKLQHELEAEERADIAESQVNKL RVKSREVHT KIISEE 136
Bison MADDEEYEEVVEYYTEETVYKEVPGETITGFYETTTTTRTSDYEQSETSRPAL
A0A6P3H861 AQPVPEKPVERKRVIRKKVDPSKFMTPTYIAHSQKMQNLFSTNKYKENYEK
(Nebulin isoform
AKGKPYAITTDTPELRRIKKVQDQLSEVKYRVDGDVAKTICHVDEKAKDIE X11)
HAKKVSQQVSKVLYKQNWEDTKDKYLLPPDAPELVQAIKNTAMFSKKLY
TEDWEADKTMFYYPYNDSPELRRVAQAQKALS DIAYKKGLAEQQTQFTSLP

DPPDIEAFKVNQVSKYKWKWSETPCFEIIATARMNSNNIS
TKKYQEDFEHMKDQIYFMQTETPEYKVNKQAGVAASKVKYKQDYEKNK
GKADYNVLPASENPLLRQLKAAGDALSDKLYKENYEKTKAKSINYCETPK
FKLDTVLHNFSSDTKYKDSYLDILGHYVGSYEDPYHTHCMRVSAQNSDK
NYKAEYEEDRGKGFFPQTITQEYEAIKKLDQCKDHTYKVHPDKTKFTQVT
DSPVLMQAQVNSKQLSDLNYKAKHENEKFKCHIPPDTPAFIQHKL NAYNLS
DNIYKHDWEKTKAKKFDIKVD AIPLLA AKANTKIASDVMYKKDYEKSKGK
MIGALSINDDPKMLHSLKTAKNQSDRLYKENYEKTKAKSMNYCETPKYQL
DTLLKNFSEAKYKDSYVQNVLGHYIGSFEDPYQVHCLKISAQNSDKNYKA
EYEEDKGKCYFPQTITQEYEAIKKLDQCKDHTYKVHPDKTKFTAVTDTPL
LQAQLNNTKQLSDLNYKAKHEGEKFKCHIPADAPQFIQHRVNAYNLS DNIYK
HDWEKTKAKKFDIKVD AIPLLA AKANTKIASDVMYKKDYEKSKGKMIGAL
SINDDPKMLHSLKTAKNQSDHEYRKDYEKSKTIYTAPLDMLPLTHAKKSQ
AIASDVVDYKHLHNYSPPD SVNVDLAKKAYALQSDVEYKADYNSWMKG
CGWMPFGSLEMEKAKRASDILNEKKYRQH PDTL KFTSIEDAPIIVQSKINQA
QRSDVAYKAKGEEVIHKYSLPADLPQFIQAKVNAYNISENLYKADLKDLSK
KGYDLRIDAIPIKA AKAARQAASDVQYKKDYEKAKGKMVG FQSLQDDPKL
VHYMNVAKIQSDREYKKAYEKT KTRHNTPHDMVNIVA AKKAQDVASNV
NYKHSLLHHYTYLPDAMDLELSKNMMHIRSDNVYKEDYNNWMKGIGWIP
GSLEVEKVKKAGDALNEKKYRQH PDTL KFTSIVDSPVMVQAKQNTQQVSD
ILYKAKGEDVKHKYTMSPDLPQFLQAKCNAYNLS DVCYKRDWHDLIAKG
TNVLGDAIPITAAKASRN IASDYKYKEAYEKAKGKQVGFRSLQDDPKLVH
YMNVAKLQSDREYKKNYENTKTSYHTPGDMVSITA AKMAQDVATNVNY
KQPIHHYTYLPDALSLHIRNVNQIQSDNVYKDEYNHFFKGMGWIPIGSLEV
EKVKKAGDALNEKKYRQH PDTI KFTSVPDSMGMVLAQHNTKQLSDLNYK
VEGEKVKHKYTMDPDVPQFIQAKVNAYNMSDSHYKADWKKTLAKGYDL
RPDAIPIVAAKSSRN IASDFKYKEAYEKT KGKQIGFRSLQDDPKLVHFMNV
AKMQSDREYKKGYEAR KTKYHTPLDMLS VTAAKKSQEVATNTNYKQPFH
HYTLLPDALNVEHSRNAMQIQSDNLYKSDFTNWMKGIGWLPLESLEVEKA
KKAGEILSEKKYRQHPEKLKFTYAMDTMEQALNKS NKLIMDKRLYTEKW
NKDKTTIHVMPDTPDILL SRVNQITMSDKLYKAGWEEEEKKGYDLRPDAIS
IKAARASRD IASDYKYKQAYEQAKGRQIGFRSLEDDPKLVHFMQVAKMQS
DREYKKAYEKS KTSFHTPVDMLS VVA AKKSQEVATNANYRNVIHTYNML
PDAMSLELAKNMMQIQSDN QYKADYADFMKGIGWLPLGSLEAEKNKKA
MEIISEKKYRQH PDTL KYSTLMDSMN MVLA KNNAKIMNEHLYKQAW EAD
KTKVHIMPDI PQIILAKANAINISDKLYKLSLEEAKKKGYDLRTDAIPIKA AK
ASRD IASDYKYKHSYEKERGKMVGFRSLEDDPKLVHSMQVAKMQSDREY
KKNYEKTKTSYHTPADMLS VTAAKDAQANITNTNYKHLIHKYILLPDAMNI
QLSKNMNRIQSDNEYKQDYNEWYKGLGWSPAGSLEVEKAKKATEYASDQ
KYRQHPSNFQFTKLND SMDMVLA KQNAHTMNKYLYTIDWNKD KTKIHV
MPDTPDILQAKQNQTMYSQKLYKLGWEEALKKGYDLPVDAISVQA AKAS
RDIVSDFKYKQGYRKQIGHHIGFRSLQDDPKLVLSMNVA KMQSEREYKKD
FEKWKTKFSSPVDMLGVVLAKK CQALVSDVDYKNYLHGWTCLPDQNDVI
HAKKAYDLQSENLYKSDLEWLKGIGWSPLGSLEAEKNKRASEI ISEKKYRQ
PPDRNKFTSIPDAMDIVLAKTNAKNRSDILYREAWDKDKTQIHIMPDTPD IIL
AKANLINTSDKFYRMGYEELRKKGYDLPVDAIPIKA AKASREIASEYKYKE
GFRMQLGHHIGARNIKDDPKMMWSMHVAKIQSDREYKKDFEKWKT KFS
PVDMLGVVLAKKCQTLVSDIDYKNYLHQWTCLPDQSDVIHARRAYDLQS
DNLYKSDLQWLRGIGWLPSGSLEDEKNKRASQILSDHVYRQH PDQFKFSSL
MDSIPMVLA KNNAITMNHRLYTEAWDKDKTTVHIMPDTPEVLLAKQNQIN

YSEKLYKGLDLKADSEYFKYKEGYRQQLG
HHIGARAIHDDPKMMWSMHVAKIQSDREYKKDFEKWKTKFSSPVDMLGV
VLAKKCQTLVSDIDYKNYLHQWTCLPDQSDVIHARRAYDLQSDNVYKSDL
QWLRGIGWVPIGSLDVEKSKRASEILSDKLYRQPPDKFKFTSVTDSLEQVLA
KNNAINMKNRLYTEAWDKDKTQVHIMPDTPEITLARTNKVNYSENLKLA
HEEAKKKGYDLRSDAIPIVAASRDIIISDYKYKDG YRKQLGHHIGARDIK
DDPKMMWSMHVAKIQSDREYKKDFEKWKTKFSSPVDMLGVVLAKKCQT
LVSDIDYKNYLHQWTCLPDQSDVIHARRAYDLQSDNVYKSDLQWLRGIG
WVPIESVDVVKCKRAAEILSDNLYRQPPDTLKFTSVPDSLEQVLA KNNAIN
MKNRLYTEAWDKDKTQIHIMPDTPEIMLARQNKLN YSETLYKL ANEEAKK
KGYDLRSDAIPIVAASRDIIISDYKYKDG YRKQLGHHIGARDIKDDPKMM
WSMHVAKIQSDREYKKDFEKWKTKFSSPVDMLGVVLAKKCQTLVSDIDY
RNYLHQWTCLPDQSDVIHARQAYDLQSDNIYKSDLQWLRGIGWVPIGSVD
VVKCKRAAEILSDNLYRQRPDTLKFTSVPDSLEQVLA KNNAINMKNRLYTE
AWDKDKTQIHMPDTPEITLARQNKIN YSENL YRQAMEEAKKEGYDLRSD
AIPIVAASREIASDYKYKEAYRKQLGHHIGARAIHDDPKMMWSVHVAK
MQSDREYKKDFEKYKTRFSSPVDMLGIVLAKKCQTLVSDVDYKHPLHEWT
CLPDQNDIIHARKAYDLQSDNLYKSDLEWLKGIGWVPIGSVEVLKAKRAGE
ILSDNIYRQRPDTLKFTSVTDSPEQVLA KNNAINMKNRLYTEAWDNDKKT
HVMPDTPEIMLAKLNRINYSKLYKLAL EESRKEGYDLRLDAIPIQAAKAS
REIASDYKYKEGYRKQLGHHIGARNIKDDPKMMWSIHAGKIQSDLEYKKD
FEKWTKFSSPVDMMGLVQAKKCQILVSDIDYKNLLHEWTCLPDQNDIIQ
ARKAYDLQSDAIYKADLEWLRGIGWVPIDSVGVEHAKRAGEILSERKYRQP
AVQLKFTSITDTPEIVLAKNNALNVSKHLYTEAWDADKTSIHVMPDTPEILL
AKSNSANISHKLYTKGWDESKMKDYDLRADAIKSAKASRDIIASDYKYKE
AYEKHKGGHHIGARSVEDDPRIMCAMNAGRIQSEREYKKEFQKWTKFSSP
VDMLGILLAKKCQTLVSDIDYRNYLHHWTCLPDQNDIIQARKAYDLQSDAI
YKADLEWLRGIGWMPEGSPEVLRVKNAQHIFRDSVYRTPVVKLKYTSIVD
TPEVVLAKSNAENISIPKYREVWDKDKT SIHIMPDTPEINLARTNALNVSNK
LYREGWDEMKMSCDVRLDAIPIQAAKASREIASDYKYKLDHEKQKGHYV
GTRTARDDNKIRWALIAGKIQNEREYRLHWAKWKSKFQSPDMLSIEHSK
NSQTLVSDIDYRHYLHQWTCMPDQNDVIQARKAYDLQSDAIYKADLEWL
RGIGWMPADSVSVNHAKHMADIFNEKKYRTKIETLSFTPVDDRVDYVTAK
HSGEILNDIKYRKDWN DTKSKYTLTETPQLHTAQEAARILDQYLYKESWEK
QKATGYILPPDAVPFVHAHHS GDVQSELKYKAEHV KQKGHYVGVP TMRD
DPKLVWFEHAGQIQNDRLYKENYHKTKAKIHIPPDMVSVLAAKEGQALAS
DIDYRNYLHQWICH PDQNDVIQARKAYDLQSDNIYKADLEWLRGIGWIPL
DSVDHVRVTRNQEMMNQIKYKKDALANYPNFTSVVDPPEIVLAKINSVNQ
SDVKYKETFNKLIKGYIFSPDTPYITHSKDMEKLYSTILYKRAWDG TKAY
GYTLDERYIPIVGAKHAD FVNSELKYKETYEKLKGHYLAGKEISEFPNVVH
CLDFQKMRSLLNYRRHYEDTKANVHIPQDMMNHVLAKRCQYILSDLEYR
HYFHQWTS LPEEPNVVRVRHAQEILSDNVYKDDLNLWLKGIGCYVWDTPQI
LHAKKSYDLQSQILYTAAGKENLKNYNLVTDTPLYVTALQSGINASEVKY
KENYHQIKDKYTTVLETVDYDRIKNLKD LFSSNLYKEAWDKVKATSYILPP
NTVSLTHAKNQKYMASHIKYREEYEKFKALYTLPRSVEDDPNTARCLRVG
KFNIDRLYRSVYEKNKMKIHI VPDMMVEMVTAKDSQKKVSEIDYRLHLHEWI
CHPDLQVNSHVRKVTDQISDIVYKDDLTWLKGIGCYVWDTP EILHAKHAY
DLRNDIKYKAHV LKTRNNYKLVTDTPVYVQAVKSGKQLSDAVYHYDYVH
SIRGRVAPTTKTVDLDRALHAYKLQSENLYRKAGLHALPTGYRLPVDTPHF
KHTKDTRYMSSYFKYKEVYEHMKAYGYTLGPNDVPFVNVRVNNITSERL

YRQLYHKLKDKLHPTDTPDKLQVSELHYTKSDFFKMQGHMISLPY
TPQVLHCRYVGDITSDIKYKEDLQVLRGMGCFLYDTPDMVRSRHLRKLWS
HYLYTDKARKMRDKYKVVLDTPEYRKVQELKTHLSELVYRAAGRKQKSIF
TSVPDTPDLTRAKRGQKLQSQYLYVELATKERPHHHAGNQTALKHARHV
KDMVSENKYKIQYEKMKDKYTPVPDTPILIRAKRAYWNASDLRYKETFQK
TKGKYHTVKDALDIVYHRTVTDHVSКИKYKENYMSQLGIWRSIPDRPEHFH
HRAVTDVSDVKYKQDLTWLKGIGCYAYDTPDFTLAEKNKTLYSKYKYK
EVFERTKSNFKYVADCPINRHFKFATQLMNEKKYRADYEQRKDKYHLVVD
EPRHLLAKIAGDQISQIKYRKNYEETKDKFTSIVDTPEHLRTTKVNKQISDIL
YKLEYNKAAPRGYTTIHDTPMLLHVRKVKDEVSDLKYKEVYQRTKSNCTI
EPDAVHIKAAKDAYKVNTNLDYKKKYEATKAYWKWTPDRPDFIQAAKST
LQQSDFEYKLDREYKLGCKLSVTDDKDMVLALKNSIIESDLKYKEKHVKE
RGSCHAVPDTPQILLAKTVSSLVSENKYKSYVKKHLAQGSYTTLPETRDTIH
VKEVTKNVSDTNYKKKFVKEKGKSNYSIMLEPPDVKHAMDVAKKQSNVA
YKKDAKENLHYTTVADRPDIKKATQAAKQASEVEYRAKHRKEGSHGLSM
LGRPDIEMAKKAACLSSQVQYRENFNKEKGKTPKYNPKDSQLYKVMKDA
NTLASEVKYKADLKKLHKPVTDMKESLIMNHVLNTSHLASSYQYKKNYEK
SKGHYHTIPDNLEQLHLKEATELQSIVKYKEKYEKERGGKPMMLDFETPTYITA
KESQQMQSGKEYRKDYEESIKGRNLTGLEVTALLHVKYATKIASEKEYRK
DLEESIRGKGLSEMEDTPDMLRAKNATQILNEKEYKRDLELEVKGRLNA
MANETPDFLRARNATDIASQIKYKQSAEMEKANFTSVVDTPETIIHAQQVKN
LSSQKKYKEDAEKCMSYYETVLDTPEMQVRVRENQKNFSLQYQYDLKNS
KGKITVVQDTPEILRVKENQKNFSSVLYKEDVSPGTAIGKTPEMMRVKQTQ
DHISSVKYKEVIGQGTPIPDLPEVKRVKQTQKHISVVMYKENLGTGIPTPVT
EIERVKRNQENFSSVLYKENLGGKTPTPTPEMERVKRNQENFSSVLYKEN
MGKGTPLPVTPEMERIKHNQENISSVLYKENVGKGTPTPTPEMQRVKRNQ
ENISSVLYKENLGGATPTPTPEMERVKRNQENFSSVLYKENMRKATPTPV
TPEMERAKRNQENISSVLYSDSFRKQIQGKAAYVLDTPEMRRVRETQRHIS
TVKYHEDFEKHKGCFPTPVVTDPTITERVKKNTQDFSDICYRGIQRKVVEMEQ
KRNDQDQETITGLRVWRTNPGSVFDYDPAEDNIQSRSLHMINAQARRSRE
QSRASGLSISGGEEKSEHSEAAHLSTYSDGGVFFSAASTGYKHARTTELPPQ
QRSSSVATQQTTVSSIPSHPSTAGKIFRAMYDYMAADADEVSFKDGDIVN
VQAIDEGWMYGTVQRTGRTGMLPANYVEAI 137 Bison
MSSDQEMAIFGEAAPYLKSEKERIEAQNKPFDAKTSVFVVDPKESFVKAT A0A6P3HPK5
VQSREGGKVTAKTEAGATVTVKEDQIFPMNPPKYDKIEDMAMMTHLHEPA (Myosin-4)
VLYNLKERYAAWMIYTYSGLFCVTVNPNYKWLVPVYNAEVVTAYRGKKRQE
APPHIFSISDNAYQFMLTDRENQSILITGESGAGKTVNTKRVIQYFATIAVTG
EKKKEEPTSGKMQGTLEDQIISANPLLEAFGNAKTVRNDNSSRFGKFIRIHF
GTTGKLASADIETYLLEKSRVTFQLKAERSYHIFYQIMSNKKPELIEMLLITT
NPYDYAFVSQGEITVPSIDDQEELIATDSAIDILGFTSDERSIYKLTGAVMH
YGNLKFQKQKREEQAEPDGTEVADKAAAYLQSLNSADLLKALCYPRVKVG
NEFVTKGQTVQQVYNVAVGALAKAVYEKMFVLMVTRINQQLDTKQPRQY
FIGVLDIAGFEIFDFNSLEQLCINFTNEKLQQFFNHHMFVLEQEYKKEGIEW
EFIDFGMDLAACIELIEKPMGIFSILEEECMFPKATDTSFKNKLYEQHLGKSN
NFQKPKPAKGKAEAHFSLVHYAGTVDYNIAGWLDKNKDPLNETVVGLYQ
KSAMKTLAFLFAGGPSAEEGGGGKKGKGGKGGSSFQTVSALFRENLNKLMT
NLRSTHPPHFVRCIIPNETKTPGAMEHELVLHQLRCNGVLEGIRICRKGFPRI
LYADFKQRYKVLNASAIPEGQFIDSKKASEKLLGSIDIDHTQYKFGHTKVFF
KAGLLGILEEMRDEKLAQLITRTQAICRGFLMRVEFRKMMERRESIFCIQYN
VRAFMNVKHWPWMKLYFKIKPLLKSAETEKEMANMKEEFKTKHEELAKS

EAKRKEEKMVTLMQEKNDLQLQVQSEADGLADAEERCDQLIKTKIQLE
AKIKELTERAEDEEEINAELTAKKRKLEDECESELKKDIDDLELTLAKVEKEK
HATENKVKNLTEEMAGLDENIAKLTKEKKALQEAHQQTLDLQAEEDKV
NTLTSSKTKLEQQVDDLEGSLEQEKKLRMDLERAKRKLEGDLKLAQESTM
DIENDKQQLDEKLKKKEFEMSNLQSKIEDEQALAMQLQKKIKELQARIEEL
EEEIEAERASRAKAQKQRSDLSRELEEISERLEEAGGATSAQIEMNKKREAE
FQKMRRDLEEATLQHEATAAALRKKHADSVAELGEQIDNLQRVKQKLEKE
KSEMKMEIDDLASNMETVSKAKGNLEKMCRTLEDQLSEVKSKEEEHQRLI
NELSAQKARLHAESGEFSRQLEEKDALVSQLSRGKQAFQTQQIEELKRQLEE
ETKAKSALAHALQSARHDCDLLREQYEEEQEGKAELQRALSKANSEVAQ
WRTKYETDAIQRTTEELEEAKKKLAQRLQDAEEHVEAVNAKCASLEKTKQR
LQNEVEDLMLDVERSNAACAALDKKQRNFDKVLSEWKQKYEETQAELEA
SQKESRSLSTELFKVKNACEESLDQLETLKRENKNLQQEISDLTEQIAEGGK
HIHELEKVKKQIEQEKGELQAALAEAASLEHEEGKILRIQLELNQVKSEIDR
KIAEKDEEIDQLKRNHLRVMESTQSTLDAEIRSRNDALRIKKKMEGDLNEM
EIQLNHANRQATEAIKNLRLNTQGVCLKDTQLHLDDAIRGQDDLKEQLAMVE
RRANLMQAEIEELRASLEQTERSRRVAEQELLDASERVQLLHTQNTSLINTK
KKLETDITQIQGEMEDILQEARNAEEKAKKAITDAAMMAEELKKEQDTS
HLERMKKNMEQTVKDLQHRLDEAEQLALKGGKKQIQKLEARVRELENEV
ENEQKRNAEAVKGLRKHERRVKELTYQTEEYRKNVLRQLDLVDKLQSKV
KAYKRQAEAEQSNVNLAKFRKLQHELEEAERADIAESQVNKL RVKSR EVHTKVISEE
138 Bison MKWVTFISLLLLFSSAYSRGVFRDTHKSEIAHRFKDLGEEHFGLVLI
AFS A0A6P3HBK1 QYLQQCPFDEHVKLVLNELTEFAKTCVADESHAGCEKSLHTLFGDELCKVA
(Serum albumin)

SLRETYGNMADCCEKQEPERNECFLSHKDDSPDLPKLPDPNTLCDEFKAD
EKKFWGKYLYEIAARRHPYFYAPELLYYANKYNGVFQECCQAEDKGACLLP
KIETMREKVLTSSARQLRCASIQKFGERALKAWSVARLSQKFPKAEFVEV
TKLVTDLT KVHKECCHGDLLECADDRADLAKYICDNQDTISSKLKECCDKP
LLEKSHCIAEVEKDAPENLPPLTADFAEDKDVCCKNYQEAKDAFLGSFLYE
YSRRHPEYAVSVLLRLAKEYEATLEECCA KDDPHACYSTVFDKLKHLVDE
PQNLIKQNC DQFEKLGEYGFQNALIVRYTRKVPQVSTPTLVEVSRLGKVG
TRCCTKPESERMPCTEDYLSLILNRLCVLHEKTPVSEKVT KCCTESLVNRRP
CFSALTPDETYVPKAFDEKLFTFHADICTLPDTEKQIKKQTALVELLKHKPK
ATEEQLKTVMENFVAFVDKCCAADDKEACFAVEGP KLVVSTQTALA

[0049] An animal protein can comprise a deer protein (Table 7). A deer protein can comprise a deer muscle protein. A deer protein can comprise a deer blood protein. A deer protein can comprise a deer albumin protein. A deer protein can comprise a sequence that has at least or about 70%, at least or about 75%, at least or about 80%, at least or about 90%, at least or about 91%, at least or about 92%, at least or about 93%, at least or about 94%, at least or about 95%, at least or about 96%, at least or about 97% at least or about 98% at least or about 99% or about 100% sequence identity to any one of SEQ ID NO: 139-147.

[0050] A food product may comprise a fragment or a truncation of a deer protein. The food product may include a fragment or truncation of any one of SEQ ID Nos: 139-147. A sequence of the fragment or truncation may be identical or substantially identical to a portion of any one of SEQ ID Nos: 139-147. A sequence of the fragment or truncation may be at least about 70%, 80%, 90%, 95%, 98%, or more identical to a portion of a sequence of any one of SEQ ID Nos: 139-147. A length of the fragment or truncation may be at least about 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or more of a length of any one of SEQ ID Nos: 139-147. A length of the fragment or truncation may be less than or equal to about 90%, 80%, 70%, 60%, 50%, 40%, 30%, 20%, 15%, 10%, or less than a length of any one of SEQ ID Nos: 139-147.

TABLE-US-000000139 Deer Proteins SEQID NO Description Amino Acid Sequence
139 Deer MLDAEDIVNTPKPDEKAIMTYVSCFYHAFAGAEQAETAANRICKVLAVNQ
A0A212DH10 ENEKLMEEYEKLASELLEWIRRTVPWLENRVGEPSMSAMQRKLEDFRDYR
(ACTN3) RLHKPPRVQEKQCQLEINFNTLQTKLRLSHRPAFMPSEGKLVSDIAKAWRGL
EQVEKGYEDWLLSEIRRLQRLQHLAEKFQQKASLHEAWTRGKEEMLSQRD
YETASLQEVRAALLRRHEAFESDLAAHQDRVEHIAALAQELNELDYHEAAS
VNSRCQAICDQWDNLGTLTQKRRDALERMEKLLETIDQLQLEFARRAAPF
NNWLDGAVEDLQDVWLVSVEETQSLVTAHDQFKATLPEADRERGAILGI
QGEIQKICQTYGLRPSSTNPYIALTPQDINTKWDTVRKLVPSRDQMLQEELT
RQQVNERLRRQFAAQANAIGPWIQGKVEEVGRLAAGMAGSLEEQMAGLR
QQEQNIINYKSNIDRLEGDHQLLQESLVFDNKHTVYSMEHIRVGWEQLLTSI
ARTINEVENQVLTRDAKGLSQEQLNEFRASFNHFDKRNGMMEPDDDFRAC
LISMGYDLGEVEFARIMTMVDPNAAGVVTFQAFIDFMTRETAETDTAEQV
VASFKILAGDKNYITAEELRRELPAEQAEYCIIRMAPYKGAGAPAGALDYV
AFSSALYGESDL 140 Deer

MSRPLTDQEKRKQISVRGLAGVENVTELKKNFNRHLHFTLVKDRNVATPR A0A212DGZ3
DYYFALAYTVRDHLVGRWIRTQQHYEYKDPKRIYYLSLEFYMGRTLQNTM (Alpha-1,4
glucan VNLALENACDEATYQLGLDMEELEEIEEDAGLGNGGLGRLAACFLDSMAT
phosphorylase) LGLAAYGYGIRYEFGIFNQKISGGWQMEEADDWLRYGNPWEKARPEFTLP
VHFYGRVEHTSQGAKWVDTQVVLAMPYDTPVPGYRNNVNTMRLWSAK
APNDFNLKDFNVGGYIQAFLDRNLAENISRVLYPNDNFFEGKELRLKQEYF
VVAATLQDIIRRFKSSKFGCLDPVAIQLNDRHPSLAPELMRILVDQERLEWE
KAWEVTVKTCAYTNHTVLPEALERWPVHLIETLLPRHLQIIYEINQRFLNRV
AAAFPGDVDRLLRMSLVEEGAVKRINMAHLCIAGSHAVNGVARIHSEILKK
TMTSKSWSLINSRIRPMGSPLGAGWLCRIGEEYIADLDQLRKLLSYVDDES
IRDVAKVKQENKLFSAYLEKEYKVHINPNSLFDIQVKRIHEYKRQLLNCL
HAAPGYHMAKMIKLITAIGDVVNHDVPVGDRLRVIFLENYRVSLAEKVIP
AADLSEQISTAGTEASGTGNMKFMLNGALTIGTMDGANVEMAEAGEENF
FIFGMRVEDVERLDQRGYNAQEYYDRIPELRHVIDQLSSGFFSPKQPDLFKD
IVNMLMHHDRLFVADYEEYVKCQERVSAKYKNPREWTRMVIRNIATSGK
FSSDRTIAQYAREIWGVEPNRQRMPPAPDEKI 141 Deer

MAQRKPEGSGFHMTCLSMALAYSFRPDANIQPHPQLGNTQKQTELGKELT A0A212CZ49
GTMPHQYPALTPEQKKELCDIAHRIVAPGKGILAADESTGSIKRLQSIGTE (Fructose-
NTEENRRFYRQLLLTADDRVNPCIGGVILFHETLYQKADDGRPFPPQVIKAK bisphosphate
GGVVDKGVVPLAGTNGETTTQGLDGLSERCAQYKKDGADFAKWRCVLKI aldolase)
GEHTPSSLAIMENANVLARYASICQQNGIVPIVEPEILPDGDHDLKRCQYVT
EKVLAAYYKALSDHHIYLEGTLLKPNMVTPGHACTQKYSHEEIAMATVTA
LRRTVPPAVPGITFLSGGQSEEEASINLNAINKCPLLKPWALTFSYGRALQAS
ALKAWGGKKENLKAQEEYVKRALANSLACQGKYTPSGKAGAAASESLFI SNHAY 142

Fragment of Deer

PAPAPPKEEKIDLSAIKIEFSKQQQDEFKEAFLFDRTGECKITLNQVGDVLR A0A212D3R8
ALGTNPTNAEVKKVLGNPSNEEMNAKKIEFEQFLPMLQAISNNKDQGTYE (MYL1)
DFVEGLRVFDKEGNGTVMGAELRHVLATLGEKMKEEEVEALMAGQEDSN
GCINYEAFVKHIMSN 143 Deer

MFVMGVNHEKYNNNTLKIVSNASCTTNCLAPLAKVIHDHFGIVEGLMTTVH A0A212CCI7
AITATQKTVDGPSGKLWRDGRGAAQNIIPASTGAAKAVGKVIPELNGKLTG (GAPDH)
MAFRVPTPNVSVVDLTCRLEKPAKYDEIKKVVKQASEGPLKGILGYTEDQV
VSCDFNSDTHSSTFDAGAGIALNDHFVKLISWYDNEFGYSNRVVDLMVHM ASKE 144
Deer MRTLRAMAMQKIFAREILDSRGNPTVEVDLHTAKGRFRAAVPSGASTGIYE
A0A212D927 ALELRDGDKSRYLGKGVLKAVEHINKTLGPALLEKASLGRLYLLPPAKFGA

(Phosphopyruvate
 NAILGVSLAVCKAGAAEKGVPYLRHIADLAGNPELILPVPAFNVINGGSHA hydratase)
 GNKLAMQEFMILPVGASSFKEAMRIGAEVYHHLKGVKAKYKDATNVG
 DEGGFAPNILENNEALELLKTAIQAAGYPDKVVIGMDVAASEFYRNGKYDL
 DFKSPDDPARHISGEKLGELYKNFIKNYPVVSIEDPFDQDDWATWTSFLSGV
 NIQIVGDDLTVTNPKRIAQAVEKKACNCLLLKVNQIGSVTESIQACKLAQSN
 GWGVMVSHRSGETEDTFIADLVVGLCTGQIKTGAPCRSERLAKYNQLMRIE
 EALGDKAVFAGRKFRNPKAK 145 Deer
 MFDQTQIQEFKEAPKKAKRRRAAAEGGSSSVFSMFDQTQIQEFKEAFTVIDQ A0A212CY74
 NRDGIIDKEDLRDTFAAMGRLNVKNEELDAMMKEASGPINFVFLNMFGE (MYLPF)
 KLKGADPEDVITGAFKVLDPGKGTIKKQFLEELLTTQCDRFSQEEMRNM
 WAAFPDPVGGNVDYKNICYVITHGDAKDQE 146 Deer
 MCEEDSTALVCDNGSGLCKAGFAGDDAPRAVFPISVGRPRHQGVMMVGM A0A212CNJ3
 GQKDSYVGDEAQSKRGILTLKYPIEHGIITNWDDMEKIMFETFNVPAMYVA (Actin)
 IQAVLSLYASGRTTGIVLDSGDGVTHNVPIYEGYALPHAIMRLDLAGRDLT
 DYLMKILTERGYVSFVTTAEREIVRDIKEKLCYVALDFENEMATAASSSSLEK
 SYELPDGQVITIGNERFRCPETLFQPSFIGMESAGIHETTYNSIMKCDIDIRKD
 LYANNVLSGGTTMYPGIADRMQKEITALAPSTMKIKIIPPERKYSVWIGGS
 ILASLSTFQQMWISKQEYDEAGPSIVHRKCF 147 Deer
 MKWVTFISLLLLFSSAYSARGVFRRDTHKSEIAHRFKDLGEDNFQGLVLIAFS A0A212D5P0
 QYLQQCPFDEHVKLVKELTDFAKECVADESHAGCEKSLHTLFGDELCKVA (ALB)
 SLRETYGDMADCCEKQEPERNECFLKHKDDNPDLPKLKPDPDTLCAEFKA
 DEKKFWGKYLVEVARRHPYFYAPELLYYANKYNGVFQECCQADDKAACL
 LPKIDTMREKVLTSARQLKCASIQKFGERALKAWTVALLSQKFPKADFA
 EVTKIVTDLTKVHKECCHGDLLECADDRADLAKYICDNQDTLSSKLKECC
 DKPVLEKSHCIAEIEKDAVPENLPPLTADFAEDKEVCKNYQEAKDVFLGTF
 LYEYARRHPEYAVSVLLRVAKEYEATLEDCCA KDDPHACYATVFDKLQHL
 VDEPQNLIKQNCDLFEKVGEYGFQNALIVRYTKKAPQVSTPTLVEVSRNLG
 RVGTCCTKPESERMSCTEDYLSVLNLRLCVLHEKTPVSEKVTKCCTESLV
 NRRPCFSALTLDETYVPKAFDEKLFTFHADICTVPDAEKQIKKQTALVELLK
 HKPKATDEQLKTMENFVAFVDKCCA AVDKEACFAVEPTLRVRGKKEK

[0051] An animal protein can comprise a chicken protein (Table 8). A chicken protein can comprise a chicken muscle protein. A chicken protein can comprise a chicken blood protein. A chicken protein can comprise a chicken albumin protein. A chicken protein can comprise a sequence that has at least or about 70%, at least or about 75%, at least or about 80%, at least or about 90%, at least or about 91%, at least or about 92%, at least or about 93%, at least or about 94%, at least or about 95%, at least or about 96%, at least or about 97% at least or about 98% at least or about 99% or about 100% sequence identity to any one of SEQ ID NO: 148-161.

[0052] A food product may comprise a fragment or a truncation of a chicken protein. The food product may include a fragment or truncation of any one of SEQ ID Nos: 148-161. A sequence of the fragment or truncation may be identical or substantially identical to a portion of any one of SEQ ID Nos: 148-161. A sequence of the fragment or truncation may be at least about 70%, 80%, 90%, 95%, 98%, or more identical to a portion of a sequence of any one of SEQ ID Nos: 148-161. A length of the fragment or truncation may be at least about 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or more of a length of any one of SEQ ID Nos: 148-161. A length of the fragment or truncation may be less than or equal to about 90%, 80%, 70%, 60%, 50%, 40%, 30%, 20%, 15%, 10%, or less than a length of any one of SEQ ID Nos: 148-161.

TABLE-US-00008 TABLE8 Chicken Proteins SEQ ID NO Description Amino Acid
 Sequence 148 Chicken KCRM
 MPFSSTHNKHKLKFS AEEFDPDL SKHNNHMAKVLTPELYKRLRDKETPSGF (Creatine

kinase TLDDVLIQTHGVDNPGYDVFVKDLFDVQDRHGGYKP M-type)
TDKHRTDLNHNENLKGGDDLDPKYVLSSRVRTGRSIKGYSLPPHCSRGERRA
VEKLSVEALNSLEGEFKGRYYPLKAMTEQEQQLIDDHFLFDKPVSPLLLA
SGMARDWPDARGIWHNDNKTFLVWVNEEDHLRVISMEKGGNMKEVFRRF
CVGLKKIEEIFKKAGHPFMWTEHLGYILTCPSNLGTGLRGGVHVKLPKLSQ
HPKFEEILHRLRLQKRGTGGVDTAAVGAVFDISNADRLGFSEVEQVQMVV
DGVKLMVEMEKKLEQNQPIDDMIPAQK 149 Chicken (ACTN2
MNSMNQIETNMQYTYNYEEDEYMTQEEEWDRDLLLLDPAWEKQQRKTFTA Alpha-actinin-
2) WCNSHLRKAGTQIENIEEDFRNGLKLMLLLEVISGERLPKPDRGKMRFHKI
ANVNKALDYIASKGVKLVSIGAEIIVDGNVKMTLGMWTHILRFAIQDISVE
ETSAKEGLLLWCQRKTAPYRNVNIQNFHLSWKDGLGLCALIHRHRPDLIDY
SKLNKDDPIGNINLAMEIAEKHLDIPKMLDAEDIVNTPKPDERAIMTYVSCF
YHAFAGAEQAETAANRICKVLAVNQENERLMEEYERLASELLEWIRRTIPW
LENRTPEKTMQAMQKKLEDFRDYRRKHKPPKVQEKQCLEINFNTLQTKLRI
SNRPAFMPSEGKMOVSDIAGAWQRLEQAEKGYEEWLLNEIRRLERLEHLAE
KFRQKASTHEQWAYGKEQILLQKDYESASLTEVRAMLKHEAFESDLAAH
QDRVEQIAAIAQELNELDYHDAASVNDRCQKICDQWDSLGTLTQKRREAL
ERTEKLLLETIDQLHLEFAKRAAPFNNWMEGAMEDLQDMFIVHSIEEIQSLIS
AHDQFKATLPEADGERQAILSIQNEVEKVIQSYSMRISASNPYSTVTVEEIRT
KWEKVKQLVPQRDQSLQEELARQHANERLRRQFAAQANVIGPWIQTKME
EIARSSIEMTGPLEDQMNQLKQYEQNIINYKHNDKLEGDHQLIQEALVFDN
KHTNYTMEHIRVGWELLTTIARTINEVETQILTRDAKGITQEQMNDFRASF
NHFDRRKNGLMDHDDFRACLISMGYDLGEAEFARIMSLVDPNGQGTVTTFQ
SFIDFMTRETADTDTAEQVIASFRILASDKPYILADELRRELPPPEQAQYCIKR
MPQYTGPGSVPGALDYTSFSSALYGESDL 150 Chicken E1BSN7
MAAPLSDGERRKQISVRGIAGLGDVAEVRKSFNRLHFTLVKDRNVATPR (Alpha-1,4
glucan DYFFALAHTVRDHLVGRWIRTQQHYERDPKRIYYLSLEFYMGRTLQNTM
phosphorylase) VNLGLQNACDEAIYQLGLDLEEELEEIEEDAGLGNGGLGRLAACFLDSMATL
GLAAYGYGIRYEFGIFNQKIIDGWQVEEADDWLRYGNPWEKARPEYMLPV
HFYGRVEHSPEGVKWVDTQVVLAMPYDTPVPGYKNNTVNTMRLWSAKA
PNDFNLQEFNVGDYIEAVLDRNLAENISRVLYPNDNFFEGKELRLKQEYFV
VAATLQDIIRRFKSSKFGCQDPVRTSFETFPDKVAIQLN DTHPALSIPELMRIL
VDVEKVDWDKAWAITKQTCAYTNHTVLPEALERWPVSMFEKLLPRHLEII
YALNQMHLD RVAALYPGDVDRLRRMSVIEEGDCKRINMAHLCVIGSHAV
NGVARIHSDIVKNTVFKDFYELEPEKFQNKTNGITPRRWLLLCNPGGLADVIA
EKIGEDFVTDLSQLKKLLDFINNETFIRDVAKVKQENKLFKSAYLEERYKVK
INPSSMFDIQVKRIHEYKRQLLNCLHAITLYNRIRSDPSKSFVPRTIMIGGKA
APGYHMAKMIKLITSIGEVVNNDPYVGDKLKVIFLENYRVSLAEKVIPAAD
LSQQISTAGTEASGTGNMKFMLNGALTIGTMDGANVEMAEAGEENLFIFG
MRVEEVEALDRRGYNAREYYDRIPELRQAIDQISSGFFSPRDPGCFRDVVN
MLMHDRFKVFADYEAYIKCQGQVDQLFMDPREWTKKVIRNIACSGKFSS
DRTITEYAREIWGVEPSATKIPPNLPRD 151 Chicken R4GM10
MTHQYPALTAEQKKELSDIALRIVAPGKGILAADES VGSMARLNQIGVEN (Fructose-
TEENRRLYRQILFSADSRVKKCIGGVIFFHETMYQKADDGTPFVQMIKDKGI bisphosphate
VVGIKVDKGVVPLAGTDGETTTQGLDGLSERCAQYKKDGADFAKWRCVL aldolase)
KISDNTPSALAIMENANVLARYASICQQNGIVPIVEPEILPDGDHDLKRCQY
VTEKVLA AVYKALSDHHVYLEGTLLKPNMVTTPGHSCPTKYSPEEIAMATV
TALRRTVPPAVPGVTFLSGGQSEEEASINLNAINTCPLVRPWALTFSYGRAL
QASALSAWRGQRDNANAATEEFVKRAEVNGLAALGKYEGSGDDSGAAGQ SLYVANHAY
152 Chicken MLE1

MAPKADVKKAAAAAPAPAPAPAPAKPAIDLKSIKIEFSKEQQD (Myosin light
DFKEAFLLFDRGTGDAKITLSQVGDIVRALGQNPTNAEINKILGNPSKEEMNA chain 1)
KKITFEEFLPMLQAAANNKDQGTFFEDFVEGLRVFDKEGNGTVMGAELRHV
LATLGEKMTREEVEELMKGQEDSNGCINYEAFVKHIMSV 153 Chicken G3P
MVKVG VNGFGRIGRLVTRAAVLSGKVQVVAINDPFIDLNYMVYMFKYDST
(Glyceraldehyde-
HGHFKGTVKAENGKLVINGHAITIFQERDPSNIKWADAGAEYVVESTGVFT 3-phosphate
TMEKAGAHKGGAKRVIISAPSADAPMFVMGVNHEKYDKSLKIVSNASCT dehydrogenase)
TNCLAPLAKVIHDNFGIVEGLMTTVHAITATQKTVDGPSGKLWRDGRGAA
QNIIPASTGAAKAVGKVIPELNGKLTGMAFRVPTPNVSVVDLTCRLEKPAK
YDDIKRVVKAADGPLKGILGYTEDQVVSCDFNGDSHSSTFDAGAGIALND
HFVKLVSWYDNEFGYSNRVVDLMVHMASKE 154 Chicken TPM1
MDAIKKMQMLKLDKENALDRAEQAEADKKAEEERSKQLEDELVALQK (Tropomyosin
KLKGTEDELDKYSESLKDAQEKLELADKKATDAESEVASLNRRIQLVEEEL alpha-1 chain)
DRAQERLATALQKLEEAEKAADESERGMKVIENTRAQKDEEKMEIQEIQLKE
AKHIAEEADRKYE EVARKLVIIEGDLERAEEAESESKCAELEELKTVTN
NLKSLEAQAEKYSQKEDKYEEEIKVLTDKLKEAETRAEFAERSVTKLEKSID
DLEDELYAQKLKYKAISEELDHALNDMTSI; 155 Chicken MYPC2
MPEPSKAAPKKEAKKKEEKKEEKKEAPPQEHKDEAPDDVHPPETPDPEGL (Myosin-
binding protein C) FLSKPQNMVESGRDVTVSARVAGAALPCAPAVKWFKGKWAELGDKSAR
C) CRLRHSVDDDKVHTFELTITKVAMGDRGDYRCEVTAKEQKDSCSFSIDVEA
PRSEGNVLQAFKRTGEGKDDTAGELDFSGLLKKREVQVEEKKKKKDEDD
QFPPEIWELLKGVTKKSEYERIAFYGITDLRGMLKRLKKVHVEPKKSEAFI
RKLDPAYQVDKGNKIKLVVELSDPDLPLKWKYKNGQLLKPSTKYVFENVGL
KRILTIHKCSLADDAAYECRVNDEKCFTEVFVKEPPVTTVVRGLEDQQVVVG
DRVVLEAEVSEEGAQVMWLKDGVDVTRDDAFKYRFKKDGKKHFLINEA
ELSDSAHYKIMTNGGESEAELSVEEKQLEVLQDMADLTVKASEQAVFKCE
VSDEKVTGRWFRNGVEVKPSKRIHISHNGRFHKLVIDDVPEDEGDYTFIPD
GYALSLSAKLNFLEIKVEYVPKQEPPKIHLD CSGKAAENTIVVVAGNKVRL
DVPISGEPAPT VTWKRGDQLFTATEGRVHIDSQADLSSFVIESAERSDEGRY
CITVTNPVGEDSATLHVRVVDVDPDPQSVRVTSVGEDWAVLSWEAPPFDG
GMPITGYLMERKKKGSMRWMLNFEVFPDTTYESTKMIEGVFYEMRVFA
VNAIGVSQPSLNTQPFMPIAPTSEPTHVLEDVTDTTATIKWRPPERIGAGG
VDGYLVEWCREGSNEWVAANTELVERCGLTARGLPTGERLLFRVISVNMA
GKSPPATMAQPVTIREIVERPKIRLPRHLRQTYIRRVGEQVNLVIPFQGKPRP
QVTWSREGGALPAEVQTRTSDVDSVFFIRSAARPLSGNYEMRVRIDNMEDC
ATLRLRVVERPGPPQAVRVMEVWGSNALLQWEPPKDDGNAEISGYTVQK
ADTRTMEWFTVLEHSRPTRCTVSELMGNEYRFRVYSENVCGTSQEPATS
HNTARIAKEGLTLKMVPYKERDLRAAPQFLTPLVDRSVVAGYTVTLNCAV
RGHPKPKVTWLKNSVEIGADPKFLSRHGLGVLSLLIRRP GPFDDGGTYGCRA
VNEMGEATTECRLDVRVPQ 156 Chicken MPSF
MSSVAVPFYQRRHKHFDQSYRNIQTRYVLEEYAARKAASRQAAHYESTGL (M-protein)
GKTTTCRLCARRARSLAHEAMQESRKRTHEQKSHASDEKRIKFASELSSLER
EIHMARHHAREQLDRLAIQRMVEENMALERHVVEEKISR APEILVRLRSHT
VWEKMSVRLCFTVQGFPSPVVQWYKNEELITPASDPAKYSVENKYGVHVL
HINRADFDDSATYSAVATNIHGQASTNCAVVVRRFRESEEPHPAGIMPFHLP
LSYDVCFTHFVDVQFLEKFGVTFATEGETLTLKCSVLVTPELKRLRPRAEWY
RDDVLIKDSKWKLYFGEGQAALSFTHLNKDDEGLYTLRMVTKGGVNECS
AFLFVRDADALIAGAPGAPMDVKCHDANRDYVIVTWKPPNTTSQNPVIGY
FVDKCEVGLENWVQCNDAPVKICKYPVTGLYEGRSYIFRVRAVNSAGISRP

SRVSEPVADPVAQVDPVQVDDLEGDIQIPGPPTNVHAS
EISKTYVVLSWDPPVPRGREPLTYFIEKSMVGSWSQVRVNAQVAVKSPRYA
VFDLAEGKPYVFRVLSANKHGIDPSEITEPIQPQDIVVVPSAPGRVVATRNT
KTSVVVQWDKPKHEENLYGYIDYSVVGSNQWEPANHKPINYNRFVVHG
LETGEQYIFRVKAVNAVGFSENSQESEAIVQAALTCPSYPHGITLLNCDGH
SMTLGWKAPKYSGGSPILGYIDKREANHKWNHEVNSSVISRTIYTVEDLT
EDAFYEFKIAAANVVGIGHPSDPSEHFCKAWTMPEPGPAYDLTVCEVRNT
SLVLLWKAPVYEGKSPITGYLVDYKEVDTEWITANEKPTSHRYFKVTDLH
QGHTYVFKVRAVNDAGVGKSSEISEPVFVEASPGTKEIFSGVDEEGNIYLG
ECKEATDASHFLWGKSYEEIEDSDKFKIETKGDHSLYFKHPDKSDLGTYCI
SVSDTDGVSSSFVLDEEELERLMTLSNEIKNPTIPLKSELAYEVLDKGEVRF
WQAESLSPNSTYRFVINDKEVENGDHRHKISCDHSNGIEMVMDKFTIDNEG
TYTVQIQDGKAKNQSSLVLIGDAFKAILAESELQRKEFLRKQGPHFSEFLYW
EVTEECEVLLACKIANTKKETVFKWYRNGSGIDVDEAPDLQKGECHLTVPK
LSRKDEGVYKATLSDDRGHVSTLELSGKVYNDIILALSRVSGKTASPLKIL
CTEEGIRLQCFLKYNEEMKVTVSHRESKISSGEKMKIGGGEDVAWLQITE
PTEKDKGNYTFEIFSDKESFKRTLDSLGSQAFDDALTEFQRLKAAFAEKNR
GKVIGGLPDVVTIMDGKTLNLTCTVFGNPDPEVWVFKNDKALELNEHYLV
SLEQGGKYASLTIKGVTSEDSGKYSIYVKNKYGGETVDVTVSVYRHGEKIPE
VNQGQLAKPRLIPSSST 157 Chicken ENOB

MSIQKIHAREILDSRGEPTVEVDLHTAKGHFRAAVPSGASTGIHEALEPRDG (Beta-enolase)
DKKRFLGKGVLKAVEHINKTIGPALIEKKISVVEQEKIDKVMIEMDGTENKS
KFGANAILGVSLAVCKAGAAEKGVPLYRHIADLAGNTELILPVPAFNVING
GSHAGNKLAMQEFMVLPGAASFHDAMRVGAEVYHSLKGVIAKAKYKGD
ATNVGGEGGFAPNILDNHEALELLKAAIAQAGYTDKVVIGMDVAASEFCR
DGRYHLDFKSPHTKRYITGEQLGEIYRGFIKDYPVVSIEDPFDQDDWEAW
KRFVFHVDIQVVGDDLTVTNPKRIAHGAEQHACNCLLLKVNQIGSVTESIQ
ACKLAQSHGWGVMVSHRSGETEDTFIADLVVGLCTGQIKTGAPCRSERLA
KYNQLMRIEALGDKAKFAGRKFRNPKAK 158 Chicken MYL11

MAPKKAKRRAAEGSSNVFSMFDQTQIQEFKEAFTVIDQNRDGIIDKDDLRE (Myosin
TFAAMGRLNVKNEELDAMIKEASGPINFVFLTMFGEKLGADPEDVIMG regulatory
light AFKVLPDGDGKSGIKKSFLLELLTTQCDRFTPEEIKNMWAAFPDVAAGNVDY chain 11)
KNICYVITHGEDKEGE 159 Chicken MYH1B

MATDADMAIFGEAAPYLKSEKERIEAQNKPFDAKSSVFVHAKESYVKST (Myosin-1B)
IQSKESGKVTVKTEGGETLTVKEDQIFSMNPPKYDKIEDMAMMTHLHEPAV
LYNLKERYAAWMIYTYSGLFCVTVNPNYKWLPVYNPEVVLAYRGKKRQEA
PPHIFSISDNAYQFMLTDRENQSILITGESGAGKTVNTKRVIQYFATIAASGD
KKKEEQPAGKMQGTLEDQIISANPLLEAFGNAKTVRNDNSSRFGKFIRIHFG
ATGKLASADIETYLLEKSRVTFQLKAERSYHIFYQIMSNKKPELIEMLLITTN
PYDYQYVSQGEITVPSINDQEELMATDSAIDILGFTPDEKTAIYKLTGAVMH
YGNLKFQKQKREEQAEPGGTEVADKAAAYLMGLNSADLLKALCYPRVKVG
NEYVTKGQTVQQVYNSVGALAKSVFEKMFVLMVVRINQQLDTKQPRQYF
IGVLDIAGFEIFDFNSLEQLCINFTEKLLQQFFNHHMFVLEQEEYKKEGIEWE
FIDFGMDLAACIELIEKPMGIFSILEEECMFPKATDTSFKNKLYDQHLGKSNN
FQKPKPGKGAEAHFSLVHYAGTVDYNITGWLEKNKDPLNETVVGLYQKS
SLKTLALLFASVGGAEAESGAGGKKGGKGGKSSSFQTVSALFRENLNKLMS
NLRSTHPPHFVRCLIPNETKTPGAMEHELVLHQLRCNGVLEGIRICRKGFPRI
LYADFKQRYKVLNASAIPEGQFIDSKKASEKLLGSIDVDHTQYKFGHTKVF
FKAGLLGLLEEMRDEKLAQLITRTQARCRGFLMRVEFKKMMERRESIFCIQ
YNVRAFMNVKHWPWMKLFKIKPLLKSAESEKEMANMKEEFEKTKEELA

KSEAKRKELEEKMVSLQEKNDLQLQVQAEADGLADAEERCDQLIKTKIQ
LEAKIKELTERAEDEEEMNAELTAKKRKLEDECESELKKDIDDLLETLAKVE
KEKHATENKVKNLTEEMAALDETIAKLTKEKKALQEAHQQTLDLQAEED
KVNTLTAKTKLEQQVDDLEGSLEQEKKLRMDLERAKRKLEGDLKMTQE
STMDLENDKQQLDEKLKKKDFEISQIQSKIEDEQALGMQLQKKIKELQARIE
ELEEIEAERTSRAKAIEKHRADLSRELEEISERLEEAGGATAAQIDMNKKRE
AEFQKMRRDLEEATLQHEATAAALRKKHADSTADVGEQIDNLQRVKQKL
EKEKSELKMEIDDLASNMESVSKAKANLEKMCRSLEDQLSEIKTKEEEQQR
TINDISAQKARLQTESGEYSRQVEEKDALISQLSRGKQAFQTQQIEELKRHLEE
EIKAKKCPAHALQSARHDCDLLREQYEEEQEAKGELQRALSKANSEVAQW
RTKYETDAIQRTTEELEEAKKKLAQRLQDAEEHVEAVNSKSCASLEKTKQRL
QNEVEDLMIDVERSNAACAALDKKQKNFDKILSEWKQKYEETQAELEASQ
KESRSLSTELFKMKNAYEESLDHLETTLKRENKNLQQEISDLTEQIAEGGKAI
HELEKVKKQIEQEKSELQTALEEAASLEHEEGKILRVQLELNQVKSDIDRK
IAEKDEEIDQLKRNHLRVVDSMQSTLDAEIRSRNEALRLKKKMEGDLNEIEI
QLSHANRQAAEAQKNLRNTQGVVKDTQIHLDDALRSQEDLKEQVAMVER
RANLLQAEIEELRAALEQTERSARKVAEQELLDASERVQLLHTQNTSLINTKK
KLESDISQIQSEMEDITIQUARNAAEEKAKKAITDAAMMAEELKKEQDTS
ERMKKNLDQTVKDLQHRLDEAEQLALKGKKKQIQKLEARVRELEGEVDA
EQKRSAAVKGVRKYERRVKELTYQSEEDRKNVLRQLQDLVDKQLQMKVKS
YKRQAEAAEELSNVNLSKFRKIQHELEEAERADIAESQVKNKLRAKSREIGK KAESEE 160

Chicken R4GIG1

MATDADMAIFGEAAPYLKSEKERIEAQNKPFDAKSSVFVHAKESYVKST
(Uncharacterized
IQSKESGKVTVKTEGGETLTVKEDQIFSMNPPKYDKIEDMAMMTHLHEPAV protein)
LYNLKERYAAWMIYTYSGLFCVTVNPNYKWLVPVYNPEVVLAYRGKKRQEA
PPHIFSISDNAYQFMLTDRENQSILITGESGAGKTVNTKRVIQYFATIAASGD
KKKEEQPAGKMQGTLEDQIISANPLLEAFGNAKTVRNDNSSRFGKFIRIHFG
ATGKLASADIETYLLEKSRVTFQLKAERSYHIFYQIMSNNKKPELIEMLLITN
PYDYQYVSQGEITVPSINDQEELMATDSAIDILGFTPDEKTAIYKLTGAVMH
YGNLKFQKQKREEQAEPDGTEVADKAAYLMGLNSADLLKALCYPRVKVG
NEYVTKGQTVQQVYNSVGALAKSVFEKMFLLWMVVRINQQLDTKQPRQYF
IGVLDIAGFEIFDFNSLEQLCINFTEKLLQQFFNHHMFVLEQEEYKKEGIEWE
FIDFGMDLAACIELIEKPMGIFSILEEECMFPKATDTSFKNKLYDQHLGKSNN
FQKPKPGKGKAEAHFSLVHYAGTVDYNITGWLEKNKDPLNETVVGLYQKS
SLKTLALLFASVGGAEAESGAGGKKGGKGGKSSSFQTVSALFRENLNKLMS
NLRSTHPPHFVRCLIPNETKTPGAMEHELVLHQLRCNGVLEGIRICRKGFP
SRI LYADFKQRYKVLNASAIPEGQFIDSKKASEKLLGSIDVDHTQYKFGHTKVF
FKAGLLGLLEEMRDEKLAQLITRTQARCRGFLMRVEFKKMMERRESIFCIQ
YNVRAFMNVKHWPWMKLFFKIKPLLKSAESEKEMANMKEEFEKTKEELA
KSEAKRKELEEKMVSLQEKNDLQLQVQAEADGLADAEERCDQLIKTKIQ
LEAKIKELTERAEDEEEMNAELTAKKRKLEDECESELKKDIDDLLETLAKVE
KEKHATENKVKNLTEEMAALDETIAKLTKEKKALQEAHQQTLDLQAEED
KVNTLTAKTKLEQQVDDLEGSLEQEKKLRMDLERAKRKLEGDLKMTQE
STMDLENDKQQLDEKLKKKDFEISQIQSKIEDEQALGMQLQKKIKELQARIE
ELEEIEAERTSRAKAIEKHRADLSRELEEISERLEEAGGATAAQIDMNKKRE
AEFQKMRRDLEEATLQHEATAAALRKKHADSTAEELGEQIDNLQRVKQKLE
KEKSELKMEIDDLASNMESVSKAKANLEKMCRSLEDQLSEIKTKEEEQQR
INDISAQKARLQTESGEYSRQVEEKDALISQLSRGKQAFQTQQIEELKRHLEE
IKAKNALAHALQSARHDCDLLREQYEEEQEAKGELQRALSKANSEVAQWR

TKYETDAIQRTEEAQKLAQRLQDAEEHVAVNSKCSLEKTKQRLQ
NEVEDLMIDVERSNAACAALDKKQKNFDKILSEWKQKYEETQAELEASQK
ESRSLSTELFKMKNAYEESLDHLETLKRENKNLQQEISDLTEQIAEGGKAIH
ELEKVKKQIEQESELQAALAEAEASLEHEEGKILRVQLELNQVKSDIDRKI
AEKDEEIDQLKRNHLRVVDSMQSTLDAEIRSRNEALRLKKKMEGDLNEIEI
QLSHANRQAAEAQKNLRNTQGVLKDTQIHLDDALRSQEDLKEQVAMVER
RANLLQAEIEELRAALEQTERSARKVAEQELLDASERVQLLHTQNTSLINTKK
KLESDISQIQSEMEDTIQEARNAEKAKKAITDAAMMAEELKKEQDTSABL
ERMKKNLDDQTVKDLQHRLDEAEQLALKGKKQIQKLEARVRELEGEVDA
EQKRSAAVKGVRKYERRVKELTYQSEEDRKNVLRQLDLVDKQLQMKVKS
YKRQAEAAEELSNVNLSKFRKIQHELEAAEERADIAESQVKNKLRAKSREIGK KAESEE 161
Chicken ALBU

MKWVTLISFIFLFSSATSRNLQRFARDAEHKSEIAHRYNDLKEETFKAVAMI (Albumin)
TFAQYLQRCSYEGLSKLVKDVVDLAQKCVANEDAPECSKPLPSIILDEICQV
EKL RDSYGAMADCCSKADPERNECFLSFKVSQPDFVQPYQRPASDVICQY
QDNRVSFLGHFIYSVARRHPFLYAPAILSFAVD FEHALQSCCKESDVGACLD
TKEIVMREKAKGVSVKQQYFCGILKQFGDRVFQARQLIYLSQKY PKAPFSE
VSKFVHDSIGVHKECCEGDMVECMDDMARMMSNLCSQQDVFSGKIKDCC
EKPIVERSQCIMEAEFDEKPADLPSLVEKYIEDKEVCKSFEAGHDAFMAEFV
YEYSRRHPFEFSIQLIMRIAKGYESLLEKCKCTDNPAECYANAQEQLNQHIKE
TQDVVKTNCDLLHDHGEADFLKSILIRYTKKMPQVPTDLLLETGKKMTTIG
TKCCQLGEDRRMACSEGYSIVIHDTCRKQETTPINDNVSQCCSQLYANRR
PCFTAMGVDTKYVPPPFNPDMFSFDEKLCSAPAEEREVGQMKLLINLIKRK
PQMTEEQIKTIADGFTAMVDKCKQSDINTCFGEEGANLIVQSRATLGIGA
[0053] An animal protein can comprise a turkey protein (Table 9). A turkey protein can comprise a turkey muscle protein. A turkey protein can comprise a turkey blood protein. A turkey protein can comprise a turkey albumin protein. A turkey protein can comprise a sequence that has at least or about 70%, at least or about 75%, at least or about 80%, at least or about 90%, at least or about 91%, at least or about 92%, at least or about 93%, at least or about 94%, at least or about 95%, at least or about 96%, at least or about 97% at least or about 98% at least or about 99% or about 100% sequence identity to any one of SEQ ID NO: 162-174.

[0054] A food product may comprise a fragment or a truncation of a turkey protein. The food product may include a fragment or truncation of any one of SEQ ID Nos: 162-174. A sequence of the fragment or truncation may be identical or substantially identical to a portion of any one of SEQ ID Nos: 162-174. A sequence of the fragment or truncation may be at least about 70%, 80%, 90%, 95%, 98%, or more identical to a portion of a sequence of any one of SEQ ID Nos: 162-174. A length of the fragment or truncation may be at least about 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or more of a length of any one of SEQ ID Nos: 162-174. A length of the fragment or truncation may be less than or equal to about 90%, 80%, 70%, 60%, 50%, 40%, 30%, 20%, 15%, 10%, or less than a length of any one of SEQ ID Nos: 162-174.

TABLE-US-00009 TABLE9 Turkey Proteins SEQ ID NO Description Amino Acid
Sequence 162 Turkey G1NL16

MPFSNSHNLLKMKHSADDEFDPDLSAHNNHMAKVLTL DLYKKLRDRQTSSG (Creatine
kinase) FTLDDVIQTGVDNPGHPFIMTVGCVAGDEESYEVFKELFDPVIEDRHGGYK
PTDEHKTDNLNADNLQGGDDLDPNYVLSSRVRTGRSIRGFCFPHCSRGERRA
IEKLSVEALGSLGGDLKGKYYALRNMTDAEQQLIDDHFLFDKPVSPLLLA
SGMARDWPDARGIWHNDNKTFLVWINEEDHLRVISMQKGGNMKEVFTRF
CTGLTQIETLFSKKNYEFMWNPHLG YILTCPSNLGTGLRAGVHIKLPNLGK
HEKFGEVLKRLRLQKRGTTGGVDTA AVGGVFDVSNADRLGFSEVELVQMV
VDGVKLLIEMEKRLEKQSIDDLMPAQK 163 Turkey

MNSNQIETNMQYTYNYEEDSYMTEQEEEWDRDLLDPAWEKQQRKTFTA A0A803Y9V3
WCNSHLRKAGTQIENIEEDFRNGLKMLLLEVISGERLPKPDRGKMRFHKKI (Actinin
alpha 2) ANVNKALDYIASKGVKLVSIGAEIIVDGNVGMTLGMIIWTLRFAIQDISVE
ETSAKEGLLLWCQRKTAPYRNVNIQNFHLSWKDGLGLCALIHRHRPDLIDY
SKLNDKDDPIGNINLAMEIAEKHLDIPKMLDAEDIVNTPKPDRAIMTYVSCF
YHAFAGAEQAETAANRICKVLAVNQENERLMEEYERLASELLEWIRRTIPW
LENRTPEKTMQAMQKKLEDFRDYRRKHKKPPKVQEKQLEINFNTLQTKLRI
SNRPAFMPSEGKMOVSDIAGAWQRLEQAEKGYEEWLLNEIRRLERLEHLAE
KFRQKASTHEQWAYGKEQILLQKDYESASLTEVRAMLRKHEAFESDLAAH
QDRVEQIAAIAQELNELDYHDAASVNDRCQKICDQWDSLGLTLTQKRREAL
ERTEKLLLETIDQLHLEFAKRAAPFNNWMEGAMEDLQDMFIVHSIEEIQSLIS
AHDQFKATLPEADGERQAILSIQNEVEKVIQSYSMRISASNPYSTVTVEEIRT
KWEKVKQLVPQRDQSLQEELARQHANERLRRQFAAQANVIGPWIQTKME
EIARSSIEMTGPLEDQMNQLKQYEQNIINYKHNDKLEGDHQLIQEALVEDN
KHTNYTMEHIRVGWELLTTIARTINEVETQILTRDAKGITQEQMNDFRASF
NHFDRRKNGLMDHDDFRACLISMGYDLGEAEFARIMSLVDPNGQGTVTQFQ
SFIDFMTRETADTDTAEQVIASFRILASDKPYILADELRRELPPPEQAQYCIKR
MPQYTGPGSVPGALDYTSFSSALYGESDL 164 Turkey

MAAPLSDGERRKQISVRGIAGLDVAEVRKSFNRHLHFTLVKDRNVATPR A0A803Y8V3
DYFFALAHTVRDHLVGRWIRTQQHYERDPKRIYYLSLEFYMGRTLQNTM (Alpha-1,4
glucan VNLGLQNACDEAIYQLGLDLEEELEEIEEDAGLGNGGLGRLAACFLDSMATL
phosphorylase) GLAAYGYGIRYEFGIFNQKIVDGWQVEEADDWLRYGNPWEKARPEYMLP
VHFYGRVEHSPEGVKWVDVTQVVLAMPYDTPVPGYKNNTVNTMRLWSAK
APNDFNLQEFNVGDYIEAVLDRNLAENISRVLYPNDNFFEGKELRLKQEYF
VVAATLQDIIRRFKSSKFGCQDPVRTSFETFPDKVAIQLNDTHPALSIPELMR
ILVDVEKVDWDKAWAITKQTCAYTNHTVLPALERWPVSMFEKLLPRHLEI
IYALNQMHLDRAALYPGDIDRLRRMSVIEEGDCKRINMAHLCVIGSHAVN
GVARIHSDIVKNTVFKDFYELDPEKFQNKTNGITPRRWLLLCNPGLADVIAE
KIGEDFVTDLSQLKKLLDFIDDETFRDVAKVQKQENKLFKFSAYLEEYKVKI
NPSSMFDIQVKRIHEYKRQLLNCLHAITLYNRKSDPSKSFVPRTIMIGGKAA
PGYHMAKMIKLITSIGEVVNDPYVGDKLKVIFLENYRVSLAEKVIPAADL
SQQISTAGTEASGTGNMKFMLNGALTIGTMDGANVEMAEAGEENLFIFG
MRVEEVEALDRQGYNAREYYDRIPELRQAIDQISSGFFSPRDPGCFRDVVN
MLMHHDREFKVFADYEAYIKCQGQVDQLFMDPREWTKKVIRNIACSGKFSS
DRTITEYAREIWGVEPSGDVFPWEA 165 Turkey GIN2A4

MTHQYPALTAEQKKELSDIALRIVAPGKGILAADESVGSMARLNQIGVEN (Fructose-
TEENRRLYRQILFSADSRVKKCIGGVIFFHETMYQKADDGTPFVQMIKDKGI bisphosphate
VVGIKVDKGVVPLAGTDGETTTQGLDGLSERCAQYKKDGADFAKWRCVL aldolase)
KISDNTPSALAIMENANVLARYASICQQNGIVPIVEPEILPDGDHDLKRCQY
VTEKVLAAVYKALSDHHVYLEGTLLKPNMVTPGHSCPTKYSPEEIAMATV
TALRRTVPPAVPGVTFLSGGQSEEEASINLNAINTCPLVRPWALTFSYGRAL
QASALSAWRGQRDNANAATEEFVKRAEVNGLAALGKYEGSGDDSGAAGQ SLYVANHAY
166 Turkey Q6W5H1

MAPKKDVKKPAAAAAPAPAPAPAPAPAKPKPEAIDLKSIKIEFSKEQQD (Myosin
essential DFKEAFLFDKTDGAKITLSQVGDIVRALGQNPTNAEMNKILGNPSKEEMN
light chain AKKITFEEFLPMLQAAANNKDQGTYEDFVEGLRVFDKEGNGTVMGAELRH
isoform 1) VLATLGEKMTREEVEELMKGQEDSNGCINYEAFVKHIMSV 167 Turkey
G1NMR6 MVKVG VNGFGRIGRLVTRAAVLSGKVQVVAINDPFIDLNYMVYMFKYDST
Glyceraldehyde-
HGHFKGTVKAENGKLVINGHAITIFQERDPSNIKWADAGAEYVVESTGVFT (3-phosphate

MEAKAGAHLAGKAGVAGVGNHKEKYNMGVNMHGVNDEKSLKIVSNASCT dehydrogenase)
TNCLAPLAKVIHDNFGIVEGLMTTVHAITATQKTVDGPSGKLWRDGRGAA
QNIIPASTGAAKAVGKVIPELNGKLTGMAFRVPTPNVSVVDLTCRLEKPAK
YDDIKRVVKAADGPLKGILGYTEDQVVSCDFNGDSHSSTFDAGAGIALND
HFVKLVSWYDNEFGYSNRVVDLMMVHMASKE 168 Turkey
MEAIKKKMQLKLDKENALDRAEQAEAEQKQAEERSKQLEDELAAMQK A0A803XNQ2
KLKGTEDELDKYSEALKDAQEKLELAEKKAADAEAEVASLNRRILVEEE (Tropomyosin
3) LDRAQERLATALQKLEEAEKAADESERGMKVIENTRALKDEEKMELQEIQ
KEAKHIAEEADRKYEEVARKLVIIEGDLERTEERAELAESKCSLEEEELKNV
TNNLSLEAQAEKYSQKEDKYEIEIKILTDKLKEAETRAEFAERSVAKLEK
TIDDLEDELYAQKLKYKAISEELDHALNDMTSI 169 Turkey
MCLQLFVFSELPPPSFLKNKELTFVYCLHKALMVECWCHSICSSPTLADQI A0A803YPL8
REVVGLESLVSSGKCILERIKHWLIIDIFLFLCISVWSIGESPPEETEKHDDSQ (Myosin-
binding RSTLFIEKPQSGSVSVGGNITFIAKVEAKDLLRKPNVKWFKGKWMDLASKA
protein C) GKHLQLKESFERHTKIHTFEMHIIQAKENYAGNYRCEVSYKDKFDSCSFDL
EVTESSQAAPSIDIRSAFKRSGDGQDDAGELDFSGLLKRREIKQQEEEPVD
VWELLKNANPSEYEKIAFQYGITDLRGMLKRLKRMREVKKSAAFSKGLD
PAYQVDKGGKVRFMVELADPTVELKWKYKNGQEIRPSAKYIFEHKGNGRIL
FINNCTMTDDARYYYVTAGDEKCESTELFVRDPPILVTGLEDNTYVGERVE
LSCEVSEDDANVKWFKNGVELTNEPKSRYRIKVEGKKHTLIIEEAAKNDNA
TYSVMTTGGQSEAKLSVDLRPLKISLALEDQTVRLGQEIHLKCEISENVEGK
WYKNGQLIEASDRVKLYHKGRIHRFIIASA AVDDDEGEYMFVPDAYNINIPC
KVHVVDPPKLHLDGLGENNTVTVVAGTKLRLEIPITGEPTPKVMWSKGD
WITDSGRIRAETYSOSSCLVIDTAEREDSGPFRITLKNEAGEDSALINIKVVD
VPDPPQAPNVTEVGEDWCVMTEPPANDGGSPILGYFIERKKKQSSRWMR
LNFELCKETTFEPKKMIEGVAYEVRVFAVNAIGTSKPSMPSKSFVPLAVTSP
PTLLAVDSVTDTSVTMKWRPPDHIGAAGLDGYVVEYCFEGTDEWIVANPE
LTDKTKFTINGLPTGSKILVRVKAVNAAGESEPRYHPQPILVKEVIEPPKIRL
PRHLKQTYTRRVGETVNLVIPFQVVRTKNTDRKTLNTHFSYATASYLLFLYV
NLGYWTAFAVNILHPEGKSHDKASGIIHSYLGTKIGHYHMSSQSN 170 Turkey G3URS1
MNVKGEISAYASLVVKRYKGEIDASHLHAGTFSMPLSLGVTPHGFASRFEI (Myomesin 1)
KFIDKFEVMFGREGETMSLGCTVVISPDIKRFKPDIEWYRNGVLITPSKWVQ
MHWGTGERASLTTHANKEDGLYTLRVVMGEYEEYSGYVFVRDADAEI
PGAPGAPLDVQCLDANKDYVIVSWKQPAVDGGNPILGYFIDRCEVGTTHW
TQCNETPVKFARFPVTGLIEGRSYIFRVRAVNRAVSLPSRVSEAVAALDPA
DRARLRSHPSAPWTGQIIVTEEEPAEGVIPGPPTDLQVIEATKNYVVLWSKP
PGQRGHEGIMYFVEKCIAGTEDWQRVNTEIPVKS PRFAVFDLSEGKSYCFR
VRCCNSAGIGEPSEATEATVVDDKLDIPKPPGRIMPTRNTDTSVVVTWTEPP
DAKELVGYIIESSVVGSGHWEPCNNNPVKGTRFICHGLVNGEKYIFRVRAV
NAAGLSEFSQDSEPIEVQAAIAAPSPPYDITVLESVRDSMVLGWKQPKATGG
TEITGYVNYREVIGGVPGKWREANIKAI SERAYRIENLKENMVYQFQVAA
ANLAGVGTPSLPSKPFKCEEWTIAVPGPPHDVTYTEVRKDSLVLWLWKEPVY
TGRSPIAGYYIDMKETEAKEEHWRSVNEKPLQKKFLKIAGLTEGVSYVFRV
RATNQAGVGRPSDITDPVVAETRPGTKEVVVNVDGDDGVISLNFECQDMSP
NSKFVWSKNYEPIEDDSRLDIDSKGGKSKAIFKNLGEDDLGIYSCVVTDTDG
VSSGYTIDEEEMKRLALLSHERKFPTIPLASELAVEILEKGEVRFWLQAEKLS
GNAKANFVFNDKEIFNGEKYKMKVDQRTGLVEMIMDKLEDKDEGTYTFQ
LQDGKATNQSSLVLIGDVFKKL RDEAEFQRKEWHRKQGPHFVDQLGWEV
TDECNVMLKCKVANIKKETHIVWYKDDREIMVDEEHDFKDG VCTLLISEFS
KKDTGIYEVILKDDR GKDVSELKLTDTAFADLMNEVCRRIALSATDLKIQST

AEIRLYSYFYVDDLRVGVTDYRVTDRVKTGVTGEQIWLQINEPT
PQDKGKYTMELFDGKTGHKKTVDLSGQAFDEAFAEFQRLKQAAIAEKNRA
RVLGGLPDVVTIQEGKALNLSCTVWGDPTPEVSWLKNEKSFVSDANCILKF
EAGKNVSFTISAVSTHDSGKYSLVVKNKYGTETSDVTVIVITVSGQTD TAK
HTVSISGQSQNNITVSKKKKATTMSKPQQEKEKKDIKPHVDEIIATEETS DS
KEIPRVEKEKNTENNQQTAKKTHGKAAAARTVP 171 Turkey
MSILKIQAREIFDSRGNPTVEVDLYTNKGLFRAAVPSGASTGIYEALERDN A0A803YQS2
DKTRYLGKGVSKAVEHVNKTIAPALISKNVNVVEQE KIDKLMLEMDGTEN
(Phosphopyruvate
KSKFGANAILGVSLAVCKAGAAEKGVPLYRHIADLAGNPEVILPVPAFNVI hydratase)
NGGSHAGNKLAMQEFMILPVGADNFKEAMRIGAEVYHNLKNVIKEKYGK
DATNVGDEGGFAPNILENKEALELLKTAIGKAGYSDKVVIGMDVAASEFYR
DGKYDLDFKSPDDPSRYISPDQLADLYKGFVKNYPLVSIEDPFDQDDWAA
WKKFTASVGIQVVGDDLTVTNPKRIAKAVEEEKSCNCLLLKVNQIGSVTESL
QACKLAQSNGWGVMVSHRSGETEDTFIADLVVGLCTGQIKTGAPCRSERL
AKYNQLLRIEEELGSKARFAGRNFRNPRIN 172 Turkey G1N0W1
MAPKKAKKRIEGANSNVFSMFEQAQIQEFKEAFTIMDQNRDGFIDKADLRD (Myosin
light TFAALGRLNVKNEEIDEMIKEAPGPINFTVFLTMFGEKLGADPEETILNAF chain 2)
KVFDPEGKGLKSAYIKEMLMTQGERFSQEEIDQMFAAFPPDVSGNLDYKNL
VHVITHGEEKD 173 Turkey GIMSQ0
MSSDAEMAIFGEAAPYL RKSEKERIEAQNRPFDAKAACFVVDQKQMYVKG (Myosin
heavy TIQSREGGKVTVKTYDDTTVTVKDDEVFPMNPPKYDKIEDMAMMTHLHEP chain
2) AVLYNLKERYAAWMIYTYSGLFCVTNPNYKWL PVYNPEVVSGYRGKKRQ
EAPPHIFSISDNAYQFMLTDRDNQSILITGESGAGKTVNTKRVIQYFATIAVT
GEKKKDQQPSKMQGTLEDQIISANPLLEAFGNAKTVRNDNSSRFGKFIRIHF
GTTGKLASADIETYLLEKSRVTFQLASERSYHIFYQIMSNKKPELIGEYLLLI
TTNPYDFTYVSQGEITVASIDDSEELLATDNAVDILGFSPDEKVGMYKLTGA
VMHYGNMKFKQKQREEQAEPDGTEVADKAAAYLMGLNSAELLKALCYPR
VKVGNEYVTKGQNVQQVYNSVGALAKSVYEKMFLWMVTRINQQLDTKQ
PRQHFIVGLDIAGFEIFDFNSLEQLCINFTNEKLQQFFNHHMFVLEQE EYKKE
GIDWEFIDFGMDLAACIELIEKPMGIFSILEEECMFPKATDTSFKNKLYDQHL
GKSNNFQKPKPAKGKAEAHFSLIHYAGTVDYNITGWLEKNKDPLNETVVG
LYQKSSLKLLSFLFSNYAGAETSDSGSGKKGGKKGGSFQTVSAVFRENLN
KLMTNLRSTHPPHFVRCLIPNETKTPGEMDHVLMHQLRCNGVLEGIRICRK
GFPSRILYADFKQRYKILNASAIPEGQFIDSKKASEKLLSSIDVDHNQYRFGH
TKVFFKAGLLGLLEEMRDEKLVSLITHMQAMCRGYLMRTEFKKMNARRES
IYIIQYNVRAFMNVKHWPMKLYFKMKPMLKSVESEKEMANMKEEFEKT
KEELAKSEAKRKELEEKMVSL LQEKNDLQLQVQSESENLA DAERCEGLIK
SKIQLEARIKELNERMEDEEEMNAELTAKKRKLEDECSELKKDIDDLETLA
KVEKEKHATENKVKNLTEEMAALDENISKLTKEKKALQEAHQQTLEYLQA
EEDKVSILTKIKTKLEQQVDDLEGSLEQERKQRM DLERAKRKLDGDLKMS
QDSIMDLKNDKQQTDERLKKKDFEISHLQSKIEDEQAQSSQLQKKIKELQA
RTEELEEEIEAERANCAKTEKQRADLSRELEEISERLEEAGGATAAQIEMNK
KRAAEFQKMRRDLEEATLQHEATAAALRKKHADSTAELGEQIDNLQRVKQ
KLEKEKSELKMEIDDLASNMESVSKAKSNLEKMCRALEDQYSEVRTKDDE
HMRLINDMNTQKTRLQTENGELTRQLEEKESLISQLTRGKQAITQQTEELK
RQLEENKAKNALAHGLQSARHDCDLLREQYEEELEAKGELQRALSKANS
EVAQWRTKYETDAIQCMEELEDAKKKLAQRLQDSEEQIEAVNSK CASLEK
TKQRLQGEVDDL MIAVERSNAACAAFDKKQKNFDKVM AEWKQKYQESQ
AELEAAQKESRTLSTEIFRMKNAYEEVLDQVETVRRENKNLQQEISDLTEQ

LAEAGKAGHGANKEKQTEQKQQAALQAEVSLAEHEEGKILRVQLEL
NQVKSGVDRRSAEKDEEIQQLKKNHQRVLESMQTMLEIRSRNDALRLK
KKMEGDLNDMEIQLSRANCQIAEIQKHLKGMQGGQLKDSQLHLDDALREND
DLKEQLAVVERRNNLMMTELEQMRAALEQTERARKVSEQELTDTSERVQI
LHSQNTSLLNTKKKLEVDITHLQNEVEDSIQEASNAEEKAKKAITDAAMMA
EELKKEQDTS AHLERMKRNLEQMVKDLQHRLDEAEQLTLKGGKKQLQKL
EARIHelenELDieQKRGTESLKGARKYERRLKElTYQSEEDKKNILRLQNL
VDKLQLKVKAYKKQAEAEQASTNLSRCRKTQHELEEAERADIAECQV
NKLRAKSRDTGRQKSGLSSELGDNTFLPVIWHNMCNIDRPETTPQVILLHQ 174 Turkey
GINCR2 MKWVTLISFIFLFSSATSRNLQVRARDAEHKSEIAHRYNDLKEETFKAVAMI
(Albumin) TFAQYLQRCSYDGLSKLVKDVVDLAQKCVANEDAPECSKSLPSIILDEICQV
EKL RDSYGAMADCCSKADPERNECFLSFKVSQPDFIQPYQRPASDVICKEY
QDNRVSLLGHFIYSVARRNPYMYAPTILSFAADYEHALQSCCQESDVGACL
DAKETAMREKAKKISVKQQYSCGILKKFGDRVFQAEKLVRLSQKYPKAPFS
EVAKLVRISKDVHKECCEGDMVECMDDMAEIINNMC SKQDVFSSKIKGCC
EKPIMERSQCIMEAEFDEKPADLPSLVEKYIQDKEVCKSFEAGHDAFMSEFV
YEYSRRHPFESTQLILRIAKGYESLLEKCKCTDNPAECYANAQDQLNQHIKE
TQDVVKTNCDLLSAHGEPDFLKSILIRYTKKMPQVSTETLLEIGKKMTAIGT
KCCQLSEDRRMACSEG YLSIVIHDMCRRQETTPINDNVSHCCSDSYAHRP
CFTAMGVDTKYVPPFPNPMFSFDEKLCSAPAEEREVGQMKLLINLIKRP
QMTEEQIKTIADGFTAMVDKCKQSDINTCFGEEGANLIVQSRATLGIGV

[0055] An animal protein can comprise a duck protein (Table 10 and Table 11). A duck protein can comprise a duck muscle protein. A duck protein can comprise a duck blood protein. A duck protein can comprise a duck albumin protein. A duck protein can comprise a protein from a mallard duck (*Anas platyrhynchos*). A duck protein can comprise a protein from a Northern mallard duck (*Anas platyrhynchos platyrhynchos*). A duck protein can comprise a protein from a Muscovy duck (*Cairina moschata*). A duck protein can comprise a protein from a domestic Muscovy duck (*Cairina moschata domestica*). A duck protein can comprise a sequence that has at least or about 70%, at least or about 75%, at least or about 80%, at least or about 90%, at least or about 91%, at least or about 92%, at least or about 93%, at least or about 94%, at least or about 95%, at least or about 96%, at least or about 97% at least or about 98% at least or about 99% or about 100% sequence identity to any one of SEQ ID NO: 175-200.

[0056] A food product may comprise a fragment or a truncation of a duck protein. The food product may include a fragment or truncation of any one of SEQ ID Nos: 175-200. A sequence of the fragment or truncation may be identical or substantially identical to a portion of any one of SEQ ID Nos: 175-200. A sequence of the fragment or truncation may be at least about 70%, 80%, 90%, 95%, 98%, or more identical to a portion of a sequence of any one of SEQ ID Nos: 175-200. A length of the fragment or truncation that is at least about 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or more of a length of any one of SEQ ID Nos: 175-200. A length of the fragment or truncation may be less than or equal to about 90%, 80%, 70%, 60%, 50%, 40%, 30%, 20%, 15%, 10%, or less than a length of any one of SEQ ID Nos: 175-200.

TABLE-US-00010 TABLE 10 Northern Mallard Duck Proteins SEQ ID NO Description
Amino Acid Sequence 175 Northern Mallard
MPFSNSHNLLKMKHSATDEFPDLSAHNNHMAKVLTLDLYKKLRDKQTPSG Duck
FTLDDVIQGTGVDNPGHPFIMTVGCVAGDEESYEVFKELFDPVIEDRHGGYK A0A493TFK1
PTDQHKTDLNAGNLQGGDDLDPNYVLSSRVRTGRSIRGFCLPPHCSRGERR (Creatine
kinase) AIEKLSVEALGSLGGDLKGKYYALRNMTDAEQQLIDDHFLFDKPVSPLLL
ASGMARDWPDARGIWHNDNKTFLVWINEEDHLRVISMQKGGNMKEVFTR
FCTGLTQIENLFKSKNHEFMWNPHLG YILTCPSNLGTGLRAGVHIKLP HLKG
HEKFGEVLKKLRLQKRGTTGGVDTA AVGGVFDVSNADRLGFSEVELVQMV

VDGVKLLIEMEKRKGQSIDDLIIPAQK 176 Northern Mallard
MNSMNQIETNMQYTYNYEEDEYMTQEEEWDRDLLLDPAWEKQQRKTFTA Duck
WCNSHLRKAGTQIENIEEDFRNGLKLMLEVISGERLPKPDRGKMRFHKI A0A493TCE5
ANVNKALDYIASKGVKLVSIGAEEIVDGNVKMTLGMWTHILRFAIQDISVE (Actinin
alpha 2) ETSAKEGLLLWCQRKTAPYRNVNIQNFHLSWKDGLGLCALIHRHRPDLIDY
SKL NKDDPIGNINLAMEIAEKHLDIPKMLDAEDIVNTPKPDERAIMTYVSCF
YHAFAGAEQAETAANRICKVLAVNQENERLMEEYERLASELLEWIRRTIPW
LENRTPEKTMQAMQKKLEDFRDYRRKHKPPKVQEKQCLEINFNTLQTKLRI
SNRPAFMPSEGKMOVSDIAGAWQRLEQAEGYEEWLLNEIRRLERLEHLAE
KFRQKASTHEQWAYGKEQILLQKDYESASLTEVRAML RKHEAFESDLAAH
QDRVEQIAAIAQELNELDYHDAASVNDRCQKICDQWDSLGLTLTQKRREAL
ERTEKLL ETIDQLHLEFAKRAAPFNNWMEGAMEDLQDMFIVHSIEEIQSLIS
AHDQFKATLPEADGERQAILSIQNEVEKVIQSYSMRISASNPYSTVTVEEIRT
KWEKV KQLVPQRDQSLQEELARQHANERLRRQFAAQANVIGPWIQTKME
EIARSSIEMTGPLEDQMNQLKQYEQNIINYKHNDKLEGDHQLIQEALVEDN
KHTNYTMEHIRVGWELLTTIARTINEVETQILTRDAKGITQEQMNDFRASF
NHFDRRKNGLMDHDDFRACLISMGYDLGEAEFARIMSLVDPNGQGTVTFQ
SFIDFMTRETADTDTAEQVIASFRILASDKPYILADELRRELPPPEQAQYCIKR
MPPYTGP GSVPGALDYTSFSSALYGESDL 177 Northern Mallard

MAAPLSDGEKRRKQISVRGIAGLDVAEVRKSFNRHLHFTLVKDRNVATPR Duck U3ISG6
DYFFALAHTVRDHLVGRWIRTQQHYERDPKRIYYLSLEFYMGRTLQNTM (Alpha-1,4
glucan VNLGLQNACDEAIYQLGLDLEEL EEEIEDAGLGNGGLGRLAACFLDSMATL
phosphorylase) GLAAYGYGIRYEF GIFNQKIVDGWQVEEADDWLRYGNPWEKARPEYMLP
VHFYGRVEHTPEGVKWVDTQVVLAMPYDTPVPGYKNNTVNTMRLWSAK
APNDFNLQEFNVGDYIEAVLDRNLAENISRVLYPNDNFFEGKELRLKQEYF
VVAATLQDIIRRFKSSKFGCRDPVRTSFETFPDKVAIQLNDTHPALSIPELMR
ILVDVEKVDWDKAW EITKRTCAYTNHTVLPEALERWPVSMFEKLLPRHLEI
IYALNQMHLD RVAALYPGDVDR LRMSVIEEGDCKRINMAHLCVIGSHAV
NGVARIHSDIVKHSVFKDFYELEPEKFQNKTNGITPRRWLLLCNPGLADVIA
EKIGEGFVTDLSQLKKLLDFIDNEMFIRDVAKVKQENKLKFSAYLEEYQYKV
KINPSSMFDVQVKRIHEYKRQLLNCLHAITLYNRIRSDPSKSFVPRTIMIGGK
AAPGYHMAKMIKLITSIGEVINNDPYVGDKLKVIFLENYRVSLAEKVIPAA
DLSQQISTAGTEASGTGNMKFMLNGALTIGTMDGANVEMAE EAGEENLFIF
GMRVEDVEALDRKGYNAL EYERLPELRQAIDQISSGFFSPRDPGCFRDVV
NMLMHHD RFKVFADYEAYIKCQGQVDQLFMDPREWTKKVIRNIACSGKFS
SDRTITEYAQEIWGV EPSSTKIPPPNIPRD 178 Northern Mallard

MTHQFPALSPEQKKALSDIAQRIVASGKGILAADESVGTMGNRLQRINVEN Duck
U3IHG8 TEENRRAFREILFSSDTSINQSIGGVILFHETLYQKDSSGKPPPAIIEKKGIVVG
(Fructose- IKLDKGTSPLAGTNGETTIEGLDGLSERCAQYKKDGVDGFWRAVLKITST
bisphosphate TPSQLAIQENANTLARYASICQQNGLVPIVEPEVLPDGDHDLQRCQYVTEK
aldolase) VLA AVYKALNDHHVYLEGTLLKPNMVTAGHACTKKYTPQDVAAATVTTL
LRTVPAAVPGICFLSGGQSEEEASLNLNAMNQSSLPKPWKLTF SYGRALQA
SALA AWLGKSENKKA AQEAFRKRAQINSLACRGQYVVSGKTDAAATQSLF TASYTY 179
Northern Mallard

MAPKKDVKKPAAAAAPAPAPAPAPAPAKPKPEAIDLKSIKIEFSKEQQD Duck U3IA79
DFKEAFLLFDRGTGDAKITLSQVGDIVRALGQNPTNAEINKILGNPSKEEMNA (Myosin light
KKITFEEFLPMLQAAANNKDQGT YEDFVEGLRVFDKEGNGTVMGAELRHV chain 1)
LATLGEKMT EEEVDEL MKGQEDSNGCINYEAFVKHILSV 180 Northern Mallard
MVKVG VNGFGRIGRLVTRAILSGKVQVVAINDPFIDLNYMVYMFKYDST Duck
HGHFRGT VKAENGLVINGNAITIFQERDPSNIKWADAGAEYVVESTGVFT A0A493TQC9

TMEKAGHLGGAKRVIISAPSADAPMFVMGVNHEKYDKSLKIVSNASCT
(Glyceraldehyde-
TNCLAPLAKVIHDNFGIVEGLMTTVHAITATQKTVDGPSGKLWRDGRGAA 3-phosphate
QNIIPASTGAAKAVGKVIPELNGKLTGMAFRVPTPNVSVVDLTCRLEKPAK dehydrogenase)
YDDIKRVVKAAADGPLKGILAYTEDQVVSCDFNGDSHSSTFDAGAGIALND
HFVKLVSWYDNEYGYSNRVLALSTLLHAANIFPSSR 181 Northern Mallard
MEAIKKKMQLKLDKENALDRAEQAEAEQKQAEERSKQLEDELAAMQK Duck
KLKGTEDELDKYSEALKDAQEKELELAEKKAADAEAEVASLNRRIQLVEEE A0A493TRM1
LDRAQERLATALQKLEEAKEKADESERGMKVNIENRALKDEEKMELQEIQ (Tropomyosin
3) KEAKHIAEEADRKYYEVARKLVIIIEGDLERTEERAELAESKCSLEEEELKNV
TNNLKSLEAQAEKYSQKEDKYEEIEIKILTDKLKEAETRAEFAERSVAKLEK
TIDDELEDELYAQKLKYKAISEELDHALNDMTSI 182 Northern Mallard
MPEPVKKTVSAFTKKPKTTEVTAGSTAVFEAETEKASVKVKWQRAGTEIT Duck U3J9W2
DSEKYVIKAEGNKHSLTINNKKEDDVYAVIAGSSKVKFELKVKEPEKSE (Myosin
binding TVTPAEAAPAPAASELPAAPAESSQNTTEVPSAGTQPEEPADPIGLFVTRPQDG
protein C3) EVTVGGNITFTAKVAGESLLKKPSVKWFKGKWMDLASKVGQHLQLHDSY
DRNNKVYTFELEIIEANMTFAGGYRCEVSTKDKFDSNENLTVNEAPLSGD
LDIRAAFRRTTEGHDEAGELNFSALLKKRGDGKTESQPDVDVWEILRKAPP
SEYEKIAFQYGITDLRGMLKRLKRIKKEKKSTAFLLKLDPAYQVDKGQKI
KLMVEVANPDADV KWLKNGQEIQVSGSRIFYEAVGNKRILTINHCSLADD
AAYECVVGEEKSFTELFVKEPPILITRPLEDQMVMVGERVEFECEVSEEGAT
VKWEKDGVELTREETFYKRFKKGDKKQYLIINESTKEDSGHYTVKTNGGV
SVAELIVQEKKLEVYQSIADLTVKARDQAVFKCEVSDENVKGIWLKNGKE
VVPDERIKISHIGRIHKLTIEDVTPGDEADYSFIPQGFAYNLSAKLQFLEV KID
FVPREEPPKIHLDCLGQSPDTIVVAGNKLRLDVPISGDPPTPTVIWQKV NKK
TDLLRSNDDSMTPSENSSDLNTDSKLLFESEGRVRVEKHEDHCVFIIEGAEK
EDEGVYRVIVKNPVGEDKADITVKVIDVPDPPEAPKISNIGEDYCTVQWQPP
KYDGGQPVLGYILERKKKKSYRWMRLNFDLLKELTYEAKRMIEGVVYEM
RIYAVNSIGMSRPSASPQPFMPIAPPSEPHTFTVEDVSDSTVALKWRPPERIG
AGGLDGYIVEYCKEGSTEWIPALPGLTERTSALIKDLVTGDKLHFRIKAINL
AGESGAIIKEPITVQEILQRPKIWLPRHLRQTLVKKVGETINIVIPFQGKPRP
KITWTKDQGSLDAKDVGIRNSNTDTILFIRKAELHHSYGAYEITLQIENMTDK
VAITIQIIDKPGPPQNIKLTDVWGFNAALEWTPPQDDGNAQILGYTVQKAD
KKTMEWYTVFDHYRRTNCVVSELIMGNEYFFRVFSENLCGLSETAATTKN
PAYIQKTGTTYKPPCYKEHDFSEPPKFTHPLVNRSVIAGYNTTLCASVRGIP
KPKIYWKYKNKVDLSGDAKYRMFSKQGVLTLEIRKPTPFDGGGYTCKAVNE
CGEAEIECKLDVRVPQ 183 Northern Mallard
MSSLAVPFYQRRHKHFDQSYRNIQTRYVLEEYAARKAASRQAAYYESSGL Duck
GKTTTCRLCARRARTLAHETMQESRKRTHEQKSHASEEKRIRFASLSALEK A0A493TRW8
EIHTARHRAREQLDRLAIQRMVEENMAMERHAVEERISRAPEILVRLRSHT (Myomesin 2)
VWEKMSVSLCFTVQGFPTPVVQWYKNEELITPASDPAKYRVESKYGVHVL
HINRADFDSDSATYSAVATNVHGQASTNCAVVVRRFREGEPPHPAGIMPFHL
PLSYDVCFTHFQFLEKFGVTFATEGETLTLKCSVLVTPELKRLRPRAEW
YRDDVLIKDSKWTCLYFGEGQAALSFTHLNKDDEGLYTLRMITKGGVNEC
SAYLFVRDADALIAGAPGAPMDVKCHDANRDYVIVTWKPPNTTSQSPVIG
YFVDKCEVGLENWVQCNDAPVKICKYPVTGLYEGRSYIFRVRAVNNAGIS
RPSRVSEPVAALDPVDLERTQTIHVDEGRKIVISKDDLEGDIQIPGPPTNVHA
SEISKTYVVLSDWPPVPRGREPLTYFIEKSMVGSWSQRVNAQVAVKSPRY
AVFDLAEGKPYVFRVVSANKHGIDPSEITEPIQPQDTVVVPSAPGRVVATR
NTKTSVVVQWDKPKHEEDLYGYYIDYSVVGSNHWEPSNHPKIKNRFFVH

GLETGEQYIFRVKAVNAVGFSENSQSEAIKVQAALTCPSYPHGITLLNCDG
HSMTLGWKAPKYSGGSPILGYMDKREANHQNWHEVNSSLITRTIYTVED
LTEDAFYEFKIAAANIVGIGQPSDPSEHFCKAWTMPEPGPAYDLTVCEVR
NTSLVLLWKAPVYEGKSPITGYLVDYKEVDKEDWITANEKPTAQRVFKVT
DLHEGHTYVFKVRAVNDAGVGKSSEVSEPVLVEAKPGTKEIFSGVDDEGNI
YLAFECKEATDASHFAWGKSYEEIEDSDKYKIETEGDHSKLYFKNPDKSDL
GTYSISVSDTDGVSSSFVMDDEELERLMAISNEIKNPTIPLKSELAYEVLEKG
EVRFWIQAESLSPDSTFRFVINDKEVENS DRHKINCDRPNGIIMVMDRFTID
NEGTYTVQIQDGKAKNQSSLVLIGDAFKAVLAESLQRKEFLRKQGP HFSE
FLYWEVTEECEVLLACKVANTKKETVFKWYKDGSGIDVDEMPDLQKGEC
HLSIPRLSRKDEGVYKATLSDDRGHVSTLELSGKVYDDIILALSRVSGKTA
SPLKILCTEEGIRLQCFLKYNEEMKISWSHRESKISSGEKMKIGGGENIAWL
QITEPTEKDKGN YTFEIFGDKESFKRTL DLSGQAFDDAYAEFQRLKAAAF AE
KNRGKVIGGLPDVVTIMDGKTLNLTCTVFGNPDPEV VWFKN DKALELNEH
YVVSLEQGKYASLTIKGVTSEDSGKFSIHVKNKYGGETVDVTVSVYRHGEK
VPEVKPGQLAKPRPIPPSEDSS 184 Northern Mallard
MPERSWIPVGTPLLRTCTHTKACFEQRSPAVPPLVSTKHWSSETTTSPVSL Duck
EKGLSVVDQE KIDNLMLEMDGTENKSKFGANAILGVSLAVCKAGAAEKDV A0A493T6U6
PLYRHIADLAGNSDLILPVP AFNVINGGSHAGNKLAMQE FMILPVGAESFRD
(Phosphopyruvate
AMRIGAEVYHNLKSVIKEKYGKDATNVGDEGGFAPNILENSEALELLKEAI hydratase)
DKAGYTDKIVIGMDVAASEFYRDGKYDLDFKSPDDPSRYISADELGDLYQS
FVRDYPVVSIEDPFDQDDWEAWSKFTANVGIQIVGDDLTVTNPKRIERA VE
EKACNCLLLKVNQIGSVTEAIQACKLAQENGWGV MVSHRSGETEDTFIADL
VVGLCTGQIKTGAPCRSERLAKYNQLMRIEEELGDEARFAGHNFRNPSVL 185 Northern
Mallard MQGENHRAPKKAKKRIEGANSNVFSMFEQAQIQEFKEAFTIMDQNRDGFID
Duck U3I9B1 KADLRDTFAALGRLNVKNEEIDEMIKEAPGPINFTVFLTMFGEK LKGADPE
(Myosin light ETILNAFKVFDPEGKGLKSAYIKEMLMTQGERFSQEEVDQMFAAFPPDVSG
chain 2) NLDYKNLVHVITHGEEKD 186 Northern Mallard
FSISDNAYQFMLTDRENQSILITGESGAGKTVNTKRVIQYFATIAASGEK KKK Duck U3J7T0
EEQSGKMQGTLEDQIISANPLLEAFGNAKTVRNDNSSRFGKFIRIHFGATGK (Myosin
motor LASADIETYLLEKSRVTFQLKAERSYHIFYQIMSNKKPELIDMLLITTNPYDY domain-
HFVSQGEITVASINDQEELMATDSAIDILGFTADEKVAIYKLTGAVMHYGN containing
LKFKQKQREEQAEPDGT EVADKAAYLMGLNSADLLKALCYPRVKVGNEY protein)
VTKGQTVQQVNNSVGALAKAVYEKMFLWMVVRINQQLDTKQPRQYFIGV
LDIAGFEIFDFNSLEQLCINF TNEKLQQFFNHMFVLEQEEYKKEGIEWEFID
FGMDLAACIELIEKPMGIFSILEEECMFPKATDTSFKNKLYDQHLGKSSNFQ
KPKPAKGKVEAHFSLVHYAGTV DYNITGWLEKNKDPLNETVIGLYQKSSV
KTLALLFANYGGAEINPQKIIKDTFKGSSSQTVSALFRENLNKLMTNLRSTH
PHFVRCIIPNETKTPGAMEHELVLHQLRCNGVLEGIRICRKGFP SRVLYADF
KQRYKVLNASAIPEGQFIDSKKASEKLLGSIDVDHTQYKFGHTKVFFKAGL
LGLLEEMRDEKLAQLITRTQARCRGFLMRVEYQRMVERRESIFCIQYNVRA
FMNVKHWPMKLF FFIKPLLKSAESEKEMANMKEEFEKTKEELAKSEAK
KKELEEKMVKL VQEKN DLQLQVQAEADALADAEERCDQLIKTKIQLEAKI
KEVTERAEDEEEINAELTAKKRKLEDECS ELKKDIDDL ELTLAKVEKEKHA
TENKVKNLTEEMAALDETI AKLTKEKKALQE AHQQTLDDLQAEEDKVNTL
TKAKTKLEQQVDDLEGSLEQEKKL RMDLERAKRKLEGD LKLSQDSIMDLE
NDKQQ LDEKLKKKDFEISQIQSKIEDEQALGMQLQKKIKELQARIEELEEEIE
AERTSRAKAEKHRADLSRELEEISERLEEAGGATAAQIEMNKKREAEFQKM
RRDLEEATLQHEATAAALRKKHADSTAELGEQIDNLQRVKQKLEKEKSEL

KMEIDDLASNSAKANSEIKMCRITLEDQLSEIKTKEEEHQRMINDLTA
QRARLQTESGEYSRQVEEKDSQISQLSRSKQAFTQQIEELKRHLEEEIKAKN
ALAHALQSARHDCDLLREQYEEEQEAKGELQRALSKANSEVAQWRTKYE
TDAIQRTEELEEAKKKLAQRLQDAEEHVEAVNAKCASLEKTKQRLQNEVE
DLMIDVERSNAACAALDKKQKNEDKILAEWKQKYEETQAELEASQKESRS
LSTELFKMKNAYEESLDHLETLKRENKNLQQEISDLTEQIAEGGKAIHELEK
VKKQIEQEKSEIQAALAEAASLEHEEGKILRLQLELNQVKSEIDRKIAEKDE
EIDQLKRNHLRIVESMQSTLDAEIRSRNEALRLKKKMEGDLNEMEIQLSHA
NRQAAEAQKNLRNTQGVLKDTQIHLDDALRTQEDLKEQVAMVERRANLL
QAEIEELRAALEQTERSARKVAEQELMDASERVQLLHTQNTSLINTKKKLET
DITQIQSEMEDAIQEARNAAEKAKKAITDASMMAEELKKEQD TSAH 187 Northern
Mallard MITFAQYLQRCSYDGLSKLVKDVVDLAQKCVANEDAPECSKSLPSIFLDEIC
Duck U3IU92 QVEKLRDSYGAMADCCAKSDPERNECFLSFKIPQPDFVQPYQRPASDVICK
(Albumin) EYEDNRVSFLGHFIYSVARRNPFLYAPTILGVAADYEHALKSCCKESDVGA
CLDGKETGIREKVKKISVKQQYSCGILKKFGDRIFQADKLALLSQKYPKASF
AEISKLIHDVKDVYKECCEGDMVECMDDRAELVNYMCSKHEAFSSKIKDC
CEKPIVERSQCIMEAEFDEKPADLPSLVEKYIQDKEVCKSF EAGHDAFMSEF
VYEYSRRHPEFSTQLILRIAKGYETLLEKCCKTDNPAECYANAQDQLNEHIK
ETQDVVKTNCDLLTTHGEADFLKALLIRYTKKMPQVSTETLLEIGKKMTGV
GTKCCKLPEDRRLPCSEGYLSMVIQDMCRRQETSPINDNVSHCCSDSYAYR
RPCFSAMGVDTKYVPPPFNPDVFNFDEKLCSAPAEERAEGQMKLLINLIKR
KPQMTEEQIKTIADGFTAMVDKCKQSDIETCFGEEGANLIVQSRATLGIGA
TABLE-US-00011 TABLE 11 Domestic Muscovy Duck Proteins SEQ ID NO
Description Amino Acid Sequence 188 Domestic
MCCLKCQQGFWDFCAPAVLARGPCFGCGALVTAVGSAMPFSNSHNLLKM Muscovy
Duck KHSATDEFPDLSAHNNHMAKVLTLDLYKKLRDKQTPSGFTLDDVIQTGVD
A0A8C3BBL6 NPGHPFIMTVGCVAGDEESYEVFKELFDPVIEDRHGGYKPTDQHKTDLNAG
(Creatine kinase)
NLQGGDDLDPNYVLSSRVRTGRSIRGFCLPPHCSRGERRAIEKLSVEALGSL
GGDLKGKYYALRNMTDAEQQLIDDHFLFDKPVSPLLLASGMARDWPDA
RGIWHNDNKTFVLWINEEDHLRVISMQKGGNMKEVFTRFCTGLTQIENLFK
SKNHEFMWNPHLGYILTCPSNLGTGLRAGVHIKLPHLGKHEKFGEVLKKLR
LQKRGTTGGVDTA AVGGVFDVSNADRLGFSEVELVQMVVDGVKLLIEMEK
RLEKGQSIDDLIPAQK 189 Domestic
RAGDKPQDEYMTQEEEWDRDLLLDPAWEKQQRKTFTAWCNSHLRKAGT Muscovy Duck
QIENIEEDFRNGLKLM LLEVISGERLPKPDRGKMRFHKIANVNKALDYIAS A0A8C3BEA7
KGVKLVSIGAEIVDGNVKMTLGM IWTIILRFAIQDISVEETSAKEGLLLWC (Actinin
alpha 2) QRKTAPYRNVNIQNFHLSWKDGLGLCALIHRHRPDLIDYSKLNKDDPIGNI
NLAMEIAEKHLDIPKMLDAEDIVNTPKPDERAIMTYVSCFYHAFAGAEQAE
TAANRICKVLAVNQENERLMEEYERLASELLEWIRRTIPWLENRTPEKTMQ
AMQKKLEDFRDYRRKHKPPKVQEK CQLEINFNTLQTKLRISNRPAFMPSEG
KMVSDIAGAWQRLEQA EKG YEEWLLNEIRRLERLEHLAEKFRQKASTHEQ
WAYGKEQILLQKDYESASLTEVRAML RKHEAFESDLAAHQDRVEQIAAIA
QELNELDYHDAASVNDRCQKICDQWDSL GTLTQKRREALERTEK LLETIDQ
LHLEFAKRAAPFNNWMEGAMEDLQDMFIVHSIEEIQSLITAHQDFKATLPE
ADGERQAILS IQNEVEKVIQSYSMRISASNPYSTVTVEEIRTKWEKVKQLVP
QRDQSLQEELARQHANERLRRQFAAQANVIGPWIQTKMEEIARSSIEMTGP
LEDQMNQLKQYEQNIINYKH NIDKLEGDHQLIQEALVFDNKHTNYTMEHIR
VGWELLTTIARTINEVETQILTRDAKGITQE QMNDFRASFNFDRRKNGL
MDHDDFRACLISMGYDLGEAEFARIMSLVDPNGQGT VTFQSFIDFMTRETA

DTDTAEQVIALADSKPYILADRELPEQAQYCIKRMPPYTGP GSVP
GALDYTSFSSALYGESDL 190 Domestic
MAAPLSDGEKRKQISVRGIAGLDVAEVRKSFNRHLHFTLVKDRNVATPR Muscovy Duck
DYFFALAHTVRDHLVGRWIRTQQHYERDPKRIYYLSLEFYMGRTLQNTM A0A8C3BW27
VNLGLQNACDEAIYQLGLDLEEEIEEDAGLGNGGLGRLAACFLDSMATL (Alpha-1,4
glucan GLAAYGYGIRYEFGIFNQKIVDGWQVEEADDWLRYGNPWEKARPEYMLP
phosphorylase) VHFYGRVEHNPEGVKWVDTQVVLAMPYDTPVPGYKNNTVNTMRLWSAK
APNDFNLQEFNVGDYIEAVLDRNLAENISRVLYPNDNFFEGKELRLKQEYF
VVAATLQDIIRRFKSSKFGCRDPVRTSFETFPDKVAIQLNDTHPALSIPELMR
ILVDVEKVDWDKAWAITKRTCAYTNHTVLPEALERWPVSMFEKLLPRHLEI
IYALNQMHLDRAALYPGDVDRLRRMSVIEEGDCKRINMAHLCVIGSHAV
NGVARIHSDIVKHSVFKDFYELEPEKFQNKTNGITPRRWLLLCNPGLADVIA
EKIGEGFVTDLSQLKKLLDFIDNKMFIIRDVAKVQENKLKFSAYLEEYQYKV
KINPSSMFDVQVKRIHEYKRQLLNCLHAITLYNRIRSDPSKSFVPRTIMIGGK
AAPGYHMAKMIKLITSIGEVINNDPYVGDCLKVIFLENYRVSLAEKVIPAA
DLSQQISTAGTEASGTGNMKFMLNGALTIGTMDGANVEMAEAGEENLFIF
GMRVEDVEALDRKGYNALYYERLPELRQAIDQISSGFFSPGDPGCFRDVV
NMLMHHDRFKVFADYEAYIKCQGQVDQLFMDPWewTKKVIRNIACSGKF
SSDRtITEYAQEIWGVESSTKIPPNIPRD 191 Domestic
MTHQFPALSPEQKKALSDIAQRIVASGKGILAADESvGTmGNRLQRINVEN Muscovy
Duck TEENRRAFREILFSSDTSINQSIGGVILFHETLYQKDSSGKPFPAIIEKEGIVVG
A0A8C3CD90 IKLDKGTSPLAGTNGETTIEGLDGLSERCAQYKKDGVDFGKWRAVLKITST
(Fructose- TPSQLAIQENANTLARYASICQQNGLVPIVEPEVLPDGDHDLQRCQYVTEK
bisphosphate VLAavyKALNDHHVYLEGTLLKPNMVTAGHACTKKYTPQDVAAATVTTL
aldolase) LRTVPAAVPGICFLSGGQSEEEASLNLNAMNQSSLPKPWKLTFsyGRALQA
SALAawLGKpenKKAaQEAfrKRAQINSLACRGQYVVSGKTDAAATQSLF TASYTY 192
Domestic MVKVGvNGfGRIGRLVTRAAILSGKVQVVAINDPFIDLNYMVYMFKYDST
Muscovy Duck
HGHFRGTVKAENGKLVINGNAITIFQERDPSNIKWADAGAEYVVESTGVFT A0A8C3CJk8
TMEKAGAhLKGGAKRVIISAPSADAPMFVMGVNHEKYDKSLKIVSNASCT
(Glyceraldehyde-
TNCLAPLAKVIHDNFGIVEGLMTTVHAITATQKTVDGPSGKLWRDGRGAA 3-phosphate
QNIIPASTGAakAVGKVIPELNGKLTGMAFRVPTPNVSVVDLTCRLEKPAK dehydrogenase)
YDDIKRVVKAAADGPLKGILAYTEDQVVSCDFNGDSHSSTFDAGAGIALND
HFVKLVSwyDNEyGYSNRVVDLMVHMASKE 193 Domestic
MDAIKKKMqMLKLDKENALDRAEQAEADKKAaEERSKQLEDELVALQK Muscovy
Duck KLKGTEDELdKYSESlkDAQEKLELADKKATDAESEVASLNRRIQLVEEEL
A0A8C3GR18 DRAQERLATALQKLEEAekAAADESERGMKVIENRAQKDEEKMEIQEIQLKE
(Tropomyosin 1)
AKHIAEEADRKYEEVARKLVIIESDLERAEEraELSESKCAELEEEELKTVTN
NLKSLEAQAEKYSQKEDKYEEEIKVLTDKLKEAETRAEFAERSVTKLEKSID
DLEDELYAQKLKYKAISEELdHALNDMTSM 194 Domestic
YPTTKPKTTEVTAGSTAVFEATEKAGVKVKWQRAGTEITDSEKYVIKAEG Muscovy
Duck NKHSLTINNvVKEDDVTYAVIAGSSKVKFELKVKEPEKSETVTPAEAAPAP
A0A8C3CSW2 AASELPAPPAESSQNTEVPSAGTQPEEPVDPIGLFVTRPQDGEVTVGGNITFT
(Myosin binding
AKVAGESLLKKPSVKWfKGKWMDLASKVGQHLQLHDSYDRNNKVYTFE protein C3)
LEIIEANMTFAGGYRCEVSTKDKFDSSNFNLTVNEIYHLCHDIVSFPFPSKTN
YIFLLLRSTEGHDEAGELNFSALLKKRGDGKTESQPDVDVWEILRKAPPSEY
EKIAFQYGITDLRGMLKRLKRIKKEEKKSTAFLKKLDPAYQVDKGQKIKLM

VEVANPDVADKNGQIEIQVSGRNGKRLITINHCSLADNAAEYC
VVGEEKSFTELFVKEPPILITHPLEDQMVMVGERVEFECEVSEEGATVKWE
KDGVELTREETFKYRFFKKDGKKQYLIINESTKEDSGHYTVKTNGGVSVAEL
IVQEKKLEVYQSIADLTVKARDQAVFKCEVSDENVKGIWLKNGKEVVPDE
RIKISHIGRIHKLTIEDVTPGDEADYSFIPQGFAYNLSAKLQFLEVKIDFVPRE
EPPKIHLDCLGQSPDTIVVVAGNKLRLDVPISGDPTPTVIWQKVNKKTDLR
SNDDSMTPSENSSDLNTDSKLLFESEGRVRVEKHEDHCVFIIEGAKEKEDEGV
YRVIVKNPVGEDKADITVKVIDVPDPPEAPKISNIGEDYCTVQWQPPKYDG
GQPVLGYILERKKKKSYRWMRLNFDLLKELTYEAKRMIEGVVYEMRIYAV
NSIGMSRPSPASQPFMPIAPPSEPTHTFTVEDVSDSTVALKWRPPERIGAGGLD
GYIVEYCKEGSTEWIPALPGLTERTSALIKDLVTGDKLHFRIKAINLAGESGA
AIIKEPITVQEILQRPKIWLPRHLRQTLVKKVGETINIVIPFQGKPRPKITWTK
DGQSLDAKDVGIRNSNTDTILFIRKAELHHSWAYEITLQIENMTDKVAITIQII
DKPGPPQNIKLTDVWGFNAALEWTPPQDDGNAQILGYTVQKADKKTMEW
YTVFDHYRRTNCVVSELIMGNEYFFRVFSENLCGLSETAATTKNPAYIQKT
GTTYKPPCYKEHDFSEPPKFTHPLVNRSVIAGYNTTLSCAVRGIPKPKIYWY
KNKMDLSGDAKYRMFSKQGVLTLEIRKPTPFDGGFYTCNAVNERGEAEIEC
KLDVRGNHMSIMHNTNTAVK 195 Domestic

MSSLAVPFYQRRHKHFDQSYRNIQTRYVLEEYAARKAASRQAAYYESSGL Muscovy
Duck GKTTCRLCARRARTLAHETMQESRKRTHEQKSHASEEKRIRFASELSALEK
A0A8C3CXN6 EIHTARHRAREQLDRLAIQRMVEENMAMERHAVEERISRAPEILVRLRSHT
(Myomesin 2) VWEKMSVSLCFTVQGFPTPVVQWYKNEELITPASDPAKYRVESKYGVHVL
HINRADFDDSATYSAVATNVHGGASTNCVVRRRFREGEEPHPAGIMPFHL
PLSYDVCFTHFHDVQFLEKFGVTFATEGETLTLKCSVLVTPELKRLRPRAEW
YRDDVLIKDSKWTCLYFGEGQAALSFTHLNKDDEGLYTLRMITKGGVNEC
SAYLFVRDADALIAGAPGAPMDVKCHDANRDYVIVTWKPPNTTSQSPVIG
YFVDKCEVGLENWVQCNDAPVKICKYPVTGLYEGRSYIFRVRAVNNAGIS
RPSRVSEPVAALDPVDLERTQTIHVDEGRKIVISKDDLEGDIQIPGPPTNVHA
SEISKTYVVLSSWDPPVPRGREPLTYFIEKSMVGSWSQVRVNAQVAVKSPRY
AVFDLAEGKPYVFRVVSANKHGISDPSEITEPIQPQDTVVVPSAPGRVVATR
NTKTSVVVQWDKPKHEEDLYGYYIDYSVVGSNHWEPSNHKPIKYNRFVVH
GLETGEQYIFRVKAVNAVGFSENSQESEAIKVQAALTCPSYPHGITLLNCDG
HSMTLGWKAPKYSGGSPILGYMDKREANHQNWHEVNSSLITRTIYTVED
LTEDAFYEFKIAAANIVGIGQPSDPSEHFKCKAWTMPEPGPAYDLTVCEVR
NTSLVLLWKAPVYEGKSPITGYLVVDYKEVDKEDWITANEKPTAQRYPKVT
DLHEGHTYVFKVRAVNDAGVGKSSEVSEPVLEAKPGTKEIFSGVDDEGNI
YLAFECKEATDASHFAWGKSYEEIEDSDKYKIETEGDRSKLYFKNPDKSDL
GTYSISVSDTDGVSSSFVMDDEELERLMAISNEIKNPTIPLKSELAYEVLEKG
EVRFWIQAESLSPDSTFRFVINDKEVENS DRHKINCDRPNGIEMVMDRFTID
NEGTYTVQIQDGKAKNQSSLVLIGDAFKAVLAESELQRKEFLRKQGPHFSE
FLYWEVTEECEVLLACKVANTKKETVFKWYKDGGSGIDVDEMPDLQKGEC
HLSIPRLSRKDEGVYKATLSDDRGHVSTLELSGKVYDDIILALSRVSGKTA
SPLKILCTEEGIRLQCFLKYNEEMKISWSHRESKISSGEKMKIGGGENIAWL
QITEPTEKDKGNYTTFEFGDKESFKRTLDSLGSQAFFDDAYAEFQRLKAAAF
KNRGKVIGGLPDVVTIMDGKTLNLTCTVFGNPDPEVVWFKNDKALELNEH
YVVSLEQGKYASLTIKGVTSEDSGKFSIHVKNKYGGETVDVTVSVYRHGEK
VPEVKPGQLAKPRPIPPSEGSS 196 Domestic

MVFKMSILKIHAREIFDSRGNPTVEVDLYTNKGLFRAAVPSGASTGIYEAL Muscovy
Duck LRDNDKTRYMGKGVSKAVEHVNKTIAPALISKNVNVVEQEKEIDKLMLDM
A0A8C3CT67 DGSENKSKFGANAILGVSLAVCKAGAAEKGVPLYRHIADLAGNPEVILPVP

(Phosphorylurate

AFNVINGGSHAGNKLAMQEFMILPVGADSFKEAMRIGAEVYHNLKNVIKE hydratase)

KYGKDATNVGDEGGFAPNILENKEALELLKTAIGKAGYSDKVVIGMDVAA

SEFYRDGKYDLDFKSPDDPSRYISPDQLADLYKGFVKNYPVVSIEDPFDQD

DWAAWKKFTGSVGIQVVGDDLTVTNPKRIAKAVEEEKSCNCLLLKVNQIGS

VTESLQACKLAQSNGWGVMVSHRSGETEDTFIADLVVGLCTGQIKTGAPC

RSERLAKYNQLLRIEEEELGSKARFAGRNFRNPRIN 197 Domestic

MAPKKAKKRIEGANSNVFSMFEQAQIQEFKEAFTIMDQNRDGFIDKADLRD Muscovy

Duck TFAALGRLNVKNEEIDEMIKEAPGPINFVFLTMFGEKLGADPEETILNAF

A0A8C3BXQ0 KVFDPPEGKGLKSAYIKEMLMTQGERFSQEEVDQMFAAFPPDVSGNLDYKN

(Myosin light LVHVITHGEEKD chain 2) 198 Domestic

MTSPDAEMAAFGGEAPYLRKSEKERIEAQNKPFDAKSSVFVAHPKESFVKG Muscovy

Duck TIQSKESGKVTVKTEGGETLTVKEDQIFSMNPPKYDKIEDMAMMTHLHEPA

A0A8C3BJT4 VLYNLKERYAAWMIYTYSGLFCVTVNPYKWLPVYNPEVVLAYRGKKRQE

(Uncharacterized

APPHIFSISDNAYQFMLTDRENQSILITGESGAGKTVNTKRVIQYFATIAASG protein)

EKKKEEQSGKMQGTLEDQIISANPLLEAFGNAKTVRNDNSSRFGKFIRIHFG

ATGKLASADIETYLLKSRVTFQLKAERSYHIFYQIMSNKKPELIDMLLITTN

PYDYHFVSQGEITVASINDQEELMATDSIDILGFTADEKVAIYKLTGAVMH

YGNLKFQKQKREEQAEPDGTEVADKAAAYLMGLNSADLLKALCYPRVKVG

NEYVTKGQTVQQVNNSVGALAKAVYEKMFLLWMVVRINQQLDTKQPRQY

FIGVLDIAGFEIFDFNSLEQLCINFTEKLLQFFNHMHMFVLEQEEYKKEGIEW

EFIDFGMDLAACIELIEKPMGIFSILEEECMFPKATDTSFKNKLYDQHLGKSS

NFQKPKPAKGKAEAHFSLVHYAGTVDYNITGWLEKNKDPLNETVIGLYQK

SSVKTLALLFANYGGAEAEASGGGASGGKKGKGGKSSSFQTVSALFRENL

NKLMTNLRSTHPHFVRCIIPNETKTPGAMEHELVLHQLRCNGVLEGIRICRK

GFPSRVLYADFKQRYKVLNASAIPEGQFIDSKKASEKLLGSIDVDHTQYKFG

HTKVFFKAGLLGLLEEMRDEKLAQLITRTQARCRGFLMRVEYQRMVERRE

SIFCIQYNVRAFMNVKHWPWMKLFFKIKPLLKSAESEKEMANMKEEFEKT

KEELAKSEAKRKELEEKMVKLVEKNDLQLQVQAEADALADAEERCDQLI

KTKIQLEAKIKEVTERAEDEEEINAELTAKKRKLEDEECSELKKDIDDLTL

AKVEKEKHATENKVKNLTEEMASLDETIAKLTKEKKALQEAHQQTLDLQ

AEEDKVNTLTAKTKLEQQVDDLEGSLEQEKKLRMDLERAKRKLEGDLKL

SQDTIMDLENDKQQLDEKLKKKDFEISQIQSKIEDEQALGMQLQKKIKELQ

ARIEELEEEIEAERTSRAKAEKHRADLSRELEEISERLEEAGGATAAQVEMN

KKREAEFQKMRRDLEEATLQHEATAATLRKKHADSTAEELGEQIDNLQRVK

QKLEKEKSELKMEIDDLASNMESVSKAKANLEKMCRTLEDQLSEIKTKEEE

HQRMINDLTAQRARLQTESGEYSRQVEEKDSQISQLSRSKQAFQTQQIEELKR

HLEEEIKAKNALAHALQSARHDCDLLREQYEEEQEAKGELQRALSKANSE

VAQWRTKYETDAIQRTEEELEAKKKLAQRLQDAEEHVEAVNAKCASLEKT

KQRLQNEVEDLMIDVERTNAACAALDKKQKNFDKILAEWKQKYEETQAE

LEASQKESRSLSTELFKMKNAYEESLDHLETCLKRENKNLQQEISDLTEQIAE

GGKAIHELEKVKKQIEQEKESEIQAALAEAEASLEHEEGKILRLQLELNQVKS

EIDRKIAEKDEEIDQLKRNHLRIVESMQSTLDAEIRSRNEALRLKKKMEGDL

NEMEIQLSHANRQAAEAQKNLRNTQAVLKDTQIHLDDALRTQEDLKEQVA

MVERRANLLQAEIEELRAALEQTERSARKVAEQELMDASERVQLLHTQNTSL

INTKKKLETDIVQIQGEMEETIQEARNAAEEKAKKAITDASMMAEELKKEQD

TSAHLERMKKNLQDQTVKDLQHRLDEAEQLALKGGKKQIQKLEARVRELEG

DVDAEQKRSAAEAVKGVRKYERRVKELTYQSEEDRKNILRLQDLVDKLQM

KVKSYKRQAEAEELSNVNLSKFRKIQHELEAEERADIAESQVNKLRAKS

REFHGGKKIEEEE 199 Domestic
MTSPDAEMAAFGGEAAPYLKSEKERIEAQNKPFDAKSSVFVAHPKESFVKG Muscovy
Duck TIQSKESGKVTVKTEGGETLTVKEDQIFSMNPPKYDKIEDMAMMTHLHEPA
A0A8C3BJT4 VLYNLKERYAAWMIYTYSGLFCVTVPYKWLVPVYNPEVVLAYRGKKRQE
(Uncharacterized
APPHIFSISDNAYQFMLTDRENQSILITGESGAGKTVNTKRVIQYFATIAASG protein)
EKKKEEQSGKMQGTLEDQIISANPLLEAFGNAKTVRNDNSSRFGKFIRIHFG
ATGKLASADIETYLLEKSRVTFQLKAERSYHIFYQIMSNKKPELIDMLLITTN
PYDYHFVSQGEITVASINDQEELMATDSADILGFTADEKVAIYKLTGAVMH
YGNLKFQKQKQREEQAEPDGTEVADKAAYLMGLNSADLLKALCYPRVKVG
NEYVTKGQTVQQVNNSVGALAKAVYEKMFLLWMVVRINQQLDTKQPRQY
FIGVLDIAGFEIFDFNSLEQLCINFTEKLQQFFNHHMFVLEQEEYKKEGIEW
EFIDFGMDLAACIELIEKPMGIFSILEEECMFPKATDTSFKNKLYDQHLGKSS
NFQKPKPAKGKAEAHFSLVHYAGTVDYNITGWLEKNKDPLNETVIGLYQK
SSVKTLALLFANYGGAEAEASGGGASGGKKGKGGKSSSFQTVSALFRENL
NKLMTNLRSTHPHFVRCIIPNETKTPGAMEHELVLHQLRCNGVLEGIRICRK
GFPSRVLYADFKQRYKVLNASAIPEGQFIDSKKASEKLLGSIDVDHTQYKFG
HTKVFFKAGLLGLEEMRDEKLAQLITRTQARCRGFLMRVEYQRMVERRE
SIFCIQYNVRAFMNVKHWPWMKLLFFKIKPLLKSAESEKEMANMKEEFEKT
KEELAKSEAKRKELEEKMVKLQVEKNDLQLQVQAEADALADAEERCDQLI
KTKIQLEAKIKEVTERAEDEEEINAELTAKKRKLEDECSELKKDIDDLTL
AKVEKEKHATENKVKNLTEEMASLDETIAKLTKEKKALQEAHQQTLLDDLQ
AEEDKVNTLTAKTKLEQQVDDLEGSLEQEKKL RMDLERAKRKLEGDLKL
SQDTIMDLENDKQQLDEKLKKKDFEISQIQSKIEDEQALGMQLQKKIKELQ
ARIEELEEEIEAERTSRAKAEKHRADLSRELEEISERLEEAGGATAAQVEMN
KKREAEFQKMRRDLEEATLQHEATAATLRKKHADSTAEELGEQIDNLQRVK
QKLEKEKSELKMEIDDLASNMESVSKAKANLEKMCRTLEDQLSEIKTKEEE
HQRMINDLTAQRARLQTESGEYSRQVEEKDSQISQLSRSKQAFTQQIEELKR
HLEEEIKAKNALAHALQSARHDCDLLREQYEEEQEAKGELQRALSKANSE
VAQWRTKYETDAIQRTEELEEAKKKLAQRLQDAEEHVEAVNAKCASLEKT
KQRLQNEVEDLMIDVERTNAACAALDKKQKNFDKILAEWKQKYEETQAE
LEASQKESRSLSTELFKMKNAYEESLDHLETLKRENKNLQQEISDLTEQIAE
GGKAIHELEKVKKQIEQEKSEIQAALEEAASLEHEEGKILRLQLELNQVKS
EIDRKIAEKDEEIDQLKRNHLRIVESMQSTLDAEIRSRNEALRLKKKMEGDL
NEMEIQLSHANRQAAEAQKNLRNTQAVLKDTQIHLDDALRTQEDLKEQVA
MVERRANLLQAEIEELRAALEQTERSARKVAEQELMDASERVQLLHTQNTSL
INTKKKLETDIVQIQGEMEETIQEARNAEKAKKAITDASMMAEELKKEQD
TSAHLERMKKNLDQTVKDLQHRLDEAEQLALKGGKKQIQKLEARVRELEG
DVDAEQKRSAEAVKGVKRYERRVKELTYQSEEDRKNILRLQDLVDKLQM
KVKSYSKRQAEAEELSNVNLSKFRKIQHELEEAERADIAESQVNKLRAKS
REFHGGKKIEEEE 200 Domestic

MKWVTLISFIFLFSATSRLQRVARDADHKSEIAHRYNDLKEETFKAVAMI Muscovy
Duck TFAQYLQRCSYDGLSKLVKDVVDLAQKCVANEDAPECSKSLPSIFLDEICQ
A0A8C3CNB8 VEKL RDSYGAMADCCA KSDPERNECFLSFKIPQPDFVQPYQRPASDVICKE
(Albumin) YEDNRVSFLGHFIYSVARRNPFLYAPTILGVAADYEHALKSCCKESDVGAC
LDGKETAIKREKVKKISVKQQYSCGILQKFGDRIFQADKLAFLSQKYPKASFT
EISKL IHDVKDVYKECCEGDMVECMDDRAELVNYICSKHESFSSKIKDCCE
KPIVERSQCIMEAEFDEKPADLPSLVEKYIQDKEVCKSFEAGHDAFMSEFVY
EYSRRHPEFSTQLILRIAKGYETLLEKCKTDNPAECYANAQDQLNEHIKET
QDVVKTNCDLLTTHGEADFLKALLIRYTKKMPQVSTETLLEIGKKMTGVG

TKCKCEDRLRSPEDYSLMPCQDTSPIQDMSHCCSDSYAYRR
PCFSAMGVDTKYVPPPFNPEMFNFDEKLCSAPAEERAAGQMKLLVNLIKRK
PQMTEEQIKTIADGFTAMVDKCKQSDIETCFGEEGANLIVQSRATLGIGA

[0057] An animal protein can comprise a salmon protein (Table 12). A salmon protein can comprise a salmon muscle protein. A salmon protein can comprise a salmon blood protein. A salmon protein can comprise a salmon albumin protein. A salmon protein can comprise a sequence that has at least or about 70%, at least or about 75%, at least or about 80%, at least or about 90%, at least or about 91%, at least or about 92%, at least or about 93%, at least or about 94%, at least or about 95%, at least or about 96%, at least or about 97% at least or about 98% at least or about 99% or about 100% sequence identity to any one of SEQ ID NO: 201-216.

[0058] A food product may comprise a fragment or a truncation of a salmon protein. The food product may include a fragment or truncation of any one of SEQ ID Nos: 201-216. A sequence of the fragment or truncation may be identical or substantially identical to a portion of any one of SEQ ID Nos: 201-216. A sequence of the fragment or truncation may be at least about 70%, 80%, 90%, 95%, 98%, or more identical to a portion of a sequence of any one of SEQ ID Nos: 201-216. A length of the fragment or truncation may be at least about 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or more of a length of any one of SEQ ID Nos: 201-216. A length of the fragment or truncation may be less than or equal to about 90%, 80%, 70%, 60%, 50%, 40%, 30%, 20%, 15%, 10%, or less than a length of any one of SEQ ID Nos: 201-216.

TABLE-US-00012 TABLE 12 Salmon Proteins SEQ ID NO Description Amino Acid
Sequence 201 Salmon B5DGP0

MPFGNTHNNFKLNFKVEEEYPDLTKHNNHMAKVLTKDMYAKLRDKQTPSG (Creatine
kinase) FTLDDVIQTGVDNPGHPFIMTVGCVAGDEESYEIFKDLLDPIISDRHSGYKPT
DKHKTDLNFENLKGGDDLPNYVLSSRVRTGRSIKGYTLPPHNSRGERRAV
EKLSVEALNTLDGEFKGKYPLNKM TDAEQEQLIADHFLFDKPVSPLLLGA
GMARDWPDARGIWHNDAKSFLVWVNEEDHLRVISMEKGGNMKEVFRRFC
VGLKRIEETFKKH NHGFMWNEHLGYVLTCP SNLGTGLRGGVHV KLPKLSTH
PKFEEILTRLRLQKRGTGGVDTASVGGVFDISNADRLGSSEVDQVQMVVDG
VKLMVEMEKKLEKGEAIDGMIPAK 202 Salmon

MTAIETQVQYGSYMTSEQVYMTQEDDWDRLDLLDPAWEKQQRKTFTAWC A0A1S3RGX6
NSHLRKAGTQIENIEEDFRNGLKLM LLEVISGERLPKPDKGKMR FHKIANV (Alpha-
actinin-3) NKALDFICSKGVKLVSIGAEIVDGNVKMTLGM IWTIILRFAIQDISVEETSAK
EGLLLWCQRKTAPYRNVNVQNFHISWKDGLALCALIHRHRPDLIDYSKLRK
DDPMGNLNTAFEVAEKYLDIPKMLDAEDIVNTPKPDEKAIMTYVSCFYHAF
AGAEQAETAANRICKVLAVNQEN EKLMEEYEKLASELLEWIRRTIPWLENR
VAEQTMRAMQQKLED FRDYRRVHKPPKVQEK CQLEINFNTLQTKLRLSNRP
AFMPSEGKMVSDIANAWKGLEQVEKGYEEWLLTEIRRLERLDHLAEKFKQK
SSLHESWTSGKV ELLSMKDYESASLMEIRALMRKHEAFESDLAAHQDRVEQ
IAAIAQELNELDYHDAVTINARCQGICDQWDNLGTLTQKRRDSLERVEKLW
ETIDQLYLEFAKRAAPFNNWMDGAMEDLQDMFIVHSIEEIQSLITAHDQFKA
TLPEADKERVATMGIQNEIVKIAQTYGIKLSGVNPNYTNLSPQDITDKWDAVK
HLVPLRDQMLQEEVARQQSNERLRRQFAAQANIIGPW IQTKMEEISHVSVDI
AGSLEEQMSNLKQYEQNIINYKCNIDKLEGDHQLSQESLIFDNKHTNYTMEH
VRVGWEQLLTTIARTINEVENQILTRDAKGISQEQLNEFRASFNFDRKRNG
MMDPDDFRACLISMGYDLGEVEFARIMTLVDSNNTGVVTFQAFIDFMTRET
AETDTADQVMASFKILASDKTYITVEELRREL PPEQADYCISRMTSYIGSGAP
PGALDYISFSSALYGESDL 203 Salmon

MPKPLTDQDKRKQISVRGLAGVENVSDLKTNFRHLHFTLVKDRNVATKR C0H8W5
DYYFALANTVRDHLVGRWIRTQQHYYEKDPKRVYYLSLEFYMGRTLQNTM (Alpha-1,4
VNLALENACDEATYQLGLDMEELQDIEEDAGLGNGGLGRLAACFLDSMAS glucan

LGLAAYGIRYEFGIFNQKIVNGWQVEEADDWLRYGNPWEKARPEYMRP
phosphorylase)

VHFYGRVEHTPEGVKWVDTQVVLALPYDTPVPGYRNNVVNTMRLWSAKA
PCDFNLKDFNVGGYIQAVLDRNLAENISRVLYPNDNFFEGKELRLKQEYFVV
AATLQDIVRRFKSSKFGSTEVVRVDLSTLPDKVAIQLNDTHPAMAIPELMRIL
LDTEHQTWEEKAWDICTRTCAYTNHTVLPEALERWPTDLLQNLLPRHLEIIYE
INRRHLERIAKLYPGDHDRLRRMSLVEEGDQKKINMAHLCIVGSHAVNGVA
RIHSDILKATLFKDFYEVDPHKFQNKTNGITPRRWLVMCNPGLAEVIAERIGE
DYIRDLQQLKLLTFVDDDSLIIHDIKVKQENKLKFSAYLEEHYKVKINPNS
MFDVQVKRIHEYKRQLLNCLHIITLYNRIKKEPNKNWTPRTIMIGGKAAPGY
HTAKLIIRLITAIGEIVNHDTVVGDRLLKVIFLENYKVTLAEKIIPASDLSEQIST
AGTEASGTGNMKFMLNGALTIGTMDGANVEMAEAGEENLFIFGMRVDDV
DAMDKSGYDAMDYYNRIPELKQAMDQIAGGFFSPDQHDLFKDIVNMLLHH
DRFKVFADYEAYIKCQEKVSALYKNPKEWTKMVIHNIAGCGKFSSDRTISQY
AREIWGMEPSLERIPAPDDPRQ 204 Salmon C0H911

MPHSFPFLSTEQKKELSDIAQKIVAPGKGILAADESTGSVAKRFQSINAENTE (Fructose-
ENRRLYRQLLFTADRVPCIGGVIFFHETLYQKTDDGKLFPQLIKERDMVVGI bisphosphate
KVDKGVVPLAGTNGETTTQGLDGLYERCAQYKKDGADFAKWRCVLKITDT aldolase)
TPSELAIKENANVLARYASICQMHGIVPIVEPEILPDGDHDLKRCQYVTEKVL
AAVYKALSDHHVYLEGTLKPNMVTTPGHSCSQKYSNQEIAMATVTALRRT
VPPAVPGITFLSGGQSEEEASINLNAMNQCPLHRPWAITSYGRALQASALK
AWGGKPENGKACQEEFIKRALNNSKACQGKYVSSGDKGAAGGESLYVANH AY 205
Salmon B5DH12

MAPKKDAKAPAKKAEPKAAAPAPAPEPDVVAAPPPLDLSTVKVEFTPDQQ (Myosin
light EDYKEAFGLFDRVGDAAKVAYNQVADIMRALGQNPTNKEVRKVLGNPSDED chain 1-
1) MAAKRLEFEAFPLMLQHIVNDPNKGTEDDYVEGLRVFDKEGNGTVMGAEL
RIVLGTLGKEMSEAEIDALMQGQEDENGSI NFEAFVKHIMSI 206 Salmon
MVKVG VNGFGRIGRLVTRAAFHSSKKGVEIVAINDPFIDLDMVYMFKYDST B5DGR3
HGRFHGEVKAEGGKLVIDGHKITVFHERDPANIKWGDAGATYVVESTGVFT
(Glyceraldehyde-
TIEKASTHLKGGAKRVVISAPSADAPMFVMGVNHEKYDNSLKVVSNASCTT 3-phosphate
NCLAPLAKVIHDNYHIIIEGLMSTVHAVTATQKTVDGPSGKLWRDGRGASQN
dehydrogenase)

IIPASTGAAKAVGKVIPELNGKITGMAFRVPTPNVSVVDLTVRLEKPASYDAI
KKVVKAAADGPMKGILGYTEQQVVSSDFNGDTHSSIFDAGAGIALNDHFVK
LVTWYDNEFGYSNRVIDLMAHMATKE 207 Salmon Q91472

MDAIKKKMQLKLDKENALDRAEGAEGDKKAAEDKSKQLEDDLVALQKK (Fast
myotomal LKGTEDELDKYSESLKDAQEKLEVAEKTATDAEADVASLNRRIQLVEEELD
muscle RAQERLATALTKLEEAKEAADESERGMKVIENRASKDEEKMELQDIQLKEA
tropomyosin) KHIAEEADRKYEEVARKLVIIESDLERTEERAELSEGKCSELEEELKTVTNNL
KSLEAQAEKYSQKEDKYEEEIKVLTDKLKEAETRAEFAERSVAKLEKTIDDL
EDELQAQKLKYKAISEELDNALNDMTSI 208 Salmon

MPEAKKPAAKTDKAAAAPADKEARSAIEEGSECRDELPDDETVPGGQSELT A0A1S3RC27
GLFVERPESTTVIKGKNVTFVAKVDSSDLLRKPNMKWLKGKWL DLGSKAG (Myosin-
binding KHVQFKEAYDRNSKVYTYEMSIIKVVEGDAGGYRCEVTSKDKCDSCTFEVT
protein C) VEAVQEEQPANILDAFKRSGDAGEDAGDLDFSALLKKREKKARDEPKEDVD
VWEILKDAKPCDYEKIAFDYGITDLRGLLKRLKKMKKVEPKKSDAFLKKLE
QCYSVDK GKRTQMHVELHDPNVQVKWLKNGVEIKPSAKYVFECVGNKRTL
TINKCNLSDDAAYECVVGEEKSFTEVFVKEPPVTITKLLDDVHV VVGEKVEF
ECEVSEEGANVKWMKDGVELTKDGKIRIKKDGKKHTLVINEATIEDIGMY

VYTNNGEGSEVEAKEVLAQSVKACEQAMFKCEVSDKCEVVGK
WFKDGVVEVKPGNRIKMSHIGRIHKLTIDDVKPQDEGNYTFVPEGYALSLSAK
LNFIEIKIEYVPRQDPPKIHLDTSSTGSKNTIVVVAGNKLRLDVEITGEPVPTV
CWMKGDTVISEAEGRRVETRTTLSSFVIEGAERPDEGQYSIIVTNPAGEDRA
ELTVKIVDVPNPPENVRCMGVGEDTATITWDPKFDGGAPIKGYLMERKKQ
GSSRWTKLNFEVFESTTYEAKKMIEGVFYEMRVFAVNGIGISQPSGNSKPFM
PIAPTSEPTRLTVEDVTDSTCALKWRPPERVGAGGVDGYIIEWCKEGEDNWV
AANKEPVDKNTYRVKGLPTGEKLLFRVVAMNIAGRSPPTMKQSVTIREIM
EYPKIRLPRQLRTKFIRKVGEKINLVIPFQGKPRPVVNWVKDGEPLENKSXGI
RTSDFDTILFIRSAERDHSPTYTSLSVQIDNMQDKADIHIQIVDKPGPPINVMVT
DVWGFNAALEWKPPKDTGNTDITGYTIQKADKKTQGWFTVYEHNRRPSCT
ASDLVMGNEYSFRVFSENICGLSDEVAFSKNTAIIGKTDLEYNKPFAKERDM
SSSPKFTAPLVDRVVAGYSTAISC AVRAYPKAKIVWMKNKMIIGEDPKFLM
QNNQGVLTNLIRKPGQFDGGKYSCKAINDLGEDEVECRLEVRVLKEKKEGD EEKK 209
Salmon MATKVMHFNQKKHVSHVSHHSSHTTKHVVKEQSSRKSSSKSVNQVQHHA
A0A1S3L6A0 TEAMAEPAYTVPAFRQRSADIEEYQRVSSHVGKGLASIQKELHRMRVVTK
(Myomesin-2 TQVENIAIKREVQEMMHKKMTLSTETDKAPDFMVALRPHTVWEKTPVKLF
isoform X1) CTVDGHPRPVVKWYKGGKPVDPLSAPGKYKIESKYGVHSLIISRCMVSDTAE
YSAVATNSHGSTTSKASVIVKRPAGGAYGSCQLFGQVPHLTVIHHSKLEITM
LDRFGVSFGTEGGSISLVCTMVVVPDLPNVPPLAQWYRDDKLLKAGKLAEI
KVGGGAATLTLPHLAKDDEGLYTLRMWTKDGTTEHSAYLFVKDAAPSVAG
APGAPISVKAFDINS DYVLVAWKPPNTTNEAAITGYFVDKRESGSATWSQCN
DAPVQICKYPVHGLNVGHAYHFRVRAVNSAGISRPSRES DKVTALDPAERE
RLQVIKLDGKHEVVIKDDDLEGYVTLPGEPTDMHASEIFKSYIVLSWKPPSPR
GRAPLWYIIEKSVSGTGAWQRINTAVKLQSPRYPVFDLEAGKKYQFRVYSE
NLYGASEASKPTDPIEKVDEHGVPSAPGQVVATRNTKSSVFVQWDPPKHPN
HLMGYIIDASLVGSKTWAPCNHKPYKNTRFVVHGLTPGETYVFRVQAVNV
YGLSDESQESTPIAVEPALTPSAPHGITMLSCDGASMIIAWNSPKHRGGSKI
NAYYVDKRDADSLAWKEVNSAPTNTRTYTVDGLTEGTFYEFKVQAGNLAG
VGVPAPSTPLKCEAWTSAPVGPAYDLAFREVRGDSLVLWKAPVYTGASA
VTGYIVEMAKKGHHYHPLNEEAIGHCYLQVTGLEAGAEYTFKVKAUNAEG
VGKASQTSEPVFAKALPGTQEIQCGVDEDTGDIFLSFEACEISETSNFAWSKN
YKEISDCTRVSIETKGKHSKLTFFVNPDVSDLGAFSVHVTDTDGVSASHTVTE
DALNTMLELSYNIRHPIVPLKHQLNIEVLEKGHVRFWLQCLKMSPAVNYRF
IVNDKEVTSSDSHKISHDVNTGVIQMTVDHFSKASEATYTVQIHDGRAKNQS
SLVLVGDFVKAALKEAEFQRSEHIRKQGPFAEYLSYHVDDDCTVLLVCKV
ANVKKETVFWHFKDEEEIVPETPPNVMSGVALPISLFSRKDQGHYKAKLSD
DRGKDTSVFDISGQVFDDIINAIAHAGSSASELVLQCTPEGIRLQCYMKYYT
EEIRTAWLHKDSKISSSEKMRIAGTAEMAWMQIREPSEKEKGHYSIQIMDAT
KSHTRTFDLGQAYTDAYEEYLRLLKAAAFKAKNRGRVVGGLPDVVTIMEQ
KTLSTCTVWGDPSPEVTWFKNENEVVSTEHAKITLEANKFASLTITAVTSE
DSGKYSINVRNKYGGFVEITVSVYKQGESPEPKMGGRGATPAPLASKTPA
KTPTPAQTPTPSLKSPTPSLKSPTPSLRSPAKSPTPTPKSPTPPRRVKSPSPARSL KSPTPPRK
210 Salmon ENOB

MSITKIHAREILDSRGNPTVEVDLYTAKGRFRAAVPSGASTGVHEALELRDG (Beta-enolase)
DKSRYLGKGTVKAVDHVNKDIAAKLIEKKFSVVDQEKIDHFMLELDGTENK
SKFGANAILGVSLAVCKAGAAEKGVPLYRHIADLAGHKDVILPCPAFNVING
GSHAGNKLAMQEFMILPIGASNFHEAMRIGAEVYHNLKNVIKAKYKDATN
VGDEGGFAPNILENNEALELLKTAIEKAGYPDKIIIIGMDVAASEFYKAGKYD
LDFKSPDDPARYITGDQLGDLYKSFIKGYPVQSIEDPFDQDDWAAWTKFTAA

[illegible]

YSDSPELRRVLAQEVLPNDMELAKVNTSLAVPNDEMEKAKNFEH
RSDLKYHEDYNKNVKGHWCEPTYIEVAVARVAMDNL SNKMYTAAYEDIK
DQINFMQTETPEYKQHKKAGHAASSIKYKKDYENSKSVADYNVLPATDNPL
LRQLRYAGTILSDKVYKASYEKS RGS SINYCDTPKFKMDSVVQQFSDAHYK
DKYENEVKGHYIGSYEDIYMQHCKKMEEIKSEQNYKADYEDIKTRCFYPQT
MSPEYEAHKKVAQCKDKVYRQHPD TVKFTQVTDSPVQLQA AINAKQLSDL
NYHAAYELDKFKCALPPDY PFFIQSRANAYNLSDSCYRYDWEKV KAKQFEV
KGDAISILAA RAHTNIASDVKYKKEFEKNKGKMGV GALS IHDDPKILHSMHV
AKIQSDREYKKNYEKTKTKYITTLDMMSNTLAKKSQGIASMTGYRTIHNRY
LLPHDAMSLDLAKKANIIQSDNEYHSDYNNSTKGTGWVAFGSMEVEKAKK
AGDILNERKYRQHPDTPFTSIADHPVMMQAQV NQIMRSDLHYKKGLEEIH
QKYS LPHNDPQFLQAKCNAYNISKNY YKQSWEE TIAKGYDMRPDAISVIHA
KHGRHIASDIQYKKDYVKGKGKQIGCRNLQDDPLL VH YMEVARLQSDKNY
KKDYHKS KLKYHSPVDMMSV VHAKHASTVQTFAGYRKIQHHYSLLP GAM
NLELARTMNTNASDVNYKADWENS VKGTGWIPIGSIPVELAKQAGAVLDEH
KYRQHPENFKFSSKMDGMN MELCKHNQKIMNVQEYKASGEKFLHTYNLPA
DAPEFMQAKCNAVNISKNY TYAHRQDILKGHV KADAIPI LAARACTNIAS
NYKYKLAYEKGRGKHVGFRNVKDDPLL VH YMEVAKMQSEQNYKKDYHK
SKLKYHSPVDMMSLVHAKHASAAQTFAGYRKSLHNYTLLPEAMPVQLART
MMEIQSNNAYKSDYNNMCRGSGWVPIGSMQVELAKTAMAARDEQGYRVH
PSTLKFTSLTGQMN MALAMTNTKLLDDASYRASGEKFKHTYNLPADSPQFL
QAKCNAQTMSESHYKHKWVEDIAKGYDIRPDAISVIHAKHGRHIASDIQYK
KDYVKGKGKQIGCRNLQDDPLL VH YMEVAKMQSDKNYKKDYHKS KLKYH
SPVDMMSLEHAKHASTVQTFAGYRKIQHHYTLTADSMNLALARTMNASSS
DYEYKSDWNNSVRGTGWIPIGSLPVELAKVGGDIQSEQKYRTHPSKFQFKKL
MDSMDIVLATDNNKIMNKQEYTKQWHNDKLT VHVM PDAPEIVLAKANAIN
MSQKTYKAGLAKMYKNGYDLKVDAIAIQA AKQGTKIASDYKYKLSHRMN
RGKQIGCRNLQDDPLL VH YMEVARLQSDKNYKKDYHKS KLKYHSPVDMM
SVVHAKHASTVQTFAGYRKIQHHYTVLADAMNVELAKQMQTIASDNLYKS
DYETSVKGTGWIPIGSLDVEKAKTAGLIGSEKLYRQHPSSEFKFTKDMHSMDV
VLANANNAIMNKQTYVDAWNKD KTSVHIMPDSMEVVLARANKINCSATLY
KLANEQDKKKGYDLRND AIAILAAKQGT SIASDFKYRAGYRKQVGHHIGAR
CVEEDPLLMLAINSAKIASDALYKKDFHQSKTKFHLPLDMIAFELAKKNQIQ
VNHAKYVTRLHNWTCLPDSTDV VHARHVYDIASDAVYKADLKCLQGIGW
VPIGSLDVEKAKKAGEVLSEKLYRQHPSNFKFTSSVQDIGMV LAKQNNDNA
NKQAYTEAWETDKTKNHIMP DAMDIVLANANKSNYSWNQYRKAMVEDK
KKGYDMRND AISIIAAKHGREIISDHKYKSSYRKQVGHHIGARCVEDDPLL
LALNSAKIASDALYKKDFHQSKTKFHLPLDMIAFELCKKNQIQVNDFN YRTH
LHNWTCLPDSTDV VHARHVYDIASEAVYRADMKWLRGIGWVPTGSLDVEK
AKKAGEILSEKKYRQHPSTLDFKSPLDGMDIVLAKHNGQIMNKRQYVEQW
DADKIKIHVGNELPELV LAKANA ITMSNKLYKQGFDETIKKGY YLPVDAISV
KAAKHGREIISDYKYKAGYSKQVGHHIGARCVEDDPLLMLAINSAKIASDAL
YKKDFHQSKTKFHLPLDMIAFELCKKNQIQVNDFN YRTHLHNWTCLPDSTD
VVHARRVYDLHSDAVYRADMEFIRGTGWVPIGSLDV LKAKKAAEILSDRLY
KQKPD SLKYTTDMHSM PMVLAKANA EIINKGN YVAQWEADKIKIHVGNEL
PELV LAKANAINNSYKLYKAELEELYKKGYNLKSDCIAIKA AKHGREIISDY
KYKAGYSEQVGHHIGARCVEDDPLIMLALNSAKIASDALYKKDFHQSKTKF
HLPVDMLTIELAKK CQM QVNDFN YRTRLHNWTCLPDSSDVVQARKVYALH
SDAVYRADMEWLKGTGWVPIGSIDVEKAKKAGEILSERKYREHPSNFKFTA
TTDDMPYALATANAQIMDKRQYVQQWEADKIQIHVGNELPELV LAKANA I

TMSKLYRENEMIQGYHLDKAKKSREIISDYKYGKAGFRNQ
GHHIGAHCVEDDPLLVLAAANAARISSDINYKKDFHLTKTKFSIPLDMMSFDL
AKKSQFQIQEFTYRTKLHNWTCMPDNTDVVRARKVYSLHSDAVYRADLEW
LRGCGWLPQGSVDVIKAKNAQNILNERLYRQPPSSIKFTSVVDSPEVVLKQ
NADILSDRKYREVWDTEKTNIHLNNDLPSLVLSKANAIASNKLYTKDWD
DKAKGYHIKEDAIKARKSRDIISDYKHEGYRKQVGGHIGARCVEEDPL
IMLALNSAKIASDALYKKDFNKSSTKTFHLPVDMILTIELAKKCQVQVNDNFY
RTHLHNWTCPLDSTDVVQARKTYDLQSDAVYKADLEWL R GCGWNPHEV
EVLKVKAHAQAVLADRGYRVKLDQKQFTAPVDRIDMVCANAAEILNEANY
RALWHTTEKTKYTITDTPGLATAREVAKNLHPKLYTKDWDKVKATGFFMPG
DAVPIKQCIHNNKVQSDHKKYKADYRKQAGHHIGARCVEDDPLIMLAINSAKI
ASDALYKKDFNKSSTKTFHLPVDMILTIELAKKCQVQVNDHDHYRTRLHNWTC
LPDSTDVTQARKAYDLQSDVVYKADLEWL R GCGWVAADCVDHVKVRNA
QKIMNERLYKKDAKDNFAKYTHIVDRPEILATINAFNLSDLKYKETFNAEK
GKYIGSEDTPQLAHSREVAKVISEKYYKLAWDESKATGYQLDAHYPVVG
KKTREIVSDVKYHDTHEKAKGHYLAGTLVDFPEVMRCGEQEKQKGLRDYH
KEYNTNKAKNHIPADIIPNMVAKRCQDMLSDVLYRTLHNWTCHPDQQDII
RARKNNEILSDVYYKDDLNLWIKGIGCYAWDTPELVRAKNAHLLQSDLKYK
EEGKKEFNYSIVTDPVYVTALLGHTWASDLNYREAYHKDKHMYTTILDT
YNYAHNATLKDIIYSTKTYTSAWDKIKAKDYTLPSDSHALKHAKQKQVILSN
VKYREDYKFKSLYTLPKSLEDDPQTLRCLKAGKMGLDRLYKADYEKTKA
KNHIPDMLDVL SARQTQNYVSEIGYRKHLHQWICMPDMQVYVQARKVNE
QLSDIFYKDDLNLWLRGIGCYAWDTPELRVKAADKLQSDNKYRAKGIEAFK
DYSVVIDTPTYETAKLSSQNLSDLHYRNDYATNVKGTNSSPAVTVDTERAR
LANFNQSDNLYKEANKNFMTGYS LPHDNPLMKQAKQNSIVASSVKYKND
YEKSKATNYTMDPDGVSFVNIRNANKVTNPRLYRELYEKKGKGIHSTYDTP
EIRQVKMTQEAVSDLCYKEKYFANRGTMIPMGVTPQMIHCNHVNEITSCLR
YKEDLLWLRGIGCFLYDTPMSTVRNITKFRSNYEVEAKKNHANFTVVLDT
PEYQRVTELKTHMSDLMYRAQGNVEKTKCTTTLDSDLVKRVKWAQQLSN
KMYMTDLASKERAHFTPEKDTPISSHARAMKVVFSEIKYKDQYEKMKHKY
TSIHDTPILVRAKKAYLQSSDLRYKESFERSKGHYHSVKDALDIVCHRRVTD
DISEVKYREKYINSLGTWRSMPDRPEFFHSRIIGENVSDIKYKEDLDWLKGIG
CYVWDRPEMVQAEKNKVLFSERHYKETFEKNRHRFKYTC DTPFFQAAKKA
SVIINDRHYRAAYEKT KD KYSSVLDDPRNLLAKENAKNSQWKYREQYERA
KNKFTSILETPEHESHKRSKAIGDNVYKMDYNKNKAKGYTLGHDTPHSTHM
KKVKEITSVLKYKELYEKSQAQINMDPESFEIRAAKEAYKNISNLDYRKKYQ
ATKNKWIWTVDRPDFIQA A KANFQQSDIEYTYDKEMMKGCVISIVDDKLAV
LAKMNADLSNVKYREKFEKAKGQYKTVLDTPTNLHAKSVRILASESKYKLA
GKMEQQTGQFTTLPTTRDTIHAKDMGKLVSEKQYKQKFEKEKGKSKYNSM
RVPPDVQHAMDVAKNQSNFAYRQEAKAKLNYTSVAMRPDIMRATQASKL
ASNIGYREKARQEAAHGGSLVHRPDIALATAVSKLNSQVQYKAKYDKEKGL
RPKYDMKESVYKTKQDAHQLASEVKYKGD LKKLHKPVTDMAESLSMQH
NLSTSKLSSDYQYKKKHEDSKGHYLM IADSPEQLHHKEASELQSHVKYKEK
YEKAKGKAMLD FETPTYVTAKEAQHMQSAKD YHKEYEESIKGKNLSGLEV
TPAMLHVRHATKIASERDYRRDLEEGVKGKGLTVFEETPELLRARNATAILS
QKEYKKTLETEIKGKGMLALATDTPDFMRARNATDILSQQKYKQTAEADRA
SYTSVIDTPDILHAQNIRNMVSQKKYREEAEKCM SHYVPVLDTPEMMRVRE
NQKNFSTIKYKDLLCRCTAISDLPEVKRVREM QKNISSVLYKDSSAKGTPQV
FTP EMERAKRNQE QISSVLYSDSFRKQVQGKA AFVLDTPEMRRVKETQRNIS
GVKYHQDFEKS KGSFTPTTSDPVMDRVKKNTQEFSDINYRGIQRRV VEME K

RRVEHEQETVTDNLQSRSLHMMSSVQAQRSS
 KEHSRSTSAMSGVGDEKSEVSEDPAAHHVSLYSNGFATSSMGYTHTKTVELQ
 QRSSSVATQQTTVSSVPSHPSTTGKTVRAMYDYSSADMDEVSFKDGDIVVN
 VQSIDEGWMYGTVQRSGKTGMLPANYVEAI 214 Salmon Q78BU2
 MCDDDETTALVCDNGSGLVKAGFAGDDAPRAVFPISVGRPRHQGVMVGM (Actin
 alpha 1-1) GQKDSYVGDEAQSKRGILTLYPIEHGIITNWDDMEKIWHHTFYNELRVAPE
 EHPTLLTEAPLNPKANREKMTQIMFETFNVPAMYVAIQAVLSLYASGRTTGI
 VLDSGDGVTHNVPIYEGYALPHAIMRLDLAGRDLTDYLMKILTERGYSFVTT
 AEREIVRDIKEKLCYVALDFENEMATAASSSSLEKSYELPDGQVITIGNERFR
 CPETLFQPSFIGMESAGIHETAYNSIMKCDIDIRKDLYANNVLSSGGTTMYPGI
 ADRMQKEITALAPSTMKIKIIPPERKYSVWIGGSILASLSTFQAMWITKQEY
 DEAGPSIVHRKCF 215 Salmon ALBU2
 MQWLSVCSLLVLLSVLSRSQAQNICTIFTEAKEDGFKSLILVGLAQNLDPST (Albumin 2)
 LGDLVPLIAEALAMGVKCCSDTPPEDCERDVADLFQSAVCSSETLVEKNDLK
 MCCEKTAAERTHCFVDHKAKIPRDL SLKAELPAADQCEDFKKDHKA FVGRF
 IFKFSKSNPMLPPHVLAIAKGYGEVLTTCCGEAEAQTCFDTKKATFQHAIA
 KRVAELKSLCIVHKKYGDRVVKAKKLVQYSQKMPQASFQEMAGMVDKIV
 ATVAPCCSGDMVTCMKERKTLVDEVCADES VLSRAAGLSACCKEDAVHRG
 SCVEAMKPD PKPDGLSEHYDVHADIAAVCQTFTKTPDVAMGKLVYEISVRH
 PESSQQVILRFAKEAEQALLQCCDMEDHAECVKTALAGSDIDKKITDETDYY
 KKMCAAEEAAVSDDNFEKSMMVYYTRIMPQASFDQLHVMSETVHDVLHAC
 CKDEPGHFVLP CAEEKLTDAIDATCDDYDPSSINPHIAHCCNQSYSMRRHCIL
 AIQPDTEFTPELDASSFHMGP ELCTKDSKD LLLSGKKLLYG VVRHKTTITED
 HLKTISTKYHTMKDKCCAAEDQAACFTEEAPKLVSESAELVKV 216 Salmon ALBU1
 MQWLSVCSLLVLLSVLSRSQAQNICTIFTEAKEDGFKSLILVGLAQNLDPST (Albumin 1)
 LGDLVPLIAEALAMGVKCCSDTPPEDCERDVADLFQSAVCSSETLVEKNDLK
 MCCEKTAAERTHCFVDHKAKIPRDL SLKAELPAADQCEDFKKDHKA FVGRF
 IFKFSKSNPMLPPHVLAIAKGYGEVLTTCCGEAEAQTCFDTKKATFQHAV
 MKRVAELRSLCIVHKKYGDRVVKAKKLVQYSQKMPQASFQEMGGMVDKI
 VATVAPCCSGDMVTCMKERKTLVDEVCADES VLSRAAGLSACCKEDAVHR
 GSCVEAMKPD PKPDGLSEHYDIHADIAAVCQTFTKTPDVAMGKLVYEISVR
 HP ESSQQVILRFAKEAEQALLQCCDMEDHAECVKTALAGSDIDKKITDETDY
 YKKMCAAEEAAVSDDSF EKSMVYYTRIMPQASFDQLHVMSETVHDVLHA
 CCKDEQGHFVLP CAEEKLTDAIDATCDDYDPSSINPHIAHCCNQSYSMRRHC
 ILAIQPDTEFTPELDASSFHMGP ELCTKDSKD LLLSGKKLLYG VVRHKTTIT
 EDHLKTISTKYHTMKEKCCAAEDQAACFTEEAPKLVSESAELVKV

[0059] An animal protein can comprise a cod protein (Table 13). A cod protein can comprise a cod muscle protein. A cod protein can comprise a cod blood protein. A cod protein can comprise a cod albumin protein. A cod protein can comprise a sequence that has at least or about 70%, at least or about 75%, at least or about 80%, at least or about 90%, at least or about 91%, at least or about 92%, at least or about 93%, at least or about 94%, at least or about 95%, at least or about 96%, at least or about 97% at least or about 98% at least or about 99% or about 100% sequence identity to any one of SEQ ID NO: 217-231.

[0060] A food product may comprise a fragment or a truncation of a cod protein. The food product may include a fragment or truncation of any one of SEQ ID Nos: 217-231. A sequence of the fragment or truncation may be identical or substantially identical to a portion of any one of SEQ ID Nos: 217-231. A sequence of the fragment or truncation may be at least about 70%, 80%, 90%, 95%, 98%, or more identical to a portion of a sequence of any one of SEQ ID Nos: 217-231. A length of the fragment or truncation may be at least about 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or more of a length of any one of SEQ ID Nos: 217-231. A length of the fragment

or truncation may be less than or equal to about 90%, 80%, 70%, 60%, 50%, 40%, 30%, 20%, 15%, 10%, or less than a length of any one of SEQ ID Nos: 217-231.

TABLE-US-00013 TABLE 13 Cod Proteins SEQ ID NO Description Amino Acid Sequence 217 Cod

MAKNCHNDYKMKFSGDEEFPDLSLHNNHMAKVLTCKDIYNKLRGKSTPSGF A0A8C4Z9R4
TVDDIIQTGVDPNPGHPFIMTVGCVAGDEESYEVFKDLLDPIISDRHSGYKPT (Creatine
kinase) DKHRTDLNFENLKGDDLDPAYVLSSRVRTGRSIKGFTLPPHNSRGERRTIE
KLSIEALATLSGEFKGKYYP LNGM TDKEQDQLINDHFLFDKPVSPLLTCAG
MARDWPDARGIWHNDAKTFLVWVNEEDHLRVISMQQGGNMKEVFRRFC
TGLLKIEETFKKH NHGFMWNEHLGYVLTCP SNLGTGLRGGVHV KLPKLST
HAKFEEILTRLRLQKRGTGGVDTASVGGVFDISNADRLGSSEVDQVQMVV
DGVKLMVAMEKKLEKSESIDDMIPAQK 218 Cod

FHKLLQTNGKGMIMLAYMTQEEGWDRDLLDPAWEKQQRKTFTAWCNS A0A8C5D617
HLRKAGTQIENIEEDFRNGLKLM LLEVISGERLPKPKDGKMRFHKIANVN (Actinin
alpha 3b) KALDYICSKGVKLV SIGAEEIVDGN AKMTLGMIWTHILRFAIQDISVEETSAK
EGLLLWCQRKTAPYRNVNVQNFHISWKDGLALCALIHRHRPDLIDYSKLR
KDDPIGNLNTAFEVAEKFLDIPKMLDAEDIVNTPKPDEKAIMTYVSCFYHAF
AGAEQAETAANRICKVLAVNQEN EKLMEEY EKLASELLEWIRRTIPWLENR
AAEQTMRAMQQKLEDFRDYRRVHKPPRVQEK CQLEINFNTLQTKLRLSNR
PAFMPSEGKMVSDIANAWKGLEQVEKGYEEWLLTEIRRLERLDHLAEKFK
QKCSLHETWTSGKEELLSQKDFESASLMEIRALMRKHEAFESDLAAHQDR
VEQIAAIAQELNELDYHDAVTVNARCQGICDQWDNLGTLTQKRRDSL ERV
EKLWETIDQLYLEFAKRAAPFNNWMDGAMEDLQDMFIVHSIEEIQSLITAH
DQFKATLPEADKERMATMGIQSEILKIAQTYGIKLSGINPYTNLSPQDISNK
WDAVKHLVPLRDQMLQEEVARQQANERLRRQFAAQANIIGPWIQTKMEEI
GHVSADITGSLEDQMNNLKQYEQNIINYKSNIDKLEGDHQLSQESLIFDNKH
TNYTMEHIRVGWEQLLTTIARTINEVENQILTRDAKGISQEQLNEFRASFNH
FDRKRNGMMDPDDFRACLISMGYDLGEVEFARIMTLVDANNTGVVTFQAF
IDFMTRETAETDTAEQVMASFKILASDKSYITVDELRRRELPLEQAEYCISRM
TRYVGADGTTGALDYISFSSALYGESDL 219 Cod

MAKPLTDQEKRKQISVRGLAGVENVTDIKS NFNRHLHYTLVKDRNVATKR A0A8C5D074
DYYFALANTVRDHLVGRWIRTQQHYEYKDPKRVYYLSLEFYMGRTLQNT (Alpha-1,4
glucan MVNLALENACDEATYQLGLDMEELQDIEEDAGLGNGGLGRLAACFLDSM
phosphorylase) ASLGLAAYGYGIRYEF GIFNQKIVD GWQVEEADDWLRYGNPW EKARPEY
MRPVHYYYGRVEHTPGGVK WVD TQVVLALPYDTPVPGYRNNVVNTMRLW
SAKAPCDFNLKDFNVGGYIQAVLDRNLAENISRVLYPNDNFFEGKELRLKQ
EYFVVAATLQDIIRRFKSSKFGSTEFVRMDLSTLPDKVAIQLNDTHPALAIPE
LMRILMDLEELPWEKAWDIVVRTCAYTNHTVLPEALERWPIDLFQNLPRH
LDIVYEINRRHLERVAQQFPGDEGRLLRMSLIEEGDHKKINMAHLCIVGSH
AVNGVARIHSEIIKDTVFKDFYEMDPEKFQNK TNGITPRRWLVMSNPGLAE
VIAERIGDDYIRDLDQLKNLLEFVDDDSLIRDVAKVKQENKMKFSAYLEEH
YKVKINPESMFDVQVKRIHEYKRQLLNCLHIITFYNRIKKQPNKAWTPRTIM
IGGKAAPGYHTAKMIIKLITAIGDIVNHDAVVGDR LKVIFLENYKVTLAEKII
PASDLSEQISTAGTEASGTGNMKFMLNGALTIGTMDGANVEMAE EAGEEN
LFIFGMRVDDVKAMDQTGYDAKSY YDRIPEL KQAMDQISGGFFSPGQPDLF
KDIVNMLMNHDRFKVFADYEEYIKCQDKVSALYKDPKEWTKMVIRNIGSS
GKFSSDRTIAQYAREIWGMEPSLEKIPAPDDPRD 220 Cod

MPHAYPFLSTEQKKELSDIAKKIVAPGKGILAADESTGSVAKRFQGINAENT A0A8C5F6L1
EENRRRLYRQLLFTADDRVAPCIGGVIFFHETLYQKTDDGKLPQLIRERGQV (Fructose-
VGIKVDKGV MPLAGTDGETTTQGLDGLYERCAQYKKDGADFAKWRCVLK bisphosphate

ITENTPSELAIMENANVLARYASICQMVRRHHGIVPIVEPEILPDGDHDLQRTQ aldolase)
YVTEKVLAAVYKALSDHHVYLEGTLLKPNMVTAGHSCTRKYSAQEVA
TVTALRRTVPPAVPGVTFLSGGQSEEQASINLNAINQCPLGRPWALTFSYGR
ALQASALKAWGGKKENGKACQEEYIKRALNNNLASQGKYVSSGDKGAAG
GESLYVANHAY 221 Cod
MAPKKDVKAPAAAAKKAEPAKKAEPPEPVAVPAPKTVDL SAVKVEFTPD A0A8C4YT70
QMEDYREAFGLFDRVGDNKVCYNQIADIMRALGQNPTNKEVKAILGNPSD (Myosin,
light EDMNSKRVD FEGFLPMMQTIVNSPNKGTLD DYVEGLRVFDKEGNGTVMG chain
1, alkali; AELRIVLSTLGEKMTEAEIDALMQGQEDENG CINYESFVKHIMSI skeletal, fast)
222 Cod Q8AWX8
MVKVGINGFGRIGRLVTRAGFQSKKVEIVAINDPFIDLEYMVYMFKYDSTH
(Glyceraldehyde-
GRFHGEVKIEGDKLVIDGHKISVFHERDPANIKWGDAGA QYVVESTGVFTT 3-phosphate
IEKASAH LKGGAKRVIISAPSADAPMFVMGVNHEKYENSLQVVS NASCTTN
dehydrogenase) CLAPLAKVIHDNFGIVEGLMSTVHAITATQKTVDGPSGKLWRDGRGASQNI
IPASTGAAKAVGKVIPELNGKITGMAFRVPTPNVSVVDLTVRLEKPAKYDDI
KKVIKAASEGPMKGILGYTEDQVVSTDFNGDIRSSIFDAGAGIALNDHFVKL
VTWYDNEFGYSNRVIDLMAHMSCKE 223 Cod
MDAIKKKMQMLKLDKENALDRAEQAEGDKKAAEDRSKQLEDDLVALQK A0A8C5CXA2
KLKGTEDEL DKYSEALKDAQE KLELA EKKAADAESDVASLNRRIQLVEEEL
(Tropomyosin 1,
DRAQERLATALTKLEEA EKAADESERGMKV IENRAMKDEEKMELQE IQLK alpha)
EAKHIAEEADRKYEEVARKLVIIESDLERTEERAELSEGK CSELEEELKTVT
NNLKSLEAQAEKYSQKEDKYEEEIKVLTDKLKEAETRAEFAERSVAKLEKT
IDDLEDELYAQKLKYKAISEELDHALNDMTSI 224 Cod
MPEAADAGELPTAEEDSGDEEGQGT ELTGLFVERPQENVVIIAGADV VITA A0A8C5B6F3
KVDSTTLTRKPTVKWLKGKWLDLASKAGKHLQFKETYDRETKIYTYEMK (Myosin
binding VIKAKVGAAGAAGAYRCEVT SKDKFDSASFEITVEAAQQSDDVDILSSFKR
protein Cb) ADAGEDDGNLDFSALLKAAKKKKRPEDDKPVIDVWELLKSAHPSEYEKIA
FEYGITDLRGMLKRLKKMKAE PKPSEAF LKRMESCYSVEKGKKIVLTCEVF
DPNVQVKWLKNGQEIKGSAKYIMETIGNIRKLTIMKTTLADDAAYECVVG
EDKCFTEVFVKEPPVTITKLMDDYHVVVGERVEFEVEVSEEGANAMWFFE
DIEIFKDTNPAKYRIKKDGRKHMLIILEAGLEDIGMYHCWTNGGHTKGELE
VEEKELLVLQDIADLTVKATDQAIFKCEVSDEKVTGKWFKDGV E VLPDER
HKMTHIGRFHKLLIDNVK PQDAGDYTFIPNGYALSLSAKLNFLEIKIDYVPR
QDPPKIHLDTTGNMVSQNTIIVVAGNKLRLDVEITGEPAPTVVWSKGEEVIT
ATEGRVRVDNRKTLSSFVIEGAEREDEANYTICVTNPAGEDKAKL FVKIVD
VPDPPQNVKCLSVGEDSSSMVWEPPAFDGGAPLKG YLMERKKKGSSRW
T KL NFDVYEATTYEAKRMIEGVYYEMRVFAINSIGMSQPSITSKPFMPLAPTS
EPTRLLVHDVTDGTCTLKWMAPERIGAGGLDGYIIEYCKEGETEWV VANK
ELCERQGYVVRGLPVGEKINFRVFAKNLAGVSP PATLAQPVTVREIVDLPKI
RLPRDLRTKYIRKVGDKINLVIPFQGKPRPVVKWYKD GAPVDEKIVGVRNS
TVDSILFIRSAERDHS GKYELVVQIENMEDRATIDLRVVERPGPPTKV K VTE
VWGFNAALEWEAPKDDGNCEITGYTIQKADKKTMEWYTIYEHNRPSCTS
SDLIMGNEYMFRVYSENLVGPSEEPGQSKNTAVIGKTGLAYNPPEFKEKDV
SCIPTFTQPLLD RSVVAGYSTAMSCSVKGFPKPKIVWMKNKMILGGDPKYI
MLNNQGVLT LNIRKPSTFDGGRYSCMAVNDLGQALVECKLEIRIPTDPTAE VK 225 Cod
A0A8C5F718 MAMRSGNYNHEYYHKQSKYVVKEYSSSES VDDYQHKA AVRIQGVVRK
(Myomesin 2b)
KMVEKYVREEPMIRGPKFVVRLRSHTVFENSPIKLFCTVEGFPVPIVTWYKD

GVIHSGHAGVYDQVDDTADQYSAVASNIHGQASSQA
SIIVKRFRQEDESCLYAWLPYTVPMPLPSIQYTKINITFTEKFGPVFASEGGSSAS
LTATMHLEPNLANLQPD SQWYRDDTRLVDSKWVKINTGRGTSTLTIPNLFK
DDEGLYTLRLVTKGGTAEHS AFISVSDGPPP VVGAPGAPMDIRIHDANRDY
VIVSWKPPNTTTEGPILGYFVDKCEVG TENWAQCNDHPIKICKYPVPGLFEG
HSYYFRVR VAVNARGISKPSRMSDPIAALDPTEFERLHALKLGGKLDVVTYD
DDMEAEGKPPGVSSGIYASETDRTYVVL SWKAPAYSSKAPMWYYIEKCQV
GTA AWQRVNTQVPIRSPRYAVFDLAEGKEYQFRVLSANMYGTSEPSEPSKL
IQTQALKGVP SAPGQVISTRETDTSL LIQWAPPKEPNNLIGYYIDQCVKGSK
DWTSANHKPHSSTKFVVSGLTTGETYVFRVQAINELGLSDESQESSPLSVKA
ALTSPTAPYDVALLGCDGQSMVLNWKKPLHTGGTAINAYFVDKRRSGAT
MWREVHIPAVTERVYQVEGLTQGA IYQFRVYAANLAGLGKSSKASADFSC
EAWTMPEPGPAYDLTASEVRDTS LVLEWKKPVYYGSGPITGYHVEYAKKG
SSDWTRANETAVSHRYLKVEGLEEGAKYVFRVRGVNEAGVGMASLPSDS
VVAKALAGAEVSCSVDEKSGDIVFAFESCQMNVGSEFLWKKNYKEITEFS
SSIVVHTEGNLCKLMFKNAV KEDFGTYSVSVSHTDGVSSSCSISAEELQKM
LELSYSIRHPIPLKSELAYKILERGRMRFWLQAE EISSKV TYKFFANNKELS
GADPNKMAHDVSTGVIEFILDHFTEDSEGTYTCQITDGGGKAQSSLV LIGDA
FKAALAEADYQRREYIRVKEGPHFSEFLT VKVGEDCSVTLVCKVANLK KES
EFHWYKEDTEIIPDV KPD LGSGVCKLPIHLFSRKTTGVYKATISDDRGRKDMS
QVDISGKV FDDVINKINQLAGASAAELVINCTATGIQLQCHMKYYTSEMNI
HWMHGESKVG HSEKCVIGGTPAMATMEVIEPNDDKDKGVYSIVITDLENTN
KRTL DLSGDVYDKAYA QFQKLKAEAYAEMNRGKVMGGLPDVVTIMEKK
TLSLTCTVCGDPKPLVSWLKN GNDVEPDNQYVVSLDQGKFACLTINGIGM
DDSGKYTMIVQNKYGGECVDIVSVYRHGDTIPVAKPNLTAKTIIPPKMPIE
LPQPKTAAPSPAPPASPS PAPPKAVGGRGMKSPTPTRRK 226 Cod
MKS NPLTSNTGEHAS YLLHVNGVMSITKIHAREILDSRGNPTVEVDLWTAK A0A8C5C5F8
GLFRAAVPSGASTGIHEALELRDGD KTRYLGKGTQKAVDHVNKD ISPKLIE
(Phosphoryruvate
KNFSVVDQE KIDKFMLELDGTENKSKFGANAILGVSLAVCKAGAAEKGVP hydratase)
LYRHIADLAGNKDVILPVPAFNVINGGSHAGNKLAMQEFMILPVGAANFHE
AMRIGAEVYHNLKNVIKAKY GKDATNVGDEGGFAPNILENNEALELLKSAI
EKAGYPDKIIIGMDVAASEFYRSGKYDLDFKSPDDPARHISGKELGDLYKTF
IANYPVQSIEDPFDQDDWANWADFTA AVDIQVVGDDLTVTNPKRIQQAVE
KKACNCLLLKVNQIGSVTESIQACKLAQSNGWGMVSHRSGETEDTFISDL
VVGLCTGQIKTGAPCRSERLSKYNQLMRIEEELGD KAKFAGKDYRHPKIN 227 Cod
MAPKKAKRRAGGGDSGSSNVFSMFEQSQIQEYKEAFTIIDQNRDGIISKDDL A0A8C4ZUF5
RDVLASMGQINVKNEELEAMVKEASGPINF TVFLTMFGEKLGADPEDVIV (Myosin
light AAFKILDPEATGSIKKEFLQELLTTQCDRFSPEEIKNMWSAFPPDVAGNVDY chain)
KNICYVITHGEEKEE 228 Cod
IMGDGEMQCFGEAAVYL RKP ERERLEAQTSPFDAKTAFFVSEPKEMY LKG A0A8C5FJN9
VLQSRDGGKATVKTLC DKTLTVKEEEIYPMNPPKYDKIEDMAMMTHLNEP (Myosin,
heavy TVLYNLKERYAAWMIY TYSGLFCVTVNPNYKWLPVYDSSV VNAYRGKKRIE chain
b) APPHIFSISDNAYQAM L TDRENQSILITGESGAGKTVNTKRVIQYFATIAVSG
PKKTEQVAGKMQGSLEDQIIAANPLLEAYGNAKTVRNDNSSRFGKFIRIHF
ATSGKLASADIETY LLEKSRVTFQLSAERSYHIFYQLMTGHKPELIDALLITK
NPFDYHMISQGEITVKSID DVEEFIATDDAIDILGFTADEKIGIYKLTGAVLH
HGNMKFKQKQREEQAEPDGN EADKIGYLMGLNSADVLKALCYPRVKVG
NEYVTKGQTC PQVHNSVMALCKSVYEKMFLWMVVRINEMLDTKQPRQFF
IGVLDIAGFEIFNYSLEQLCINFTNEKLQQFFNH HMFVLEQEEYKKEGIEW

EFIDFGMDLAACIELIEKPMGIFSIILEEEMFPAKASDITTFKKNLHDQHLGKTK
AFEKPKPAKGKAEAHFSLVHYAGTVDYNITGWLDKNKDPLNDSVIQLYQK
SSVKLLALLYASHALPDADAASKKGAKKKGGSFQTVSALFRENLGKLMTN
LRSTHPHFVRCLIPNESKTPGLMENFLVIHQLRCNGVLEGIRICRKGFPSRVL
YGDFKQRYKVLNASVIPEGQFIDNKKASEKLLGSIDVDSTQYKFGHTKVFF
KAGLLGALEEMRDEKLAALVTLTQALCRGFNMNRKEFVKMMERREAVFTI
QYNIRSFMNVKTWPWMKVYFKIKPLLKSAETEKEMAQMKETFGKTKEDL
AKALSKKKELEEKMVSLQEKNDLLQVQSEAETLSLSEERCEGLIKSKIQL
EARLKETSERLEDEEEINSELTAKKRKLEDECESELKKDIDDLELTAKVEKE
KHATENKVKNLTEEMSSQDEIIAKLTKEKKALQEAHQQTLDLQAEEDKV
NTLSKAKCKLEQQVDDLEGSLEQEKRLRMDLERAKRKLEGDLKLAQESIM
DLENDKQQSDEKIKKRDFETSQLLNKIEDEQTLAQFQKKIKELQARIEELE
EEIEAERAAARAKVEKQRADLSRELEEISERLEEAGGATSLQIEMNKKREAEF
QKLRRDLEESTLQHEATSAAALRKKQADSVAELGEQIDNLQRVKQKLEKEK
SECKMEIDDLSSNMEAVAKSKANLEKLARTLEDQLSELKSKNDEHVRQLN
DLSVORARLQTENGEFGRQQEKEALISQLTRGKQAYTQQIEELKRHIEEEV
KAKNALAHSVQSARHDCDLLREQFEEEQEAKAELQRGMSKANSEVAQWR
SKYETDAIQRTEELEEAKKKLAQRLQDAEEAVEATNSKSCASLEKTKQRLLG
EVEDLMIDVERANALAAANLDDKKQRNFDKVLAEWKQKYEESQAELEGAQK
EARSLSSTEMFKLKNSYEEESLDQLETCLKRENKNLQQEISDLSEQMGETGKTIH
ELEKGKKTVEVEKSELQTSLEEAEATLEHEESKILRVQLELTQIKGEVDRKM
AEKDEEMEQIKRNSLRVIDSMQSNLDSEIRSRNDALRIKKKMEGDLNEMEI
QLSHANRQASEAQKQLRNVQGLKDAQLHLDDALRGQEDMKEQVAMVE
RRNNLMLAEIEELRAALEQTERGRKVAEQELVDASERLGLLHSQNTSLINT
KKKLETDLIQIQGEVEDTVQEARNAEEKAKKAITDAAMMAEELKKEQDTS
SHLERMKNLELTVKDLQHRLDEAESLAMKGGKKQLQKLEARVRELETEC
EAEQRRGSEAIKGVKRYERKVKELSYQCDEDKKNICRLQDLVDKLQLKMK
AYKRQAEESSEEQANTHLSRYRKVQHELEEAQERADVSESQVNKLRTKSRN GITKIIN 229
Cod MATAGMPTRKVRKRVAVDTSKFMTPLYLAHSMKMLEQYSHNMYRKAYDL
A0A8C5F7Y0 SRGTPPVHIADTPMVRIRKAQEQLSEVKYRMENKARAGSLYDGEAKEIA
(Nebulin) HVKHMSSELLSKVLYRQKWDETCDRYLLPPDAPELVAVKNAANYSKKLY
TEDWDLEKASFLPYSDSPELRRVAKAQEALSDIQYKKGQYDQRKLKFTSLAD
RPDVELAKKVFTQRSCLKYKEDYNKNVKGQWCETPYFDVATARVAMENL
SSRRYTQDWEAAKDQINFMQTETPVYDTNKKAGTSASSVKYRKDYETK
DKSDYNLTPATENPLLRLRLRYAGTILSDKVYKKNYEKSRGFSINYCDTPKF
KMDSVLKKFTDTHYQDKYKNEVQGHYIGSYEDIYMLHCQKVEGFKSEQN
YKAEYEDIKTRCFFPQTITPEYEVDDKLAQCKDKVYRQHPDKVKFTQVTDS
PVQVQAALNAIQLSDINYKAKYEQTCTKNSLPADYPFFIQSRVNAFNLSENC
YKYNWEKEKAKRFEMKGDASILAAARAHTNIASDVKYKKEYEKNRGRMV
GALSIHDDPKILHSVHVAKIVSNREYTKDYEKVKTKYHTSLDMMSVTLAK
KSQAIASMTGYKKISNRYLMPYDAILVDLAKKANISQSDNEYKSDYNQYTK
GSPWITFGSMDVELAKKAGEILSEKKYRKHPDTPVFTAIDDHPVMLQARVN
QLQRSDFYKKGLEEIHQKYSPLIDMPQLARANSYNISKNYKAAWEET
ISKGYDLKLDAISIKAAKAAARHAASDIQYKKAYEKARGHHVGFNRNIKDDPL
LVHYMQVAKMQSEKNYKKDYHKSLLKYHSPVDMMSLVHAKQASKAQT
YAGYRKIQHHYSLMPDAMNLVLARKMNTDASDCEYKSEWNNYVRGTGW
VPIGSIGVETAKIGGQIQSENKYRTHPSTFKFSKLMDSMDELATQAYTSAW
EKDKRTIHVMPDSPEIVLAKANAITMSDVREYYYKLHQNGYHLKGDAIPIV
AAKLGTKIASNYKYKLDYEKARGHHVGFNRNIKDDPLLHVHYMQVAKMQSE
KNYKKDYEKAKLKYHSPVDMMSLVHAKEASKAQTAYAGYRKIPHTYFLLP

DNMQQLCRLMQEIQSDNEYKSEYNNLYKGTGWVPIGSDVELAKTANSA
LREHGYSRQHPSSFKFTSKTDSMDMVLALANSKIIDQAAYKASGEKVKHTY
HLPADAPEFIQAKFNAQTMSKSHYKQKWVEHIAKGYDMRPDAISILHAKQ
GRHIASDIQYKKAYEKARGHHVGFERSIKDDPLLHVHYMQVAKMQSEKNYK
KDYEKAKLKYHSPVDMMSLVHAKEASKAQTYAGYRKIQHHYNLMPDAM
NLVLARKMNTDASDCEYKSEWNNYVRGTGWVPIGSIGVETAKIGGEIQSE
NKYRTHPSTFKFSKLMDSMDLELATANNKIMNKDKRTIHVMPDSPEIVLAK
ANAITMSDKLYKAGVMEMAKKGYHLKGDAIPIVAAKLGTKIASNYKYKL
DYEKARGHHVGFERNIKDDPLLHVHYMQVAKMQSERNYTKDYQKSKLKYHS
PVDMMMSLVHAKEASKAQTYAGYRKIEHHYNLMPDAMNLVLARKMNTDA
SDCEYKSEWNNYVRGTGWVPIGSIGVETAKIGGEIQSENKYRTHPSTFKFSK
LMDSMDLELATANNKIMNKQAYTSAWEKDKRTIHVMPDSPEIVLAKANAI
TMSDKLYKAGVMEMAKKGYHLKGDAIPIVAAKLGTKIASNYKYKLDYEK
ARGHHVGFRCIQDDPLLHVHYMQLAKQQSEKNYKKDYEKAKLKYHSPVDM
MSLVHAKDATKAQTYAGYRKIQHHYSLPDAMNLVLARNMQVNISDNQY
KSDYESYCKGIGWPIGSLDVEKAKIAGKIQSEKLYRQHPSNLKFTKDMFSM
DLTLATANNQIMNKVLLALYCVDYTLMNILSSGNNLTNNFVVSFSKLYKL
DNELSKKKGHNVRKDAIAILAAKNCTTIASDYKYKAGYRKQVGGHHIGARC
VEDDPLLVLALNSARIASDALYKKDFNKSSTKFNLPVDMMAFELAKKNQI
QVNHANYITRLHNWTCLPDSTDVVNARRAYEIQSDNYYKEDQKLMQGIG
WVPIGSLDVEKAKKAGEILSEHLYRKHPSNYKFTSKCDAMPLVLAKANAQI
TNKKAYHDAWDKEKSNIHIMPDAMDVVLAKANRANYSIKRYKQANEDAK
KKGYDLRNDIAIGIIAARAQRNIISDIKYKAGYRKQVGGHHIGARCVEDDPLL
MLALNSARIASDYLYKKDFNKSSTKFNLPVDMMAFELAKKNQIQVNHAN
YITRLHNWTCLPDSTDVVNARRAYEIQSDNYYKEDQKLMQGIGWVPIGSL
DVEKAKKAGEILSEHLYRKHPSNYKFTSKCDAMPLVLAKANAQITNKKAY
HDAWDKEKSNIHIMPDAMDVVLAKANRANYSIKGYKQANEDAKKKGYD
LRNDIAIGIIAARAQRNIISDIKYKAGYRKQVGGHHIGARCVEDDPLLVLALNS
ARIASDYLYKKDFNKSSTKFNLPVDMMAFELAKKNQIQVNHANYITRLHN
WTCLPDSTDVVNARRAYEIQSDNYYKEDQKLMQGIGWVPIGSLDVEKAKK
AGQILSEHLYRKHPSNYKFTSKCDAMPLVLAKANAQITNKKAYHEAWDKE
KSNIHIMPDAMDVVLAKANRANYSIKKYKEANEESSKKGYDLRNDIAISIIA
ARAQRNIVSDYKYKAGYRKQTGGHHIGARCVEDDPLLVLALNSARIASDAL
YKKDFNKSSTKFNLPVDMMAFELAKKNQIQVNHANYITRLHNWTCLPDST
DVVHARHVYEIQSDAVYKADLKWLQGMGWVPIGSLDVEKTKKAGEILSE
RKYRQPASSHKFTSPLDAMEIVLAKNNALTMNQRLYTEAWEKDKTKLHIT
SETPEILSQQNAINMSNKKYKQGYEDTIKKGYLPGDAISVKSASHSRNIIS
DYKYKAGYRTQVGGHHIGARCVEDDPLLMLALNSARIASDYLYKKDFNKS
TKFNLPVDMMAFELAKKNQIQVNHANYITRLHNWTCLPDSTDVVNARKA
YDLQSDAVYKADMEFMKGIGWVPIGSLDVVKAKHAGKILSDHLYKQKTD
KLKYTSDTKAMPMLLAKANAEMNKKSYVAAWEKDKIKTHLIPDLPEILL
SRSNAYNLSNKMYPEDVVKMFRKGYHLKGDAIAIVAACKGREIISDIKYK
AGYRKQVGGHHIGALSVRDDPLIMLALNSARIASDALYKKDFNQSKTKFHLP
VDMLTIELAKKCQVQVNDNFNYRTRLHNWTCLPDSTDVVQARKAYDLQSD
AVYKADMEWIKGQGWWVPIGSLDVEKAKKAAEILSERKYREHPSKFKFTAK
TDDMPYALAKANAEMNKQSYVAAWESDKTKLHIMPDTPEILGQQNKLN
NSRKYYREAYEETLKKGYLPGDAISVLAARASRNIISDYKYKAGFRKQCG
HHIGALSVQDDPLLMMAANAAKIFSANLYKQDFHKTCTKYKLPVDMVAF
DLAKTAQSHMKEVTYRTRLHNWTCPDNTDVVHARKVYDLHSDAVYRA
DMEWIRGCGWSPQDSVDVVKARNAQKILNENLYRQAPSTVKFTSVVDLPA

IVLSKQNAENIAXENYKDKLTHLFDIPSVQLAKANAININSLYT
KAWDNQKAKAYHIKEDTIAVLKARASRDIISDYKYKTGYRKQVGGHHIGAL
SVRDDPLIMLALNSARIASDALYKKDFNQSKTKFHLPVDMMLTIELAKKCCQIQ
VNDFNYRTRLHNWTCLPDSTDVQARKAYDLQSDAVYKADMEWIRGCG
WVPHECVDVVKVHAQKILADRGYRVKLDLDTQKFTAPVDRMDFVCAKNA
AEVLNDAKYREAWHTDKSKFSLVDTPGLATSREVAKIMHPKLYTKDWDK
VKATGYFMPGDAVPIKQCTQTTKVQSSHKYLEGYRKQVGGHHIGALSVFDD
PLIMLAMNSAKIASEALYKKDFNKSSTKTKFHLPVDMMLTIELAKKCCQIQVNDF
NYRTRLHNWTCLPDSTDVVNARKAYDLQSDFVYKADLEWIRGCGWVPAE
SVHHVKVRKAQEILNERLYKKGAIENFSKYTHVVDPELILARVAAANLSD
LKYKETFNNEKGHYIGSEDTPQLAHSRECAKNISEAKYHESHEKAKGHYM
ANTLVDFPEVVRCGEQEKAKALRVYQKEYHESKTNNHIPSIIANQVAKRC
QDMLSDVIYRTHLHQWTCPPDQLDTVRARKGNEILSDVFYKEDLTWMKG
MGCYVWDTPEIVRAKKSSELQSELKYKEAGKKEFNNSIVTDTPVYVTAIL
GSTWASELNYREAYHKEKHLYTTILDTHDYARCHDFKHFFSNKIYTAAWD
KIKAKGYKIPHDSHAMMHAKQQKVILSGVKYKEDYEKFKSLYSLPKCLQD
DPATARCIKAGKLNLDRLYKEDYEKTKAKNHIPADMMLDIVSGREVQRNVS
EINYRKRLHQWVCLPDMQVQVQARKVNDQLSDIFYKDDLNYLRGIGCYA
WDTPEILRVKTAGDQQSENKYKAKAVEAYKDYTVVLDTPTYETAKQNAV
NLSDLHYRLDYNANVKGTNTAPAVTMETERARLANYNQSDNLYKEANKN
SMATGYLMTNDTPVVKQVKNNSLVASSVKYKAAYEMMKATGYTLHPEG
VTFVNTRKVNQVINPRLYREQYEKDKDKIHSTLDTDPDIRQVKMNQEHLSL
CYREKYNNTKGQLINMPMTPQLMHCQHVNQITSDLRYKEDLLWLRGLGCF
MYDTPPEMVNVRNITKFKNNYQVEAKKNHNSKFSVVLDTPPEYQRVTELKSH
MSMVSPTSNTSLNTLDSVDIQRSKWAQQLSNKFLYTDLASKERAHFSQEVE
TPAMGLARTMKVTNSYNKYKEQYEKMKHRYTSIADTPILIRAKKAYLQSS
DLRYKETFERSKGHYHTVKDALDILCHRRVTDDISEVKYREKYINSLGTWK
SIPDRPEFFFNRIVGENVSSVKYKEDLEWMKGIGCFVWDTPEMVQADINKT
LYSGRLYKASFENRYNFKYTCDTPFHAAKNASSLINERSYRSSYEKNKD
KFTITDDPRYQLARENKMSQAKYKSRAKELLKSGCNELHRPDILTALYQ
SVLGSKWKYREQYERVKSKFTSILETPEYESHKRSKKIGDIVYRMEYNKNK
AKGYSLPYDTPHQNHMCKVQALTSNLKYREVYEKAKSQINMDPDAFEIRA
AKEAYKNTTNLDYKKKYEATKNKWIWTVDRPDFISHSRNTLQQSDVEYKY
DKEMMKGCVIPIVDDKITLLAKKGCESMASDIKYKEKYEKAKGHYNPVADS
PQILHAKAVRTLASESKYKEASKKEMQSGSFTTLPETRDTVHSKEVNKLISG
KLYKSKFEKEKGQSKYNRMIVPPDVQHAMDVAKSQSSYAYKRDAKDNLH
YTTTADRPPDIRKATQAAKLISDVGYREKARAEASRGGSALTYPDIALATEVS
KLTSLVDAKSALHGAASYDTLQMRHSSKMSALTSNVKYHEKFDKEMKGGK
KPKYDLKDSKIYQTLKDATNLASEVKYKGDLLKHLHPVTDMAESLSMQH
NLSTSKLSSDYRYKKKFKDSKGHYLVIADTPEQLHHKEATELQSSVSGPSLC
GVHVCKDYKKDYEEGMRGKNLCGLEVTPAMIHVRHATKIASEREYRKDL
EEGVKKGKGLSAPEETPEHLRAKNATHILSEREYKKSLELEIKGKGMLALAT
DTPDFMRARNATDILSQT KYKHTAELDRASFTSVIDTPDIIHAQQMRNIISQN
KYKAEAGKTM SHYVPVFDTPEMQRVRENQKNFSTLQYQADHKIIKGKVLT
VKDTPPEMLRVKENTKNFSSVLYKDSKAKGTPVVFTPEMERVKRNQEHISSV
LYSDSFRKQVQGKA AFVLDTPEMRRVKESQLNISGVRYHQDHEKSKGSFT
AMASYPEMERVKKNNTQEFSDINYRGIQRRVVEMERRRVVEHEQETTTDLR
VWRTNPGSVFDYDPAEDNIQSRSLHMMSVHAQRRSKEHSRSTSAMSGPAD
EKSEMSHEGDHMHMSTYSNGFGASSVGYQTTKTVELQQRSSSVATQQTTVS
SIPSHPSTTGKTVRAMYDYGAADNDEVTFKDGDVIVNVQSIDEGWMYGTV

QRTGKTGMLPANYVEAV 230 Cod

MSTDAEMAVYGQAAYLRKPEKERLEAQSKPFDAKSAAYVTDVKELYLKC A0A8C5ARQ5
TILKKEGGKVTVKVHDTEEEKTVKDDDDVTPMNPYPKYDKIEDMAMMTHLN (Myosin
heavy EASVLYNLKERYAAWMIYTYSGLFCATVNPYKWLPVYDQHVVCAYRGKK chain,
fast skeletal RMEAPPHIFSVSDNAYQFMLTDRENQSVLITGESGAGKTVNTRKVIQYFATI
muscle-like) SVGGDKKKDIGAGKIQGSLEDQIIAANPLLEAYGNAKTVRNDNSSRFGKFIR
IHFGTSGKLSSADIETYLLKSRVTFQLPAERGYHIFYQMMTNHKPEIEMTL
ITTNPYDFPTCSMGQIAVASIDDKEELVATDTAIDILGFNNDEKMGIIYKLTG
AVLHHGNMFKKQKQREEQAEPEGTEADADKVAYLLGLNSADMLKGLCYPR
VKVGNEYVTKGQTVPPQVLNSVSALAKSIYERMFLWMVIRINQMLDTKQPR
SSFIGVLDIAGFEIFDYNSEQLCINFTEKLLQFFNHHMFVLEQEEYKKEGI
IWEFIDFGMDLAACIELIEKPMGIFSILEEECMFPKASDVTFKAKLYDQHLGK
TKAFEKPKPAKGKAEAHFSLVHYAGTVDYNITGWLDKNKDPLNDSVVQL
YQKSGVKLLPVLYPPVASEDSTKKGGGKKGGSMQTVSSQFRENLGKLMTN
LRSTHPHFVRCLIPNESKTPGLMENFLVIHQLRCNGVLEGIIRICRKGFPRIY
ADFKQRYKVLNASVIPEGQFIDNKKASEKLLGSIDVNHEEYKFGHTKVFFK
AGLLGTLEEMRDEKLATLVGMIQALCRGFQMRKEFKKMMERRHAVYTIQ
YNIRSFMNVKTPWPMKVYYKIKPLLKSAETEKELSAMKENYEKMKKDLA
TALAKKKEEEKMVSLLEKNDLSLQVASEGENLNDAEERCEGLIKSKIQL
EARLKETTERLEDEEEMNADLTAKKRKLEDECESELKKDIDDLTLAKVEK
EKHATENKVKNLTEEMASQDESVAKLSKEKKALQEAHQQTLDLQAEED
KVNTLSKAKTKLEQQVDDLEGSLEQEKKLMDLERSKRKLEGDLKLAQES
IMDLENDKQQSDEKIKKKDFETSQLLSKIEDEQSLAIQLQKKIKELQARIEEL
EEEIEAERAARAKVEKQRADLSRELEEISERLEEAGGATSAQIEMNKKREAE
FQKLRRDLEESTLQHEATAAALRKKQADSVAELGEQIDNLQRVKQKLEKE
KSEYKMEIDDLSSNMETVAKAKGNLEKICRTLEDQLSEIKAKSDENSQRIND
IGAQRARLLTENGEFGRQLEEKEALVSQLTRGKQAYTQQIEELKRQIEEEVK
AKNALAHGVQSARHDCDLLREQFEEEQEAKAELQRGMSKANGEVAQWRS
KYETDAIQRTTEELEESKKKLAQRLQEAEEQIEAVNSKASLEKTKQRLQGE
VEDLMVDVERANGLAANLDKKQRNFDKVLAEWKQKYEEGQAELEGTQK
EARSSTELFKMKNSYEEALDQLETLKRENKNLQQEISDLSEQIGETGKSIH
ELEKSKKQVETEKTEIQTALAEAEGTLEHEESKILRVQLELNQIKGEVDRKL
AEKDEEMEQIKRNSLRITDSMQSTLDSEVRSRNDALRIKKKMEGDLNEMEI
QLSHANRQAAEAQKQLRNVQGQLKDAQLHLDDAVRGGEDLKEQAAMVD
RRNGLMVAEIEELRAALEQTERGRKVAEQELIDASERVGLLHSQNTSLLNT
KKKLETDLVQVQGEVDDTVQEARNAEEKAKKAITDAAMMAEELKKEQDT
SSHLERMKKNLEVTVKDLQHRLDEAENLAMKGGKKQLQKLESRVRELETE
VEGEQRRGVDAVKGVRYERRVKELTYQTEEDKKNGARLQDLVDKLQLK
VKAYKRQAEAEQANSYLSKCRKVQHELEEAERADIAETQVKNLRSKS
RDGGKGKDPAE 231 Cod Q78AY8

MCDDEETALVCDNGSGLVKAGFAGDDAPRAVFPISVGRPRHQGVMMVGM (Actin alpha
1) GQKDSYVGDEAQSKRGILTLKYPIEHGIITNWDDMEKIWHHTFYNELRVAP
EEHPTLLTEAPLNPKANREKMTQIMFETFNVPAMYVAIQAVLSLYASGRTT
GIVLDSGDGVTHNVPVYEGYALPHAIMRLDLAGRDLTDYLMKILTERGYSF
VTTAEREIVRDIKEKLCYVALDFENEMATAASSSSLEKSYELPDGQVITIGN
ERFRCPETLFQPSFIGMESAGIHETAYNSIMKCDIDIRKDLYANNVLSGGTT
MYPGIADRMQKEITALAPSTMKIKIIPPERKYSVWIGGSILASLSTFQQMWI
SKQEYDEAGPSIVHRKCF

[0061] An animal protein can comprise a tuna protein (Table 14). A tuna protein can comprise a tuna muscle protein. A tuna protein can comprise a tuna blood protein. A tuna protein can comprise

a tuna albumin protein. A tuna protein can comprise a sequence that has at least or about 70%, at least or about 75%, at least or about 80%, at least or about 90%, at least or about 91%, at least or about 92%, at least or about 93%, at least or about 94%, at least or about 95%, at least or about 96%, at least or about 97% at least or about 98% at least or about 99% or about 100% sequence identity to SEQ ID NO: 232.

[0062] A food product may comprise a fragment or a truncation of a tuna protein. The food product may include a fragment or truncation of any one of SEQ ID Nos: 232. A sequence of the fragment or truncation may be identical or substantially identical to a portion of any one of SEQ ID Nos: 232. A sequence of the fragment or truncation may be at least about 70%, 80%, 90%, 95%, 98%, or more identical to a portion of a sequence of any one of SEQ ID Nos: 232. A length of the fragment or truncation that is at least about 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or more of a length of any one of SEQ ID Nos: 232. A length of the fragment or truncation may be less than or equal to about 90%, 80%, 70%, 60%, 50%, 40%, 30%, 20%, 15%, 10%, or less than a length of any one of SEQ ID Nos: 232.

TABLE-US-00014 TABLE 14 Tuna Proteins SEQ ID NO Description Amino Acid Sequence 232 Tuna MCDDDETTALVCDNGSGLVKAGFAGDDA XP_042264497.1 PRAVFPSIVGRPRHQGVGMVGMGQKDSYV (actin alpha 1, GDEAQSQRGILTLKYPIEHGIITNWDDM skeletal EKIWHHTFYNELRVAPEEHPTLLTEAPL muscle b) NPKANREKMTQIMFETFNVPAMYVAIQA VLSLYASGRTTGIVLDAGDGVTHNVPVY EGYALPHAIMRLDLAGRDLTDYLMKILT ERGYSFVTTAEREIVRDIKEKLCYVALD FENEMATAASSSSLEKSYELPDGQVITI GNERFRCPETLRFQPSFIGMESAGIHETA YNSIMKCDIDIRKDLYANNVLSGGTTMY PGIADRMQKEITALAPSTMKIKIIPPE RKYSVWIGGSILASLSTFQQMWISKQEY DEAGPSIVHRKCF

[0063] An additional gene that is not a gene of interest can be a gene that modulates protein expression. A gene that modulates protein expression can enhance protein expression, decrease protein expression, stop protein expression, or maintain protein expression.

[0064] A gene that modulates protein expression can increase protein expression by at least about 5%, at least about 10%, at least about 20%, at least about 30%, at least about 40%, at least about 50%, at least about 60%, at least about 70%, at least about 80%, at least about 90%, at least about 100%, at least about 150%, at least about 200%, at least about 300%, at least about 400%, at least about 500%, at least about 600%, at least about 700%, at least about 800%, at least about 900% at least about 1,000%, at least about 5,000%, at least about 10,000% or more. A gene that modulates protein expression can increase protein expression by at most about 10,000%, at most about 5,000%, at most about 1,000%, at most about 900%, at most about 800%, at most about 700%, at most about 600%, at most about 500%, at most about 400%, at most about 300%, at most about 200%, at most about 150%, at most about 100%, at most about 90%, at most about 80%, at most about 70%, at most about 60%, at most about 50%, at most about 40%, at most about 30%, at most about 20%, at most about 10%, at most about 5%, or less.

[0065] A gene that modulates protein expression can increase protein expression by at least or up to about 1-fold, at least or up to about 2-fold, at least or up to about 3-fold, at least or up to about 4-fold, at least or up to about 5-fold, at least or up to about 6-fold, at least or up to about 7-fold, at least or up to about 8-fold, at least or up to about 9-fold, at least or up to about 10-fold, at least or up to about 20-fold, at least or up to about 30-fold, at least or up to about 40-fold, at least or up to about 50-fold, at least or up to about 60-fold, at least or up to about 70-fold, at least or up to about 80-fold, at least or up to about 90-fold, at least or up to about 100-fold, at least or up to about 500-fold, at least or up to about 1,000-fold, at least or up to about 5,000-fold, or at least or up to about 10,000-fold. A gene that modulates protein expression can increase protein expression by at most or less than about 10,000-fold, at most or less than about 5,000-fold, at most or less than about 1,000-fold, at most or less than about 500-fold, at most or less than about 100-fold, at most or less than

about 90-fold, at most or less than about 80-fold, at most or less than about 70-fold, at most or less than about 60-fold, at most or less than about 50-fold, at most or less than about 40-fold, at most or less than about 30-fold, at most or less than about 20-fold, at most or less than about 10-fold, at most or less than about 9-fold, at most or less than about 8-fold, at most or less than about 7-fold, at most or less than about 6-fold, at most or less than about 5-fold, at most or less than about 4-fold, at most or less than about 3-fold, at most or less than about 2-fold, or at most or less than about 1-fold.

[0066] A gene that modulates protein expression can decrease protein expression by at least about 5%, at least about 10%, at least about 20%, at least about 30%, at least about 40%, at least about 50%, at least about 60%, at least about 70%, at least about 80%, at least about 90%, at least about 100%, at least about 150%, at least about 200%, at least about 300%, at least about 400%, at least about 500%, at least about 600%, at least about 700%, at least about 800%, at least about 900% at least about 1,000%, at least about 5,000%, at least about 10,000% or more. A gene that modulates protein expression can decrease protein expression by at most about 10,000%, at most about 5,000%, at most about 1,000%, at most about 900%, at most about 800%, at most about 700%, at most about 600%, at most about 500%, at most about 400%, at most about 300%, at most about 200%, at most about 150%, at most about 100%, at most about 90%, at most about 80%, at most about 70%, at most about 60%, at most about 50%, at most about 40%, at most about 30%, at most about 20%, at most about 10%, at most about 5%, or less.

[0067] A gene that modulates protein expression can decrease protein expression by at least or up to about 1-fold, at least or up to about 2-fold, at least or up to about 3-fold, at least or up to about 4-fold, at least or up to about 5-fold, at least or up to about 6-fold, at least or up to about 7-fold, at least or up to about 8-fold, at least or up to about 9-fold, at least or up to about 10-fold, at least or up to about 20-fold, at least or up to about 30-fold, at least or up to about 40-fold, at least or up to about 50-fold, at least or up to about 60-fold, at least or up to about 70-fold, at least or up to about 80-fold, at least or up to about 90-fold, at least or up to about 100-fold, at least or up to about 500-fold, at least or up to about 1,000-fold, at least or up to about 5,000-fold, or at least or up to about 10,000-fold. A gene that modulates protein expression can decrease protein expression by at most or less than about 10,000-fold, at most or less than about 5,000-fold, at most or less than about 1,000-fold, at most or less than about 500-fold, at most or less than about 100-fold, at most or less than about 90-fold, at most or less than about 80-fold, at most or less than about 70-fold, at most or less than about 60-fold, at most or less than about 50-fold, at most or less than about 40-fold, at most or less than about 30-fold, at most or less than about 20-fold, at most or less than about 10-fold, at most or less than about 9-fold, at most or less than about 8-fold, at most or less than about 7-fold, at most or less than about 6-fold, at most or less than about 5-fold, at most or less than about 4-fold, at most or less than about 3-fold, at most or less than about 2-fold, or at most or less than about 1-fold.

[0068] An additional gene of interest can be a gene that modulates protein secretion. A gene that modulates protein secretion can enhance protein secretion, decrease protein secretion, stop protein secretion, or maintain protein secretion.

[0069] A gene that modulates protein secretion can increase protein secretion by at least about 5%, at least about 10%, at least about 20%, at least about 30%, at least about 40%, at least about 50%, at least about 60%, at least about 70%, at least about 80%, at least about 90%, at least about 100%, at least about 150%, at least about 200%, at least about 300%, at least about 400%, at least about 500%, at least about 600%, at least about 700%, at least about 800%, at least about 900% at least about 1,000%, at least about 5,000%, at least about 10,000% or more. A gene that modulates protein secretion can increase protein secretion by at most about 10,000%, at most about 5,000%, at most about 1,000%, at most about 900%, at most about 800%, at most about 700%, at most about 600%, at most about 500%, at most about 400%, at most about 300%, at most about 200%, at most about 150%, at most about 100%, at most about 90%, at most about 80%, at most about 70%,

[0073] An expression cassette can comprise at least one regulatory sequence. A regulatory sequence can comprise a promoter, a repressor, an enhancer, a terminator, a 5' UTR, a 3' UTR or a combination thereof. A regulatory sequence can be part of a coding sequence. Alternatively, a regulatory sequence can be a non-coding sequence. An expression cassette can comprise at least 1, at least 2, at least 3, at least 4, at least 5, at least 6, at least 7, at least 8, at least 9, at least 10 or more regulatory sequences.

[0074] A promoter can be an endogenous *P. pastoris* promoter or variant thereof. Alternatively, a promoter can be a yeast promoter from a different species (e.g., *S. cerevisiae*) or variant thereof. Alternatively, a promoter can be a synthetic or novel promoter. A promoter can be an AOX1 promoter, an ADH3 promoter, a TEF1 promoter, a PGK1 promoter, a TPI1 promoter, a PHO89 promoter, a THI11 promoter, an AOD promoter, or a GAP promoter.

[0075] An enhancer can be an endogenous *P. pastoris* enhancer or variant thereof. Alternatively, an enhancer can be a yeast enhancer from a different species (e.g., *S. cerevisiae*) or variant thereof. Alternatively, an enhancer can be a synthetic or novel enhancer. An enhancer can be an AOX1 enhancer or a GAP enhancer.

[0076] A repressor can be an endogenous *P. pastoris* repressor or variant thereof. Alternatively, a repressor can be a yeast repressor from a different species (e.g., *S. cerevisiae*) or variant thereof. Alternatively, a repressor can be a synthetic or novel repressor. A repressor can be an AOX1 repressor or a GAP repressor.

[0077] A terminator can be an endogenous *P. pastoris* terminator or variant thereof. Alternatively, a terminator can be a yeast terminator from a different species (e.g., *S. cerevisiae*) or variant thereof. Alternatively, a terminator can be a synthetic or novel terminator. A terminator can be an AOX1 terminator, a RPL15A terminator, a RPL41B terminator, a DAS1 terminator, a VMA2 terminator, a SRP54 terminator, a FUR4 terminator, a PLB3 terminator, a GIC1 terminator, a FBP1 terminator, a CYC1 terminator, a IDP1 terminator, a PRC1 terminator, a RPL3 terminator, a PGK1 terminator, a YPT31 terminator, an ASP3-1 terminator, a TPO4 terminator, a RPS28A terminator, a VMA16 terminator, a LSC2 terminator, a DIT1 terminator, a MIG2 terminator, an ADH2 terminator, an ADH3 terminator, an ICY2 terminator, a PHO89 terminator, a THI11 terminator, an AOD terminator, or a GAP terminator.

[0078] A UTR can be a 5' UTR or a 3' UTR. A UTR can be an endogenous *P. pastoris* UTR or variant thereof. Alternatively, a UTR can be a yeast UTR from a different species (e.g., *S. cerevisiae*) or variant thereof. Alternatively, a UTR can be a synthetic or novel UTR. A UTR can be an AOX1 UTR, a RPL15A UTR, a RPL41B UTR, a DAS1 UTR, a VMA2 UTR, a SRP54 UTR, a FUR4 UTR, a PLB3 UTR, a GIC1 UTR, a FBP1 UTR, a CYC1 UTR, an IDP1 UTR, a PRC1 UTR, a RPL3 UTR, a PGK1 UTR, a YPT31 UTR, an ASP3-1 UTR, a TPO4 UTR, a RPS28A UTR, a VMA16 UTR, a LSC2 UTR, a DIT1 UTR, a MIG2 UTR, an ADH2 UTR, an ADH3 UTR, a TEF1 UTR, a PGK1 UTR, a TPI1 UTR, a PHO89 UTR, a THI11 UTR, an AOD UTR, or a GAP UTR.

[0079] An expression cassette can comprise a signal sequence. A signal sequence (e.g., a secretion sequence or a nuclear localization sequence) can aid with and/or induce protein secretion. An expression cassette can comprise at least 1, at least 2, at least 3, at least 4, at least 5, at least 6, at least 7, at least 8, at least 9, at least 10 or more signal sequences.

[0080] A signal sequence can be an endogenous *P. pastoris* signal sequence or variant thereof. Alternatively, a signal sequence can be a yeast signal sequence from a different species (e.g., *S. cerevisiae*) or variant thereof. Alternatively, a signal sequence can be a synthetic or novel promoter. A signal sequence can be an amylase signal sequence, an inulinase signal sequence, an invertase signal sequence, a lysozyme signal sequence, an albumin signal sequence, a glucoamylase signal sequence, or a killer protein signal sequence, a *S. cerevisiae* Ost1 signal sequence, a *S. cerevisiae* alpha mating factor, a partial *S. cerevisiae* alpha mating factor, or a *Pichia* Exg1p signal sequence.

[0081] A regulatory sequence can modulate protein yield. A regulatory sequence can enhance

500-fold, at least or up to about 1,000-fold, at least or up to about 5,000-fold, or at least or up to about 10,000-fold. A regulatory sequence that modulates protein yield can decrease protein yield by at most or less than about 10,000-fold, at most or less than about 5,000-fold, at most or less than about 1,000-fold, at most or less than about 500-fold, at most or less than about 100-fold, at most or less than about 90-fold, at most or less than about 80-fold, at most or less than about 70-fold, at most or less than about 60-fold, at most or less than about 50-fold, at most or less than about 40-fold, at most or less than about 30-fold, at most or less than about 20-fold, at most or less than about 10-fold, at most or less than about 9-fold, at most or less than about 8-fold, at most or less than about 7-fold, at most or less than about 6-fold, at most or less than about 5-fold, at most or less than about 4-fold, at most or less than about 3-fold, at most or less than about 2-fold, or at most or less than about 1-fold.

[0086] A regulatory sequence can modulate protein secretion. A regulatory sequence that modulates protein secretion can enhance protein secretion, decrease protein secretion, stop protein secretion, or maintain protein secretion.

[0087] A regulatory sequence that modulates protein secretion can increase protein secretion by at least about 5%, at least about 10%, at least about 20%, at least about 30%, at least about 40%, at least about 50%, at least about 60%, at least about 70%, at least about 80%, at least about 90%, at least about 100%, at least about 150%, at least about 200%, at least about 300%, at least about 400%, at least about 500%, at least about 600%, at least about 700%, at least about 800%, at least about 900% at least about 1,000%, at least about 5,000%, at least about 10,000% or more. A regulatory sequence that modulates protein secretion can increase protein secretion by at most about 10,000%, at most about 5,000%, at most about 1,000%, at most about 900%, at most about 800%, at most about 700%, at most about 600%, at most about 500%, at most about 400%, at most about 300%, at most about 200%, at most about 150%, at most about 100%, at most about 90%, at most about 80%, at most about 70%, at most about 60%, at most about 50%, at most about 40%, at most about 30%, at most about 20%, at most about 10%, at most about 5%, or less.

[0088] A regulatory sequence that modulates protein secretion can increase protein secretion by at least or up to about 1-fold, at least or up to about 2-fold, at least or up to about 3-fold, at least or up to about 4-fold, at least or up to about 5-fold, at least or up to about 6-fold, at least or up to about 7-fold, at least or up to about 8-fold, at least or up to about 9-fold, at least or up to about 10-fold, at least or up to about 20-fold, at least or up to about 30-fold, at least or up to about 40-fold, at least or up to about 50-fold, at least or up to about 60-fold, at least or up to about 70-fold, at least or up to about 80-fold, at least or up to about 90-fold, at least or up to about 100-fold, at least or up to about 500-fold, at least or up to about 1,000-fold, at least or up to about 5,000-fold, or at least or up to about 10,000-fold. A regulatory sequence that modulates protein secretion can increase protein secretion by at most or less than about 10,000-fold, at most or less than about 5,000-fold, at most or less than about 1,000-fold, at most or less than about 500-fold, at most or less than about 100-fold, at most or less than about 90-fold, at most or less than about 80-fold, at most or less than about 70-fold, at most or less than about 60-fold, at most or less than about 50-fold, at most or less than about 40-fold, at most or less than about 30-fold, at most or less than about 20-fold, at most or less than about 10-fold, at most or less than about 9-fold, at most or less than about 8-fold, at most or less than about 7-fold, at most or less than about 6-fold, at most or less than about 5-fold, at most or less than about 4-fold, at most or less than about 3-fold, at most or less than about 2-fold, or at most or less than about 1-fold.

[0089] A regulatory sequence that modulates protein secretion can decrease protein secretion by at least about 5%, at least about 10%, at least about 20%, at least about 30%, at least about 40%, at least about 50%, at least about 60%, at least about 70%, at least about 80%, at least about 90%, at least about 100%, at least about 150%, at least about 200%, at least about 300%, at least about 400%, at least about 500%, at least about 600%, at least about 700%, at least about 800%, at least about 900% at least about 1,000%, at least about 5,000%, at least about 10,000% or more. A

regulatory sequence that modulates protein secretion can decrease protein secretion by at most about 10,000%, at most about 5,000%, at most about 1,000%, at most about 900%, at most about 800%, at most about 700%, at most about 600%, at most about 500%, at most about 400%, at most about 300%, at most about 200%, at most about 150%, at most about 100%, at most about 90%, at most about 80%, at most about 70%, at most about 60%, at most about 50%, at most about 40%, at most about 30%, at most about 20%, at most about 10%, at most about 5%, or less.

[0090] A regulatory sequence that modulates protein secretion can decrease protein secretion by at least or up to about 1-fold, at least or up to about 2-fold, at least or up to about 3-fold, at least or up to about 4-fold, at least or up to about 5-fold, at least or up to about 6-fold, at least or up to about 7-fold, at least or up to about 8-fold, at least or up to about 9-fold, at least or up to about 10-fold, at least or up to about 20-fold, at least or up to about 30-fold, at least or up to about 40-fold, at least or up to about 50-fold, at least or up to about 60-fold, at least or up to about 70-fold, at least or up to about 80-fold, at least or up to about 90-fold, at least or up to about 100-fold, at least or up to about 500-fold, at least or up to about 1,000-fold, at least or up to about 5,000-fold, or at least or up to about 10,000-fold. A regulatory sequence that modulates protein secretion can decrease protein secretion by at most or less than about 10,000-fold, at most or less than about 5,000-fold, at most or less than about 1,000-fold, at most or less than about 500-fold, at most or less than about 100-fold, at most or less than about 90-fold, at most or less than about 80-fold, at most or less than about 70-fold, at most or less than about 60-fold, at most or less than about 50-fold, at most or less than about 40-fold, at most or less than about 30-fold, at most or less than about 20-fold, at most or less than about 10-fold, at most or less than about 9-fold, at most or less than about 8-fold, at most or less than about 7-fold, at most or less than about 6-fold, at most or less than about 5-fold, at most or less than about 4-fold, at most or less than about 3-fold, at most or less than about 2-fold, or at most or less than about 1-fold.

[0091] A food product or substance can be comprised of the recombinant proteins described herein. The food product or substance may comprise recombinant proteins described herein provided in an isolated or purified form. Alternatively, or in addition to, the recombinant proteins described herein may be provided to the food product or substance as whole cells comprising the recombinant proteins. A food product or substance can be a food for animals. A food product or substance can be food for a pet (e.g., companion animal), a farm or agricultural animal, a wild animal, or a human. A food product or substance can be food for an obligate carnivore (e.g., a feline), which requires animal protein as a part of the diet.

[0092] A food product can be a wet puree. Alternatively, a food product can be a dry pellet. A food product can replace an animal or human diet. Alternatively, a food product can supplement an animal or human diet.

[0093] A food product can contain additional ingredients such as, but not limited to, animal-based meat, amino acid supplements (e.g., taurine), vitamins (e.g., vitamin E), fats, nutritional yeast, *spirulina*, agar, xanthan gum, kelp powder, squash powder, sweet potato powder, *papaya* powder, flaxseed oil, coconut oil, probiotics, cranberry powder, wheat grass powder, turmeric, coconut flour, chitosan, pumpkin puree, inulin powder, guar gum, green tea extract, koji, stabilizers, vegetables, fruits, microbes, fillers, or flavoring compounds.

EXAMPLES

Example 1: Recombinant Protein Expression

[0094] In this experiment, a *Pichia pastoris* cell was transfected with a DNA sequence, resulting in recombinant protein expression.

[0095] Host cells were chosen based on predetermined characteristics such as, but not limited to, doubling time, ability to properly fold and make appropriate post-translational modifications on recombinant proteins, protein secretion, ability to grow on cost-effective and/or easily available media/feedstock, or genetic tractability. Host cells that do not efficiently secrete proteins may be lysed to release the target recombinant proteins.

[0096] *Pichia* cells were grown overnight in YPD media in baffled flasks at 30° C. to an optical density at 600 nm wavelength (OD 600) of about 1 to 2.6. Cells were collected by centrifugation and resuspended at room temperature for 30 min in 100 mM lithium acetate (LiOAc), 10 mM dithiothreitol (DTT), 0.6M sorbitol, and 10 mM Tris-HCl pH 7.5, at 1e8 cells/ml. Cells were pelleted and washed four times in 1 mL 1M sorbitol. Cells were resuspended in 1M sorbitol at a final concentration of 1e9 cells/ml. 40 µL of cell suspension was mixed with 1 uL (100-1 ug) of DNA, transferred to a 0.2 cm gap cuvette, and incubated on ice for 5 minutes.

[0097] The expression cassette used comprised a promoter, a 5' untranslated region, a signal sequence for protein secretion, an open reading frame (ORF) of an animal muscle protein, a 3' UTR, and a transcriptional termination and 3' end processing sequence. The construct used a *P. pastoris* AOX1 promoter, a *P. pastoris* 5' UTR, a *P. pastoris* 3' UTR, a *S. cerevisiae* alpha mating factor, a rabbit serum albumin ORF, and a *P. pastoris* terminator.

[0098] The expression cassette was transfected into the *P. pastoris* host cell using electroporation at 1.5 kV, 200 ohms, and 25 uF, where the expression cassette was integrated into the genome of *P. pastoris*. Electroporated cells were diluted in 1 mL 1M sorbitol. Cells were transferred to 15 ml round-bottom tubes with 1 ml YPD media and incubated while shaking at 30° C. for 1 hour.

[0099] Cells were plated on YPDS plates and incubated at 30° C. Once the cells' endogenous protein expression machinery expressed the protein of interest, the cells secreted the protein of interest into the media, which was harvested and purified for further use.

Example 2: Analysis of Secreted Proteins

[0100] In this experiment, genes of interest were cloned into plasmids containing the GAP1 promotor, *Saccharomyces cerevisiae* alpha-Mating factor signal sequence, and a zeocin resistance gene. Plasmids were linearized and transformed into *Pichia pastoris* cells using electroporation. Transformed cells were spread onto YPDS plates containing 100 micrograms per milliliter (ug/mL) zeocin and incubated at 30° C. for 3-4 days. Colonies were restruck to single colony twice. To produce secreted protein, cells were resuspended in 4 milliliters (mL) of buffered complex glycerol medium (BMGY) (rich glycerol) and cultured shaking at 250 revolutions per minute (rpm) and 30° C. for 4 days. Every 24 hours, 200 microliters (uL) of culture was removed and centrifuged to separate cells from supernatant. A 10 µL aliquot of supernatant was used to analyze secreted proteins on a sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE) gel.

[0101] FIG. 1 shows a Coomassie stained SDS-PAGE gel of an example media from *Pichia pastoris* cells including a construct for secreted recombinant rabbit albumin or wildtype cells. Wildtype cells lack the construct for secreted recombinant rabbit albumin. Lane 1 **101** shows a band at 66 kilo Daltons (kDa), which corresponds to the recombinant rabbit albumin. Lane 2 **102**, which correspond to the wildtype media, lacks a band at 66 kDa. Lane 3 **103** shows a bovine serum albumin marker with a band at 66 kDa.

[0102] FIG. 2 shows a Coomassie stained sodium SDS-PAGE gel of an example media from *Pichia pastoris* cells including a construct for secreted recombinant rabbit Tropomyosin alpha-1 chain or wildtype cells. The wildtype cells lack the construct. Lane 2 **201** shows the wildtype expressed protein from wildtype cells. Lanes 3-5 **202** show secreted protein expression for Tropomyosin alpha-1 chain. Lanes 3-5 show a band **203** at 33 kDa corresponding to the Tropomyosin alpha-1 chain.

[0103] For proteins produced in a bioreactor, cells were resuspended in YPG medium and grown in shake flasks at 30° C. for 16 hours. The cells were transferred to seed fermenters and grown at 30° C. for another 16 hours. The cultures were transferred to a 100 liter (L) fermenter and batch fed a glycerol-based media for 92 hours at 30° C. The culture was then centrifuged to separate solid and liquid phases, filtered, and spray dried to produce the final protein powder.

[0104] FIG. 3 shows a Coomassie stained SDS-PAGE gel of an example recombinant rabbit albumin. Lane 1 **301** shows an example recombinant rabbit albumin (rALBU-RABIT) grown from a bioreactor. The rabbit albumin peptides were purified and spray dried. The protein powder was

resuspended and run on an SDS-PAGE gel. Lane 2 **302** shows a bovine serum albumin marker.

Example 3: Cat Food with Rabbit Proteins

[0105] In this example, cat food is made using rabbit proteins. Rabbits are naturally hunted by felines in the wild, and rabbit proteins are hypoallergenic for cats. A recombinant rabbit protein is produced and used as an ingredient in both wet and dry cat food, snacks, and/or treats. In some formulations, the protein is in dried powder form. In some formulations, the protein is in a solution. For dry food, ingredients are mixed and extruded.

[0106] Twenty-five rabbit proteins are produced using the recombinant protein expression method of Example 1. The twenty-five rabbit proteins tested are found in Table 2.

[0107] The rabbit proteins are harvested from the recombinant protein expression method and produced into a wet pureed cat treat. Additionally, the rabbit proteins are harvested from the recombinant protein expression method and produced into a dry extruded cat food kibble.

Example 4: Pet Food with Cow Proteins

[0108] In this example, pet food is made using cow proteins. A recombinant cow protein is produced and used as an ingredient in both wet and dry pet food, snacks, and/or treats. In some formulations, the protein is in dried powder form. In some formulations, the protein is in a solution. For dry food, ingredients are mixed and extruded.

[0109] The cow proteins are produced using the recombinant protein expression method of Example 1. The cow proteins tested are found in Table 3.

Example 5: Pet Food with Sheep Proteins

[0110] In this example, pet food is made using sheep proteins. A recombinant sheep protein is produced and used as an ingredient in both wet and dry pet food, snacks, and/or treats. In some formulations, the protein is in dried powder form. In some formulations, the protein is in a solution. For dry food, ingredients are mixed and extruded.

[0111] The sheep proteins are produced using the recombinant protein expression method of Example 1. The sheep proteins tested are found in Table 4.

Example 6: Pet Food with Pig Proteins

[0112] In this example, pet food is made using pig proteins. A recombinant pig protein is produced and used as an ingredient in both wet and dry pet food, snacks, and/or treats. In some formulations, the protein is in dried powder form. In some formulations, the protein is in a solution. For dry food, ingredients are mixed and extruded.

[0113] The pig proteins are produced using the recombinant protein expression method of Example 1. The pig proteins tested are found in Table 5.

Example 7: Pet Food with Bison Proteins

[0114] In this example, pet food is made using bison proteins. A recombinant bison protein is produced and used as an ingredient in both wet and dry cat food, snacks, and/or treats. In some formulations, the protein is in dried powder form. In some formulations, the protein is in a solution. For dry food, ingredients are mixed and extruded.

[0115] The bison proteins are produced using the recombinant protein expression method of Example 1. The bison proteins tested are found in Table 6.

Example 8: Pet Food with Deer Proteins

[0116] In this example, pet food is made using deer proteins. A recombinant deer protein is produced and used as an ingredient in both wet and dry pet food, snacks, and/or treats. In some formulations, the protein is in dried powder form. In some formulations, the protein is in a solution. For dry food, ingredients are mixed and extruded.

[0117] The deer proteins are produced using the recombinant protein expression method of Example 1. The deer proteins tested are found in Table 7.

Example 9: Pet Food with Chicken Proteins

[0118] In this example, pet food is made using chicken proteins. A recombinant chicken protein is produced and used as an ingredient in both wet and dry pet food, snacks, and/or treats. In some

formulations, the protein is in dried powder form. In some formulations, the protein is in a solution. For dry food, ingredients are mixed and extruded.

[0119] The chicken proteins are produced using the recombinant protein expression method of Example 1. The chicken proteins tested is found in Table 8.

Example 10: Pet Food with Turkey Proteins

[0120] In this example, cat food is made using turkey proteins. A recombinant turkey protein is produced and used as an ingredient in both wet and dry pet food, snacks, and/or treats. In some formulations, the protein is in dried powder form. In some formulations, the protein is in a solution. For dry food, ingredients are mixed and extruded.

[0121] The turkey proteins are produced using the recombinant protein expression method of Example 1. The turkey proteins tested are found in Table 9.

Example 11: Cat Food with Duck Proteins

[0122] In this example, cat food is made using duck proteins. Ducks are naturally hunted by felines in the wild, and duck proteins are hypoallergenic for cats. A recombinant duck protein is produced and used as an ingredient in both wet and dry cat food, snacks, and/or treats. In some formulations, the protein is in dried powder form. In some formulations, the protein is in a solution. For dry food, ingredients are mixed and extruded.

[0123] The duck proteins are produced using the recombinant protein expression method of Example 1. The duck proteins tested are found in Table 10 and Table 11.

Example 12: Pet Food with Salmon Proteins

[0124] In this example, pet food is made using salmon proteins. A recombinant salmon protein is produced and used as an ingredient in both wet and dry pet food, snacks, and/or treats. In some formulations, the protein is in dried powder form. In some formulations, the protein is in a solution. For dry food, ingredients are mixed and extruded.

[0125] The salmon proteins are produced using the recombinant protein expression method of Example 1. The salmon proteins tested are found in Table 12.

Example 13: Pet Food with Cod Proteins

[0126] In this example, pet food is made using cod proteins. A recombinant cod protein is produced and used as an ingredient in both wet and dry pet food, snacks, and/or treats. In some formulations, the protein is in dried powder form. In some formulations, the protein is in a solution. For dry food, ingredients are mixed and extruded.

[0127] The cod proteins are produced using the recombinant protein expression method of Example 1. The cod proteins tested are found in Table 13.

Example 14: Pet Food with Tuna Proteins

[0128] In this example, pet food is made using tuna proteins. A recombinant tuna protein is produced and used as an ingredient in both wet and dry pet food, snacks, and/or treats. In some formulations, the protein is in dried powder form. In some formulations, the protein is in a solution. For dry food, ingredients are mixed and extruded.

[0129] The tuna proteins are produced using the recombinant protein expression method of Example 1. The tuna protein tested is found in Table 14.

Example 15: Food Product for Human Consumption

[0130] In this example, a food product for human consumption is made using any of the proteins described in Tables 2-14. A recombinant protein from Tables 2-14 is produced and provided in a food, beverage, or supplement. In some formulations, the protein is in dried powder form. In some formulations, the protein is in a solution.

[0131] While preferred embodiments of the present invention have been shown and described herein, it will be obvious to those skilled in the art that such embodiments are provided by way of example only. It is not intended that the invention be limited by the specific examples provided within the specification. While the invention has been described with reference to the aforementioned specification, the descriptions and illustrations of the embodiments herein are not

meant to be construed in a limiting sense. Numerous variations, changes, and substitutions will now occur to those skilled in the art without departing from the invention. Furthermore, it shall be understood that all aspects of the invention are not limited to the specific depictions, configurations or relative proportions set forth herein which depend upon a variety of conditions and variables. It should be understood that various alternatives to the embodiments of the invention described herein may be employed in practicing the invention. It is therefore contemplated that the invention shall also cover any such alternatives, modifications, variations or equivalents. It is intended that the following claims define the scope of the invention and that methods and structures within the scope of these claims and their equivalents be covered thereby.

Claims

1.-41. (canceled)

42. A method for producing a pet food product, the method comprising: (a) contacting a plurality of host cells with a recombinant nucleic acid, wherein the recombinant nucleic acid encodes an albumin protein of an animal selected from the group consisting of a mammal, a bird, and a fish; (b) expressing the albumin protein encoded by the recombinant nucleic acid in the plurality of host cells; and (c) generating a pet food product comprising the albumin protein expressed in the plurality of host cells.

43. The method of claim 42, wherein the albumin protein is a rabbit albumin protein.

44. The method of claim 42, wherein host cells of the plurality of host cells comprise one or more members selected from the group consisting of *Pichia* cells, *Saccharomyces* cells, *Schizosaccharomyces* cells, *Trichoderma* cells, *Aspergillus* cells, *Escherichia* cells, *Bacillus* cells, and *Lactococcus* cells.

45. The method of claim 42, wherein the albumin protein comprises an amino acid sequence that is at least 70% identical to any one of SEQ ID NOs: 16, 78, 93, 108, 123, 138, 161, 174, 187, 200, 215, or 216.

46. The method of claim 45, wherein the amino acid sequence is at least 70% identical to SEQ ID NO 78.

47. The method of claim 45, wherein the amino acid sequence is at least 90% identical to any one of SEQ ID NOs: 16, 78, 123, 161, 174, 187, 200, 215, or 216.

48. The method of claim 45, wherein the amino acid sequence is at least 95% identical to any one of SEQ ID NOs: 16, 78, 108, 123, 161, 174, 187, 200, 215, or 216.

49. The method of claim 42, further comprising isolating the albumin protein from the plurality of host cells.

50. A method for producing a pet food product, the method comprising: (a) contacting a plurality of host cells with a first recombinant nucleic acid encoding an albumin protein and a second recombinant nucleic acid that encodes a protein different from the albumin protein; (b) expressing the albumin protein encoded by the first recombinant nucleic acid and the protein encoded by the second recombinant nucleic acid in the plurality of host cells; and (c) generating a pet food product comprising the albumin protein and the protein expressed in the plurality of host cells.

51. The method of claim 50, further comprising isolating the albumin protein and the protein from the plurality of host cells.

52. The method of claim 50, wherein the albumin protein or the protein is a rabbit protein.

53. The method of claim 52, wherein the albumin protein and the protein are rabbit proteins.

54. The method of claim 50, wherein the albumin protein comprises an amino acid sequence that is at least 70% identical to any one of SEQ ID NOs: 16, 78, 93, 108, 123, 138, 161, 174, 187, 200, 215, or 216.

55. The method of claim 54, wherein the amino acid sequence is at least 70% identical to SEQ ID NO 78.

- 56.** The method of claim 50, wherein the protein comprises an amino acid sequence that is at least 70% identical to any one of SEQ ID NOs: 1-15, 17-77, 79-92, 94-107, 109-122, 124-137, 139-160, 162-173, 175-186, 188-199, 201-204, or 217-232.
- 57.** The method of claim 56, wherein the amino acid sequence is at least 70% identical to any one of SEQ ID NOs: 4, 54, 57, 58, 59, 60, 63, 64, 67, 74, or 77.
- 58.** A method for producing a pet food product, the method comprising: (a) contacting a plurality of host cells with a recombinant nucleic acid; (b) expressing a protein encoded by the recombinant nucleic acid; and (c) generating a pet food product comprising the protein expressed in the plurality of host cells, wherein the protein comprises an amino acid sequence that is at least 70% identical to any one of SEQ ID NOs: 5, 8, 9, 11, 14, 16, 19-21, 23, 24, 26-29, 32-35, 37-41, 44-49, 51-54, 56, 57, 59, 65-79, 81, 82, 84, 93, 94, 96, 97, 99, 108, 109, 111, 112, 114, 117, 123, 124, 126, 127, 129, 138-141, 143, 147, 148, 150, 151, 153, 161, 162, 164, 165, 167, 169, 170, 172, 174, 175, 177, 178, 180, 184, 185, 187, 188, 190-192, 197, 200, 201, 203-206, 208, 209, 213, 215, 216, 217, 219-222, 224, 225, or 229.
- 59.** The method of claim 58, wherein the protein is a rabbit protein.
- 60.** The method of claim 58, wherein the protein is an albumin protein.
- 61.** The method of claim 58, wherein the amino acid sequence is any one of SEQ ID NOs: 5, 8, 9, 11, 14, 16, 19-21, 23, 24, 26-29, 32-35, 37-41, 44-49, 51-54, 56, 57, 59, 65-79, 81, 82, 84, 93, 94, 96, 97, 99, 108, 109, 111, 112, 114, 117, 123, 124, 126, 127, 129, 138-141, 143, 147, 148, 150, 151, 153, 161, 162, 164, 165, 167, 169, 170, 172, 174, 175, 177, 178, 180, 184, 185, 187, 188, 190-192, 197, 200, 201, 203-206, 208, 209, 213, 215, 216, 217, 219-222, 224, 225, or 229.
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